
SERIAL AUTOCORRELATION AND TRENDS

IN LINEAR DERIVATIVES

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JUNE 2026

Thesis defense on September 2, 2026

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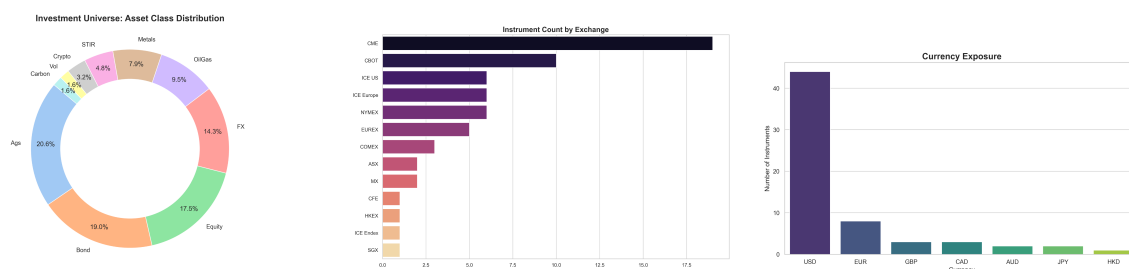
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Abstract

Modern economic theory holds that no return above the market rate should be possible, the Markowitz efficient frontier being the most efficient portfolio achievable.¹ The underlying assumption of this proposition is the random character of price returns which by essence cannot be predicted. Bachelier, in 1900, in his thesis “Théorie de la Spéculation”, would be one of the first to take interest in the statistics of returns and would initiate the principle, later developed, of the random walk of prices, positing that they are not purely random but can be modeled to a certain extent.² Within the multitude of studies in the literature that Bachelier inspired, a specific field has focused on observing the phenomena of directional persistence of stock prices, called in the jargon: *momentum*, *trend*, *directional strength*. It is in this field that our study will take place, where we will attempt to take up the existing literature, which will be composed of academic and practitioner texts, before advancing the study by hypothesis testing. Our objective is to contribute to the research on this theme by demonstrating the existence, the properties and the exploitability of trends in futures prices with a data sample ending on January 2, 2026. The recent data will also allow us to propose answers to the hypotheses according to which the markets would no longer be subject to these phenomena, in particular: how can CTA funds subsist if pure trend exploitation is no longer possible post-2010? Our thesis is that a systematic continuous long-short CTA strategy, augmented with three complementary sources of alpha (*carry*, *skew*, *vol-of-vol*) at equal weighting, restores the performance of pre-2010 CTAs on the post-2010 sub-period, where pure *trend-following* remains durably degraded. More precisely, the multi-alpha architecture works not because the four pillars contribute uniformly, but because the dominant alpha rotates with the monetary, volatility and credit regime. It is this rotation that restores performance more than a simple mechanical addition.

Method

The instrument of choice for the study will be *futures*. The latter allows broad and diverse coverage, from equity indices to commodities. The choice of data provider is made under budget constraint, the realization of the study being self-financed. Bloomberg and CSI Data are two sources widely cited in the literature for their quality, but their prohibitive prices led us to choose a provider with lesser quality and quantity of data at a more modest price: NorgateData. The sample remains very satisfactory, with 62 instruments retained (from a Norgate pool of 105 symbols), from 1987 until January 2, 2026, with a great diversity of asset classes, exchanges, currencies, and economic sectors. Once the data reprocessing is done, the study sample breaks down as follows:

**Exhibit 1.** Distribution by economic sector.**Exhibit 2.** Distribution by exchange.**Exhibit 3.** Distribution by trading currency.**Exhibit 4.** Exhaustive table of the sample.

Norgate code	Description	Asset class	Currency	Exchange
6A	AUDUSD currency	FX	USD	CME
6B	GBPUSD currency	FX	USD	CME
6C	CADUSD currency	FX	USD	CME
6E	EURUSD currency	FX	USD	CME
6J	JPYUSD currency	FX	USD	CME
6M	MEXUSD currency	FX	USD	CME
6N	NZDUSD currency	FX	USD	CME
6S	CHFUSD currency	FX	USD	CME
AFB	Eastern Australia Feed Barley	Ags	EUR	ASX
AWM	Eastern Australia Wheat	Ags	EUR	ASX
BRN	Brent Crude Oil	OilGas	USD	ICE Europe
BTC	Bitcoin	Crypto	USD	CME
CC	Cocoa NY	Ags	USD	ICE US
CGB	Canadian 10 year bond	Bond	CAD	MX
CL	Light sweet crude Winter	OilGas	USD	NYMEX
CRA	3 Month CORRA	STIR	CAD	MX
CT	Cotton #2	Ags	USD	ICE US
DC	Washington DC Housing Index	Housing	USD	CME
DX	US Dollar Index	FX	USD	ICE US
EMD	E-mini S&P Midcap 400 Futures	Equity	USD	CME
ES	US equity index S&P500	Equity	USD	CME
ETH	Ether	Crypto	USD	CME
EUA	ECX EU carbon emissions allowance	OilGas	EUR	ICE Europe
FBTP9	Italian 10 year bond BTP	Bond	EUR	EUREX
FCE	French equity index CAC40	Equity	EUR	Euronext
FDAX9	DAX 30 Index	Equity	EUR	EUREX
FESX9	EURO STOXX 50	Equity	EUR	EUREX
FGBL9	Euro-Bund	Bond	EUR	EUREX
FGBM9	Euro-Bobl	Bond	EUR	EUREX
FGBS9	Euro-Schatz	Bond	EUR	EUREX
FGBX	German Buxl 15–30 year bond	Bond	EUR	EUREX
FOAT9	French 10 year bond OAT	Bond	EUR	EUREX
FSMI	Swiss equity index SMI	Equity	CHF	EUREX
FTDX	TecDAX	Equity	EUR	EUREX
GAS	Dutch TTF Natural Gas Futures	OilGas	EUR	ICE Endex
GC	Gold	Metals	USD	COMEX
GD	SGX USD in Singapore Dollar Futures	FX	SGD	SGX

Exhibit 4. Exhaustive table of the sample.

Norgate code	Description	Asset class	Currency	Exchange
GF	Feeder cattle	Ags	USD	CME
GWM	UK Natural Gas	OilGas	AUD	ICE Europe
HE	Lean hogs	Ags	USD	CME
HG	Copper	Metals	USD	COMEX
HO	Heating oil	OilGas	USD	NYMEX
HSI	Hong Kong equity index Hang Seng	Equity	HKD	HKEX
HTW4	MSCI Taiwan Index	Equity	HKD	HKEX
KC	Coffee	Ags	USD	ICE US
KE	Wheat ICE	Ags	GBP	ICE Europe
KOS	Korean equity index KOSPI	Equity	KRW	KRX
LBR	Lumber	Ags	USD	CME
LCC	London Cocoa	Ags	GBP	ICE Europe
LE	Live cattle	Ags	USD	CME
LEU9	Euribor	STIR	EUR	ICE Europe
LFT9	FTSE 100	Equity	GBP	ICE Europe
LLG	Long Gilt	Bond	GBP	ICE Europe
LRC	Robusta Coffee 10 Tonne	Ags	USD	ICE Europe
LSU	White Sugar	Ags	USD	ICE Europe
LWB	Feed Wheat	Ags	USD	ICE Europe
M2K	Micro E-mini Russell 2000	Equity	USD	CME
MBT	Micro Bitcoin	Crypto	USD	CME
MES	Micro E-mini S&P 500	Equity	USD	CME
MET	Micro Ether	Crypto	USD	CME
MHI	Hang Seng mini	Equity	HKD	HKEX
MNQ	Micro E-mini Nasdaq-100	Equity	USD	CME
MWE	Hard Red Spring Wheat	Ags	USD	MGEX
MYM	Micro E-mini Dow Jones	Equity	USD	CBOT
NG	Natural gas US	OilGas	USD	NYMEX
NIY	Nikkei 225	Equity	JPY	CME
NKD	Nikkei 225 Dollar	Equity	JPY	CME
NQ	US equity index Nasdaq	Equity	USD	CME
OJ	Orange Juice (FCOJ-A)	Ags	USD	ICE US
PA	Palladium	Metals	USD	NYMEX
PL	Platinum	Metals	USD	NYMEX
RB	RBOB Gasoline	OilGas	USD	NYMEX
RS	Rapeseed (Canola)	Ags	CAD	ICE US
RTY	E-Mini Russell 2000 Index	Equity	USD	CME
SB	Sugar #11	Ags	USD	ICE US
SCN4	FTSE China A50 Index	Equity	USD	SGX
SI	NYMEX Silver Index	Metals	USD	COMEX
SJB	Japanese Govt Bond Mini	Bond	JPY	SGX
SNK4	Nikkei 225 SGX	Equity	JPY	SGX
SO3	3-Month SONIA	STIR	GBP	ICE Europe
SR3	Secured Overnight Financing Rate 3M	Bond	USD	CME
SSG4	MSCI Singapore Index	Equity	SGD	SGX
SXF	Canadian equity index TSX composite	Equity	CAD	MX
TN	Ultra 10-Year US Treasury Note	Bond	USD	CBOT
UB	US 20 year bond	Bond	USD	CBOT
VX	Vol US equity VIX	Vol	USD	CFE
WBS	WTI Crude Oil	OilGas	USD	ICE Europe

Exhibit 4. Exhaustive table of the sample.

Norgate code	Description	Asset class	Currency	Exchange
YAP4	ASX SPI 200	Equity	AUD	ASX
YIB4	ASX 30 Day Interbank Cash Rate	STIR	AUD	ASX
YIR4	ASX 90 Day Bank Accepted Bills	STIR	AUD	ASX
YM	Mini Dow Jones Industrial Average	Equity	USD	CBOT
YXT4	ASX 10 Year Treasury Bond	Bond	AUD	ASX
YYT4	ASX 3 Year Treasury Bond	Bond	AUD	ASX
ZB	Ultra Treasury Bond	Bond	USD	CBOT
ZC	Corn	Ags	USD	CBOT
ZF	US 5 year bond note	Bond	USD	CBOT
ZL	Soybean oil	Ags	USD	CBOT
ZM	Soybean Meal	Ags	USD	CBOT
ZN	US 10 year bond note	Bond	USD	CBOT
ZO	Oats	Ags	USD	CBOT
ZQ	30 Day Fed Funds	Bond	USD	CBOT
ZR	Rough Rice	Ags	USD	CBOT
ZS	Soybean	Ags	USD	CBOT
ZT	US 2 year bond note	Bond	USD	CBOT
ZW	Wheat	Ags	USD	CBOT

We follow the method practiced in the literature, consisting in building our sample with as the primary criterion the volume of instruments, that is, their liquidity. Once the sufficiently liquid instruments are evaluated, we attempt to construct a diversified sample, with the 4 largest positions being the FX market (14.3%), the equity market (17.5%), the debt market (19%) and agricultural commodities (20.6%). Combined, these 4 sectors represent 71.4% of the universe in number of instruments. The construction of the sample could have been different with more available data, but we deem the sample sufficiently valid for our study.

More quantitatively, our 62 instruments compare favorably to the academic benchmark samples: Moskowitz, Ooi & Pedersen (2012) employ 58 liquid futures contracts; Asness, Moskowitz & Pedersen (2013) cover eight asset classes with several tens of instruments; Hurst, Ooi & Pedersen (2017) operate on a universe of comparable size. Our asset-class coverage is moreover broader: we include STIR, carbon and crypto-currencies (post-2017 additions that earlier references did not have available), while voluntarily restricting duplicate *micro* contracts (whose fragmented liquidity would bias the measurement).

Concerning the construction of our continuous time series, we follow the methodology of Robert Carver (2023) using the Panama method.³ The treatment applied to individual contracts comprises three successive steps:

1. **Per-contract liquidity filter.** For each individual contract, three minimum thresholds are checked over its entire history: cumulative volume ≥ 100 , average daily volume ≥ 1 , and number of days with strictly positive volume ≥ 5 . Contracts that fail (typically zombie contracts linked to seasonality, particularly on commodities) are discarded.

2. **Targeted restoration for Panama continuity.** If two consecutive retained contracts have no common trading day, we search among the previously discarded contracts for an intermediate contract whose data window fills the gap; this contract is re-imported. This step precedes adjustment and minimizes interpolation fallbacks.
3. **Panama adjustment.** Construction of the continuous series starting from the most recent contract (2026) and going back to the oldest. At each transition, the roll date is determined by priority order: (1) first day of overlap where the volume of the following contract exceeds that of the current contract; (2) failing that, first day where the *open interest* of the following exceeds that of the current; (3) failing that, five days before the last available day of overlap between the two contracts. The price gap on the roll day is added to a cumulative adjustment that is applied retroactively to the prices of earlier contracts, so that the price differences are preserved exactly.

By construction of the cumulative adjustment, certain instruments present negative prices at the beginning of the sample (when the sum of the prior roll gaps exceeds the *spot* price of the time). This phenomenon, with no impact on returns since these are computed in difference and reported on the non-adjusted price (cf. formula of Exhibit 7), is widely accepted in the literature for our case of study. The instruments concerned are listed in Exhibit 5.

Inst.	Series start	Min Close	Total obs.	$n_{<0}$	%
6M	1995-04-25	-0.075	7720	1212	15.70
BRN	1988-06-23	-30.770	9594	3769	39.28
HG	1978-03-13	-99.600	12019	2645	22.01
HO	1978-11-15	-0.199	11819	2074	17.55
HSI	1996-04-01	-2086.000	7335	108	1.47
LLG	1988-02-23	-10.730	9568	242	2.53
YAP4	1987-01-14	-178.800	9929	86	0.87
ZB	1978-03-06	-31.781	12041	2510	20.85
ZN	1982-05-03	-33.469	10993	1342	12.21
ZS	1977-11-21	-199.250	12122	2516	20.76

Exhibit 5. Audit of the Panama-adjusted series by instrument: frequency of negative prices (concerned instruments only).

The transition from the Norgate pool (105 symbols) to the retained sample (62 active instruments) rests on a mechanical procedure: deduplication of *micro* contracts already covered by their *full size* version, exclusion of regional duplicates and insufficiently liquid contracts. The detail of the 42 discarded instruments is given in Exhibit 6. This selection is frozen for the entirety of the study.

Exhibit 6. Instruments dropped from the Norgate pool (105 to 62). Deduplication reasons.

Code	Description	Reason
MES	E-mini S&P 500 micro	Micro contract (duplicate of full-size)
M2K	E-mini Russell 2000 micro	Micro contract (duplicate of full-size)
MNQ	E-mini Nasdaq-100 micro	Micro contract (duplicate of full-size)
MYM	E-mini Dow micro	Micro contract (duplicate of full-size)
MBT	Micro Bitcoin	Micro contract (duplicate of full-size)
MET	Micro Ether	Micro contract (duplicate of full-size)
WBS	WTI Crude Oil (ICE)	Regional duplicate of CL
KE	Wheat (ICE)	Regional duplicate of ZW
LCC	London Cocoa	Regional duplicate of CC
LRC	Robusta Coffee 10T	Regional duplicate of KC
LSU	White Sugar	Regional duplicate of SB
LWB	Feed Wheat	Regional duplicate of ZW
MWE	Hard Red Spring Wheat	Regional duplicate of ZW
NKD	Nikkei 225 Dollar	Regional duplicate of NIY
SNK4	Nikkei 225 (SGX)	Regional duplicate of NIY
SSG4	MSCI Singapore Index	Regional duplicate (SGX liquidity)
SCN4	FTSE China A50 Index	Regional duplicate (SGX liquidity)
HTW4	MSCI Taiwan Index	Regional duplicate (SGX/HKEX liquidity)
FCE	CAC40	Regional duplicate (FESX9)
FSMI	SMI	Regional duplicate (FESX9)
FTDX	TecDAX	Regional duplicate (FDAX9)
FBTP9	BTP Italy 10y	Regional duplicate (FGBL9)
FOAT9	OAT France 10y	Regional duplicate (FGBL9)
FGBX	Buxl 15–30y	Regional duplicate (FGBL9)
MHI	Hang Seng mini	Regional duplicate of HSI
DC	DC Housing Index	Illiquid
AFB	Eastern Australia Feed Barley	Illiquid
AWM	Eastern Australia Wheat	Illiquid
OJ	Orange Juice	Illiquid
ZO	Oats	Illiquid
ZR	Rough Rice	Illiquid
LBR	Lumber	Illiquid
CRA	3M CORRA	Illiquid STIR (3 STIR retained: SO3, SR3, LEU9)
YIB4	ASX 30d Interbank Cash	Illiquid STIR
YIR4	ASX 90d BAB	Illiquid STIR
GWM	UK Natural Gas	Low-liquidity contract
KOS	KOSPI	Norgate sampling issues
GD	SGD/SGX FX full-size	Regional FX duplicate
TN	Ultra 10Y US T-Note	Synthetic extension rejected
ZQ	30d Fed Funds	Synthetic extension rejected
YYT4	ASX 3Y T-Bond	Synthetic extension rejected
YM	Mini Dow Jones \$5	Size duplicate (YM $\equiv$ MYM/Dow)

References

The financial literature, particularly that touching market finance, aims to be applicable. Although theories and hypotheses can be put to the test against *backtests* and batteries of statistical tests

following regressions of all kinds, the argument of authority remains its capacity to produce real profits. Our references thus count renowned and established practitioners in addition to classical academics. It is notably with Andreas F. Clenow (2013), then (2023), that the idea of this study was born, and then with Robert Carver (2023) that it materialized, 2 widely cited practitioners. The academic bases on which the practitioners rely and that we will also take up are multiple and are the object of an exhaustive list at the end of the study, those most similar to our works are the following studies: Antti Ilmanen (2011); Kent Daniel, Tobias J. Moskowitz (2016); Pedro Barroso, Pedro Santa-Clara (2015); Tobias J. Moskowitz, Yao Hua Ooi, Lasse Heje Pedersen, (2012); Clifford S. Asness, Tobias J. Moskowitz, Lasse Heje Pedersen (2013); Ari Levine, Lasse Heje Pedersen (2016), Brian Hurst, Yao Hua Ooi, Lasse Heje Pedersen (2017). We will attempt to take up the methods applied to these studies to compare our results, where we will be able to use their results as subjects of regression. This will also allow us to observe the qualitative differences of data sample insofar as the universe of instruments observed in our study is systematically different from theirs.⁴

WHAT IS A TREND?

1.1 Definition of a trend: synonymy between serial autocorrelation and subtraction of rolling averages, recall of Markov processes

Any question of trend is above all a question of measurement. Financial prices can be measured from a multitude of angles – opening price, closing price, volume, open interest, *bid-ask spread* – and the trend verdict can depend on the choice made. The main price measure we will use for this study will be the daily closing prices of our instruments, this measure will allow us a fine granularity to observe the behaviour of our sample and will allow us a certain flexibility in the statistics. Intra-day movements are therefore not considered, first because they exhibit specific characteristics that require an entire study set apart, and second because our data provider does not give us access to this fineness. Once the data reprocessing is done as explained previously, we then have time series, namely a continuous sequence of daily closing prices for each instrument considered. We wish to point out that NorgateData proposes continuous time series of its instruments, but that their construction methodology is not sufficiently explicit, we have thus preferred to construct them ourselves from the raw contracts. However the prices only interest us indirectly, the movements of these are our object: we therefore need to compute their daily returns to construct distributions and observe their behaviours. The usual arithmetic or logarithmic method is however not appropriate, as illustrated by the example of Exhibit 8.

Exhibit 7. Return equations.

Arithmetic	Logarithmic	Adjusted (Panama)
$\frac{P_n - P_{n-1}}{P_{n-1}}$	$\ln\left(\frac{P_n}{P_{n-1}}\right)$	$\frac{\text{Panama}P_n - \text{Panama}P_{n-1}}{P_{n-1}}$

where

P_n = closing price at date n ,

Panama P_n = adjusted price from the Panama method.

Exhibit 8. Application of the formulas of Exhibit 7 to the ES instrument (S&P 500 E-mini, contract ES-1997Z) on the dates of 14 and 15 October 1997.

Variable	14/10/1997	15/10/1997
Raw price P_n	977.5	973.5
Arithmetic return	−0.4092 %	
Logarithmic return	−0.4100 %	
Adjusted price Panama P_n	1,643.65	1,639.65
Arithmetic return	−0.2434 %	
Logarithmic return	−0.2437 %	
Adjusted return (Panama)	−0.4092 %	

This corrected return equation directly follows the one used by Robert Carver (2023), which allows us to obtain the returns that would have been obtained in a position in the raw contract by measuring on the adjusted prices.⁵ The intuition being that we relativize the variation of the adjusted price on that of the real price. Once the daily returns are computed for the whole sample, we produce the usual statistics such as variance, standard deviation, median, skew in order to better understand the behaviour of the data. A typical statistical presentation table is presented in Exhibit 9. Our methodology still follows that of Robert Carver (2023), which we summarize quickly for the computational phase: a year is considered as 256 trading days; our annualisations are then multiplied by 256 for the mean return, by 16 for our standard deviation, by 16 also for our Sharpe ratio. By simplification, the Sharpe ratio is here modified by removing the risk-free term, so that it is simply: mean divided by standard deviation.^{6, 7} The skew follows the classical methodology. The two ratios *lower tail* and *upper tail* are composed in order to represent the kurtosis of our distributions, Carver proposes an elegant solution to better understand the meaning of these values: subtract the mean from the daily returns, rank them in percentiles, then divide the 1st by the 30th for the *lower tail*, the 99th by the 70th for the *upper tail*, then divide the two ratios obtained by 4.43.⁸ This method produces the results of Exhibit 9, which translate how many times the extreme returns of the distribution are with respect to a Gaussian distribution.

Exhibit 9. Statistical table of the ES instrument.

Instrument	ES
Description	US equity index S&P 500
Years of data	28
Mean annual return	8.16%
Annualised std. dev.	19.80%
Sharpe ratio	0.41
Skew	-0.04
Lower tail	2.12
Upper tail	1.71

A quick reading of the table goes as follows: the ES instrument, which represents the US equity index S&P500 and for which we have 28 years of data (256*28), exhibits annualised mean daily returns of 8.16%. Its annualised mean daily standard deviation is 19.80%, its Sharpe ratio thus being 0.41. The distribution of returns is slightly more negative than a Gaussian distribution, with a (daily) skew of -0.04. The distribution exhibits much fatter tails than the normal, with an extreme negative return 2.12 times larger than what we could see in a Gaussian distribution, and an extreme positive return 1.71 times.

From there, we can enter the core of the study and proceed to the measurement of trends: in a trend is a value that is superior to a prior value. So that we can say that a GDP, if it is superior to that of the previous year, is in a trend of growth. We come back once again to our initial problem of measurement, namely its relativity: is a GDP that has fallen for 10 years in a major way, and which has only risen by an extremely small percentage since last year, really in an upward trend? Tobias J. Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen (2012), test this hypothesis by observing the serial autocorrelation of annual returns of instruments, seeking to demonstrate whether the sign of the returns of the previous year is an indicator of future returns. More precisely, they compare in a monthly rolling manner whether the closing price of the last month is superior to that of 12 months ago, and take *long* or *short* positions according to the sign of the result. This study therefore defines a price as in a trend if it is currently superior or inferior to its price of last year, corroborating our previous example of GDP.⁹

Another way to measure a trend is to use averages. The simplest example would be to measure the average closing price of an instrument during its entire existence, from which we could subtract the real latent price: if the sign is positive, then the price is at a level higher than its historical average, and we could say that it is in an upward trend; and inversely if the sign is negative. To take up our GDP example again, if the average evolution of the latter was historically 1%, and the GDP of the latent year turns out to have evolved by 1.5%, then: $1.5\% - 1\% > 0$; we could say that the latter is in an upward trend. Valeriy Zakamulin (2015) demonstrates the importance of these measures, which we will exploit later in the study.¹⁰ We can thus posit 2 characteristics of trends:

- 1) The measurement of a trend is characterised by a subtraction of a latent value with a prior

relative value.

- 2) Any price, any value, is by definition in perpetual trend, upward or downward, except for the rare times when property 1 gives rise to a result equal to 0.

Moreover, we could wonder about the state of a price that does not deviate significantly from its relative value of comparison to be considered in a trend, but this question brings back to the personal choices made to consider what a trend is. We thus prefer to maintain the duality of states of a price, which allows in particular a more robust basis for studies of the kind of Markov regimes.¹¹

The Markov regime method describes a finite number of states in which a price following a stochastic process can be found, as well as the transition probabilities between these states. For our case, we would posit that the two regimes are the upward trend and the downward trend, then perform a regression to observe how each regime describes the observed returns.¹² This would give us another source of observation of a trend, but we still need to define the probabilities of regime change, which bring us to the following problem. The free estimation of a transition matrix, of the means and conditional variances per regime, immediately adds several degrees of freedom to the model : each additional parameter is an occasion of adjustment to the particularities of the historical sample without guarantee that the regimes identified *ex post* have a validity *ex ante*. This problem is precisely the object of the methodological discussion that will follow.

Sample-fitting, data-mining, overfitting: these terms are used interchangeably in the literature to describe the same common problem to all statistical studies: the optimisation of criteria on the sample.¹³

The solution commonly employed to this problem is to use neutral arbitrary values in measurements to avoid simply following the data but rather to test it.

For example, let us try to observe the profitability of a strategy that would go *long* when an instrument is in an upward trend and inversely. If we define a trend as the difference between the closing price of the day and its rolling arithmetic average of the last 256 days, we understand that it amounts to betting on the ability of the average returns of the previous year to influence future returns, hence a variant of the study observed previously. We could see in such a strategy a bet on the inefficiency of markets, against modern economic theory.¹⁴ The results we will obtain could be considered seriously since the methodological criteria are applicable in a systematic manner. The wrong method of parameterising these criteria would be to program a script that, by iteration, would optimise over a very large quantity of possible parameters to define which best definition of the trend produces the best statistics. Surely, the results would be fantastic with unbeatable returns. The problem of these results is obvious: attempting to apply this strategy in the real world will very likely lead to immediate failure. We propose to illustrate our point with the following example:

Exhibit 10. Comparison of two strategies on the ES instrument : 256-day moving average vs. parameter optimised to 370 days.

Metric	Fixed (256 d.)	Optimised (370 d.)	Difference
Annual Return	0.81%	3.18%	2.37%
Sharpe Ratio	0.041	0.162	0.121
Max Drawdown	-55.51%	-47.59%	7.92%
t-statistic	0.220	0.861	0.641
p-value	0.8259	0.3892	-0.4366
Significant (5%)	No	No	

We test the strategy of taking a *long* position when the daily price is above the moving average of the last 256 days and inversely, we maintain the position until the inverse is true and bet in the opposite direction. We test only on the ES instrument, and we compute the annual return, the Sharpe ratio, the maximum drawdown as well as the cumulative return. As we can see, the meagre results suggest the inefficacy of this strategy. However, if we iterate on the best quantity of days to employ for our average, we find that with exactly 370 days, we multiply by more than 4 our annual return, and that we accumulate a positive return this time. Can we conclude that the evolution of the price of this instrument is dictated by the average of its last 370 days? That would be absurd, as shown by the following graph:

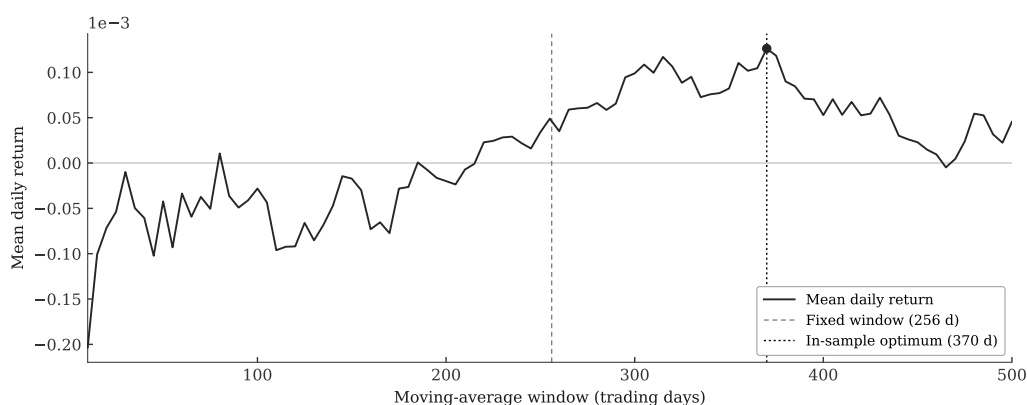


Exhibit 11. Graph demonstrating the risk of data-snooping during optimisations relating to Exhibit 10.

A large quantity of parameters between 250 and 400 days give rise to a similar distribution of daily returns, the program only finds the combination that coincides with the optimum value. This result has neither foundation, nor predictive value. The example is deliberately simplistic to demonstrate the problem, but it can easily be pushed to generate nonsensical results by playing on leverage and volatility.¹⁵

From this follows another undesirable property of our results, we cannot in any case be certain of our returns: as a consequence of the irregularity of financial data, statistical tests such as t-tests

are unfortunately futile for returns, a problem raised exhaustively by Antti Ilmanen (2011) and more quickly by Robert Carver (2023).¹⁶ We have nonetheless produced the t-statistics for our returns, but the two are not statistically significant, despite the artificial improvement produced by the optimisation. Our results corroborate those of Kent Daniel, Tobias J. Moskowitz (2016), who demonstrate that the foundation of parameters is essential to the functioning out of sample.¹⁷

Thus, we have demonstrated what a trend consists of, the basic methods to measure it as well as certain consequences of the methodologies employed to do so. It will now be a matter of observing why such trends exist and why they persist.

1.2 Properties of a trend: market inefficiency, mass phenomena or macro-economic foundations?

We propose three avenues of exploration to account for the existence and the persistence of trends on financial prices: inflation, the behaviour of actors, historical macro-economic phenomena. Inflation is the most intuitive hypothesis, indeed positive inflation in the majority of countries since 1960 should push upwards the price of impacted goods. If the correlation holds, the assets should grow in price at a rhythm similar to that of inflation. We have chosen three commodities that seem to us most propitious for this observation: oil, gold and copper. For inflation, we observe the US CPI and the Euro Area HICP:

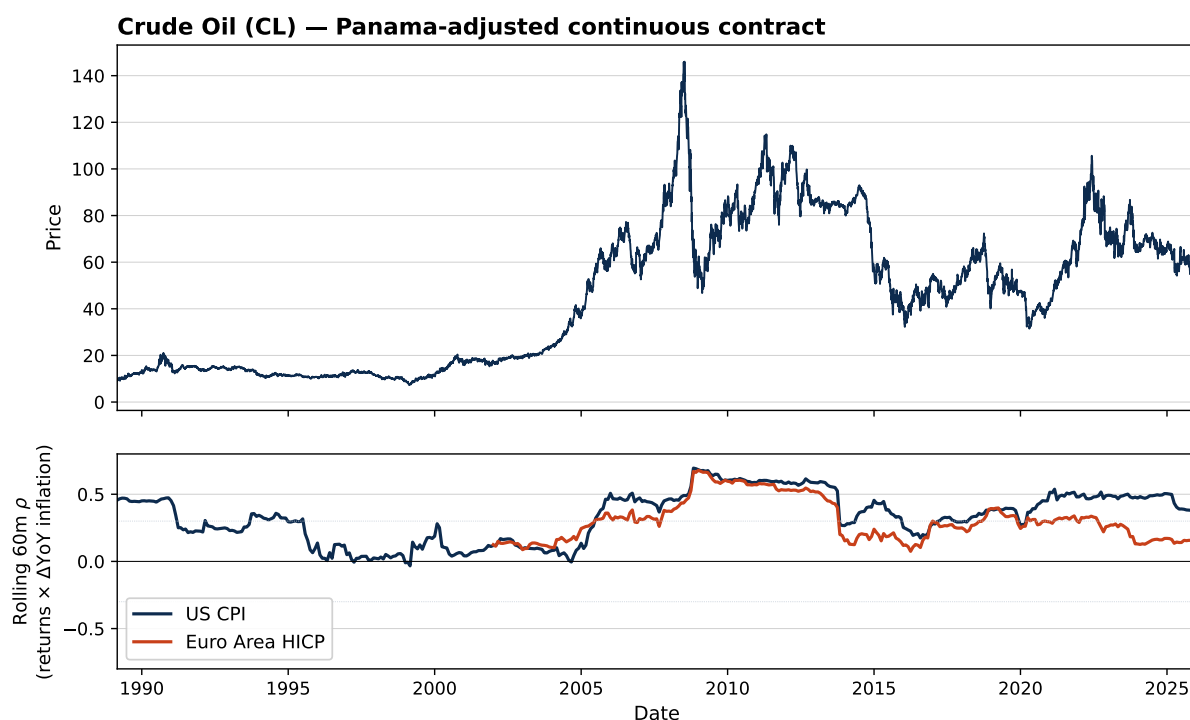


Exhibit 12. Crude oil (CL) : adjusted price (top) and 60-month rolling correlation of monthly returns with the variation of YoY inflation (bottom), for US inflation (FRED CPIAUCSL, navy blue) and the euro area (FRED CP0000EZ19M086NEST, Eurostat, orange). The dotted lines mark $\pm 0,3$.

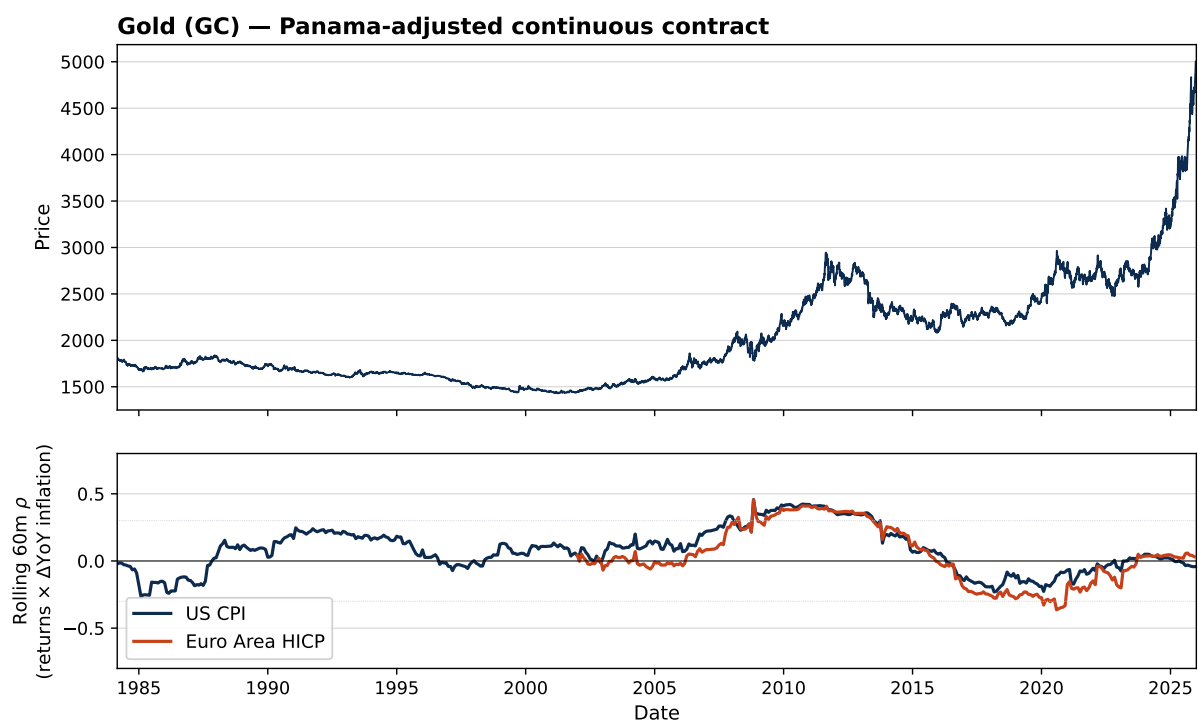


Exhibit 13. Gold (GC) : adjusted price and 60-month rolling correlation with US and euro area inflation. Same conventions as *exhibit 12*.

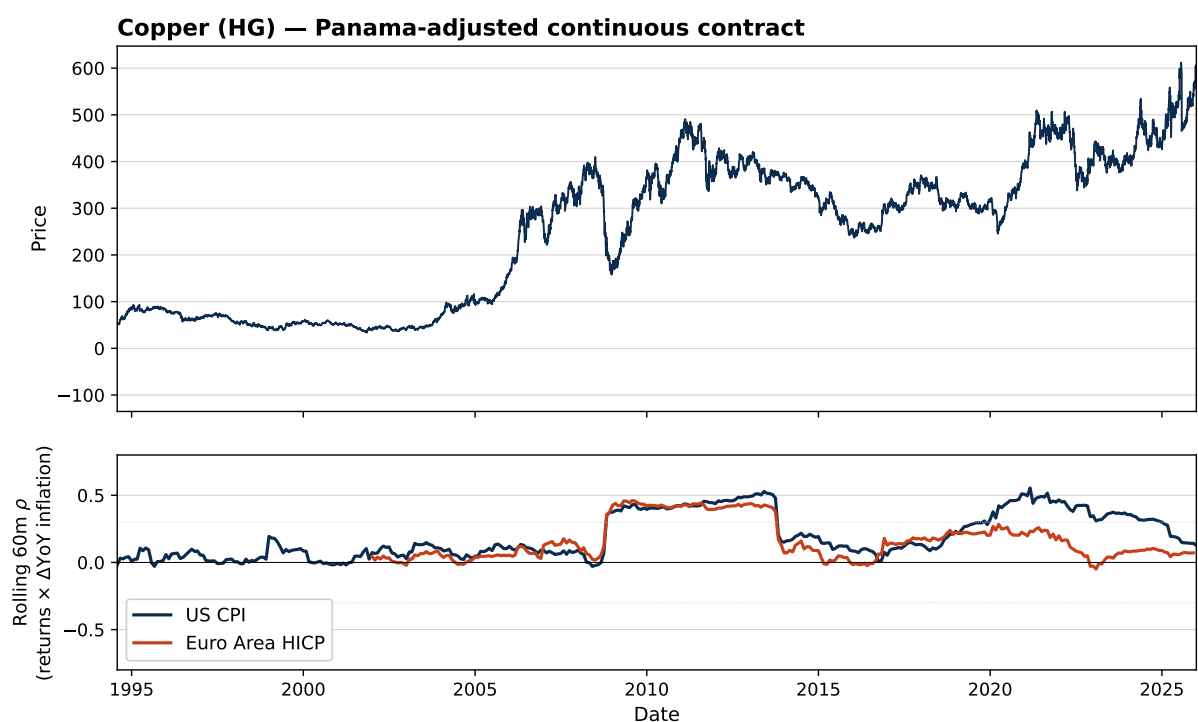


Exhibit 14. Copper (HG) : adjusted price and 60-month rolling correlation with US and euro area inflation. Same conventions as *exhibit 12*.

We measure the rolling correlation over 5 years of each instrument with each inflation measure. Gold (GC) presents a quasi-null correlation with the two inflation measures over the entire

sample (monthly Corr $\approx +0,11$ US and $+0,05$ EA ; YoY Corr $\approx +0,02$ US and $+0,02$ EA), refuting the idea that gold constitutes a mechanical hedge against inflation ; copper (HG) has a low to moderate correlation ($+0,14$ to $+0,27$), variable over time ; only crude oil (CL) displays a significant correlation ($+0,38$ monthly, $+0,56$ in YoY with US inflation), but this result is partially mechanical since the price of energy is itself a major component of the CPI. The rolling correlation moreover reveals sign switches according to monetary regimes (post-2008 crisis, post-2020), incompatible with a stable structural relationship. *In fine*, inflation does not robustly explain the prices of instruments.¹⁸

Indirectly influenced by inflation are the prices of debt instruments, whose value depends among other things on policy rates, which themselves depend on inflation. A graph of their evolution is this time immediately more intuitive, where we can observe major trends of stable upward growth over time:

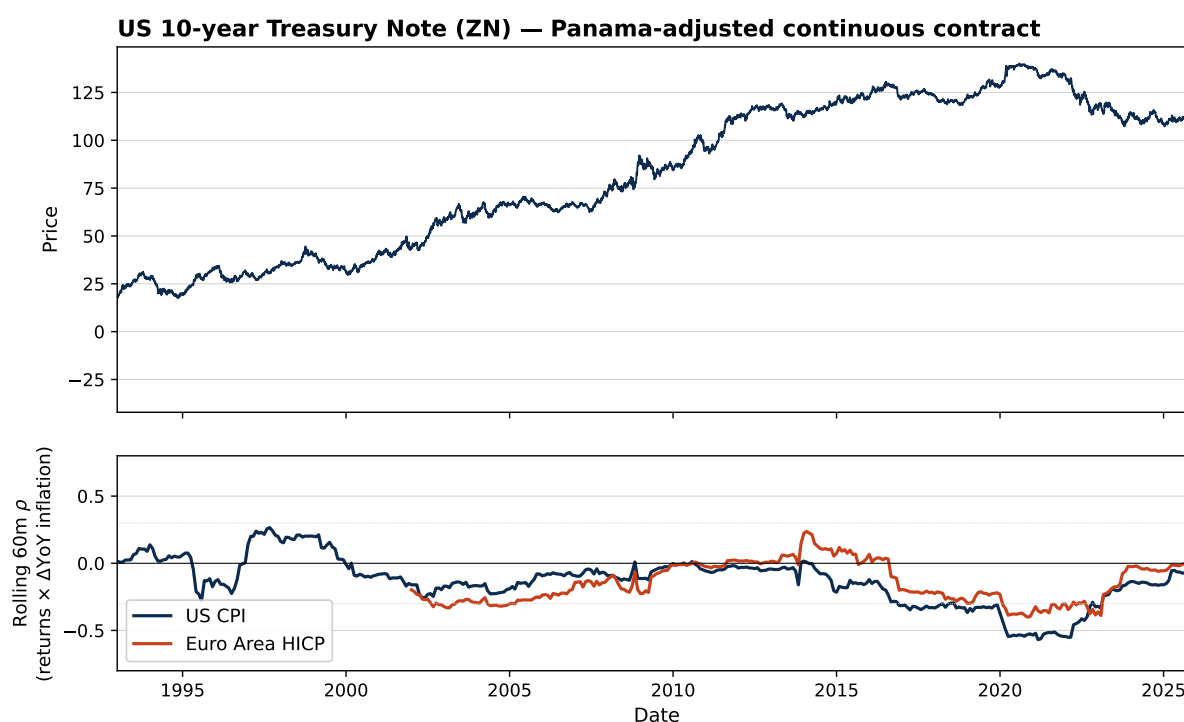


Exhibit 15. US 10Y Treasury Note (ZN) : adjusted price and 60-month rolling correlation of monthly returns with the variation of YoY inflation, for US and euro area inflation. Same conventions as *exhibit 12*. The expected correlation is negative: higher inflation = higher rates = lower bond prices.

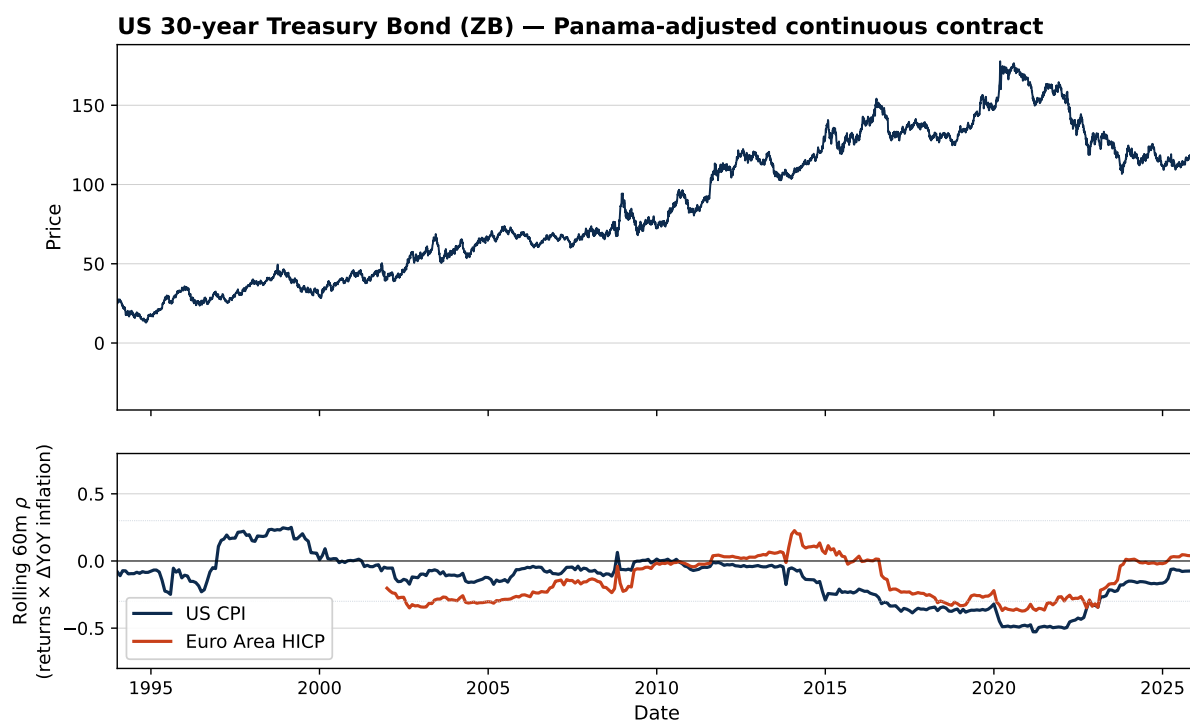


Exhibit 16. US 30Y Treasury Bond (ZB) : adjusted price and 60-month rolling correlation with US and euro area inflation. Same conventions as *exhibit 12*.

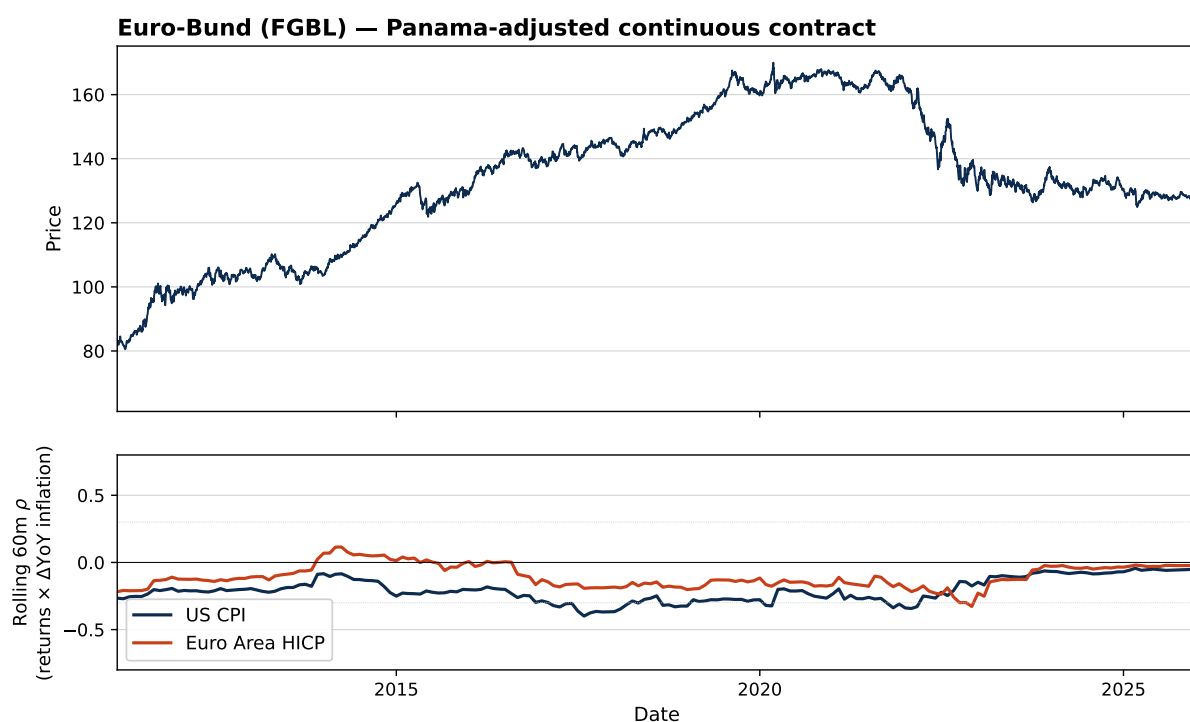


Exhibit 17. Euro-Bund (FGBL) : adjusted price and 60-month rolling correlation with US and euro area inflation. Same conventions as *exhibit 12*.

The case of US debt instruments is also explained by a continuous rate-cutting policy of the FED, going from 15% in the 1980s to reach 0% during the subprime crisis and mechanically allowing

debt instruments to grow inversely. Inflation seems to play a less important role here than a secular macro-economic phenomenon with the massive expansion of the USA, necessitating to issue ever more remunerative debt to be attractive and to fund its perpetually growing budget.¹⁹

We could repeat the exercise for all economic sectors of our sample and globally observe a non-null but weak correlation with inflation giving rise to a growth trend, with the exception of the VIX instrument due to its nature. Some instruments of our sample also present negative average annual returns, demonstrating that they can neither be explained by inflation nor by macro phenomena:

Exhibit 18. Statistical table of the 6E instrument.

Instrument	6E
Description	EURUSD currency
Years of data	27
Mean annual return	−0.60%
Annualised std. dev.	9.13%
Sharpe ratio	−0.07
Skew	0.02
Lower tail	1.37
Upper tail	1.38

Exhibit 19. Statistical table of the CC instrument.

Instrument	CC
Description	Cocoa NY
Years of data	45
Mean annual return	−0.10%
Annualised std. dev.	25.42%
Sharpe ratio	−0.00
Skew	−0.04
Lower tail	1.34
Upper tail	1.45

Exhibit 20. Statistical table of the SO3 instrument.

Instrument	SO3
Description	3-Month SONIA
Years of data	7
Mean annual return	−0.32%
Annualised std. dev.	0.93%
Sharpe ratio	−0.34
Skew	−0.05
Lower tail	1.47
Upper tail	1.68

In sum, faced with the weak correlation with economic phenomena, let us recall that the first cause of variation of a price on an exchange is the quantity of buyers and sellers. Market actors are ultimately the prime cause of appearance and persistence of trends. Our results are similar to those of Clifford S. Asness, Tobias J. Moskowitz, Lasse Heje Pedersen (2013), who demonstrate the presence of trends in all asset classes whose existence remains largely unexplained by economic foundations,²⁰ and our development on the behaviour of actors is already advanced by Kent Daniel and Tobias J. Moskowitz (2016), who demonstrate the asymmetry of trends in prices through the behaviour of actors in periods of market stress.²¹

We must suppose that the existence of these trends is due to a mixture of these propositions, but we wish to lean towards the behavioural study: the emotional instability of actors is a major explanation. The historical example of gold as a protective asset of stability is after all not founded on any rational economic foundation: the quantity traded for industrial purposes does not justify the behaviour of its price, and the appeal in the world of jewellery joins our behavioural argument. This stability proper to the asset in a historical manner has notably been flouted recently since 2024, where we observe a dizzying progression that is not supported by any obvious economic justification, theory of the flight of investments out of the dollar set aside, and on which we can only speculate on the justifications: the 5-year rolling correlation of gold to the US CPI is +0,11 over our entire sample (cf. Exhibit 13) – barely distinguishable from zero and incompatible with the thesis "gold = mechanical hedge against inflation". Over the 2024–2026 sub-period, the progression of gold coincides with the decline in the attractiveness of the dollar as a reserve currency, documented notably by IMF COFER data (passage from 71% in 2000 to 58% in 2024 of the share of the dollar in official reserves) and the net exit of foreign investors from the US Treasury in 2025. We can propose behavioural hypotheses to explain such irrationalities: actors seek to participate in an investment when they see that the latter is growing strongly, reinforcing the growth and creating a loop, not wanting to be part of those who have not profited from a trend taking place. This hypothesis fundamentally violates the theory of market efficiency, inducing the non-rationality of the behaviour of actors, but can be largely observed in the economic context leading to the crash of 1929 in the USA. Moreover, modern

finance is organised around the expected future cash flows of assets discounted, the value of an asset can easily grow strongly over a long period if it is expected that the latter distributes returns commensurately: the most contemporary example being the absolutely massive valuations in the US technology sector for assets not intrinsically generating cash-flows, and for which the future realisation of these cash-flows is still in doubt.

In fine, the existence and persistence of trends is largely inexplicable, although certain macro-economic hypotheses appear most probable, the cognitive biases of actors present on the market incontestably play a role.²²

We also wish to point out that we have here only roughly observed trends on a scale in tens of years, to illustrate historical trends. All sorts of trends can be detected on the intra-day, daily, weekly or monthly scale, with more or less important significance. Now that the definition of a trend is posed, its properties and its foundations studied, we propose to study the mechanics according to which they can be exploited.

1.3 Exploitability of a trend: similarities to a long *gamma* position

Thus, how are the returns of futures organised? Classically, the value of a forward or future contract is governed by the following equation:

$$\text{Payoff}_{\text{long}} = S_T - K, \quad \text{Payoff}_{\text{short}} = K - S_T \quad (1.1)$$

Exhibit 21. Tables of valuation equations of a forward/future.

where

S_T = price of the instrument at date T ,

K = strike price (discounting is not considered here).

Namely, the intrinsic value of a contract is the difference between which we can sell it and the value at which our counterparty must buy it from us, and inversely for a short position. Futures allow continuous pricing thanks to the exchanges and clearing houses via the *mark-to-market* process, eliminating the need to calculate the state of the portfolio for this instrument and allowing strong liquidity. Some contracts can give rise to a delivery, but we directly set aside this possibility by changing contracts well before the first day of possible delivery call begins, in continuity of the method practised in the literature. In sum, the value of a position that we take can be summarised as the difference of the price between which we entered the position and the current price as a function of the direction long or short. Following the previous equations, we illustrate with the classical payoff diagram of forwards and futures:

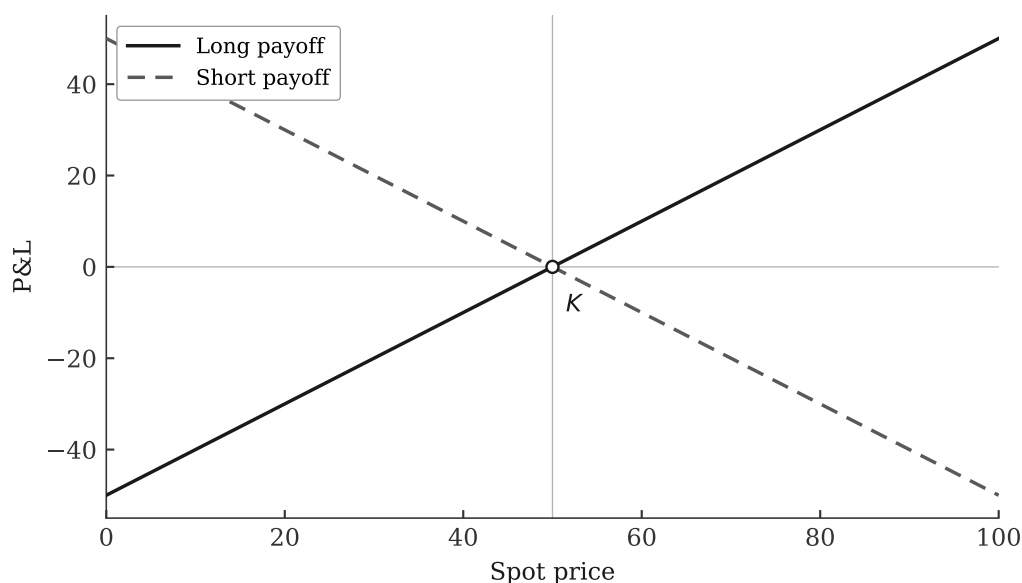


Exhibit 22. Payoff diagram of a forward, future position.

The strike price K can be considered here as the price at which we entered the position. Futures are a linear derivative instrument as we can see, so that a long position has a possible gain theoretically infinite, with a maximum loss of K , and a short position has a possible gain of K , with a loss theoretically infinite. Our study considers long-short strategies, which therefore bet upward as well as downward, just like those cited in references. In its most basic state, a strategy seeking to exploit upward as well as downward trends can take two forms: discrete or continuous. We pursue Robert Carver's approach by studying here the continuous form of this strategy, more representative of institutional portfolio management, namely: we always have a perpetual but variable position in an instrument. Although this position may at times be 0, and therefore technically closed, this case is rare and not significant. Moreover, the payoff of such a strategy follows exactly the previous graph at an instant t .

A tradition in the financial literature since 1973 with the eponymous option pricing model of Fischer Black and Myron Scholes is to compare the mechanics of financial assets with the parameters of options.²³ Kent Daniel, Tobias J. Moskowitz (2016) show that trend strategies present characteristics of selling a call option in periods of crisis, and of buying call options in periods of fluctuating markets.²⁴ Asness et al. (2013) quantify the convexity of returns in periods of crisis by illustrating it in a manner similar to the *gamma* of an option, a finding largely repeated in our references and also by Robert Carver, who will add to the point that these strategies present characteristics similar to options without the inconvenience of the *theta* of the latter.²⁵ The Γ is the second derivative of an option and represents the convexity of the latter, namely by how much the price of the option varies as a function of the variation of its *delta*, the first derivative which describes the variation of the price of the option as a function of the variation of the underlying. The Θ is a quantity or a coefficient representing the loss of value of an option over time as the latter approaches its maturity date.

Why these comparisons? Momentum strategies are intrinsically bets on the continuity of movement of the price in the direction on which we bet. In practice, these trends only materialise rarely, markets rather having the habit of being in states of fluctuation in both directions, with occasionally a prolonged detachment in one of the directions. To address this problem, it is necessary to apply this strategy to a large number of instruments. This solution allows the use of the law of large numbers which is fundamentally the source of diversification, and rests on the ability of the few winning positions to eclipse the large number of small losses incurred.

To draw the parallel, the small but regular losses incurred are comparable to the *premium* paid for an option or to the Θ according to Carver ; the rapprochement with the Γ is found in the intrinsic bet, the price must experience a large continuous directional movement to generate profits, which amounts to being long Γ . The Δ is more intuitive since futures are liquid and generally follow very closely the variations of the underlying. A link with *vega* is more problematic insofar as the impact of volatility depends strongly on the implementation of the strategy, we take up the postulate of Brian Hurst, Yao Hua Ooi, Lasse Heje Pedersen (2017) where it is demonstrated that the strategy is generally considered *long vega*, but that at a longer scale it turns out to be globally *vega-neutral* with an impact depending more on the correlation between instruments over time than the volatility itself.²⁶ Finally, to come back briefly to the Θ , futures are not victims of it in the strict sense of the latter, the Θ attributed to a *trend* strategy refers to the small continuous losses produced by the false signals generated via a fluctuating market, which is the normal state of any market. Let us specify however that by construction of the maturity scale of futures contracts, positions lose value over time without necessary movement of the underlying, we then speak of *contango*. The comparison with the Θ is not made for an argument of symmetry: some contracts experience the inverse effect of gaining in value over time without movement of the underlying, then named *backwardation*. These two terms are the source of the so-called *carry* in the exchange of futures.²⁷

More formally, a momentum strategy can be approximated to that of an option by a Taylor series development, whose formula is given by John C. Hull (2022):²⁸

$$dV \approx \underbrace{\Delta dS}_{\text{Directional Bet}} + \underbrace{\frac{1}{2} \Gamma (dS)^2}_{\text{Convexity (Trend)}} + \underbrace{\Theta dt}_{\text{Time Decay (Whipsaw)}} \quad (1.2)$$

where

V = value of the portfolio/position,
 S = price of the instrument,
 t = time,
 Δ = directional exposure (position size),
 dS = variation of the price of the instrument,
 Γ = acceleration of the exposure as a function of Δ ,
 Θ = losses due to non-trending fluctuations (*time decay*).

The first term on the right side of the equation thus represents our directional exposure, the second the state of trend or convexity, the last our cost of operation. The efficacy of such a strategy consists in maximising the first two terms and reducing the third.

Thus, the essence of these strategies can be summarised in a bet on the inertia of prices: the previous movement influences the future movement. The success of this bet manifests itself ideally in long and stable trends, allowing entry into a position at the beginning of the trend and an eventual reinforcement during the growth of the latter, allowing a large margin of benefits realised before the inflexion of this trend signifying its end. In an ideal situation, the inflexion of the trend would be realised progressively and with stability, allowing a progressive inversion of the position until having totally changed the sign of the exposure and therefore completely cashing in the benefits, in addition to being prepared to profit from the trend in the inverse direction. In its simplest state, a practical illustration resembles Exhibit 23:

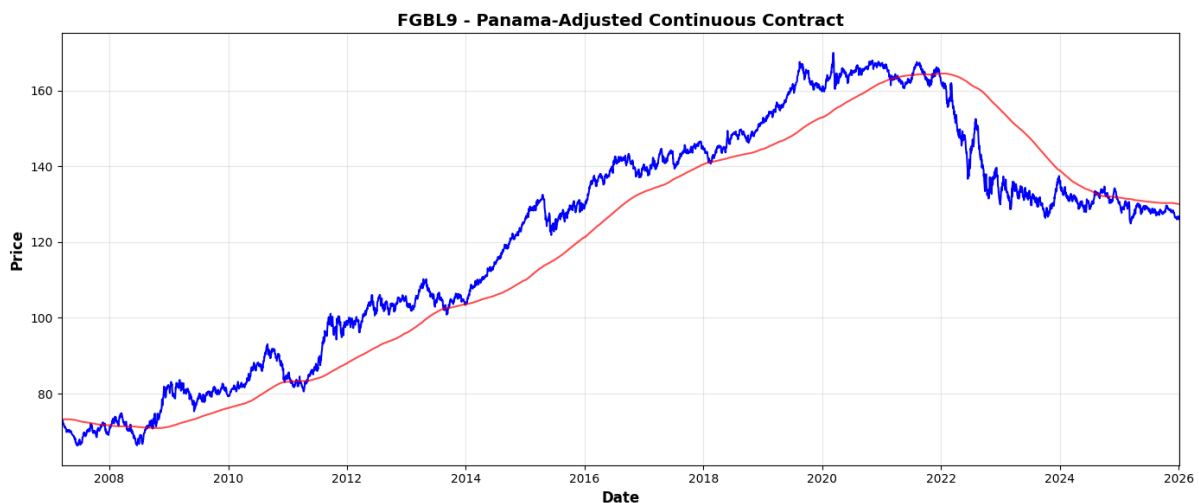


Exhibit 23. Graph of the Panama-method adjusted price of the FGBL9 instrument, in red line is represented an arithmetic moving average of the last 512 days.

Where we would be long when the price of the instrument would be above the red line, and short when the price would be below the red line, which represents here the average price of the price over the last 512 days. This example is specially constructed for illustrative purposes and has no foundation as such, but demonstrates the intuition behind trend bets: a global trend from 2008 until 2022 allows a stable following of the strategy with a few rebounds in 2011 and 2014, which

would have allowed the generation of profits. The trend reversal in 2022 turns the position short and also captures the correction. Although basic, this system can easily be implemented for a very large number of instruments without problems and has the advantage of operating upward as well as downward. The ease of use and observation coupled with the lack of explanation of its success explains the large literature surrounding this theme of financial study.

Thus, this first part of the study will have posed the concept of a trend from its measurement and definition, its properties and foundations, then its characteristics of exploitation. We wish to pursue this theoretical and explicative part with a practical and quantitative observation by observing no longer simply the trends but the strategies employed to exploit them. We have demonstrated that the returns generated by the exploitation of trends are largely insignificant at the statistical level, it may then appear surprising to want to put them to the test without the possibility of explaining fundamentally the functioning of the latter. However, the markets have the habit of transforming the impossible into the possible, as Long Term Capital Management, Amaranth Advisors, Bear Stearns, Lehman Brothers as well as the long list compiled by Sebastian Mallaby (2010) could attest.²⁹

LONG-SHORT MOMENTUM OR TREND-FOLLOWING STRATEGIES

2.1 The success of CTAs: why a price-based approach is irrational but effective

"Our magistrates have well known this mystery. Their red robes, their ermines in which they swaddle themselves like furred cats, the palaces in which they judge, the fleurs-de-lys, all this august apparatus was quite necessary; and if physicians did not have cassocks and mules and if doctors did not have square caps and robes too ample by four parts, they would never have duped the world..."³⁰

Thus, in the social order, clothes do make the man, and with good reason; it is entirely otherwise in the world of market finance. A theory can only be sustained on the condition that its application is a success, and the measure of success in this study is the return distributed to investors. A fund manager does not need to embody the American stereotype of a Wall Street trader to be perceived as excellent, he simply needs to deliver to his clients suitable profits. Taking up the challenge, the 1980s in the USA saw the birth of a specific field of the investment fund industry: the Commodity Trading Advisors. In parallel, the first academic document empirically establishing the systematic profitability of a strategy of buying *winners* and selling *losers* on the US equity universe is Jegadeesh & Titman (1993). Difficult to translate, CTAs are traders, portfolio managers, specializing in the futures market, including forwards and futures. We can broadly distinguish them in two categories: sectoral CTAs and systematic CTAs: the first specialize in an instrument, a sector, and use knowledge and skills unique to that instrument or sector to extract profits; an example would be a trader expert in oil who would only be present on that instrument. The latter consider a universe to which they will apply the same rules systematically; it is notably in this field that the exploitation of trends fits best. Investment funds specialized in this strategy notably provide us with historical data allowing us to test the theory that we have so far put forward, despite all the biases discussed in the literature concerning reporting in the hedge funds industry. We take up the list proposed by Andreas F. Clenow (2023): MSCI World TR, Millburn Diversified, Dunn World Monetary and Agriculture, Chesapeake Classic, Campbell Managed Futures, Eckhardt Evolution, Estlander Alpha Trend, Tactical Institutional Commodity, Transtrend Enhanced Risk, Mulvaney Global Markets, Alpha Quest Original.³¹ We can roughly construct an index of this list, by taking up Clenow's data, to understand the effectiveness of the global exploitation of trend phenomena: beginning between 1988 and 1999 through 2023,

with a geometric mean of returns ranging from 8.12% up to 16.34% and an average of these returns of 11.02%. Although we cannot fundamentally explain the reasons for these returns, it would seem that the exploitation of trends has borne fruit for several decades. A point shared in the literature and that we will also demonstrate later in the study is the weak correlation of the returns with a portfolio of market equities, thus making these strategies excellent additions to a more classical equity-bond portfolio. A study of the precise returns of these funds would have been an interesting angle of observation that we must leave to future research because of the high prices proposed by BarclaysHedge that start only at \$5000 for the classical database. Andreas F. Clenow (2023) however carries out a study similar and produces the following graph, which we take up for illustrative purposes to demonstrate the use and effectiveness of these strategies:³²



FIGURE 3.1 Trend-followers' performance.

Exhibit 24. Evolution graph on a logarithmic scale of the wealth of investment funds specializing in trend strategies. In the legend, the list we have previously cited. Source : “Following the Trend: Diversified Managed Futures Trading”, Andreas F. Clenow, 2023, 2nd Edition, John Wiley & Sons Ltd., page 46, figure 3.1. *Reproduced with the generous permission of Mr. Clenow himself.*

Why such an attraction? First we must admit that these strategies are not more esoteric than those of “*stat-arb*” practiced elsewhere, insofar as statistical arbitrage has evolved so as to no longer have a precise meaning, but rather all sorts of strategies relying on a quantitative approach. Already in 1994 the fund LTCM experienced a success as dazzling as it was ephemeral with a team of managers as renowned as could be, without anyone at the time being able to completely understand their strategy. Furthermore, if the *cross-arbitrage* between debt instruments at a global scale with an exploitation of implied spreads is already poorly understood, the leverage employed by the fund certainly was not by their investors. Thus, it is not necessary for an investor to understand a strategy in order to invest in it as long as it works; it is therefore less surprising that the CTAs could have experienced such success.

Next, the CTAs offer an obvious and simple service: better diversification. The most efficient possible portfolio, the most dominant, in the modern portfolio theory of Markowitz rests above all on the one offering the best diversification. The price of a percentage of return obtained is measured by its risk, illustrated in its most elementary form by a simplified ratio of Sharpe that we employ, namely the expression of the risk-free rate is removed. More formally, the classical equation of expected return of a two-instrument portfolio:

$$\mathbb{E}(R_p) = w_1 \mathbb{E}(R_1) + w_2 \mathbb{E}(R_2) \quad (2.1)$$

Exhibit 25. Classical formula of the expected return of a portfolio composed of two instruments.

where

$\mathbb{E}[R_p]$ = expected return of portfolio p ,
 w_1, w_2 = weights of the instruments,
 R_1, R_2 = returns of instruments 1 and 2.

The risk of this portfolio is measured by its volatility, namely its variance, generally standardized to its standard deviation, more formally:

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \sigma_1 \sigma_2 \rho_{1,2} \quad (2.2)$$

Exhibit 26. Classical formula of the variance of a portfolio composed of two instruments.

where

σ_p^2 = variance of portfolio p , σ_p its standard deviation,
 $\rho_{1,2}$ = correlation between instruments 1 and 2.

The *ex-post* return of the portfolio thus dispenses with the expectation, and we can therefore state the simplified equation of the Sharpe ratio³³:

$$S = \frac{R_p}{\sigma_p} \quad (2.3)$$

Exhibit 27. Formula of the simplified Sharpe ratio, where the risk-free term is removed.

The key objective here is to minimize the correlation coefficient in the construction of a portfolio, whose classical approach is a weighting between equity assets and debt assets. The CTAs propose a solution to this problem by allowing investment in new less accessible asset classes

such as metals, agriculture-related goods, industrial goods, etc. Andreas F. Clenow (2013) proposes to illustrate this point by plotting a graph of Markowitz efficient frontier composed of a portfolio with two assets: the MSCI World equities and a trend strategy composed of about twenty diversified futures.³⁴ Reproduced in Exhibit 28, we can observe that a weighting of 60-70% in MSCI World equities and 30-40% in trend strategy allows obtaining a portfolio more dominant than a classical equity portfolio, even if the latter is diversified.

Figure 5.2 Diversified futures as an enhancement to an equity portfolio

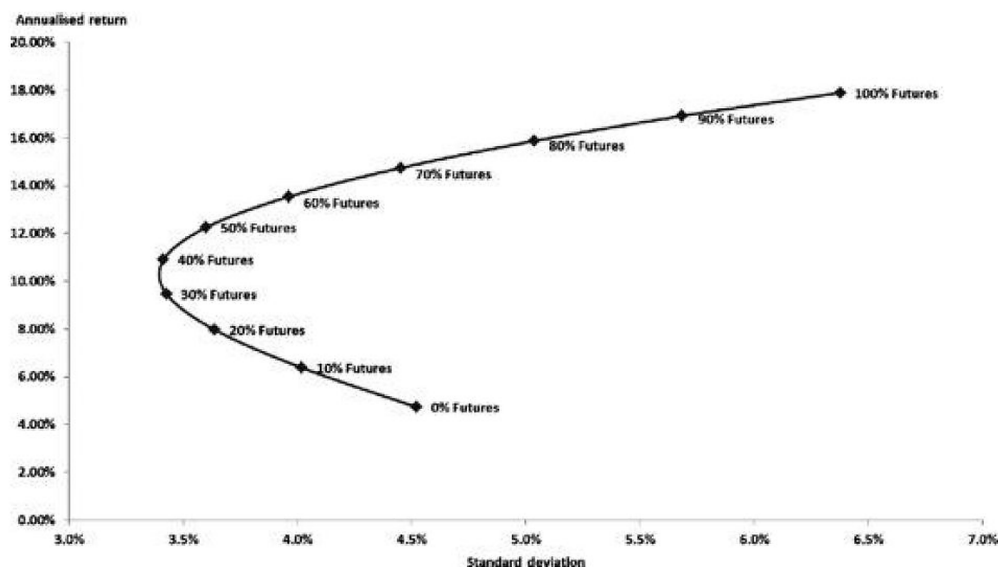


Exhibit 28. Graph of Markowitz efficient frontier with the MSCI World equities and a trend strategy of about twenty futures. Source : “Following the Trend: Diversified Managed Futures Trading”, Andreas F. Clenow, 2013, 1st Edition, John Wiley & Sons Ltd., page 81, figure 5.2. *Reproduced with the permission of Mr. Clenow.*

A verification of this proposition on our sample corroborates these results; we calculate the correlation of the instruments with two indices: the MSCI World and the ES. The presentation table Exhibit 29 shows that our sample presents an average correlation of 0.47 to the S&P500 and of 0.34 to the MSCI World, which makes this investment universe an attractive addition to a classical portfolio. The presentation is for demonstrative purposes; these correlations could be even lower if we did not take into account the indices of the equity sector of the sample. A qualitative selection can easily lead to finding around ten liquid instruments capable of supporting substantial investment sizes with correlations close to 0, allowing optimal diversification in portfolio construction. Moreover, it is also possible to employ various hedging strategies with these instruments insofar as some present a strong negative correlation, as long as the evolution of these correlations is closely monitored. The success of CTAs is therefore not a mystery; they offer a useful service to any institutional investor in addition to attractive returns.

Exhibit 29. Pearson correlations of the monthly returns of each instrument with the ES and the MSCI World, over the common available history. Sources : NorgateData (ES), Curvo (MSCI). Sorted by ρ_{MSCI} descending.

Instrument	Corr_ES	Corr_MSCI_World	Instrument.	Corr_ES.	Corr_MSCI_World.
FDAX9	0,80	0,81	LLG	-0,02	0,09
FESX9	0,80	0,80	HE	0,10	0,09
EMD	0,92	0,80	HO	0,28	0,08
NQ	0,82	0,77	SI	0,23	0,07
RTY	0,87	0,76	ZS	0,19	0,07
SXF	0,79	0,72	GAS	0,25	0,06
LFT9	0,75	0,71	YXT4	-0,15	0,05
ETH	0,67	0,67	FGBL9	-0,05	0,04
NIY	0,69	0,67	ZM	0,13	0,04
YAP4	0,73	0,61	KC	0,18	0,03
HSI	0,56	0,48	ZW	0,14	0,02
6M	0,53	0,42	ZB	-0,15	0,02
BTC	0,38	0,38	6B	0,31	0,01
RB	0,46	0,36	FGBM9	-0,10	0,01
SO3	0,32	0,32	NG	0,03	0,01
6C	0,56	0,28	CC	0,09	-0,01
6N	0,56	0,28	RS	0,05	-0,01
6A	0,55	0,25	SB	0,15	-0,02
GF	0,15	0,18	6J	0,00	-0,02
PA	0,18	0,17	LEU9	-0,13	-0,04
LE	0,17	0,17	ZN	-0,20	-0,05
HG	0,43	0,17	GC	0,06	-0,06
EUA	0,18	0,16	UB	-0,15	-0,07
PL	0,29	0,16	FGBS9	-0,19	-0,08
CT	0,32	0,15	CGB	-0,09	-0,09
DX	-0,31	0,13	SJB	-0,12	-0,15
CL	0,32	0,13	ZF	-0,20	-0,16
ZL	0,28	0,12	6S	0,11	-0,19
SR3	0,17	0,10	6E	0,28	-0,20
BRN	0,29	0,10	ZT	-0,26	-0,25
ZC	0,21	0,10	VX	-0,68	-0,66
Avg_Corr_ES	0,47				
Avg_Corr_MSCI	0,34				

Thus, why does not the entirety of financial actors practice such strategies if they are so beneficial? First because one of the factors on which the exploitation of trends rests is the weak correlation among the instruments of the portfolio. Although it is not excessively difficult to construct such a portfolio at a given moment, it is necessary to rebalance it continuously to maintain this equilibrium. Moreover, the correlations between instruments have a harmful property that must be admitted: they rise abruptly during periods of financial crisis. Kent Daniel and Tobias J. Moskowitz (2016) demonstrate how a diversified trend strategy can be heavily affected by this phenomenon, and Ilmanen (2011) proposes explanations for this phenomenon such as the “flight-to-quality” or the consequence of successive margin calls of positions with leverage effects.³⁵ In short, it would be a mistake to claim that these strategies do not present inherent risks. Relying on the postulate of correlation stability generally works, until it no longer works and catastrophic losses ensue: Nassim Nicholas Taleb (2004) corroborates this discourse more generally by specifying his point to the sale and purchase of variance through options, demonstrating that uncertain and extreme events appear much more frequently than perception leads one to imagine.³⁶ Elegant solutions have however been proposed to respond to these

problems, the general essence of which is a management of the position size scaled to the latent standard deviation. We devote a sub-part of our third part to this point, and do not insist further here.

Ceteris paribus, the exploitation of trends by the CTAs is an empirical success, and their capacity to raise funds without being able to actually explain the underlying reasons of their returns is no longer a mystery. Regarding returns and performance, a usual question in the financial literature is the foundation of the alpha generated by hedge funds: luck or talent? Andrew W. Lo, (2002) demonstrates that the returns announced by the latter are too often biased due to the serial autocorrelation of returns, implying that the study of the latter as such is futile.³⁷ However, Sebastian Mallaby (2010) and William Poundstone (2005) expose with a historical approach undeniably high returns for a large number of them, biased or not.^{38, 39} The question is then transformed, and the turn that Antti Ilmanen (2011) adopts is the one we will take up for its elegance and clarity: alpha, alternative beta or beta ?⁴⁰ The distinction is already complex in the case of classical long equity investment funds, and gains in nuances with funds using ever more sophisticated strategies. The case of CTAs is of vast intellectual richness insofar as taking a position in a trend is fundamentally not complicated, but managing to do so systematically implies a certain alpha. We propose to expose this point in our next part and to demonstrate that CTAs produce only alternative beta in the short term, by studying how the latter transforms into alpha in the long run.

2.2 Correlations of these strategies: micro and macro-trends, leverage and risk management; meager intrinsic differences between implementations

“If you engaged in a Russian roulette–type strategy with a low probability of large loss, one that bankrupts you every several years, you are likely to show up as the winner in almost all samples—except in the year when you are dead.”⁴¹

The qualitative difference between two strategies, two funds, two traders, can be measured directly by the returns generated. A more important measure however is their capacity not to implode violently. This capacity to survive is obviously only observable *ex-post*, hence the survivorship bias present in hedge fund reporting, and is therefore difficult to predict insofar as at the institutional level, no one would employ strategies having a risk of total implosion voluntarily. Being therefore unable to produce a scale of the willingness of portfolio managers to write the fateful letter to their clients, gauging a fund by its risk management is a first lead. The simplified Sharpe ratio is here a fitting illustration: 100% of return with a standard deviation of 50% is strictly equal to 1% for a standard deviation of 0.5%: 0.5. Can we then measure the qualitative difference of strategies by the volatility allowed? The answer to this question requires deep nuances, but a manager incapable of remaining within the acceptable bounds of the targeted

volatility of the portfolio risks understanding abruptly why Russian roulette is not an Olympic sport. The volatility target is however of great importance during the execution of trend-following strategies, and is capable of producing largely different impressions of two funds using exactly the same strategy.

Moreover, futures allow more or less any level of volatility thanks to the leverage authorized by the exchanges: indeed, the initial margin represents the collateral that an actor must deposit on the broker's account in order to be able to enter a position in a contract on an instrument. Although these margins are recalculated daily by the exchange, with some being clearly higher than average, for example on the VIX, this freedom can easily produce exposures with leverage that are at first sight surprising. For example, the Chicago Mercantile Exchange offers the possibility of entering a position on the micro ES contract with a margin of about \$2000 in January 2026. The exposure of a futures contract depends on its specifications: the CME informs us on the micro ES contract where one contract gives access to an exposure of \$5 times the quotation of the S&P500 index. In practice, this translates into movements in the quotation of the instrument in ticks. 4 of these ticks compose a point, which are for this instrument \$1.25, composing a point at \$5. Thus, an actor with \$30,000 can reach an exposure in USD of about 5 times 7000, which is the rounded quotation of the S&P500 index in January 2026, obtaining thus a leverage of the order of 1.16. However, another actor with only \$2000 can also enter the exact same position and obtain a comfortable and peaceful leverage of 17.5. Where the first actor can withstand a negative variation in his position, the second can receive a margin call at every tick.

Another more extreme way to illustrate the leverage inherent to futures is to calculate the potential loss of a long position in a contract on the instrument if the latter fell to 0, or even into negative values as recent history has demonstrated. Namely, our first actor, if the price of the micro futures ES fell to 0, would have a loss of 7000 multiplied by 5 in dollars: \$35,000. We can here speak of implosion, because our actor still owes \$5000 to his broker that he does not have. In the same case, our second actor will thus find himself indebted by \$33,000 to his broker. We must admit that these situations are not realistic and a price variation of the order of 10% is common for some instruments, where it is absolutely unimaginable for others. Hence we better understand how two funds using exactly the same trend strategy can have different returns by the target volatility, obtained by playing on the leverage: "...just make sure you're adequately margined to take the heat."⁴²

The second major difference between trend strategies is the temporal scale. It is not here a question of the slight differences of a few days produced by an optimization that could lead to sampling biases, but of the will to exploit short or long trends. Defining exactly the duration from which a trend is micro or macro is not the object of this study; we take up: a scale is short if it covers less than 6 months of data, and long if it covers more than 6 months. This 6 month threshold corresponds to the inflection point of the autocorrelation function documented by Moskowitz, Ooi & Pedersen (2012, Figure 1), where the average positive autocorrelation on their panel of 58

futures contracts goes from significantly positive (1–6 months) to significantly negative (12–36 months). Hurst, Ooi & Pedersen (2017) employ a conceptually identical distinction by measuring the returns of their three successive *TSMOM* (1, 3, 12 months), thus isolating the micro/macro boundary around 6 months as well. Furthermore, intra-daily scales can be considered and pushed up to a scale in seconds or even smaller measures, opening the door to the world of high-frequency trading. However, insofar as we do not have access to this type of data and where this practice stands at the paroxysm of complexity, even for specialized institutional funds, it is not considered here. Thus, two funds using exactly the same strategy can have significantly different returns due to the temporality they wish to exploit. We also put forward the argument that this property is the principal characteristic of differentiation among strategies in this register: the universe selected on which the strategy will operate as well as the parameters of long-only, short-only or long-short are also elements strongly influencing the behavior of the latter. Yet, a systematic strategy rests on an axiom that requires a strong conviction concerning the behavior of prices in a global manner, and on which we will come to build the system by selecting the mechanisms, including the investment universe and the direction of positions. Concretely, the difference is greater between a fund using data from the last 12 months and another using only the last month with identical mechanisms, than between two funds using the last 12 months but with different intrinsic mechanisms. The difference is not purely quantitative but intellectual, and reflects the attention paid to data more or less old, with the belief that this data will have an effect in the future. It manifests itself in the degree of opposition to the efficiency of markets of modern economic theory, or in the affinity with the information theory of Shannon notably discussed by William Poundstone (2005).⁴³

So how do these differences materialize? To illustrate it, we carry out 6 different experiments conducted in the literature by adapting them to our sample. It would have been interesting, again, to have a database of the investment funds of the sector, of which we could have compared the results. Andreas F. Clenow (2013) performs a similar experiment and demonstrates that the returns of these funds allow finding in a surprisingly close manner the intrinsic mechanisms of the strategies they employ.

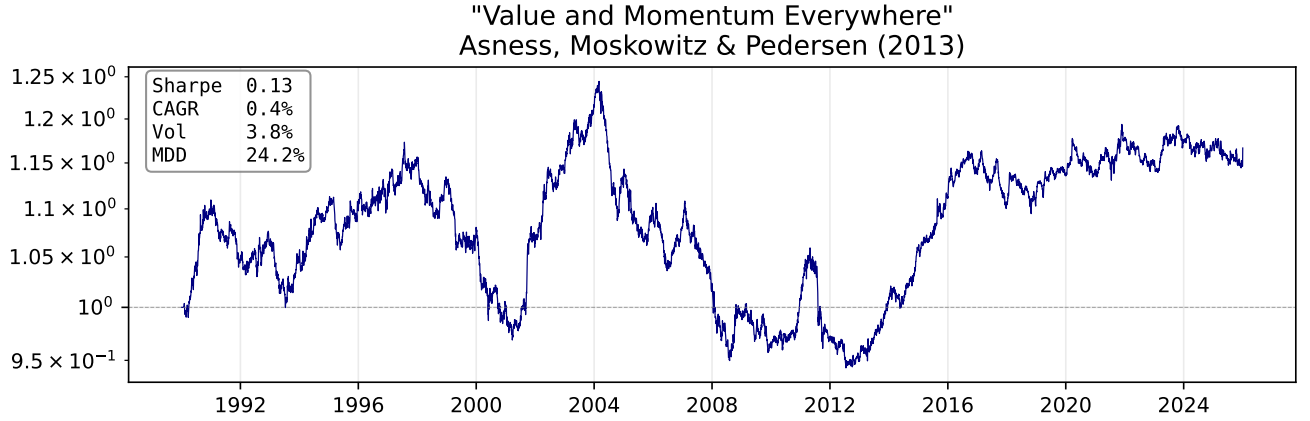


Exhibit 30. *Value and Momentum Everywhere* (Asness–Moskowitz–Pedersen 2013, §III.A). **Signals** : value = $-r_{60m}$ (long-term reversion) and momentum = r_{12m} minus 1 month (*skip-1*). **Construction** : intra-asset-class z-score, composite 0,5 z_{val} + 0,5 z_{mom} , inverse-volatility rank weighting, long-short dollar-neutral by class. **Scaling** : each asset class at $\sigma_{ex-ante} = 10\%$, equal-weighting of classes. **Rebalancing** : monthly.

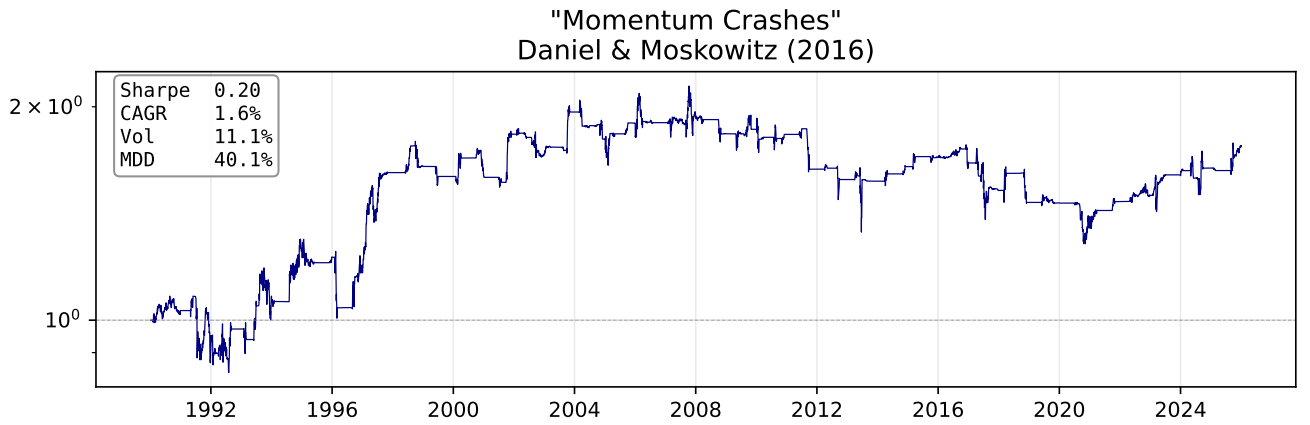


Exhibit 31. *Momentum Crashes — Dynamic WML* (Daniel–Moskowitz 2016, §4). **Signals** : r_{12m} minus 1 month. **Selection** : top tertile long / bottom short (the document uses deciles on $\sim 3\,500$ stocks ; adapted to the tertile for the universe of 62 futures). **Static WML weighting** : 6-month inverse-variance for each of the two long and short legs, the overall portfolio being kept *dollar-neutral*, $\sigma_{WML, target} = 12\%$. **Dynamic overlay** (eq. 6) : $w_t \propto \hat{\mu}_{WML,t} / \hat{\sigma}_{WML,t}^2$ estimated on a 126-day rolling window, normalized by the 5-year average of this ratio then bounded to the interval $[0, 4]$. **Rebalancing** : monthly.

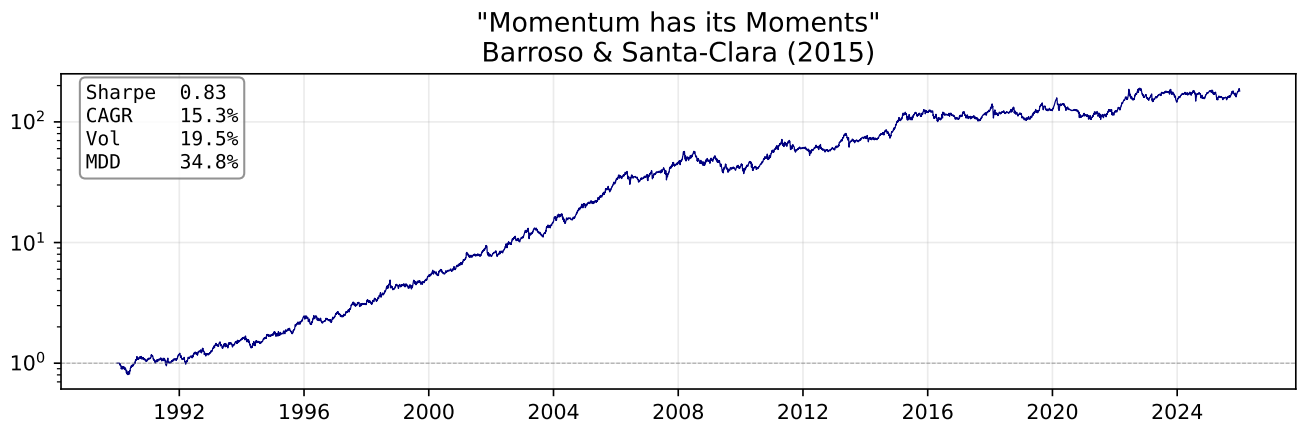


Exhibit 32. *Momentum has its Moments* (Barroso–Santa-Clara 2015, eq. 6). **Signal** : sign of r_{12m} minus 1 month (long positive, short negative). **Constant-volatility weighting** : multiplicative scaling at the WML portfolio level by $k_t = \sigma_{target} / \hat{\sigma}_{WML,t}$, where $\hat{\sigma}_{WML,t}$ is the 126-day rolling realized volatility of the WML portfolio return itself. **Target** : $\sigma_{WML} = 12\%$ annualized, with $k_t \leq 4$ to prevent over-leverage. **Rebalancing** : monthly.

"A Century of Evidence on Trend-Following Investing" Hurst, Ooi & Pedersen (2017)

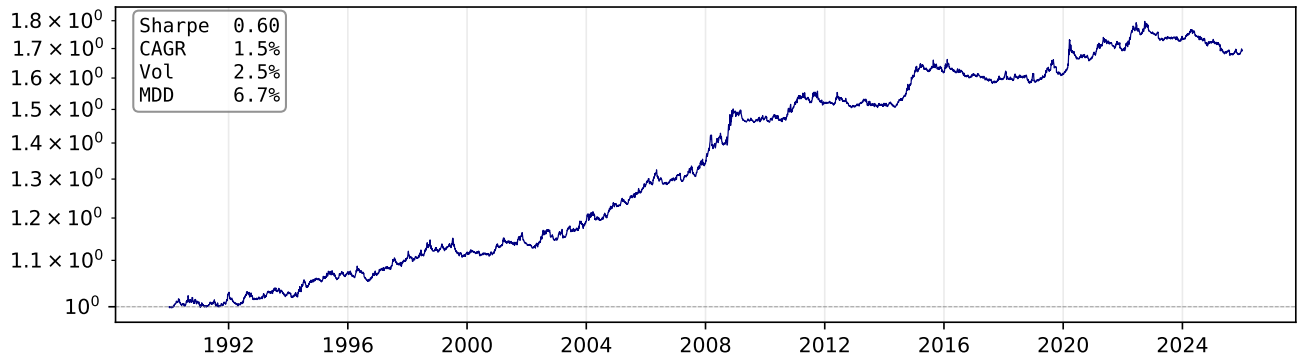


Exhibit 33. *A Century of Evidence on Trend-Following Investing* (Hurst–Ooi–Pedersen 2017, §III). **Composite signal** : arithmetic mean of the signs of the 1-month, 3-month and 12-month returns. **Weighting** : $\sigma_{\text{target}} / \hat{\sigma}_{252}$ (1-year realized volatility) applied per market, then 1/N average over the 62 markets. **Target** : 10% per market. **Rebalancing** : daily.

"Which Trend Is Your Friend?" Levine & Pedersen (2016)

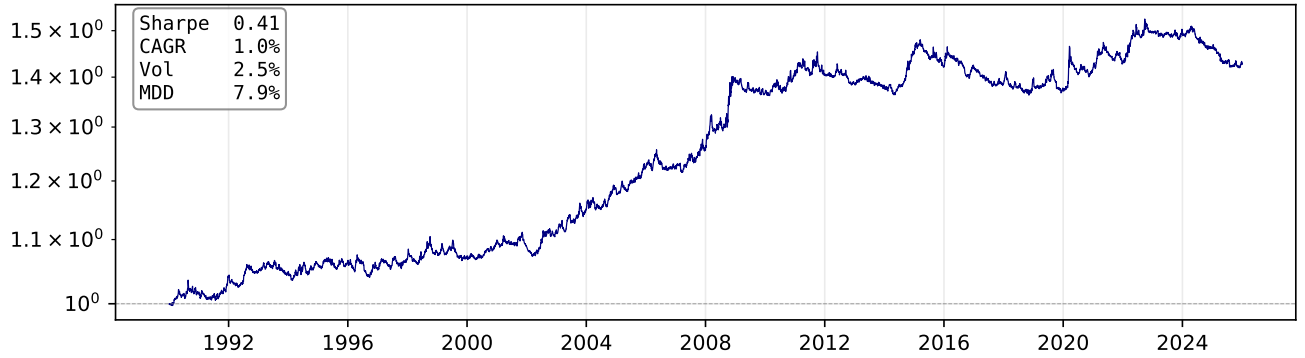


Exhibit 34. *Which Trend Is Your Friend?* (Levine–Pedersen 2016). **Signal** : MACROSS, arithmetic mean of the signs of the three exponential moving average crossovers on the log-price retained in the original document, namely $\text{sgn}(\text{EMA}_f - \text{EMA}_s)$ for the speeds $(f, s) \in \{(3, 12), (8, 32), (32, 128)\}$ (fast/slow ratio = 1:4). **Weighting** : $\sigma_{\text{target}} / \hat{\sigma}_{252}$ (1-year realized volatility) per market, then 1/N average over the active markets. **Target** : 10% per market. **Rebalancing** : daily.

"Time Series Momentum" Moskowitz, Ooi & Pedersen (2012)



Exhibit 35. *Time Series Momentum* (Moskowitz–Ooi–Pedersen 2012). **Signal** : $\text{sgn}(r_{12m})$ (sign of the cumulative return over 12 months). **Weighting** : $\sigma_{\text{target}} / \hat{\sigma}_{60}$ where $\hat{\sigma}_{60}$ is the EWM standard deviation (center-of-mass 60 days, cf. note 8 of the document) of daily returns per market, then 1/N average over the 62 markets. **Target** : $\tau = 40\%$ per market (in accordance with the document). **Rebalancing** : monthly.

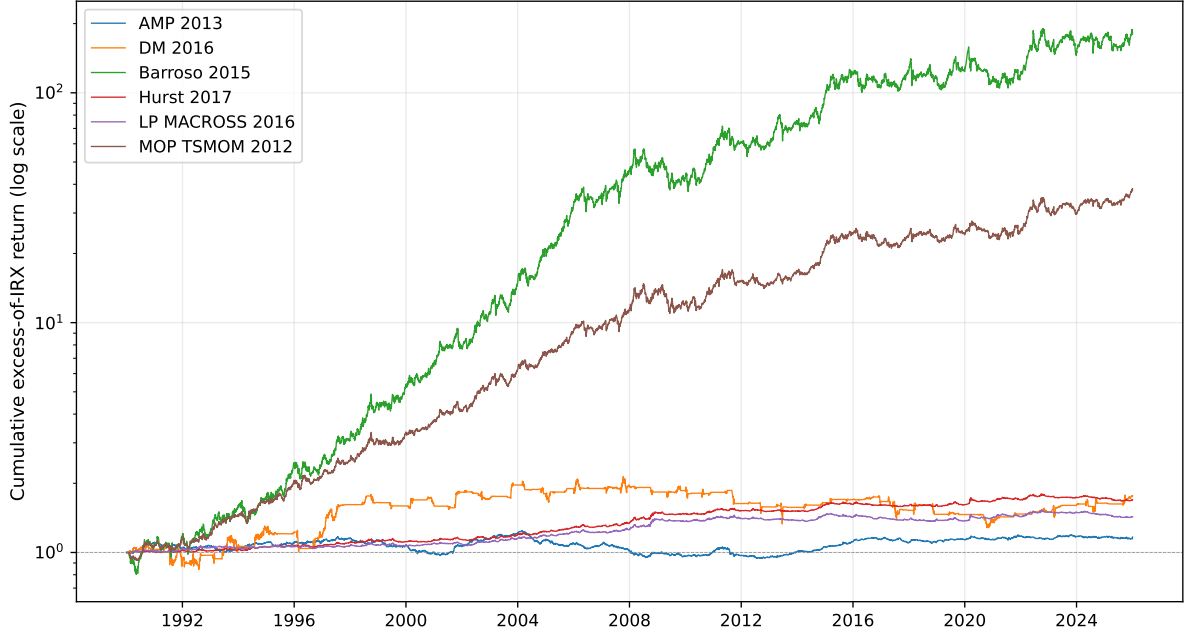


Exhibit 36. Wealth evolution graph of portfolios replicating the study strategies applied and adapted to our sample, logarithmic scale. The six strategies start on January 3, 1990.⁴⁴

We can observe that the theories on which a strategy rests heavily influence their behavior. More precisely, observation of the raw curves (Exhibits 30–35) reveals three distinct families. First, the *cross-sectional* strategies (AMP 2013, DM 2016) which select a subset of instruments by tertile/decile present the most volatile and most penalized trajectories post-2015 – the *momentum crash* of 2009 and that of 2015 are visible to the eye. Second, the strategies with *portfolio-level volatility-targeting* (Barroso 2015) smooth these shocks at the cost of a discrete modulation of the leverage that produces its own artifacts in a regime of volatility transition. Third, the strategies with *per-market volatility-targeting* (Hurst 2017, LP 2016, MOP 2012) deliver the most stable trajectories – at the cost of a mechanical dilution of the exposure that does not fully take advantage of the strongest individual convictions per instrument. The normalization to 10% that follows isolates the proper contribution of the signal definition.

$$r_{\text{scaled}} = r \times \frac{\sigma_{\text{target}}}{\sigma_{\text{realized}}} \quad (2.4)$$

where

$$\begin{aligned} r_{\text{scaled}} &= \text{adjusted monthly return,} \\ r &= \text{raw monthly return,} \\ \sigma_{\text{target}} &= 10\%, \\ \sigma_{\text{realized}} &= \text{realized monthly volatility.} \end{aligned} \quad 45$$

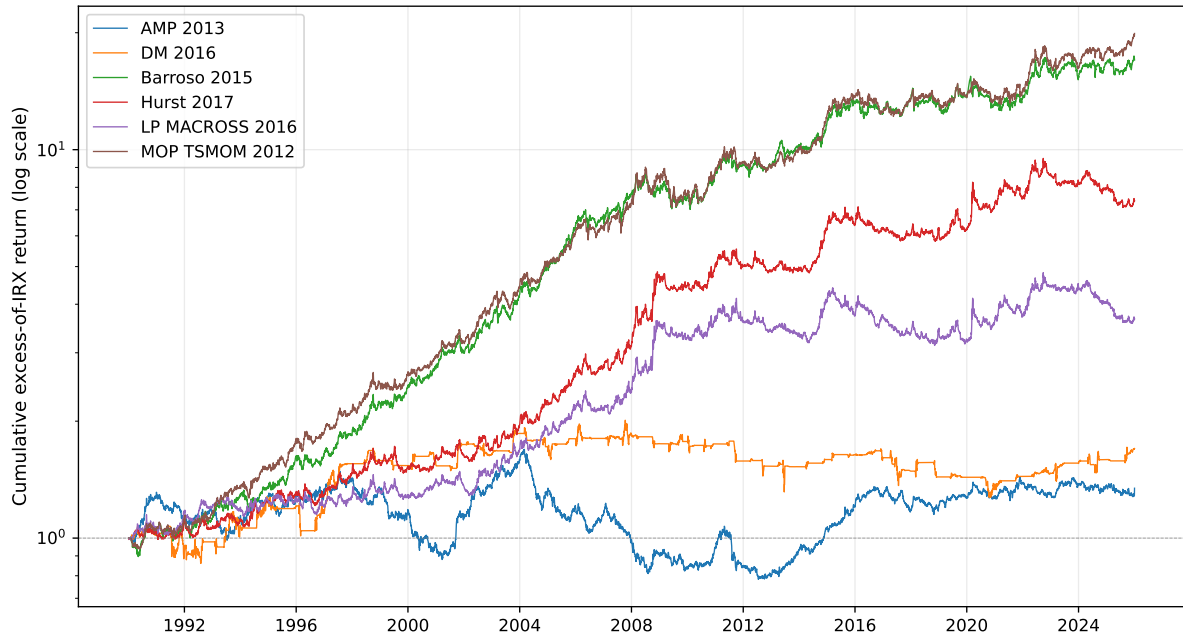


Exhibit 37. Wealth evolution graph of portfolios replicating the study strategies applied and adapted to our sample, logarithmic scale, returns weighted by the average ex-post standard deviation at 10%. The database is aggregated to monthly returns for comparison by geometric evolution. The common start date is fixed at January 3, 1990.

We can intuitively observe two points: first, although the curves evolve at different levels, some seem subject to the same shocks. Second, the strategies seem to no longer grow as strongly post-2010.

Exhibit 38. Correlation matrix of the strategies of Exhibit 36. Lower triangle only, diagonal suppressed, values $|\rho| > 0,5$ in bold.

	AMP_2013	Barroso_2015	DM_2016	Hurst_2017	LP_2016
Barroso_2015	0.1137				
DM_2016	0.0289	0.1981			
Hurst_2017	0.0961	0.5690	0.1150		
LP_2016	0.0997	0.4418	0.0601	0.8748	
Moskowitz_2012	0.1406	0.8952	0.1886	0.6589	0.5163

Thus, although the implementations can vary, we understand that trend strategies in their simplest state can have a correlation ranging to low from high. The average correlation of the table is 0.3331, with the lowest at 0.0289 and the highest at 0.8952. The median is 0.1886. The internal structure of the matrix is more instructive than these averages. Four of the six strategies form a core of strong bilateral correlations ($\rho \in [0,44, 0,90]$) : Barroso 2015, Hurst 2017, LP 2016 and MOP 2012 – exactly those that share a *time-series* momentum signal (sign of cumulative

return or moving-average crossover) with *per-market volatility-targeting*. Their coexistence at high correlations reflects that they all capture the same economic phenomenon with slightly different parameters. AMP 2013 and DM 2016 conversely appear as special cases : all their correlations to the other strategies lie below 0,20, a direct consequence of their *cross-sectional* signals in tertiles/deciles – for AMP the *value + momentum* mix that dilutes the pure momentum component, for DM the dynamic *overlay* that removes the exposure to the worst *momentum crashes*.

“They are all doing the same thing.”⁴⁶ The systematic exploitation of trends is fundamentally simple, the mechanisms at play are all strongly similar. The differences in risk target within the implementations can largely skew the results as we have shown in Exhibit 36, but these differences fade for the most part when the standard deviation is standardized as in Exhibit 37. The unexplained variance remaining between the returns can thus only come from the different informational values granted to a temporality more or less long and signals deviating more or less from a pure trend exploitation.

In fine, we have demonstrated that temporality and target volatility were the two major factors of differentiation between trend strategies, and that they remained despite these differences strongly correlated. The most decorrelated examples being strategies deviating from a pure trend exploitation. We wish to continue the study by observing the returns of these strategies since 2010, which we briefly discussed. It is usually mentioned in the literature that trend strategies do not experience the same success they could have before that date, calling into question the practical interest of the latter for the future. We will notably try to respond to this problem: are trend strategies useful if their returns are weak?

2.3 Longevity of use: market regime change since 2010, overcrowding, are trends dead?

An inherent problem of systematic strategies is their constancy. Where a generalizable property in mathematics is constantly true, a generalizable strategy is not necessarily effective *ad vitam aeternam*. Worse still, a strategy with a positive return expectation can easily lead to ruin with enough bad luck: after all, who defaults on a loan allowing one to live under a roof ? It is therefore not surprising that the capacity of adaptation of the actors on the markets is the differential character in their survival, and we must admit that in Europe at least, banking regulation as well as that of the financial markets have strongly hardened with the Basel regulation since 2008. But can we speak of a real regime change? Do stock prices behave differently? George Kleinman (2013) observes this point and demonstrates empirically that an undeniable point has strongly changed since the 1990s: volatility. More precisely, he argues that stock prices experience more violent movements in the 2010s than ever observed before. “Indeed, the price of oil now fluctuates as much in a random day as it did in an average year before 1990.”⁴⁷ How does this evolution affect trend strategies? Intuitively, we understand that a market in violent fluctuations cannot be in

trend, all the more so since the violence of this variance manifests itself on a daily scale. Another problem reflected in this volatility is the leverage present in the markets as well as the number of actors present in the same positions on the same strategies: *overcrowding*. Sebastian Mallaby (2010) exposes this principle notably through the rapid quant crash of 2007, where investment funds having exactly the same positions implode when they all try to liquidate their position simultaneously, effectively killing all liquidity and allowing the price to fall violently.⁴⁸

To measure the *overcrowding* phenomenon, we propose to construct a proxy for it with the data available on the CFTC site. The data that interest us is available only from 2006, it is therefore at that date that our sample begins. We also note that the year 2022 is not available for the two reports, and is therefore discarded. We choose two classes of available reportings: “Disaggregated Futures Only Reports” and “Traders in Financial Futures ; Futures Only Reports”. In the first, we observe the category “Managed Money”, which represents the volume of contracts held by CTAs which is the purest proxy for trend strategies. In the second, the category “Leveraged Funds” captures the same principle for CTAs and investment funds. The sample of data represents a wide range of assets that we do not cite, but which is valid for the study.

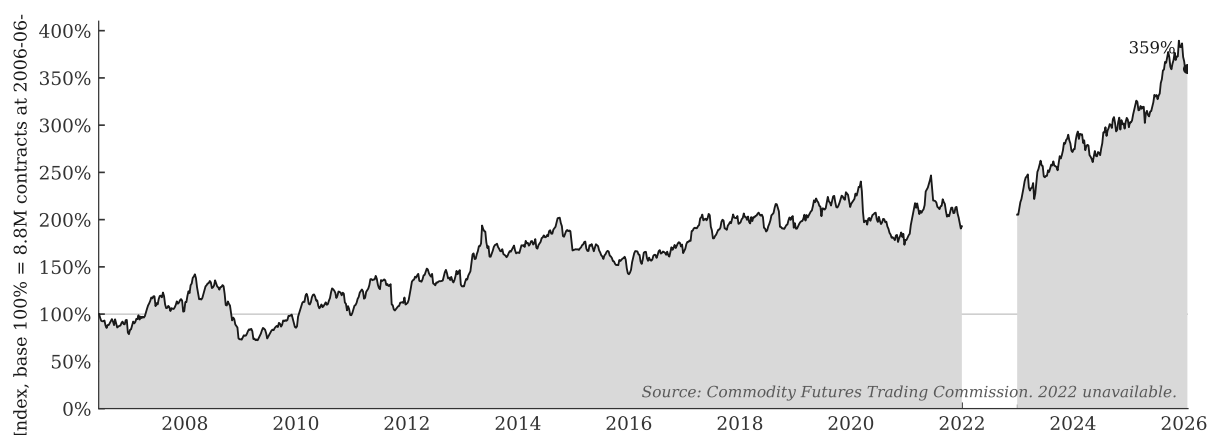


Exhibit 39. Representation graph of the number of contracts held by CTAs and hedge funds on futures, proxy for institutional trend-following strategies. The year 2022 is not considered.

We can observe that the volume of contracts held has strongly increased in 20 years: our sample has a contract volume of 8,807,697 on 13/06/2006, which is multiplied by 3.59 up to reaching 31,654,959 on 27/01/2026. This evolution demonstrates the ever-increasing size of trend strategies on various assets, strongly contributing to the *overcrowding* hypotheses. A decomposition of the data between financial and non-financial assets corroborates our hypotheses according to which these data represent trend exploitations: the TFFFOR is multiplied by 3.79 and the DFCOR by 3.08. This homogeneous evolution is proper to systematic strategies that target without proper discrimination the assets. In addition, we also propose to compute a score representative of the intensity in which the actors position themselves directionally. To do so, a qualitative reprocessing of the data is necessary, and we end up with only 36 assets sufficiently

rich in data for the analysis. We start by calculating the net weekly positioning of each instrument, which is the subtraction of the long volume relative to the short volume. If the result is positive, the actors are majoritarilly positioned upward, if it is negative they thus bet downward. More formally:

$$\text{NetPosition}_{i,t} = \text{LongContracts}_{i,t} - \text{ShortContracts}_{i,t} \quad (2.5)$$

where

i = instrument,
 t = date.

To have a standardized result, we subtract the mean of net position of the last 12, 24, 36 months from the current net position, we then divide by the standard deviation of the periods measured. Thus obtaining several Z-scores that allow us to represent the extremity of the positioning. More formally:

$$Z_{i,t,k} = \frac{\text{NetPosition}_{i,t} - \mu_{i,t,k}}{\sigma_{i,t,k}} \quad (2.6)$$

where

$k \in \{12, 24, 36\}$ = period measured (months),
 $\mu_{i,t,k}$ = arithmetic mean of the net position over k ,
 $\sigma_{i,t,k}$ = standard deviation of the net position over k .

We then take the mean of the 3 scores. These are the values we finally use for the calculations, so that:

$$Z_{i,t,k} = \frac{\text{NetPosition}_{i,t} - \mu_{i,t,k}}{\sigma_{i,t,k}}, \quad k \in \{12, 24, 36\} \quad (2.7)$$

$$Z_{i,t}^{\text{composite}} = \frac{Z_{i,t,12} + Z_{i,t,24} + Z_{i,t,36}}{3} \quad (2.8)$$

This allows us to produce graphs so as to be able to observe when the actors are positioned in an extreme manner:

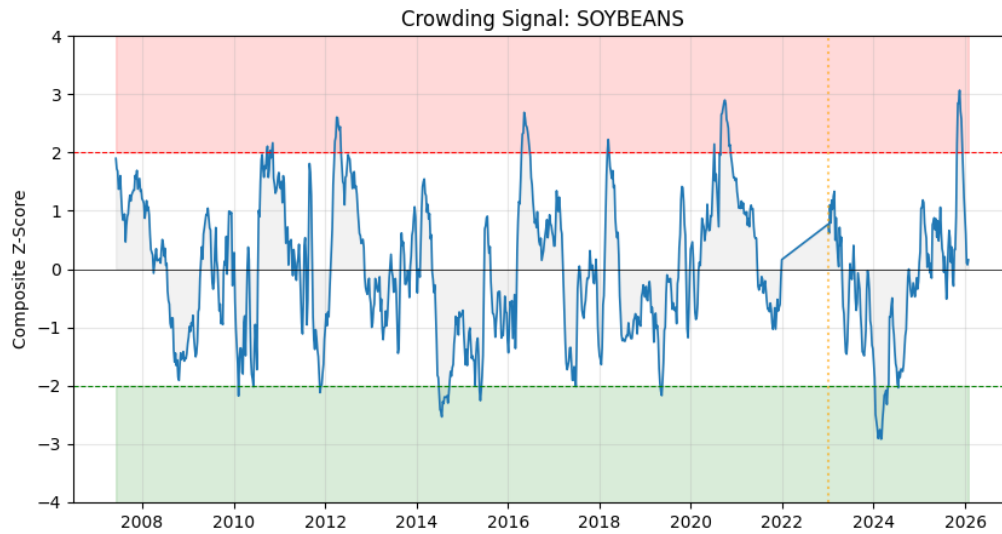


Exhibit 40. Evolution graph of the composite Z-score on soybean. By construction, -2 represents an extreme concentration of shorting, +2 an extreme concentration of long. We pass linearly from our last data point of 2021 to the first of 2023.

Thus, we can compute a proxy for the *overcrowding* phenomenon with these data by observing at what frequency scores of -2 and +2 are present in the sample. It would have been ideal to be able to compare a sample sufficiently large before and after 2010 for a more complete study. We compensate for our small quantity of data by measuring the relative frequency of extreme positionings before and after 2010:

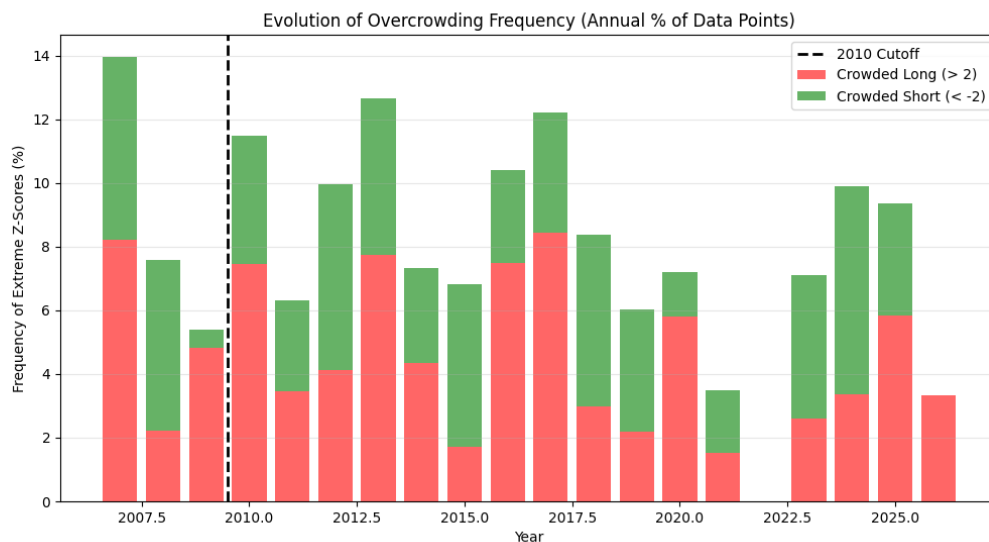


Exhibit 41. Frequency over time of extreme score values.

Period	Pre-2010	Post-2010
Total_Points	4 206	27 154
Crowded_Long_Count	192	1 250
Crowded_Short_Count	146	1 075
Freq_Long	4.56%	4.60%
Freq_Short	3.47%	3.96%
Total_Extreme_Freq	8.04%	8.56%

Exhibit 42. Table of relative frequency of extreme scores, separated between data before and after 2010.

If our data indeed seem to demonstrate an increase of the volume placed in these strategies with a multiplication of the order of 3, we do not observe extreme positionings more pronounced than previously, the gap being only 0.52%. Concretely, a growing *overcrowding* phenomenon should lead all trend strategies to experience a unified movement, all entering positions at similar timings and inversely; thereby causing more extreme movements. In the absence of concrete evidence of this phenomenon, we propose the following hypotheses:

- 1) The evolution of the size of trend strategies is a natural explanation of the compound interest of funds that invest their profits, thus leading to positions relative to their size larger
- 2) The greater volumes invested are due to a growing will of financial actors to have an exposure to these strategies for reasons of diversification. This hypothesis is consistent with the growth of the hedge fund market in general over the period (from \$1,5T in 2005 to ~\$5T in 2024 according to HFR), coupled with a stable average allocation to CTAs in proportion. The rise in volume would then reflect an effect of asset class size and not a specific concentration on *trend-following*.
- 3) The critical size of a possible *overcrowding* would not yet have been reached : if the risk exists, its manifestations have not yet materialized over the period observed.

However, what about the returns ? It follows from our first hypothesis that these strategies make sufficiently large profits to allow a reinvestment tripling the size of positions in 20 years. A study of the returns generated by the strategies of Exhibit 36 however calls this proposition into question.

Exhibit 43. Arithmetic monthly average returns of the sample of strategies (normalized to 10% of volatility), decomposed before/after 2010. Annualization: monthly $\times 12$.

Strategy	Monthly avg		Yearly avg	
	Pre-2010	Post-2010	Pre-2010	Post-2010
AMP_2013	-0.02%	0.27%	-0.20%	3.20%
DM_2016	0.29%	-0.01%	3.50%	-0.08%
Hurst_2017	0.67%	0.30%	8.05%	3.65%
Moskowitz_2012	0.89%	0.54%	10.73%	6.44%
Barroso_2015	0.88%	0.46%	10.59%	5.57%
LP_2016	0.85%	0.34%	10.17%	4.10%
Avg	0.59%	0.32%	7.14%	3.81%

The annualized arithmetic return average drops from 7.14% to 3.81% for our sample of strategies, a division by 1.87. It would thus seem that trend strategies have experienced substantial difficulties since the 2010s, which we cannot explain by an overcrowding phenomenon. A study of the standard deviation of returns demonstrates that the problem indeed resides in the profits generated and not in a volatility rising over the sample.

Exhibit 44. Standard deviation of monthly arithmetic returns of the sample of strategies (normalized to 10% of volatility), decomposed before/after 2010. Annualization: monthly $\times \sqrt{12}$.

Strategy	Monthly std		Yearly std	
	Pre-2010	Post-2010	Pre-2010	Post-2010
AMP_2013	3.27%	2.29%	11.33%	7.93%
DM_2016	3.28%	1.70%	11.37%	5.88%
Hurst_2017	3.02%	2.97%	10.45%	10.30%
Moskowitz_2012	2.85%	2.95%	9.87%	10.23%
Barroso_2015	2.79%	2.84%	9.65%	9.84%
LP_2016	3.01%	2.94%	10.44%	10.17%
Avg	3.04%	2.62%	10.52%	9.06%

Our sample shows a declining annual volatility average (10.52% to 9.06%), although modest, supporting the argument that the post-2010 drop is mainly due to returns and not to volatility.

Thus, are trends dead? We had previously posed that a price is always in trend, however the exploitation of trends is evidently no longer as profitable as in the past. We must admit that if trends are not dead, their classical exploitation is post-2010 under the prism of pure profitability: Andreas F. Clenow (2023) partially corroborates our remarks, by demonstrating in his study that apart from a few rare years after 2010, the returns of trend strategies are majoritarily disappointing; Brian Hurst, Yao Hua Ooi, Lasse Heje Pedersen (2017) also demonstrate that the decade after

2010 is the one with the weakest returns of their sample of more than 100 years.⁴⁹ Yet, these strategies continue to offer a simple and diverse exposure that makes them excellent additions to more classical portfolios, and the weak Sharpe they produce can be excused as long as they represent only a small share of a broader portfolio.

Moreover, we postulate that a paradigm change has indeed taken place, requiring fundamental alterations to trend exploitations. Long-short strategies based on a simple trend signal experience lesser efficiency since 2010, a solution to this problem is proposed in the bibliographies of Andreas F. Clenow and Robert Carver, with their two works of 2023 the most relevant for our study: more diversification.⁵⁰ It is not here a question of pushing diversification like a fund pushing leverage to appear more attractive, but of developing a diversification of strategies. The distinction is essential. A diversification of instruments runs up against an empirical wall : past a certain threshold of liquidity, the new instruments added to the universe present either prohibitive costs, or behaviors themselves adverse to trend strategies, or both. Saturation is therefore rapid. A diversification of strategies, on the other hand, exploits several sources of alpha simultaneously on the same universe, alongside the trend component.

In fine, our first part of the study will have made explicit the theoretical foundations of the exploitation of trends, our second the practical application, where we have demonstrated the institutional use of these strategies, their attractiveness, and illustrated the difficulties they encounter since 2010. Our third part will consist in the elaboration of a trend strategy, which will rest among other things on the foundations developed in our first part and will attempt to propose a solution to the problems evoked in the second part.

CONSTRUCTION OF A TREND STRATEGY: FORM OVER SUBSTANCE

“The primary reason for the failure of the chaos theory approach appears to be the fact that financial markets are extremely complex, involving a very large number of different participants, each with different objectives and different sets of information – and, above all, each of whom are human with human emotions and irrationalities. The consequence of this is that financial and economic data are usually far noisier and ‘more random’ than data from other disciplines, making the specification of a deterministic model very much harder and possibly even futile.”⁵¹

3.1 Definition and construction

“Statistical evidence suggests that the returns of a large number of financial assets exhibit a positive autocorrelation in the short term (inertia bias) and a negative autocorrelation in the long term (mean-reversion bias)[...].”⁵²

Thus, the foundation on which our study relies is the serial autocorrelation of the prices in our sample, and to be more precise of their returns. Classically, a linear regression model is said to be serially correlated when the model’s residuals have a correlation different from 0. However, the analysis of time series of prices with the dependent variable being the same price at an earlier date is characterized by an autoregressive model, which will by construction have a non-null correlation. In more concrete terms, we attempt to explain the variations in today’s price with the previous prices. Unsurprisingly, the random walk of returns fully manifests itself in this case: the autocorrelation detected in our sample for our daily returns with lags of 1 to 256 days are quasi-null, although statistically significant with p-values below 5% after a lag of 32 days according to a Ljung-Box Q test.⁵³ These results are identical to those found in the literature. We exclude the exploitation of short-term trends and concentrate on those at longer horizons by attempting to explain the return of the future month with the aggregated returns of the last 3, 6, 9, 12 months. To do this, we perform a univariate linear regression by ordinary least squares on each of our instruments with each of the observation windows. For simplicity, a month is considered as 20 trading days and the monthly return is computed by arithmetic sum of the 20 days considered. More formally:

Let i be an instrument and $k \in \{3, 6, 9, 12\}$ where k represents the number of aggregated months. Our dependent variable is the return of the following month, computed :

$$y_t = \sum_{s=t+1}^{t+20} r_s \quad (3.1)$$

Our independent variables are the cumulative returns over k months, computed :

$$x_t^{(k)} = \sum_{s=t-20k+1}^t r_s \quad (3.2)$$

For each instrument i and each horizon k , we regress independently :

$$y_t = \alpha + \beta_k \cdot x_t^{(k)} + \varepsilon_t \quad (3.3)$$

where

β_k = slope (> 0 continuation, < 0 reversal),

α = intercept,

ε_t = error term.

Thus, we avoid the multicollinearity problem with a univariate model. To address the second problem of heteroscedasticity and residual autocorrelation, we use the Newey-West estimator, Newey and West (1994), with the truncation parameter described by Chris Brooks (2019):⁵⁴

$$L = \text{INTEGER} \left[4 \left(\frac{T}{100} \right)^{2/9} \right] \quad (3.4)$$

to measure the statistical significance of our coefficient β_k :

$$\text{Var}_{\text{NW}}(\hat{\beta}) = (X'X)^{-1} \hat{S} (X'X)^{-1} \quad (3.5)$$

Where the matrix:

$$\hat{S} = \hat{\Gamma}(0) + \sum_{j=1}^L w(j) [\hat{\Gamma}(j) + \hat{\Gamma}(j)'] \quad (3.6)$$

Is composed of an autocovariance matrix:

$$\hat{\Gamma}(j) = \frac{1}{n} \sum_{t=j+1}^n (x_t \hat{\varepsilon}_t)(x_{t-j} \hat{\varepsilon}_{t-j})' \quad (3.7)$$

And Bartlett weightings, the kernel:

$$w(j) = 1 - \frac{j}{L+1} \quad (3.8)$$

Our test statistic is then:

$$t_{NW} = \frac{\hat{\beta}_k}{\hat{\sigma}_{NW, \hat{\beta}_k}} \quad (3.9)$$

Under the initial hypothesis $H_0 : \beta_k = 0$ and $H_a : \beta_k \neq 0$, our statistic follows a t distribution with $n - 2$ degrees of freedom.

We produce the regression results in Exhibit 45:

Exhibit 45. Table of univariate regression results, for each interval we produce the standard deviation, the minimum value, the maximum value and the median. In the columns: the average beta, hence the slope, our average t-statistic, our average p-value as well as the average R² coefficient.

Months	Average Bêta	Average t-stat	Average p-value	Average R ²
3	0.0003	0.07	0.44	0.0067
<i>std dev</i>	5%	138%	36%	1%
<i>min</i>	-0.1126	-2.98	0.002	0
<i>max</i>	0.131	3.11	0.9926	0.0677
<i>median</i>	0.00015	0	0.3111	0.00345
6	0.0002	0.07	0.43	0.0068
<i>std dev</i>	3%	109%	30%	1%
<i>min</i>	-0.0666	-2.41	0.0126	0
<i>max</i>	0.0675	2.52	0.9628	0.0491
<i>median</i>	0.00195	0.055	0.51885	0.00155
9	-0.0003	0.05	0.43	0.0069
<i>std dev</i>	2%	112%	32%	1%
<i>min</i>	-0.0685	-2.64	0	0
<i>max</i>	0.0563	4.38	0.9952	0.0464
<i>median</i>	0.00295	0.125	0.55225	0.0013
12	-0.0003	0.05	0.42	0.0070
<i>std dev</i>	2%	116%	31%	1%
<i>min</i>	-0.049	-2.26	0	0
<i>max</i>	0.048	4.49	0.9996	0.0537
<i>median</i>	0.00625	0.33	0.44875	0.00185

With a statistical significance level of 5%, ($p < 0.05$), our significant results break down as follows:

At the 12-month scale, only 5 instruments have a statistically significant beta: 6J, FGBS9, ZT, LFT9, ZM. (NG : $p = 0.0542$)

At 9 months, 6 instruments: NG, LFT9, SB, ZT, PA, FGBS9. (ZM: $p = 0.0557$)

At 6 months, 7 instruments: NG, ZM, LEU9, LFT9, ZT, PA, SB. (HG: $p = 0.0557$)

At 3 months, 10 instruments: NG, GAS, ZS, 6J, ZM, LEU9, LFT9, VX, ZW, ZT. (CL: $p = 0.0615$)

To interpret our results, at each observed horizon, about 10% of the coefficients of our slope, positive as well as negative, are probably different from 0. To illustrate, take the instrument ZT with a 3-month window and a beta of 0.0857, a t-statistic of 3.11 and therefore a p-value of 0.002. Theoretically, if we compute the return of the last 3 months in a rolling manner and take a position in the same direction as the return obtained until the end of the next month, and we let the number of times we try this tend to infinity, we should correctly be positioned in the same direction as the return. By conducting the experiment on our sample, we observe a success rate of 56.45% over 434 months. We conduct the experiment for all our significant results and

present the results in Exhibit 46:

Exhibit 46. Table of prediction results of the sign of aggregated returns for the upcoming month, based on the sign of the 3, 6, 9, 12 months for the instruments and intervals for which we obtain statistically significant results in Exhibit 45. In columns, the aggregation period, the instrument, the number of successes, the sample size and the ratio of the two.

Lookback	Instrument	Bêta ($p < 0.05$)	N	HitRate	Average
3M	NG	0.06	436	56%	54%
3M	GAS	0.06	462	54%	
3M	ZS	-0.06	594	53%	
3M	6J	0.05	575	51%	
3M	ZM	-0.06	591	52%	
3M	LEU9	0.13	206	64%	
3M	LFT9	-0.11	205	52%	
3M	VX	-0.07	261	49%	
3M	ZW	-0.08	592	54%	
3M	ZT	0.09	434	56%	
6M	NG	0.0442	436	56%	55%
6M	ZM	-0.0455	591	53%	
6M	LEU9	0.0675	206	61%	
6M	LFT9	-0.0666	205	50%	
6M	ZT	0.0358	434	55%	
6M	PA	0.0373	543	56%	
6M	SB	0.046	580	57%	
9M	NG	0.0319	436	56%	55%
9M	LFT9	-0.0685	205	50%	
9M	SB	0.0321	580	53%	
9M	ZT	0.0563	434	60%	
9M	PA	0.0249	543	57%	
9M	FGBS9	0.0354	252	54%	
12M	6J	0.0242	575	55%	54%
12M	FGBS9	0.0302	252	56%	
12M	ZT	0.048	434	58%	
12M	LFT9	-0.049	205	47%	
12M	ZM	-0.0343	591	52%	

To summarize, on our sample of 62 instruments and at our horizons k , about 10% of our experiments have a useful result. By attempting to exploit these results, the success rate is around 55%.

However, it must be admitted that due to our small samples, between 200 and 700 months, we cannot recognize them as statistically significant. A p-value test of a binomial distribution shows that approximately half of our “successes” are not distinguishable from noise. The lack

of statistical significance is unfortunately common in the study of contemporary returns for a simple reason: we lack data. Studying medium- and long-term phenomena requires a certain aggregation of the data to remove daily noise and identify trends, which leads us to a small number of observations.⁵⁶ Furthermore, if an exact, systematic and significant correlation existed between the autocorrelation of past returns and future returns, the economic opportunity would instantly disappear insofar as it would be exploited simultaneously by all actors. The lack of statistical evidence of this approach is precisely what leaves room for the economic opportunity of trend strategies, whose modern foundation can only rest on empirical success and a psychological rationale.

Before starting the actual construction, we describe the conditions to which the strategy must comply and more generally the architecture:

- 1) Continuous operation: positions are not taken in a discrete manner, namely we permanently have at least one position on each instrument. This constraint is imposed by our continuous *vol-targeting* framework (cf. Carver 2023) : a discrete strategy imposes exposure shocks incompatible with the variance-based sizing that we will take up in §2. Moreover, operating continuously allows us to benefit from portfolio diversification permanently and not at isolated moments.
- 2) Long-Short: being continuous, the strategy freely takes both upward and downward bets without discrimination or bias. The asymmetry of a *long-only* constraint would mechanically bias the strategy toward the classic equity *beta*, thus losing the zero correlation to the market that justifies the institutional appeal of the register (cf. chapter 2, §3). We therefore voluntarily sacrifice the gross performance gains that a *long-only* variant would allow in order to preserve the diversification property.
- 3) Systematic: the described principles, the constructed signals, are applied uniformly on all instruments of the sample without discrimination or condition. The sample is not reworked, the 62 instruments are used as such without sectoral specification, the portfolio weighting is uniform. The uniform application avoids the discretionary allocation bias by instrument or by asset class ; it also guarantees the reproducibility of the *backtest* and limits the degrees of freedom likely to be exploited via *sample-fitting*. No instrument is excluded *ex post* for its poor performance – even the STIRs, which contribute negatively (cf. appendix §30), are kept in the universe.
- 4) Purely out-of-sample: all principles, signals, are applied in a rolling manner with as information provided only that available at date t . The lock of 1,280 days per instrument eliminates any *look-ahead* in the calibration phase of the signal scalars.
- 5) Delayed execution: the signals, principles, indicating a position at date t are only executed at the opening price at $t+1$. Execution at $t + 1$ reflects operational reality : a signal computed at the close of day t is executed at the open of the following day, never simultaneously. Testing simultaneous execution would be technically more flattering but industrially false ; a test at $t + 2$ (additional one-day latency) is moreover presented in appendix §11 and does not degrade the conclusions.

6) Exchange rates: exposures and pnl are converted daily into the base currency of the portfolio: USD. The conversion is done using historical spot rates and follows principles 4 and 5. The use of historical *spot* rates preserves the absence of *look-ahead* (consistent with condition 4) and reflects a realistic USD fund that rebalances its currency exposures daily rather than freezing an arbitrary conversion at the start of the sample.

7) Commissions and fees: commissions and exchange fees are drawn from the CME, conservative slippage fees are included for each transaction. All returns and ratios are computed net of these fees. Note: these fees and commissions are slightly overestimated and held fixed for the entire sample. Holding fees fixed over the entirety of the sample (1990–2026) voluntarily ignores the secular decline in transaction costs : our 2026 costs are applied to the 1990s. A *stress-test* of the fees (multipliers from 0× to 3× baseline) is presented in appendix §20.

I. Trend — Serial autocorrelation is the foundation of our strategy, and one of its most direct uses lies in the exploitation of averages. The other elements of the system are essentially added on top of this one.

Thus, this signal looks at the direction in which the prices are according to certain windows. To do this, we use three EWMAc of different speeds. More concretely, an EWMA of span N is defined recursively such that:

$$\text{EWMA}_N(P_t) = \alpha P_t + (1 - \alpha) \text{EWMA}_N(P_{t-1}), \quad \alpha = \frac{2}{N+1} \quad (3.10)$$

Therefore an EWMAc:

$$\text{EWMAc}(f, s) = \text{EWMA}_f(P_t) - \text{EWMA}_s(P_t) \quad (3.11)$$

With f our fast EWMA and s our slow EWMA.

More visually:

EWMAC (64, 256) Trend Extraction across Macro Regimes

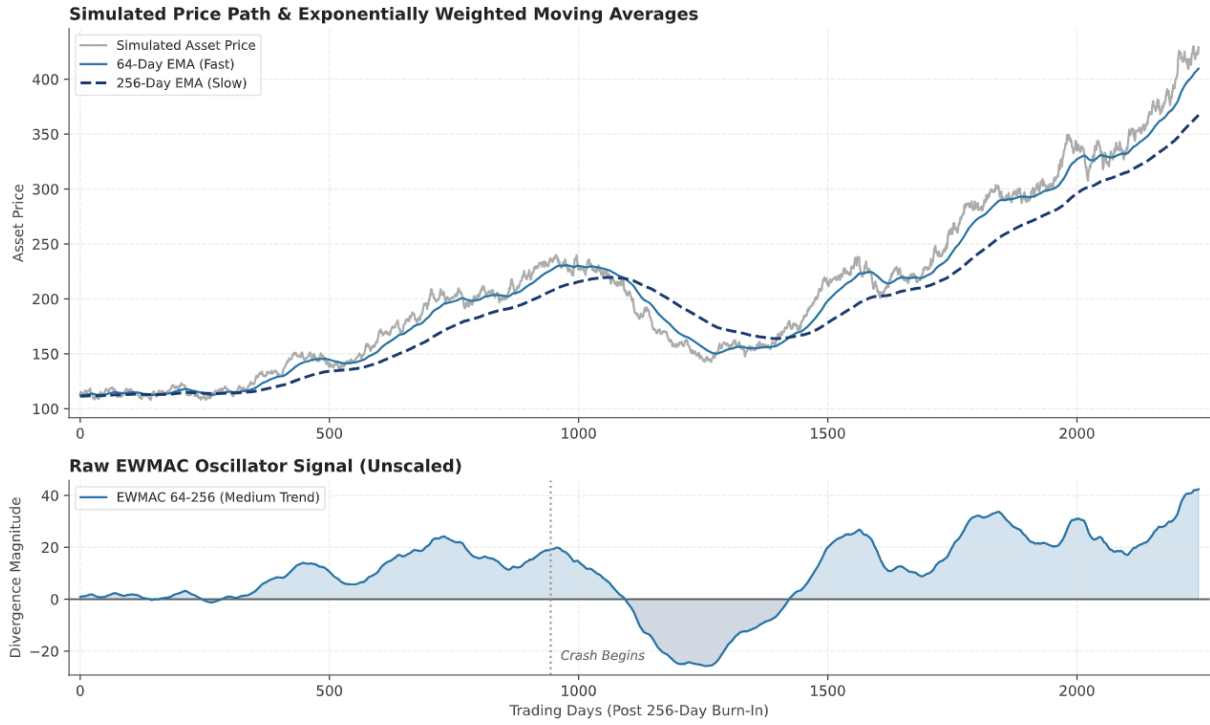


Exhibit 47. Visual illustration of the EWMAC (fast – slow).

Our 3 EWMACs are 16-64, 32-128, 64-256. We directly take these parameters from Robert Carver for the 64-256, and add a faster signal and a slower one from Ari Levine, Lasse Heje Pedersen (2016).⁵⁷ The precise values are not important, the idea is to have three opinions on the state of a price at three different levels: short, medium and long term. When the fast average is higher than the slow average, we are in a trend and take a long position; a short position when the fast average is lower than the slow average. The three EWMACs are equally weighted:

$$f_{\text{trend}}^{\text{TS}} = \frac{1}{3} \left[\text{EWMAC}(16, 64) + \text{EWMAC}(32, 128) + \text{EWMAC}(64, 256) \right] \quad (3.12)$$

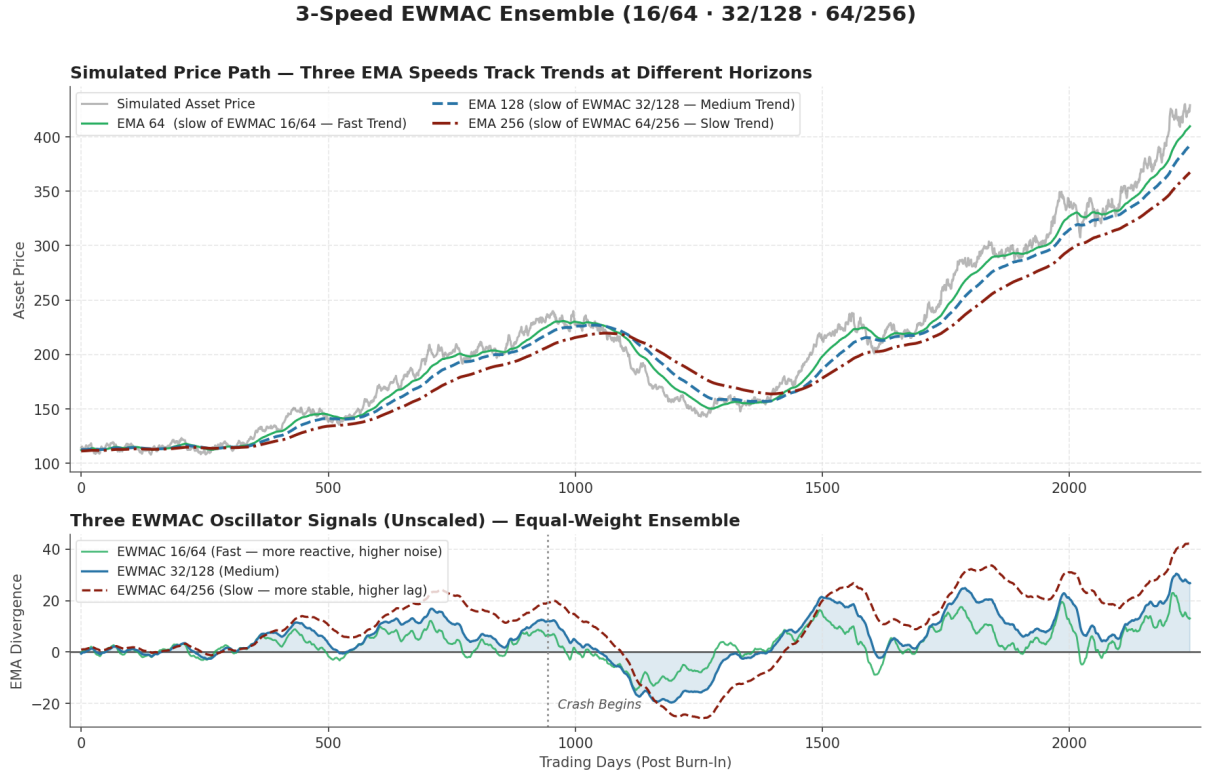


Exhibit 48. Aggregation of the three EWMACs (16/64, 32/128, 64/256).

Moreover, we wish to observe the relative trend of an instrument compared to the others, allowing us to better identify a strong conviction from a weak conviction. To do so:

$$f_{\text{trend}}^{\text{XS}}(i, t) = \frac{r_{i,t}^{256} - \bar{r}_t^{256}}{\sigma_t^{256}} \quad (3.13)$$

We attribute 50% of the signal to this second component and 50% to the first one coming from the EWMACs, allowing us to judge whether an instrument is in a trend and whether it is more so than the others: the 256-day window for the Z-score aligns this signal on the slow bound (64, 256) of the time-series component, ensuring that an instrument detected as in a trend by the EWMAC is also detected by the XS component when its trend is strong relative to the others. The 50/50 weight avoids an additional degree of freedom that would have to be justified *ex post*.⁵⁸

$$f_{\text{trend}} = 0.5 \cdot f_{\text{trend}}^{\text{TS}} + 0.5 \cdot f_{\text{trend}}^{\text{XS}} \quad (3.14)$$

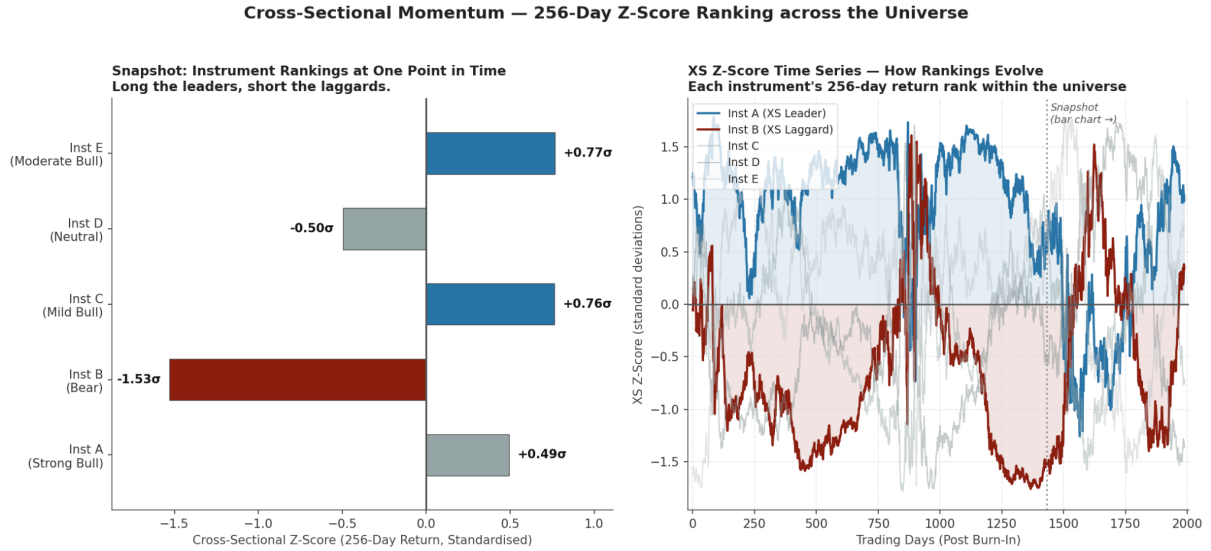


Exhibit 49. Cross-sectional momentum signal (XS) based on the z-score of relative returns.

II. Carry — This signal exploits the structural yield curve of the different futures contracts of an instrument. We adopt the most simplistic approach by comparing the price of the contract expiring earliest and the next one, then compute the annualized implied return under the condition of no change in the spot price of the instrument. The prices we observe in this case are the prices of the raw contracts and not those adjusted by the Panama method, which would distort the structural spread of the curve. Concretely:

$$f_{\text{carry}}(i, t) = \frac{F_1(i, t) - F_2(i, t)}{F_1(i, t)} \times \frac{256}{\Delta T} \quad (3.15)$$

where

$F_1(i, t)$ = price of the nearest contract (front-month) at date t ,

$F_2(i, t)$ = price of the following contract (next-month) at date t ,

ΔT = number of calendar days between F_1 and F_2 .

Thus, if the instrument is in backwardation, we bet upward, if it is in contango, downward.⁵⁹

More visually:

Real-World Carry: Extreme Commodity Term Structures

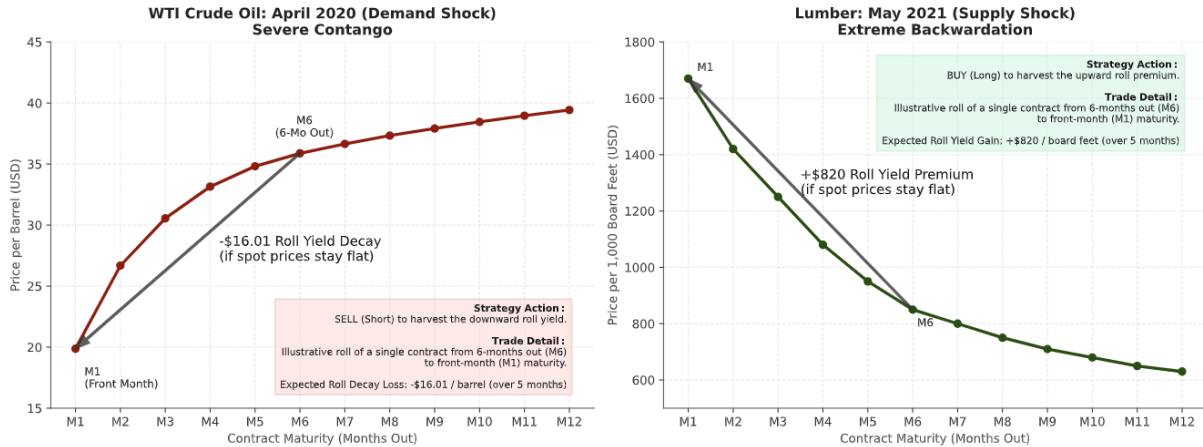


Exhibit 50. Visual illustration of the carry signal.

III. Skewness — This signal is a pure sale of insurance on the instruments. We measure in a rolling manner over 256 days the skew of the daily returns of an instrument and invert the sign:

$$f_{\text{skew}}(i, t) = -\text{Skew}(\{r_{i, \tau}\}_{\tau=t-256}^t) \quad (3.16)$$

An instrument with a strong negative skew implies recurring losses, thus requiring a premium from investors to compensate for this characteristic. We collect this premium by buying the unattractive assets and selling the most attractive ones.⁶⁰

More visually:

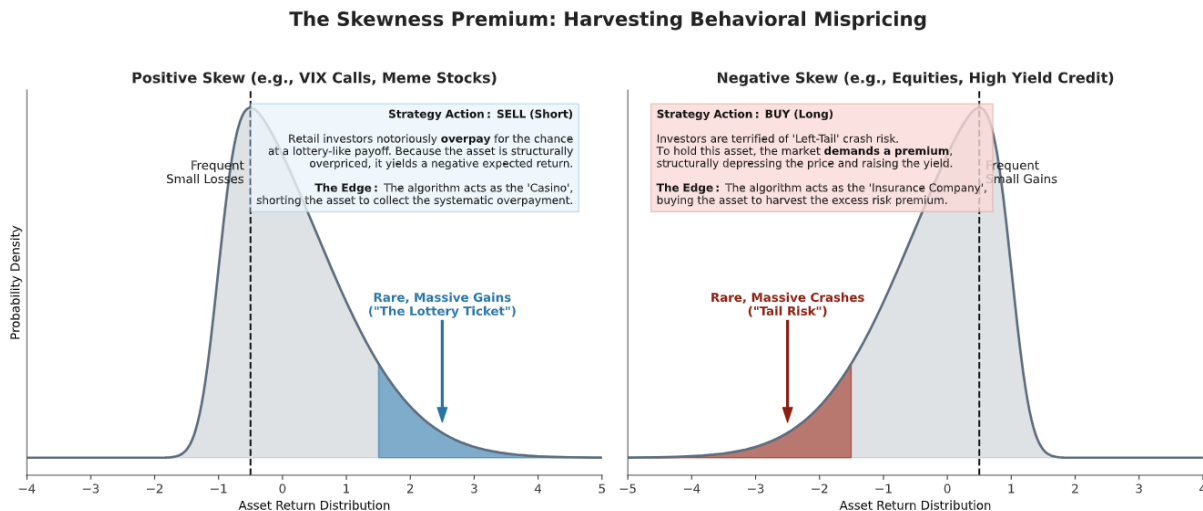


Exhibit 51. Visual illustration of the skewness signal.

IV. Volatility of volatility — This signal measures a regime and not a proper conviction.⁶¹ We compute in a rolling manner the standard deviation of the standard deviation of an instrument by

first taking the last 21 days of daily returns and computing their volatility (inner):

$$\sigma_{\text{inner}}(i, t) = \text{std}(\{r_{i,\tau}\}_{\tau=t-21}^t) \quad (3.17)$$

We then smooth this volatility over the last 3 months by computing its standard deviation over the last 64 days:

$$\text{VoV}(i, t) = \text{std}(\{\sigma_{\text{inner}}(i, \tau)\}_{\tau=t-64}^t) \quad (3.18)$$

We finally normalize by annualizing the standard deviation of the volatility for comparison:

$$\overline{\text{VoV}}(i, t) = \frac{1}{256} \sum_{\tau=t-256}^t \text{VoV}(i, \tau) \quad (3.19)$$

We finally multiply the signal by the sign of the price change of the last 64 days to obtain the direction:

$$f_{\text{vov}}(i, t) = \text{sign}(\text{pct_change}_{64}(P_{i,t})) \times \left(- \left(\frac{\text{VoV}(i, t)}{\overline{\text{VoV}}(i, t)} - 1 \right) \right) \quad (3.20)$$

Instruments in periods of high VoV underperform in terms of risk-adjusted returns, the premium demanded by investors to bear this strong ambiguity being integrated into the price before the dissipation of the shock (Baltussen, van Bakkum & van der Grient, 2018). This signal allows us to brake the strategy when these periods appear.⁶²

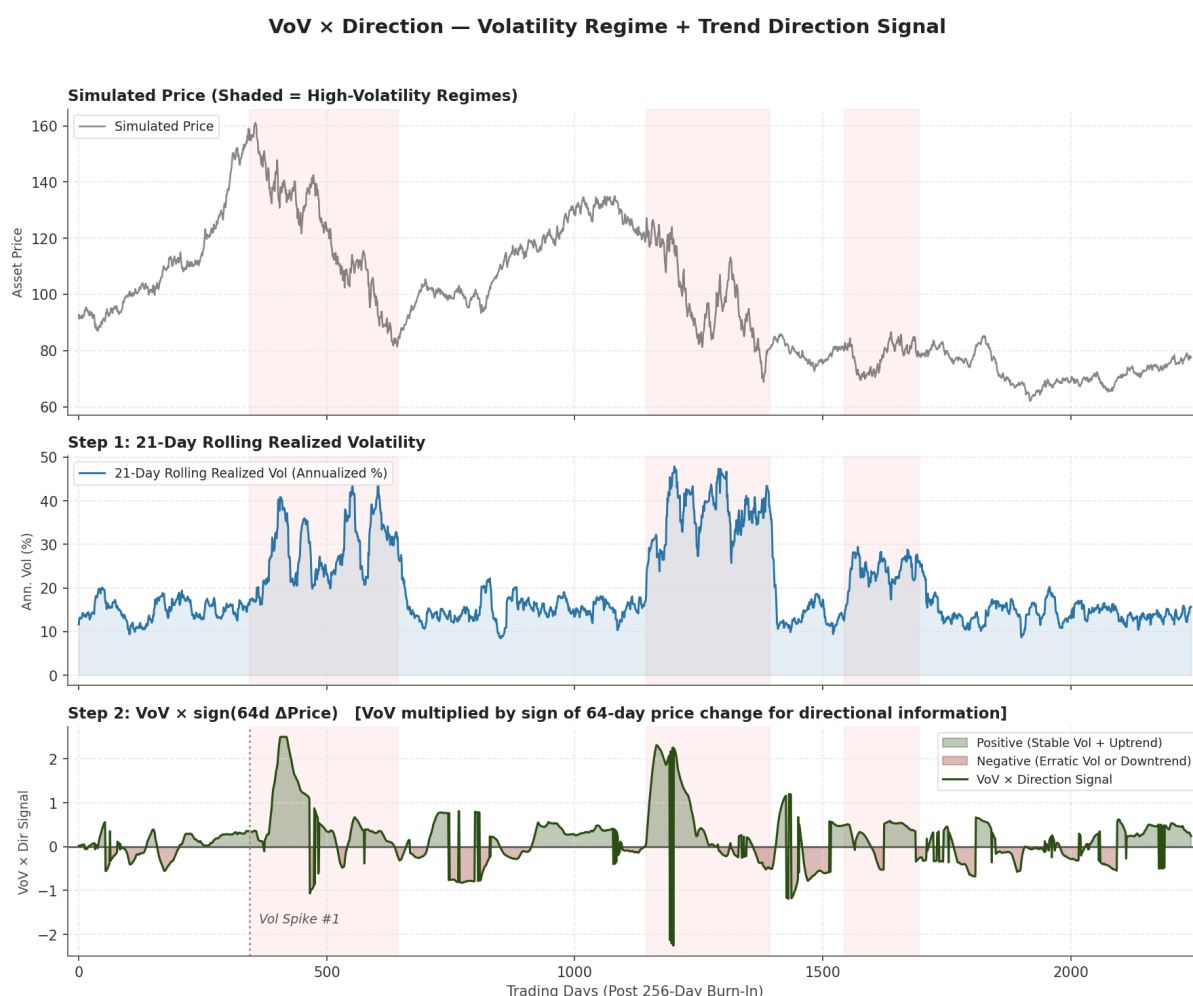


Exhibit 52. Visual illustration of the vol-of-vol signal.

These 4 signals exploit different phenomena and allow strong diversification, having an average correlation close to 0.3. We mechanically diversify the portfolio by allowing the strategy to generate profits in periods where a pure trend exploitation would experience losses.⁶³

The choice of a $1/N$ weighting lies in a concern for sample fitting,⁶⁴ drawn from DeMiguel-Garlappi-Uppal (2009) who show that the $1/N$ rule dominates *out-of-sample* the vast majority of allocations optimized on seven standard portfolio strategies. Optimizing *ex post* the weights of the four alphas on 36 years of daily data would introduce four additional degrees of freedom without expected out-of-sample gain ; we prefer the robustness of equal weighting. A Simplex optimization of these weights is moreover tested in appendix §10 and confirms that $1/N$ lies in the upper quartile of the distribution of random Dirichlet allocations.

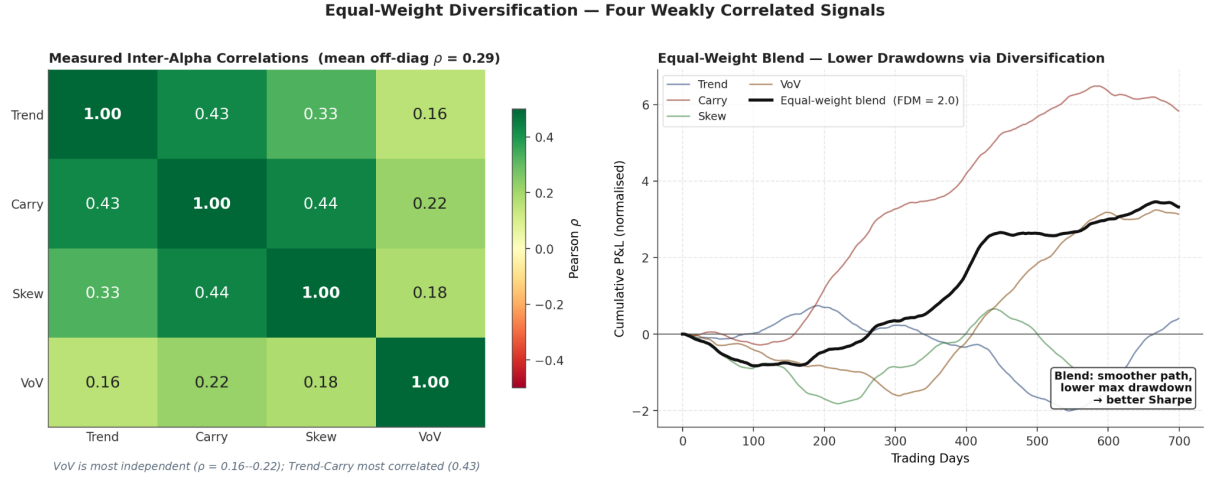


Exhibit 53. 1/N-weighted aggregation of the four signals (trend, carry, skew, VoV).

V. Normalization of signals — By construction, our signals result in values that have no meaning as such. We normalize them so that their absolute value converges to Φ over a rolling window W :

$$S_t = \frac{\Phi}{\frac{1}{W} \sum_{\tau=t-W}^t |f_\tau|}, \quad \Phi = 10, W = 1,280 \quad (3.21)$$

Thus, a signal of +10 is interpreted as a normal upward conviction; -10 a normal downward conviction; +20 a strong upward conviction and -20 a strong downward conviction. The maximum signal is 20, according to Carver's methodology, allowing to prevent a nonsensical signal from taking incoherent positions.⁶⁵ All the signals of our system thus have an equal weight and correctly reflect our weighting of 25% of the capital to each one of them.

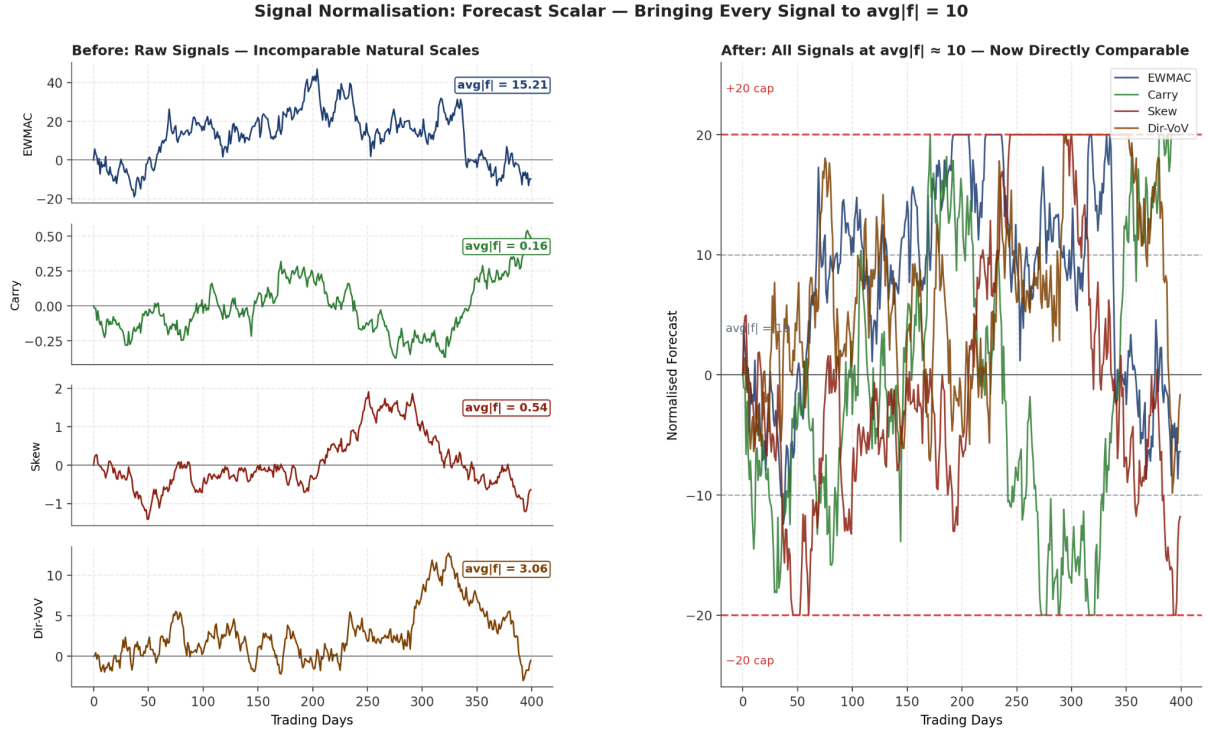


Exhibit 54. Scaling of signals to the target ± 20 (Carver).

VI. Signal aggregation and internal correlation. — Every day and for each instrument, we have an aggregated signal:

$$f_{\text{raw}} = 0.25 f_{\text{trend}} + 0.25 f_{\text{carry}} + 0.25 f_{\text{skew}} + 0.25 f_{\text{vov}} \quad (3.22)$$

However, our 4 signals being relatively decorrelated, the aggregated signal is mechanically weaker than each of the individual signals. As a result, the strategy is systematically under-engaged. To compensate for this problem, we apply a multiplier that seeks to correct this under-allocation by directly increasing the signals. This multiplier, called FDM, is directly taken from Carver but slightly modified so that its maximum is obtained analytically:

$$\text{FDM}_t = \min\left(2.0, \max\left(1.0, \frac{1}{\sqrt{\mathbf{w}^\top \mathbf{R}_t \mathbf{w}}}\right)\right) \quad (3.23)$$

where

$$\mathbf{w} = [0.25, 0.25, 0.25, 0.25]^\top,$$

$\mathbf{R}_t = 4 \times 4$ correlation matrix by exponentially weighted average, aggregated over the 62 instruments. This variable is generally computed by instrument, we compute it on the whole universe out

of concern for *overfitting*. Our limit of 2 is the theoretical maximum of 4 equally weighted uncorrelated signals : $1/\sqrt{4 \times 0,25^2} = 2,0$.⁶⁶

This variable essentially serves as leverage on the strategy. To ensure that it remains stable, we must re-normalize it:

$$f_{\text{fdm}}(i, t) = f_{\text{raw}}(i, t) \times \text{FDM}_t \quad (3.24)$$

$$S_t^{(2)} = \frac{10}{\frac{1}{W} \sum_{\tau=t-W}^t |f_{\text{fdm}}(i, \tau)|}, \quad W = 1,280 \quad (3.25)$$

$$f_{\text{master}}(i, t) = \text{clip}\left(f_{\text{fdm}}(i, t) \times S_t^{(2)}, -20, +20\right) \quad (3.26)$$

Without this second normalization, the magnitude of the signal would drift and 10 would no longer be a normal position.

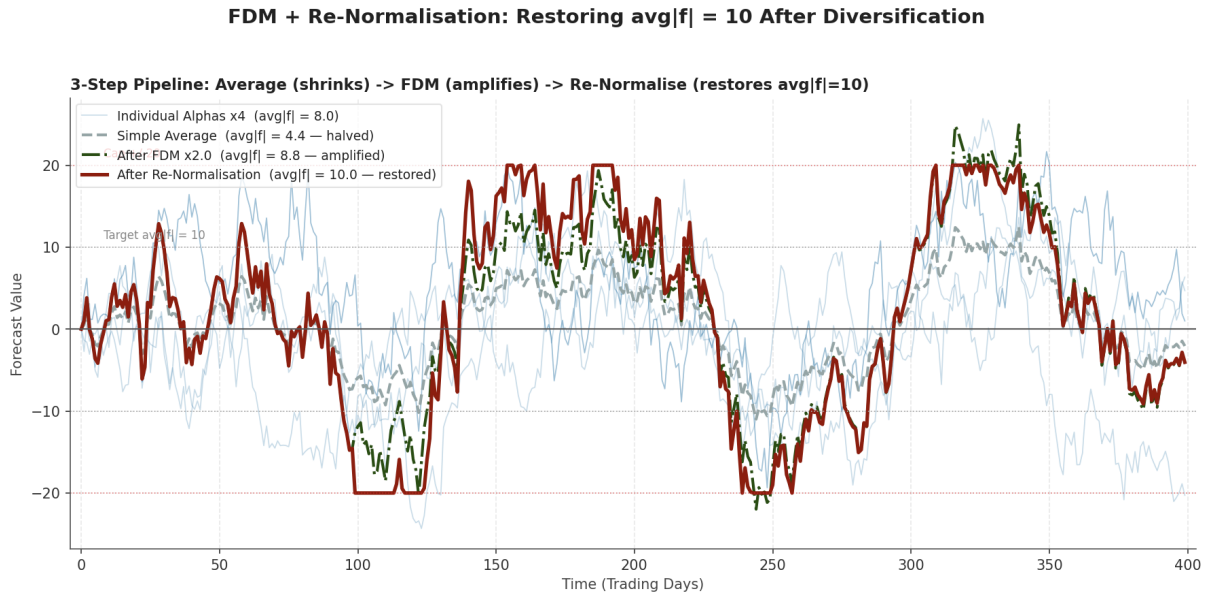


Exhibit 55. Application of the pooled FDM on the whole universe and re-normalization of the signal by a rolling scale factor.

VII. Signal smoothing — The aggregated signal we obtain is at this stage strongly volatile, we smooth it with an EWM of halflife 1 day, which is the rate at which the portfolio rebalances:

$$f_{\text{smooth},t} = \alpha f_t + (1 - \alpha) f_{\text{smooth},t-1}, \quad \alpha = 1 - e^{-\ln 2/1} = 0.5 \quad (3.27)$$

Concretely, we give our signal a short memory.

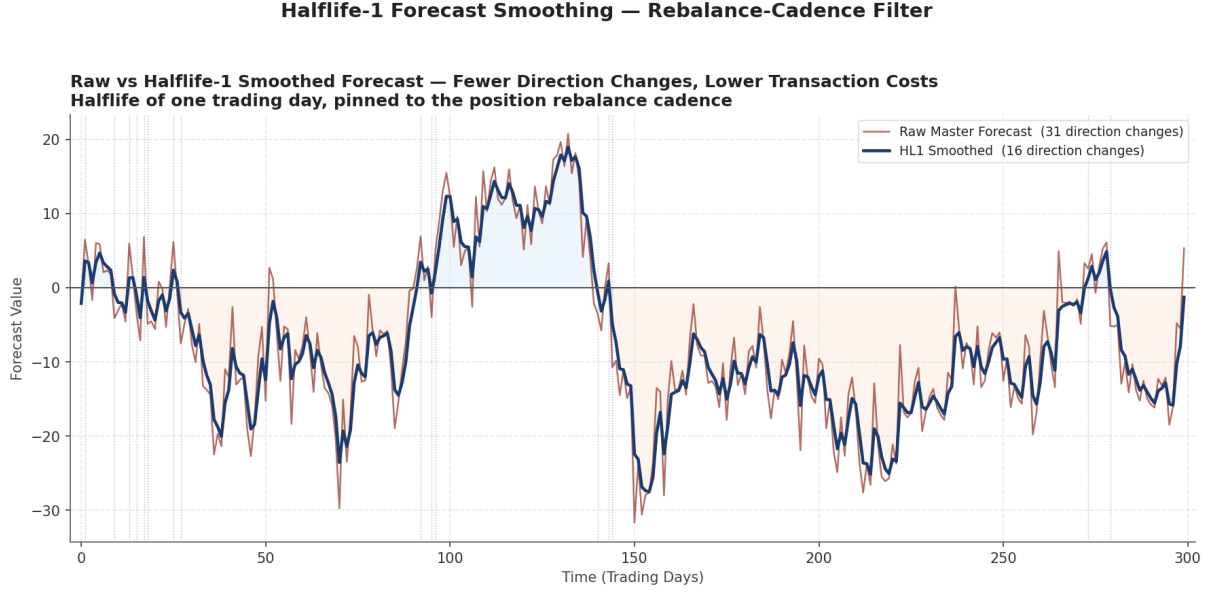


Exhibit 56. EWM smoothing (halflife = 1 day) of the aggregated signal.

VIII. Risk management — We have proposed a bounded solution in our signals, and we do the same with the volatility of the portfolio by targeting the volatility itself and the drawdown. To do so, we use a sigmoid function:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (3.28)$$

The target volatility allocated to each instrument is 20%. Although this target may seem high, our portfolio is large and diversified, effectively leaving room for Markowitz's efficiency theory and mechanically delivering a lower volatility. Thus, when the latent volatility of an instrument exceeds our target of 20%, we use the sigmoid function to continuously reduce our exposure in the instrument:

$$g_{\text{vol}}(i, t) = 1 - (1 - 0.50) \cdot \sigma\left(10.0 \cdot \left(\frac{\hat{\sigma}_{i,t}}{\sigma_{\text{target}}} - 1.0\right)\right) \quad (3.29)$$

The maximum reduction of the exposure is 50%, regardless of the extremity of the volatility. This choice is made out of concern for *overfitting*, and avoids the strategy being performant simply because it is dormant.⁶⁷

We apply the same principle to the latent drawdown of the last 64 days of the strategy on the instrument when it falls more than -10% (shifted by one day to avoid *data-snooping*) :

$$g_{\text{dd}}(i, t) = 1 - (1 - 0.50) \cdot \sigma\left(10.0 \cdot (-0.10 - \text{DD}_{i,t}^{64})\right) \quad (3.30)$$

The maximum reduction of the exposure is 50%, regardless of the amplitude of the drawdown for the same reasons as previously stated. The choice of -10% rather than -5% or -20% is derived analytically: $\text{vol_target}/2 = 20\%/2 = 10\%$. Concretely, if the instrument consumes more than half of its risk budget allocated annually over 3 months, the instrument is probably in a regime adverse to our strategy and we thus reduce our exposure. The slope coefficient of 10 in the function is also derived analytically from the target volatility: $k = 2/\text{vol_target} = 2/0.20 = 10$; allowing the sensitivity of the function to be directly linked to the target volatility of the strategy insofar as it goes from quasi-inactive to quasi-saturated around the trigger threshold: $\text{sigma}(+2) = \sim 0.88$ and $\text{sigma}(-2) = \sim 0.12$.⁶⁸ The slope k and the drawdown threshold are therefore both derived from τ : $k = 2/\tau$ and $\text{threshold} = -\tau/2$. Modifying τ automatically recalibrates both quantities. We thus multiply our two bounds by our signals to obtain our final signal:

$$f_{\text{final}}(i, t) = \text{clip}(f_{\text{smooth}}(i, t) \cdot g_{\text{vol}}(i, t) \cdot g_{\text{dd}}(i, t), -20, +20) \quad (3.31)$$

Our two bounds serve as an emergency lever, able to reduce the signal on an instrument up to 25% of its full position when the volatility is extreme and the drawdown is excessive.⁶⁹

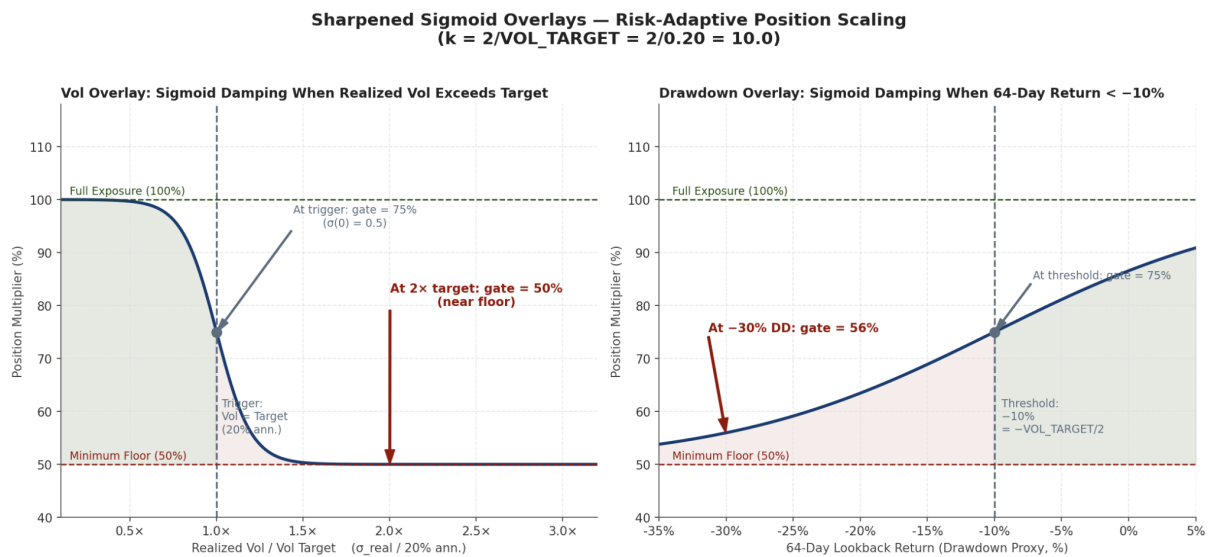


Exhibit 57. Sigmoid function attenuating positions under extreme volatility or drawdown.

The realized volatility of the strategy is therefore necessarily lower than the target, since on the target variables, the positions are already reduced to 0.75 when they are exactly at $\text{vol}=20$ or $\text{drawdown} = -10\%$.

IX. Execution in contracts — Once our signals are fully built, we need to translate them into a number of contracts. To do so:

$$N_{i,t} = \frac{f_{\text{final}}(i, t)}{10} \cdot \frac{E_t \cdot \tau \cdot w_i \cdot \text{IDM}_t}{\hat{\sigma}_{i,t}^{\$}} \quad (3.32)$$

where

E_t = NAV at date t ,
 $\tau = 0.2$ (target volatility),
 $w_i = 1/N$ (equal weighting),
 IDM_t = internal correlation measure, bounded $[1, 4]$ according to Carver, computed as follows:

$$\text{IDM}_t = \min \left(4.0, \frac{1}{\sqrt{\mathbf{w}_{\text{inst}}^\top \mathbf{C}_t \mathbf{w}_{\text{inst}}}} \right) \quad (3.33)$$

where

$\mathbf{w}_{\text{inst}}$ = vector of $1/N$ weightings on the instruments active at date t ,
 $\mathbf{C}_t = N \times N$ correlation matrix between the returns of the active instruments, estimated by EWM,
 4.0 = Carver's cap, namely $\sqrt{N}$ for $N = 16$ perfectly decorrelated instruments.

The IDM_t allows the strategy not to be permanently in under-regime due to the strong diversification of the universe. Conversely, if the correlations of the universe rise strongly, it allows to reduce the exposure to a level of 1, which is the normal level.⁷⁰

We first define $\hat{\sigma}_{i,t}$, the daily volatility of the price variations of the instrument i at date t (expressed in the local price units of the contract), computed according to Carver by mixing a short-term component (EWM 32 days) and a long-term component (rolling average 2,560 days) :

$$\hat{\sigma}_{i,t} = 0.70 \cdot \text{EWM}_{32}(\text{std}(\Delta P_i))_t + 0.30 \cdot \text{Rolling}_{2560}(\text{std}(\Delta P_i))_t \quad (3.34)$$

The *annualized* dollar volatility of a contract, used in the size $N_{i,t}$ above, is then obtained by unit conversion (multiplication by the contract's *pointsize* and the USD exchange rate of the quotation currency) and annualization by $\sqrt{256}$:

$$\hat{\sigma}_{i,t}^{\$} = \hat{\sigma}_{i,t} \cdot \text{pointsize}_i \cdot \text{fx}_{i,t} \cdot \sqrt{256} \quad (3.35)$$

This formula mixes recent volatility with long-term volatility to obtain a dynamic estimate while preserving a certain memory and thus avoiding nonsensical values.⁷¹

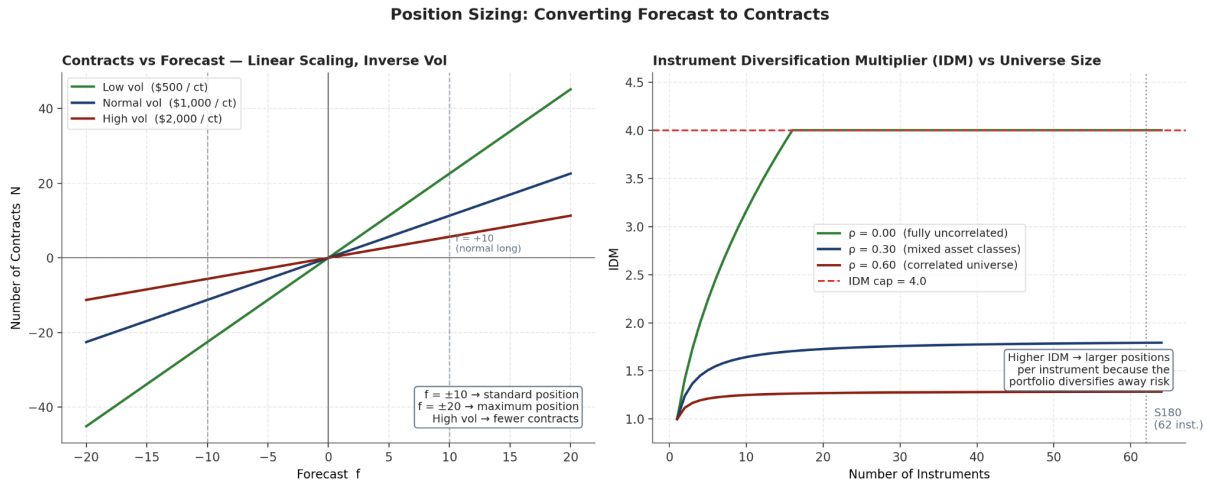


Exhibit 58. Volatility estimation by EWM + long-term blend.

X. Reduction of transactions — In its current state, the strategy is continuous and rebalances daily. The direct problem we must smooth is therefore the number of transactions, insofar as an insignificant variation of a signal purely due to noise still forces the strategy to rebalance unnecessarily. To erase these fees, we only modify a position when the latter deviates by 10% from the current position:

$$N_{i,t}^{\text{exec}} = \begin{cases} N_{i,t}^* & \text{if } |N_{i,t}^* - N_{i,t-1}| > b \cdot |N_{i,t-1}|, \\ N_{i,t-1} & \text{otherwise,} \end{cases} \quad b = 0.10 \quad (3.36)$$

The choice of this 10% buffer is directly taken from Carver, and is arbitrary enough that we keep it as such.⁷²

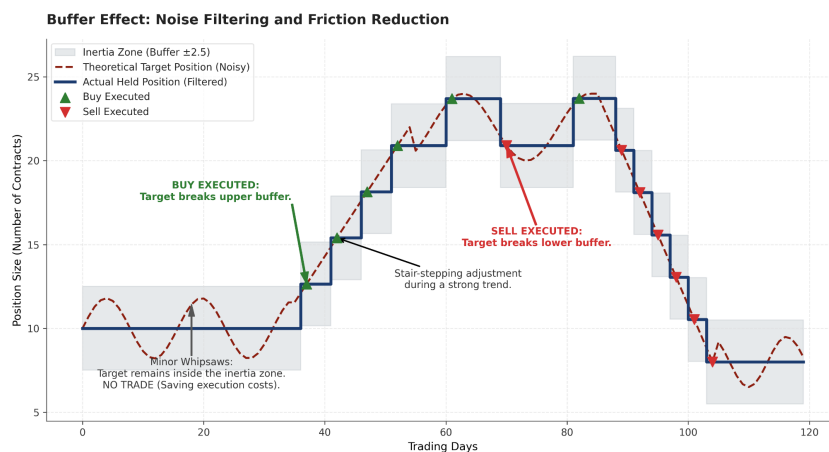


Exhibit 59. 10 % buffer on position changes to reduce transactions.

XI. Cash management — The strategy operates exclusively through futures contracts on exchange, allowing us to post the collateral by investing the capital in the US Treasury bills rate

(%IRX). The strategy thus produces interest at this spot historical rate on its NAV daily. In practice, a fund would not actually receive this return which must be reduced by the clearing house haircut, all the more so as the fund would have to keep a certain part of cash to be able to respond to potential redemptions. This simplification therefore somewhat overestimates the returns, but does not significantly influence the ratios and facilitates the construction of the architecture.

$$\text{day_cash}_t = \frac{\text{IRX}_t}{100 \times 256} \times E_t \quad (3.37)$$

where

IRX_t = annualized yield of the 13-week T-bill at date t , expressed in percentage,

E_t = total NAV of the portfolio,

100 = conversion factor percentage to fraction,

256 = *de*-annualization factor (256 trading days) of the annual yield.

All the Sharpe ratios that we will produce will therefore be correctly adjusted for this rate to reflect the trading performance and not be inflated by the risk-free yield.

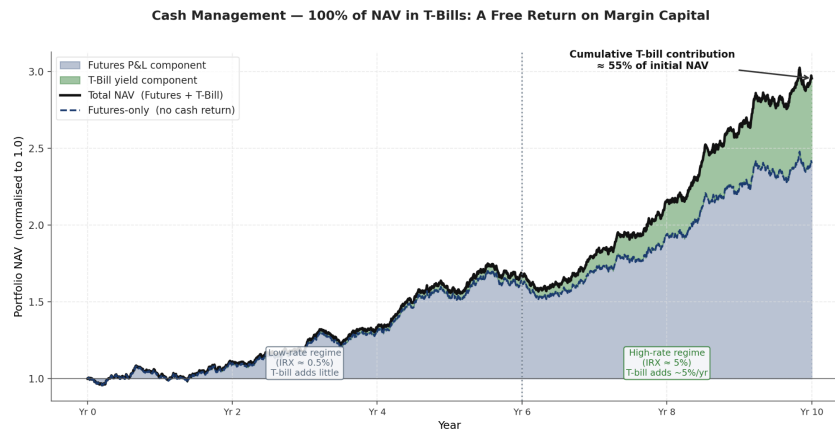


Exhibit 60. Capitalization of collateral at the IRX rate (13-week T-bill) for cash management.

XII. Summary. —

Full Strategy Pipeline

Per instrument · Per day · 62 global futures

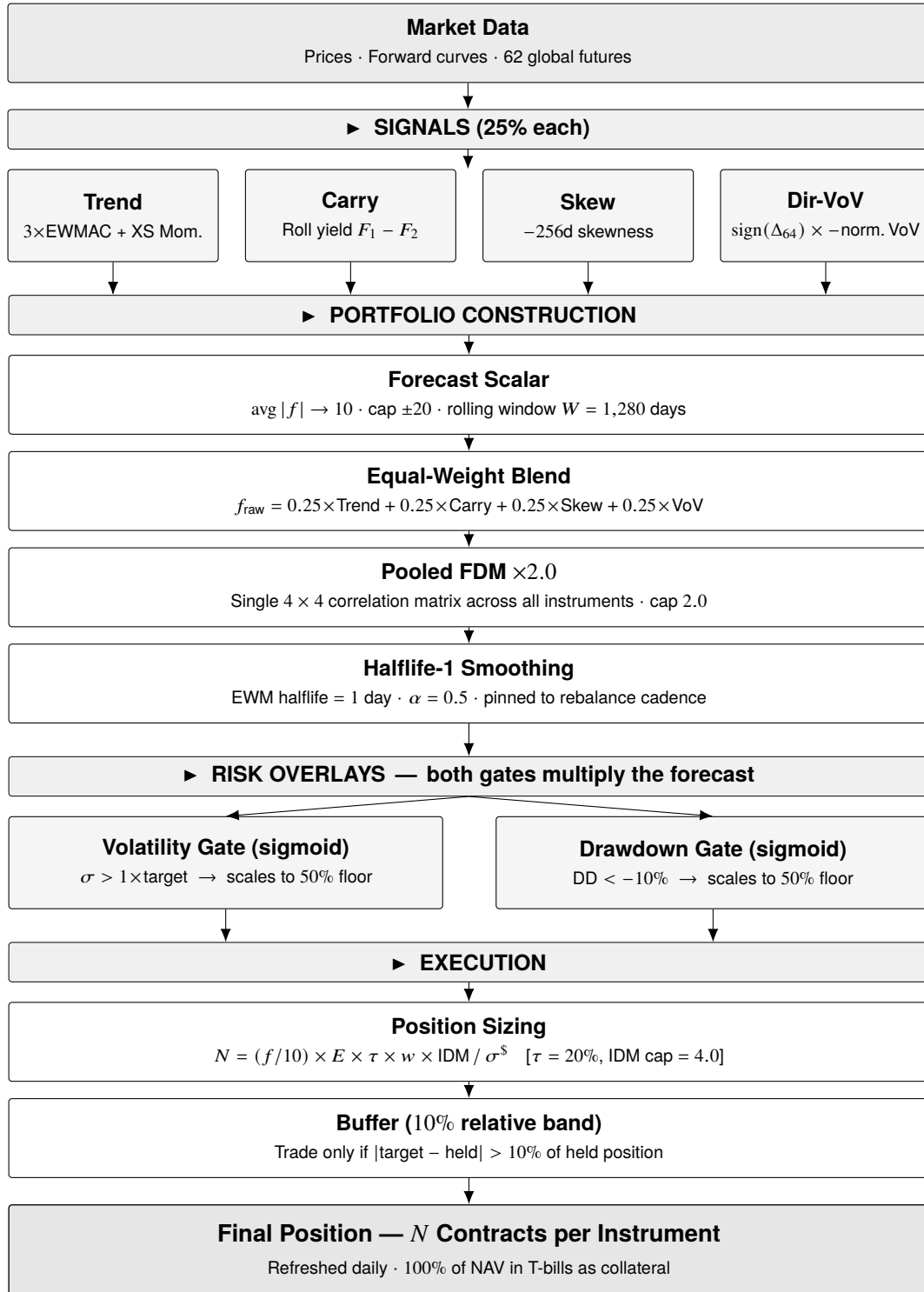


Exhibit 61. Complete architecture producing the daily position in contracts $N_{i,t}$ for each instrument i in the strategy. The market data feed four alpha signals weighted at 25% (*trend*, *carry*, *skew*, *Dir-VoV*) ; the portfolio construction normalizes the *forecasts* via a scale factor, aggregates them equally, applies the pooled 4×4 FDM and an EWM smoothing of half-life 1 ; two sigmoid risk gates (*volatility* and *drawdown*) multiply the *forecast* before execution converts it into an integer number of contracts via the vol-target sizer with a relative buffer of 10%.

The strategy being constructed, we propose to study its characteristics and comment on the performance.

3.2 Backtest

“There are three kinds of lies: lies, damned lies, and backtests.”⁷³

Thus, past returns are no guarantee of future returns. This phrase, which has an axiomatic feel, is a tautology in its own right, which we must consider without being fearful. Something that has never worked in the last 35 years may by chance work in the future. Something that has systematically worked in the last 35 years too. We nevertheless prefer to lean toward the second rather than the first.

Furthermore, we do not take any *backtest* result as an absolute truth, in this case regarding its ability to produce out-of-sample profits in the future. The objective of this part and more generally of the study is to demonstrate that trend-following CTA strategies are still viable post-2010. It follows that the ratios and performances we produce are merely ancillary; what matters is an improvement in performance, not these performances themselves.

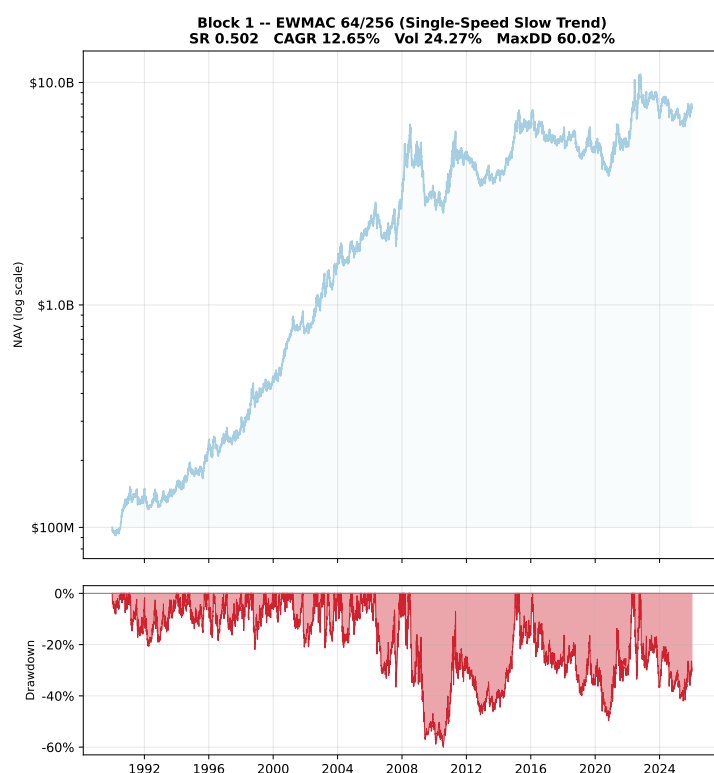
We also strive to produce the most adverse statistical tests to our knowledge in the appendix with a rigorous and conservative approach regarding the results we present. If other methods or means are available and we have not tested them, either we do not know them, or they require technical means impossible at our scale.

All the results we propose follow the methodological notes previously put forward, to which we add some details:

- 1) All strategies start on 1990-01-03. At this stage, the universe is 27 instruments, large enough to be realistically invested. Moreover, this year is the one that Andreas Clenow takes up in his first book of 2013 for a similar exercise. The possible selection bias is entirely addressed in our test appendix.
- 2) All strategies start with a capital of 100 million USD. The choice is arbitrary although it deserves a longer discussion that we will address later.
- 3) All strategies reinvest their profits and dynamically weight their positions according to the NAV.
- 4) All strategies take as target volatility variable 20%.
- 5) All SR ratios are presented systematically with the subtraction of the risk-free rate.

The foundation on which we have relied to build the strategy rests directly on Carver’s iteration number 6: a simple continuous *long-short* strategy with an EWMA of 64-256.⁷⁴

This first iteration already proposes interesting ratios and an unbeatable degree of parsimony. A Sharpe of 0.5 over 36 years with such simplicity is the most direct proof of the success of these strategies in the past. The skew of all our results is slightly biased since it is calculated on a daily basis; a weekly or monthly aggregation gives smoother results. We depart from Robert Carver on this point and adopt this method. The important point of these results, however, is the post-2010 underperformance, with ratios that systematically deteriorate. Macro-economic explanations are easily found to justify this underperformance, notably from the side of strong intervention by central banks. Nevertheless, this real underperformance translates into a prolonged absence of the long, slow trends found in the 1990s and 2000s.

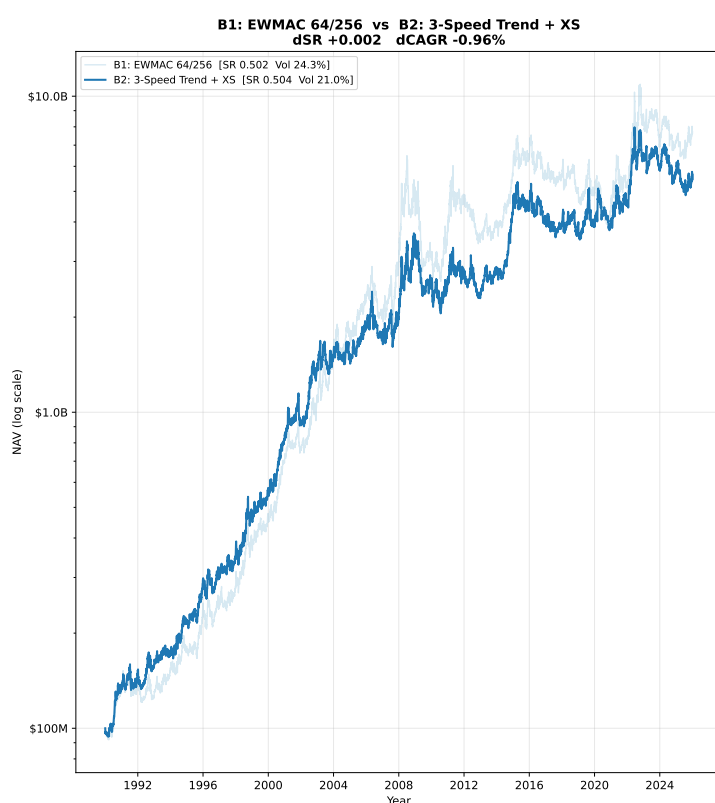


Metric	Value
Panel A: Full Sample	
Sharpe Ratio	0.50
CAGR	12.65%
Sortino Ratio	0.66
Calmar Ratio	0.21
Skew	-0.48
Lower Tail	1.27
Upper Tail	1.07
Ann. Volatility	24.27%
Total Trades/Yr	3,309
Panel B: Post-2010	
Sharpe Ratio	0.30
CAGR	4.48%
Sortino Ratio	0.38
Calmar Ratio	0.09
Skew	-0.69
Lower Tail	1.31
Upper Tail	1.01
Ann. Volatility	24.95%

Exhibit 62. Block 1 — EWMAC 64/256 (Single-Speed Slow Trend).

The first corrective approach is similar to *ensemble learning*, where we will simply diffuse information over several positions: a short-term environment, a medium-term one and a final long-term one. To do this we take up the proposals of Ari Levine, Lasse Heje Pedersen (2016) by dividing the existing EWMAC in two for the medium window and dividing it by 4 for the short window.⁷⁵ The strategy is therefore able to

detect three dimensions of trends, allowing it to be more precise in the post-2010 environment where these become rarer. Moreover, we favor instruments in stronger trends by comparing these continuously through a Z-score:



Metric	Value
--------	-------

Panel A: Full Sample

Sharpe Ratio	0.50
CAGR	11.69%
Sortino Ratio	0.67
Calmar Ratio	0.26
Skew	-0.42
Lower Tail	1.26
Upper Tail	1.09
Ann. Volatility	20.96%
Total Trades/Yr	4,658

Panel B: Post-2010

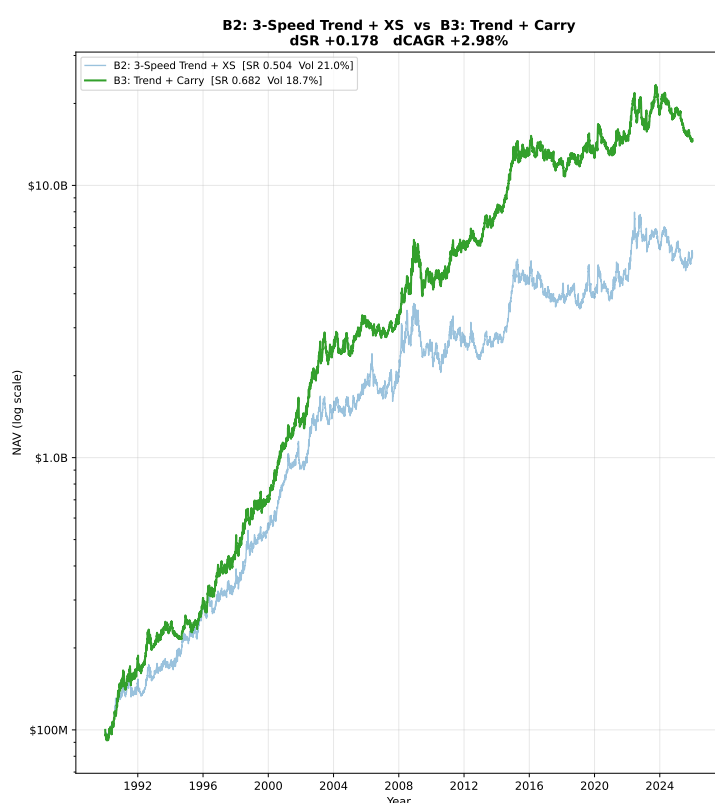
Sharpe Ratio	0.28
CAGR	3.71%
Sortino Ratio	0.36
Calmar Ratio	0.08
Skew	-0.54
Lower Tail	1.30
Upper Tail	1.05
Ann. Volatility	21.54%

Exhibit 63. Block 2 — 3-Speed EWMAC + XS Momentum.

This iteration allows better stability of the strategy with a slight improvement of the ratios, but above all a better control of volatility. Let us also note that the Sharpe ratio deteriorates slightly post-2010. The simple fact that it deteriorates only marginally despite the massive increase in the number of transactions and therefore fees demonstrates that we capture additional profits while having a more reactive management of volatility. It is however dubious to want to add such a number of parameters for such a slight improvement; we defend them by purely economic arguments: there exist economic cycles at short, medium and long scales. Manifestly, trends of the same calibers must exist; it is natural to want to measure them. Of course, we could add an infinity of these measurements to theoretically have a more precise result; we avoid this path to keep Occam's razor sharp. Moreover, events, whatever they may be, naturally lead some instruments to experience stronger and more pronounced trends than others at a given moment; it is also natural to want to measure these interactions.

Concretely, here we merely add granularity to the signal by adding the EWMACs and the Z-score.

It is however in a diversification of signals that the real improvements are found; the framework $IR \approx IC\sqrt{BR}$ of Grinold & Kahn (2000) precisely quantifies this gain by the square root of the *breadth*, and is why we propose to diversify the strategy with carry⁷⁶ :



Metric	Value
Panel A: Full Sample	
Sharpe Ratio	0.68
CAGR	14.67%
Sortino Ratio	0.94
Calmar Ratio	0.38
Skew	-0.31
Lower Tail	1.18
Upper Tail	1.05
Ann. Volatility	18.71%
Total Trades/Yr	4,296
Panel B: Post-2010	
Sharpe Ratio	0.43
CAGR	6.29%
Sortino Ratio	0.58
Calmar Ratio	0.14
Skew	-0.45
Lower Tail	1.22
Upper Tail	1.01
Ann. Volatility	18.05%

Exhibit 64. Block 3 — Trend (50%) + Carry (50%).

Carry is nothing other than the discount rate of the forward price of an instrument. Although the term is specific to the lexicon of futures, the concept is more commonly understood when the subject is the yield curve for fixed instruments: a 10-year OAT trading at 95 will at maturity be worth 100 in all cases barring default of the sovereign system, mechanically generating a gain of 5.26%. Depending on the remaining maturity of the OAT at the moment it trades at 95, we can derive the forward discount rate. The exploitation of this phenomenon is identical for our debt instruments, is called *cost of carry* or *roll yield* for our commodities and interest rate differential for currencies. The spot exploitation of the curves of these instruments requires taking into account intrinsic differences that we can entirely bypass on the futures market, the latter being systematized on exchange.

This iteration weights 50% of the capital to carry and 50% to the previous block. The extraction of a diversifying source of alpha (weakly correlated, not strictly orthogonal) strongly improves performance at all levels and is entirely justified from a parsimony point of view. The strategy

necessarily departs from a pure trend extraction to gain in stability but remains perfectly within the universe of CTA funds. The hunt for alpha sources is the core business of fund research and can be extremely pushed; to avoid the risk of *overfitting* we strive to consider only the simplest additions with the most obvious economic foundations.

Thus, the last weakly correlated signal that we wish to inject into the strategy is a sale of insurance by measuring skew⁷⁷ :



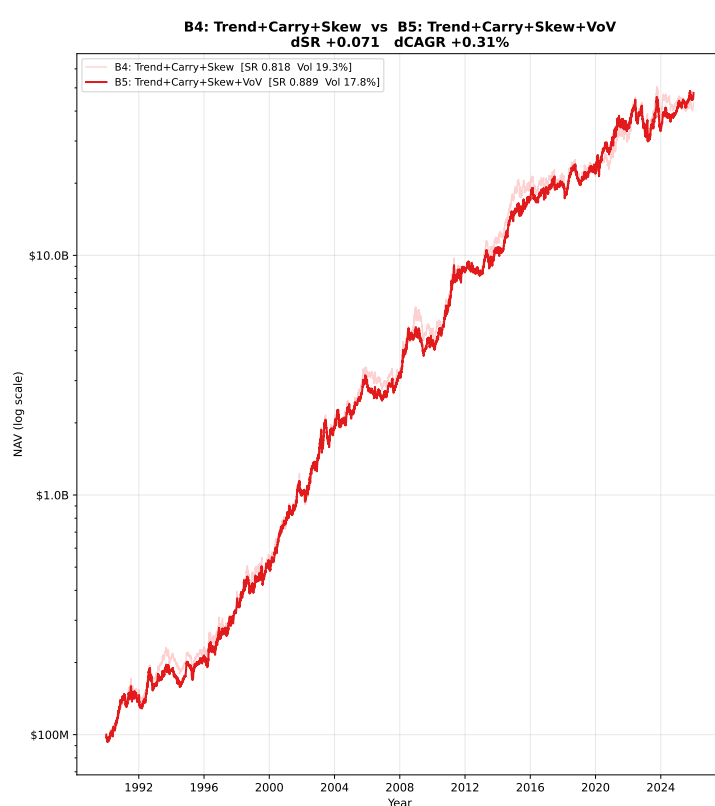
Metric	Value
Panel A: Full Sample	
Sharpe Ratio	0.82
CAGR	18.10%
Sortino Ratio	1.12
Calmar Ratio	0.57
Skew	-0.44
Lower Tail	1.18
Upper Tail	1.04
Ann. Volatility	19.34%
Total Trades/Yr	4,129
Panel B: Post-2010	
Sharpe Ratio	0.73
CAGR	13.10%
Sortino Ratio	0.96
Calmar Ratio	0.39
Skew	-0.63
Lower Tail	1.22
Upper Tail	1.01
Ann. Volatility	19.55%

Exhibit 65. Block 4 — Trend (34%) + Carry (33%) + Skew (33%).

Investors overpay for payoffs with strong positive asymmetry (lotteries, deeply out-of-the-money options) and underprice exposures to rare but severe losses (insurance). From this empirically documented imbalance (Fernandez-Perez, Frijns, Fuertes & Miffre, 2018) follows a structural *premium* for the seller of asymmetry. We propose to play the role of the *market-maker* of this asymmetry to improve the performance of the strategy. There is nevertheless a contradiction in wanting to sell positive skew when the principle of a trend strategy is to exploit this characteristic in order to have rare large gains covering the small but numerous small losses. We are aware that our proposal departs from a purely trend strategy and justify it: we still respond to our specifications on one hand; it is fundamentally not possible to significantly improve our first iteration of pure trend strategy without bringing in less pure alpha sources on the other.

This addition, weighted at $1/N$ with the two other sources, strongly stabilizes the strategy and pushes the efficiency of the portfolio, allowing a large improvement in ratios. (Weighted at 34% for trend by simplification in this iteration.) The price to pay for these improvements is however a strong degradation of the skew of the strategy. Let us also note that the difference in performance diminishes significantly at this iteration compared to the previous ones, corroborating the argument that: lacking diversification of instruments, trend strategies can be improved, even in adverse regimes, with a diversification of strategies.

The last signal we propose is purely defensive, and is based on a rationale of uncertainty not of returns, but of the volatility of these⁷⁸ :



Metric	Value
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Panel A: Full Sample	
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Sharpe Ratio	0.89
CAGR	18.41%
Sortino Ratio	1.23
Calmar Ratio	0.56
Skew	-0.45
Lower Tail	1.17
Upper Tail	1.05
Ann. Volatility	17.75%
Total Trades/Yr	4,650

Panel B: Post-2010	
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Sharpe Ratio	0.84
CAGR	14.56%
Sortino Ratio	1.11
Calmar Ratio	0.42
Skew	-0.61
Lower Tail	1.24
Upper Tail	1.04
Ann. Volatility	18.11%

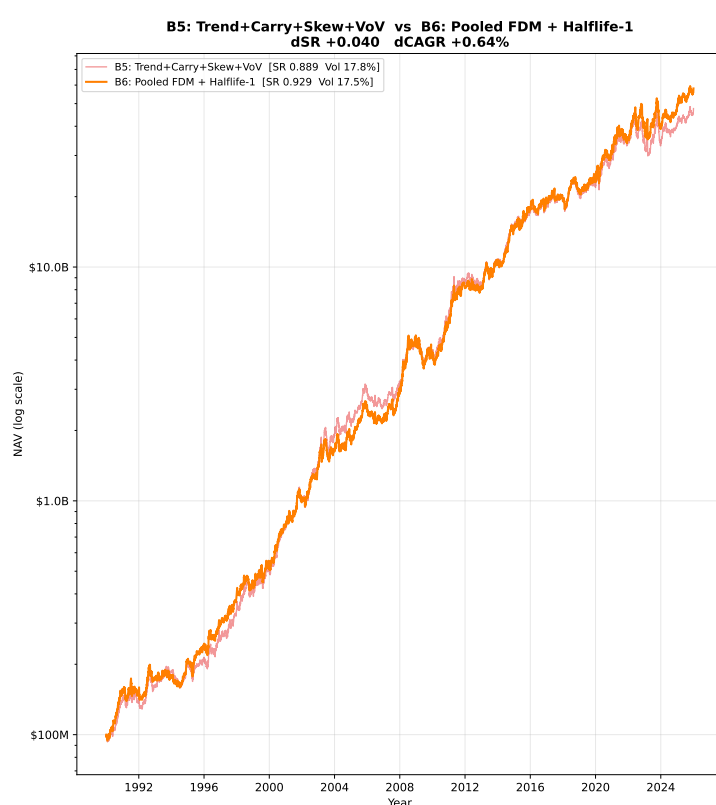
Exhibit 66. Block 5 — Quad Alpha (per-instrument FDM).

The volatility of volatility measures the uncertainty of uncertainty.⁷⁹ The economic argument supported is based on the premium demanded by investors to bear a greater risk in instruments having naturally a greater volatility. Thus, they also demand an additional premium when a volatility regime is also unstable, for example an instrument usually experiencing a strong volatility which suddenly ceases to be volatile. Our 4 signals are thus weighted at $1/N$, with each counting for 25% of the final signal.

More concretely, this signal serves as a measure of uncertainty in a dimension higher than the

volatility of pure returns, implying that periods of high VoV lead to adverse regimes for the strategy and conversely.

Furthermore, blocks 3 to 5 followed the FDM computation described by Carver, in this case instrument by instrument, where we sought to compute the 4x4 correlation matrix of the signals on each instrument. Our next iteration replaces this computation with a single 4x4 correlation matrix where each value is the average of the signal over the 62 instruments; this method smooths the values and avoids the intrinsic noise of daily returns, notably by renormalizing the FDM in a final step to ensure that the values remain stable. Moreover, the final signals are also themselves smoothed by computing an EWM with a halflife equal to our portfolio rebalancing frequency. These two corrections aim to stabilize the guidance of the strategy and make it less abrupt⁸⁰ :



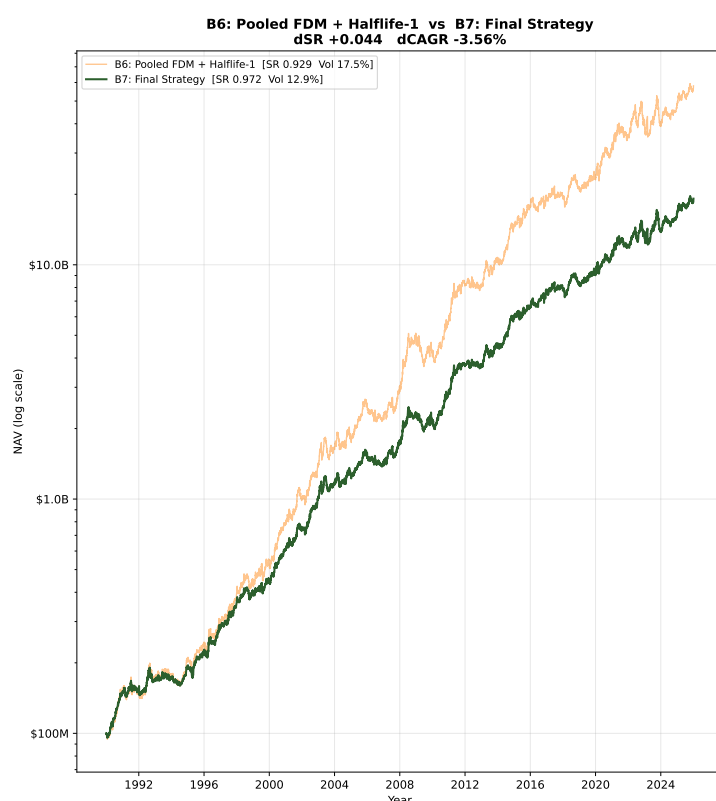
Metric	Value
Panel A: Full Sample	
Sharpe Ratio	0.93
CAGR	19.05%
Sortino Ratio	1.29
Calmar Ratio	0.64
Skew	-0.40
Lower Tail	1.17
Upper Tail	1.04
Ann. Volatility	17.54%
Total Trades/Yr	4,604
Panel B: Post-2010	
Sharpe Ratio	0.94
CAGR	16.17%
Sortino Ratio	1.25
Calmar Ratio	0.52
Skew	-0.56
Lower Tail	1.25
Upper Tail	1.02
Ann. Volatility	17.67%

Exhibit 67. Block 6 — Pooled FDM (cap 2.0) + Halflife-1 Smoothing.

This purely structural improvement allows a strong extraction of the real signals by discarding noise. One of the challenges of this strategy is that it mechanically receives 4 different signals per day, all subject to the vagaries of daily prices. Refining the filtering of signals and internal correlation allows discarding useless information and allows the strategy to maintain its performance post-2010.

Finally, the last element we propose is an emergency lever. Being *long-short* and continuous,

the strategy is perpetually in position on all the instruments at its disposal. The addition of our sigmoid function on drawdown and volatility per instrument allows strongly braking the exposure when events at more than ten standard deviations appear, or simply when an instrument decimates the pnl over a period⁸¹ :



Metric	Value
--------	-------

Panel A: Full Sample

Sharpe Ratio	0.97
CAGR	15.49%
Sortino Ratio	1.34
Calmar Ratio	0.72
Skew	-0.41
Lower Tail	1.16
Upper Tail	1.04
Ann. Volatility	12.90%
Total Trades/Yr	4,702

Panel B: Post-2010

Sharpe Ratio	0.99
CAGR	13.06%
Sortino Ratio	1.31
Calmar Ratio	0.58
Skew	-0.59
Lower Tail	1.22
Upper Tail	1.01
Ann. Volatility	13.28%

Exhibit 68. Block 7 — Final Strategy (fSigmoid Overlays).

Volatility naturally drops insofar as exposure is dynamically reduced on instruments where our conditions are unfavorable. This reduction is however what allows the strategy to display an even stronger maintenance of performance between the full sample and post-2010, by reducing exposure in instruments naturally experiencing the fewest trends. We summarize the iterations with a graph and a table :

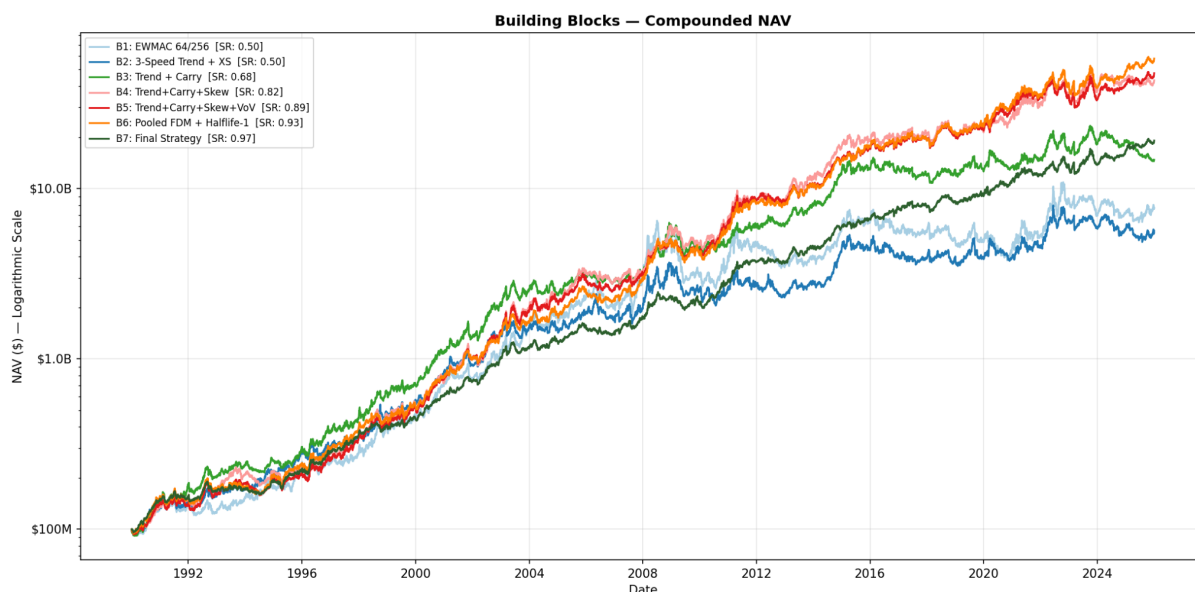


Exhibit 69. NAV curves of the 7 iterations (Blocks 1–7), logarithmic scale.

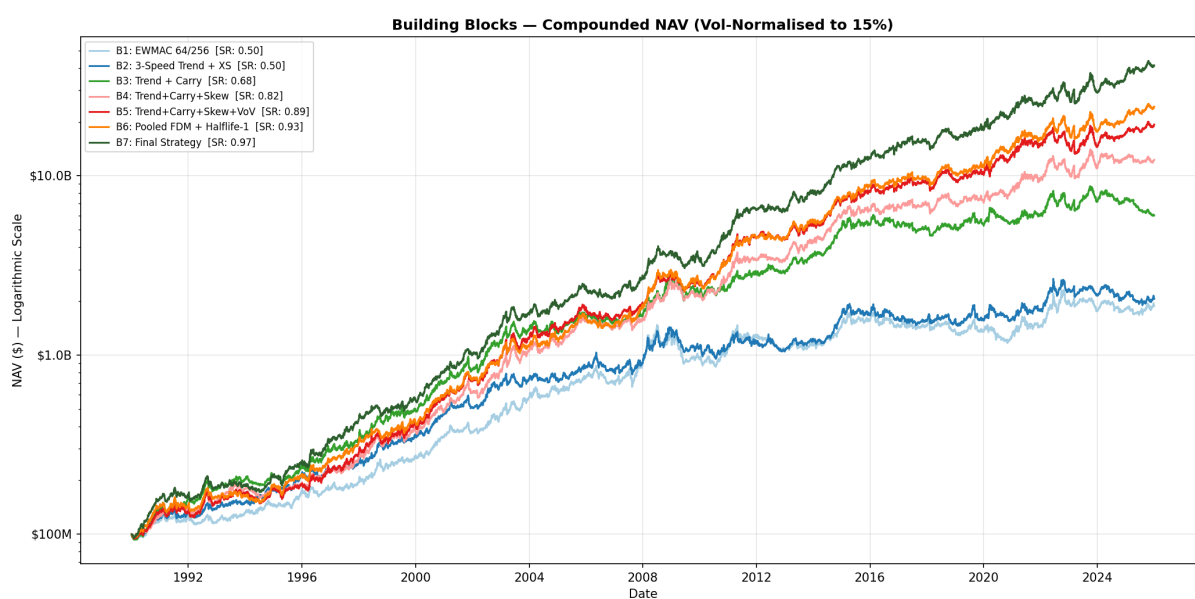


Exhibit 70. NAV curves of the 7 iterations normalized at 15 % volatility.

The diversification of strategies allows the portfolio to almost double its Sharpe ratio while decently improving the other ratios present, demonstrating that CTA strategies, if they allow themselves not to be purely trend-following, can strongly improve even in the most complex regimes. Let us also recall that our approach seeks to favor parsimony: the degree of optimization is nonexistent and the complexity of mechanics or alpha sources is voluntarily kept simple. It is not here a question of presenting the highest Sharpe but rather of demonstrating that a continuous, systematic and *long-short* strategy is capable of benefiting from Markowitz's theorem without increasing its diversification of instruments but only that of its strategies.

Exhibit 71. Strategy Building Blocks

B1: EWMAC 64/256, B2: 3-Speed Trend + XS, B3: Trend + Carry, B4: Trend+Carry+Skew, B5: Trend+Carry+Skew+VoV, B6: Pooled FDM + Halflife-1, B7: Sigmoid Overlays

Metric	Block 1	Block 2	Block 3	Block 4	Block 5	Block 6	Block 7
Panel A: Full Sample							
Sharpe Ratio	0.50	0.50	0.68	0.82	0.89	0.93	0.97
Sortino Ratio	0.66	0.67	0.94	1.12	1.23	1.29	1.34
Calmar Ratio	0.21	0.26	0.38	0.57	0.56	0.64	0.72
Skew	-0.48	-0.42	-0.31	-0.44	-0.45	-0.40	-0.41
Lower Tail	1.27	1.26	1.18	1.18	1.17	1.17	1.16
Upper Tail	1.07	1.09	1.05	1.04	1.05	1.04	1.04
CAGR	12.65%	11.69%	14.67%	18.10%	18.41%	19.05%	15.49%
Ann. Volatility	24.27%	20.96%	18.71%	19.34%	17.75%	17.54%	12.90%
Max Drawdown	-60.02%	-44.23%	-38.18%	-31.75%	-33.15%	-29.56%	-21.59%
Total Trades/Yr	3,309	4,658	4,296	4,129	4,650	4,604	4,702
Panel B: Post-2010							
Sharpe Ratio	0.30	0.28	0.43	0.73	0.84	0.94	0.99
Sortino Ratio	0.38	0.36	0.58	0.96	1.11	1.25	1.31
Calmar Ratio	0.09	0.08	0.14	0.39	0.42	0.52	0.58
Skew	-0.69	-0.54	-0.45	-0.63	-0.61	-0.56	-0.59
Lower Tail	1.31	1.30	1.22	1.22	1.24	1.25	1.22
Upper Tail	1.01	1.05	1.01	1.01	1.04	1.02	1.01
CAGR	4.48%	3.71%	6.29%	13.10%	14.56%	16.17%	13.06%
Ann. Volatility	24.95%	21.54%	18.05%	19.55%	18.11%	17.67%	13.28%
Max Drawdown	-52.41%	-47.26%	-44.31%	-33.61%	-34.98%	-30.82%	-22.50%

Note. All seven blocks share the following infrastructure, held constant throughout the incremental construction: 62-instrument global futures universe (equities, rates, STIR, bonds, commodities, FX, volatility, carbon, crypto); Carver (2015) forecast-to-position framework with FORECAST_TARGET = 10, FORECAST_CAP = ± 20 ; blended volatility estimator (70% EWM-32 + 30% rolling-2560); volatility-target sizer at $\tau = 20\%$ with equal instrument weights $w_i = 1/N_{\text{active}}$; Instrument Diversification Multiplier (IDM) from active- n correlation; 10% relative position buffer; per-instrument rolling forecast scalar (window = 1,280 days); out-of-sample gate of 1,280 days per instrument (no look-ahead); and IRX (13-week T-bill) excess-return accounting. Each block adds *only* the component named in its column header; everything else is identical across blocks.

Exhibit 72. 95 % confidence intervals on the Sharpe ratio by block, excess-of-IRX.

Politis–White stationary block-bootstrap : $B = 2\,000$ draws, average block length 21 days.

Block	Full sample		Post-2010	
	SR	95% CI	SR	95% CI
B1	0.50	[0.18, 0.84]	0.30	[-0.17, 0.79]
B2	0.50	[0.18, 0.83]	0.28	[-0.17, 0.76]
B3	0.68	[0.37, 1.01]	0.43	[-0.08, 0.91]
B4	0.82	[0.49, 1.13]	0.73	[0.26, 1.19]
B5	0.89	[0.57, 1.21]	0.84	[0.38, 1.35]
B6	0.93	[0.61, 1.25]	0.94	[0.47, 1.43]
B7	0.97	[0.65, 1.29]	0.99	[0.52, 1.51]

On the full sample, the intervals of B6 and B7 overlap very largely; the improvement from B6 to B7 is not distinguishable at the 5% threshold. In post-2010, the interval of B1 includes 0 ($[-0.17, 0.79]$) whereas that of B7 excludes 0 ($[0.52, 1.51]$): the multi-alpha strategy is statistically distinguishable from a pure trend strategy on the recent period. The confidence intervals of an annualized SR on 36 years are intrinsically wide (around ± 0.33), which confirms that no gap B_i to B_{i+1} below ≈ 0.15 SR can be considered significant in isolation.

However, the stability of performance reflects a phenomenon going beyond Markowitzian diversification. The decomposition by sub-period (Exhibit 96bis, Annex §7) shows that the contribution of each alpha component shifts from one decade to the next. Over the post-2010 sample, *trend* remains positive but erodes: its contribution within the aggregated strategy falls by 0.092 SR. On the 2024–2026 window, taken as a stress-test of the monetary regime most unfavorable to *trend*, its contribution becomes actively negative (-0.381 SR), while *skew* alone delivers $+0.718$ SR and brings the SR of the aggregated strategy to $+1.476$. If the four components mechanically added their premiums, the contributions should be parallel. What we observe is a relay: the dominant component changes with the macroeconomic regime, and the strategy benefits without discretionary intervention.

The mechanism can be read in the construction of the components. The *carry* and the *skew* are cross-sectional *long-short*: they buy and sell simultaneously on both sides of the instrument universe: *carry* finances itself on instruments with negative carry to gain exposure to instruments with positive carry, *skew* sells left asymmetry where it is overpaid and buys it where it is underpaid. The alpha comes from the cross-sectional differential, not from a net exposure to the premium, which makes each component contingent on a different macroeconomic state.

More concretely, we regress the share of Sharpe held by each component (rolling Sharpe at 756 days) on three independent state variables: real Fed funds rate (lagged by one month), 21-day realized volatility on panama-adjusted ES, BAA-DGS10 credit spread (FRED).¹¹² We take a monthly panel 1997-10 – 2026-01, $N = 340$ months, Newey-West standard errors at $L = 12$ months.

We observe that *carry* retracts with equity volatility ($\beta_{\text{vol}} = -0.0064$, $t = -2.95$) and expands

with the credit spread ($\beta_{\text{credit}} = +0.1000$, $t = +3.73$, $R^2 = 0.15$). *Skew* is similar on the credit spread ($\beta_{\text{credit}} = -0.1027$, $t = -3.23$, $R^2 = 0.20$). *Trend* is influenced by volatility, not by credit ($\beta_{\text{vol}} = +0.0050$, $t = +2.55$, $R^2 = 0.11$). *Vol-of-vol* does not relate to any of the three variables, which corresponds to its role as an event-driven component rather than a regime component. Three components out of four are therefore significant on at least one macro-economic phenomenon, and the two dominant ones are diametrically opposed on the credit spread: the rotation has a specific economic signature.

Exhibit 73. Average 3-year rolling Sharpe by regime (real rates $\times$ realised ES vol), monthly panel 1997–2026, $N = 340$ months. Panel medians as thresholds (real rate = -0.56% , realised ES vol = 11.89%). Dominant component in bold. Block-*bootstrap* CI ($\ell = 12$ months, $B = 5000$).

Regime	n_{months}	Trend	Carry	Skew	Vol-of-Vol	Dominant
Low real rates, low vol	89	0.52	0.94	0.76	0.56	Carry
Low real rates, high vol	81	0.64	0.48	0.61	0.41	Trend
High real rates, low vol	81	0.70	0.61	0.84	0.15	Skew
High real rates, high vol	89	0.83	0.84	0.64	0.42	Carry

The amplitude of the rotation remains uneven from one cell to another. It is sharp in the two low-volatility regimes: *Carry* takes the lead in low real rates ($+0.94$ against $+0.76$ for *Skew*, gap of 0.18 SR), and *Skew* in high real rates ($+0.84$ against $+0.70$ for *Trend*, gap of 0.14 SR). The two high-volatility cells are significantly tighter: in low real rates, *Trend* ($+0.64$) only leads *Skew* ($+0.61$) by 0.03 SR; in high real rates, *Carry* ($+0.84$) and *Trend* ($+0.83$) are statistically tied. The rotation is therefore readable in regimes of contained volatility, and more diffuse in high volatility where the identity of the dominant is a matter of the *bootstrap* margin rather than of a clean economic separation.

Is this rotation, however, predictable *ex ante*? At each month t , we re-estimate the regressions on $[1997, t]$ and predict the dominant component at $t + 1$ from the current macro state. Exhibit 73bis reports the two test statistics, at horizon $h = 1$ (independent monthly predictions) and at horizon $h = 12$. Both horizons remain significant at the $p < 10^{-3}$ threshold. There is therefore an exploitable informational signal in the three state variables — at 48% *hit* versus 25% for the uniform draw, the relative advantage is material but insufficient to be truly informative. Concretely, the rotation leaves a predictable trace in the macro state, although this is insufficiently sharp to serve as a discretionary tactical *overlay*.

Exhibit 73bis. *Walk-forward* test of *ex ante* predictability of the dominant component. Uniform draw over four classes (*baseline* 25%). Regressions re-estimated each month on the history $[1997, t]$, prediction h months ahead from the macro state realized at $t + h$. *Block bootstrap* ($\ell = 12$ months, $B = 5\,000$) on horizon $h = 12$ to absorb the overlap.

Horizon	n	Hit	Baseline	z	p
$h = 1$ (independent monthly)	279 targets	48.0 %	25.0 %	+8.88	$\ll 10^{-3}$
$h = 12$ (<i>block bootstrap</i>)	279 targets	38.4 %	25.0 %	+3.64	$< 10^{-3}$

But the strategy does not need to anticipate: it mechanically redistributes the risk budget toward the best-rewarded components of the moment, without additional parameter or tactical overlay. The rotation is absorbed by the weighting architecture, and this is precisely why performance is maintained where pure *trend* collapses.

3.3 Positioning, institutional appeal and implementability

"I'm very uncomfortable with black box trading where I'm dealing with algorithms I don't understand. Everything we do we could do on the back of an envelope with a pencil."⁸²

Thus, trend strategies and more generally CTA strategies exist fundamentally for 3 of their properties: their simplicity, crisis alpha and zero correlation to the market.⁸³

Our proposal is slightly more complex than some strategies that we can find in the literature, being an aggregation of several elements. However, it remains among the simplest, and can be explained in its entirety in a single page. This simplicity is a real argument explaining the size of these funds on the market: everyone understands what the strategy does, there are no esoteric elements and therefore fundraising is much easier. Moreover, this mechanical and conceptual simplicity is also accompanied by a strategy with relatively stable and attractive performances:

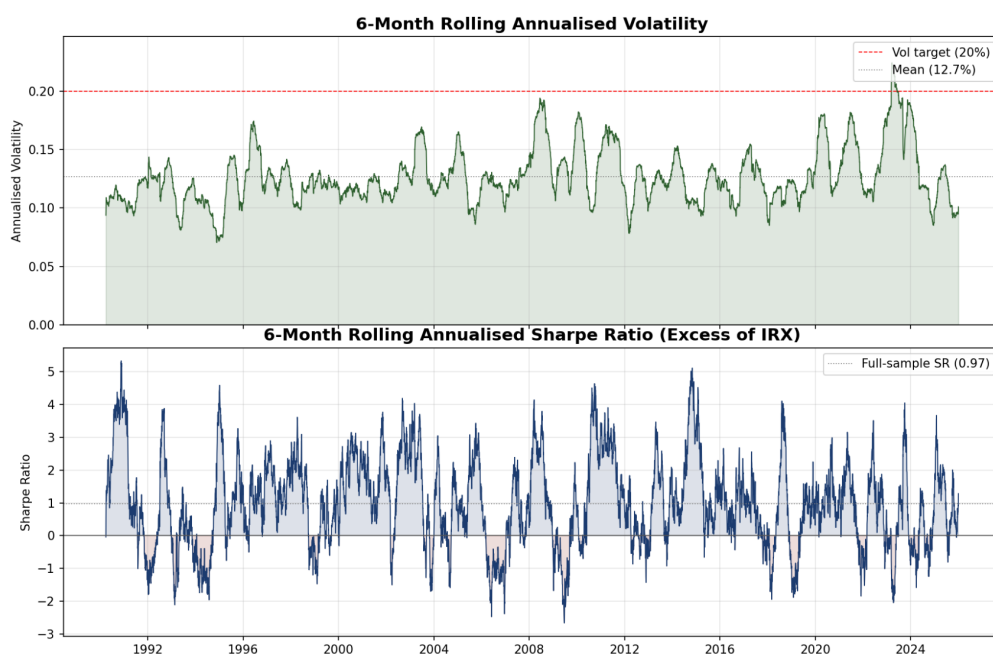


Exhibit 74. 6-month rolling average of annualized volatility and Sharpe ratio.

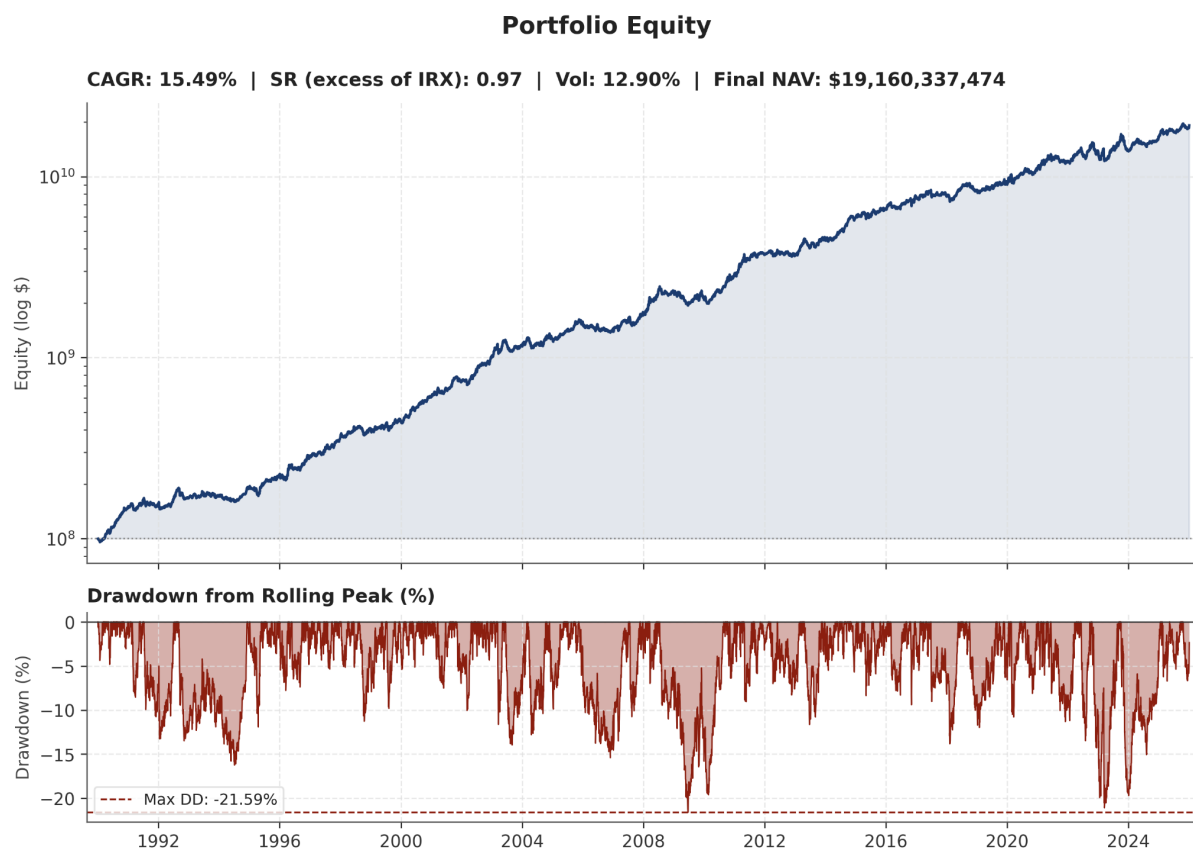


Exhibit 75. Portfolio value and drawdown.

Volatility remains globally stable with 4 different signals and the Sharpe ratio has a classic profile of CTA strategies with strong rises when trends are correctly captured and a drawdown manifesting during periods of zig-zag which constitute the tails of the distribution:

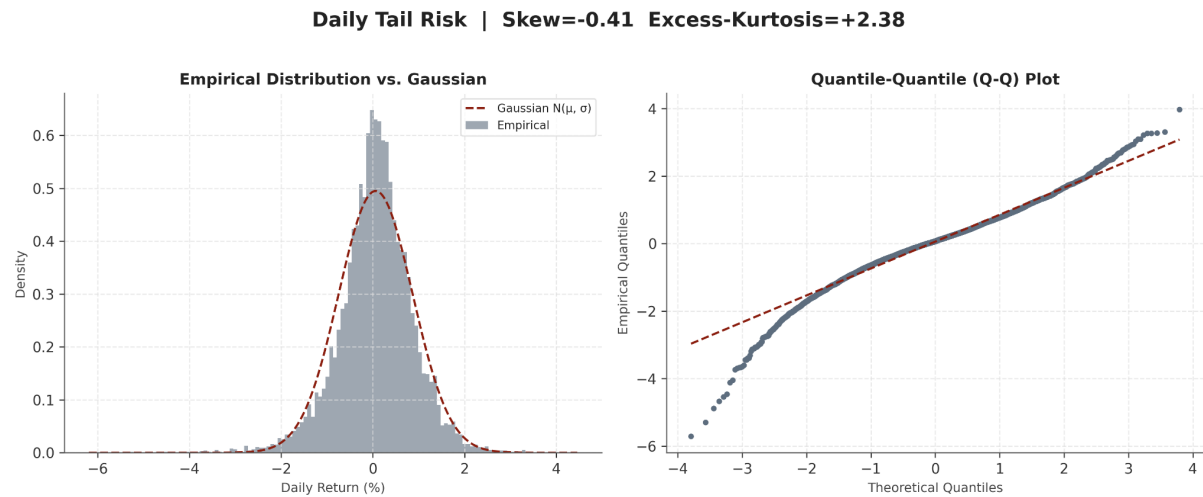


Exhibit 76. Daily distribution.

Compounded Return Profile (Weekly vs. Monthly)

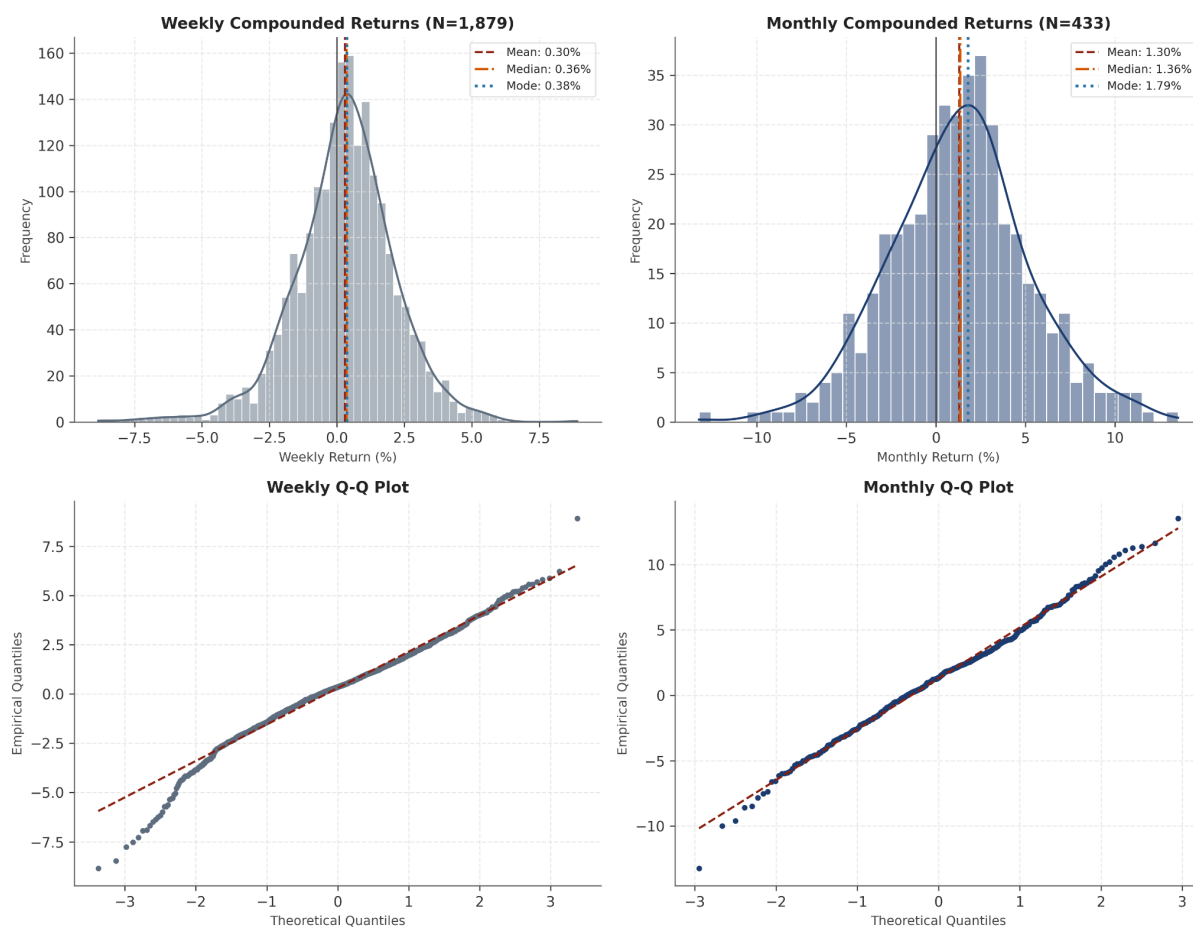


Exhibit 77. Weekly and monthly distribution.

These funds put forward the conceptual simplicity of their approach as a commercial argument: occasionally high performances produced by mechanisms explainable without advanced mathematics.

Next, the second attractive argument is crisis alpha: generating profits when the market collapses globally:

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual	RF	N
1985	+2.6	+11.9	-3.9	+8.5	+5.3	+1.1	-3.9	+1.9	+0.1	-1.2	-3.0	-8.7	+9.4	7.7	23
1986	+0.9	+5.2	-0.4	+1.1	+2.9	+6.7	+1.2	+2.6	-4.1	+1.5	+0.6	+5.5	+25.8	6.2	24
1987	+11.9	+1.5	+0.8	-8.0	-1.4	+5.2	+0.0	-0.7	+2.4	-0.8	+3.8	+3.6	+18.7	6.0	25
1988	+3.8	+2.1	+2.4	+0.8	+2.4	-9.7	+4.7	+1.5	+3.8	-2.5	+0.3	+0.7	+9.8	6.9	26
1989	+2.1	-3.0	+10.1	-1.1	+5.6	-0.4	+5.2	-0.4	+0.7	+0.6	+0.7	+1.3	+22.7	8.4	26
1990	-2.6	+2.4	+4.2	+6.7	-2.1	+7.7	+6.8	+3.3	+4.4	+5.4	+2.6	+4.2	+52.0	7.8	27
1991	+1.9	-1.2	-5.2	+5.2	+3.6	+4.3	-5.9	+2.6	+0.2	-1.5	-3.2	+3.3	+3.3	5.5	27
1992	-5.6	+1.9	+0.9	+2.3	-1.0	+6.7	+10.6	+4.3	-5.8	-5.2	-0.6	-0.0	+7.4	3.5	30
1993	+2.5	+1.8	-2.7	+0.9	+1.6	+2.6	-0.9	+0.5	-3.8	+3.0	-3.7	+1.8	+3.5	3.1	33
1994	+0.3	-3.4	+0.8	-1.6	-1.1	-2.4	+1.3	+2.2	+2.8	+2.8	+6.9	+1.8	+10.4	4.4	33
1995	-0.8	+0.3	-6.0	-0.6	+9.7	+4.8	+3.9	-1.3	+1.4	+1.1	+2.1	+2.8	+18.0	5.7	35
1996	-0.1	-3.8	+1.0	+13.5	-1.7	+1.3	-2.9	+0.5	+6.9	+4.5	+5.6	-0.5	+25.5	5.2	35
1997	+4.0	-2.3	+1.8	+3.3	-1.5	+2.0	+8.6	-4.2	+3.4	+4.2	+1.6	+4.3	+27.6	5.2	35
1998	+2.3	-1.1	+5.8	-0.4	+2.1	+2.7	+2.3	+0.2	-4.9	-4.4	+5.4	+2.6	+12.7	4.9	35
1999	-2.6	+5.1	-1.7	-0.2	+2.2	+1.0	-3.4	+0.7	+5.1	+3.0	-0.4	+0.1	+8.8	4.8	35
2000	+2.1	+6.9	-2.9	+9.5	+3.6	-0.8	+2.9	+3.2	+0.6	+2.6	+4.2	-0.0	+36.4	6.0	36
2001	+4.0	+0.4	+6.7	-4.8	-0.2	+3.9	+2.3	+4.1	+8.0	+2.9	-3.0	-2.1	+23.6	3.5	38
2002	+0.4	+0.6	-4.0	+8.4	+2.3	+6.7	+3.3	+3.5	+2.2	+0.4	-1.9	+11.6	+38.0	1.6	39
2003	+7.3	+3.6	-2.6	+2.4	+8.8	-4.4	-7.9	+0.2	+4.5	-0.2	-0.8	+2.4	+12.9	1.0	39
2004	+1.8	+7.5	-2.5	-7.4	+2.7	+1.1	-1.6	+2.2	+6.4	+3.1	+0.8	-2.8	+10.9	1.4	43
2005	+0.3	-1.2	+5.7	+0.9	+2.3	+3.2	+0.0	+1.2	+7.9	+0.1	+2.0	-0.9	+23.2	3.2	44
2006	-7.6	+2.1	+0.5	-1.9	-0.7	+0.8	-4.5	+1.6	-0.4	-1.4	-0.5	+3.2	-8.9	4.9	45
2007	+1.3	-0.9	+3.9	+5.9	+0.2	+0.1	-1.6	-0.5	-0.6	+6.1	+3.2	+1.3	+19.5	4.5	46
2008	+5.7	+8.9	+5.2	+0.3	+2.3	+10.2	+1.2	-5.0	+1.9	-3.2	+1.7	+1.8	+34.6	1.4	46
2009	-3.5	+0.1	-2.2	-1.8	-6.6	+1.0	+1.3	+3.5	+2.6	-4.2	+10.8	-9.6	-9.7	0.1	48
2010	-2.3	-2.0	+6.1	+4.8	+4.3	+0.2	+1.3	+6.0	+5.3	+4.3	-0.7	+6.8	+39.3	0.1	56
2011	+0.2	+11.4	+3.7	+11.1	-5.3	-1.9	+7.2	+2.5	-4.7	+3.1	+1.9	-2.2	+28.4	0.0	56
2012	+1.7	+1.9	-1.4	-1.9	+4.2	-4.1	+0.9	-0.5	+1.5	-2.7	+0.3	-0.7	-1.1	0.1	56
2013	-0.3	+8.0	+5.0	+7.0	-6.7	-3.3	+5.0	-4.5	+4.3	+4.9	+3.2	+0.2	+23.7	0.0	58
2014	-1.7	+2.0	-1.7	+2.7	+6.3	+1.6	-0.8	+7.8	+5.1	+6.9	+0.1	-0.8	+30.2	0.0	58
2015	+4.0	-1.7	+3.3	-3.7	+3.5	-4.6	+8.6	-4.6	+2.9	+3.2	+1.2	-0.5	+11.3	0.0	59
2016	+5.7	+2.7	-4.6	-3.2	+1.2	+2.5	+3.5	+0.8	+2.0	+0.4	-0.0	+3.6	+15.0	0.3	59
2017	-2.4	+5.0	+0.9	+4.0	+0.2	-4.9	+0.3	+3.1	-2.1	+2.2	-0.8	-2.7	+2.4	0.9	59
2018	-0.8	-2.0	+1.3	+4.0	+5.7	+2.7	+3.1	+2.0	-0.4	-2.7	-3.9	-1.4	+7.3	2.0	59
2019	-3.0	+3.1	+2.4	+2.0	-1.0	+0.8	+3.0	+2.9	+1.4	-3.0	+4.9	-4.8	+8.4	2.1	59
2020	+3.7	-1.0	+5.8	+2.3	+1.8	+1.8	+4.6	-0.2	-2.6	-3.2	+2.2	+6.6	+23.6	0.3	59
2021	-2.1	+6.4	+2.7	+3.6	+3.7	-4.0	+4.1	-0.6	-5.8	+0.5	-2.1	+1.1	+6.9	0.0	59
2022	+1.7	+1.2	+5.8	+3.7	+3.4	-6.2	-3.3	+8.4	+7.0	+2.5	-8.7	+0.6	+15.6	2.0	59
2023	-8.6	+10.3	-10.0	+2.8	+5.8	+6.3	+0.5	+1.7	+11.3	+0.5	-13.1	-3.7	+0.4	5.3	62
2024	+2.5	+8.2	+1.2	+3.2	-3.0	-2.0	-1.7	+1.4	+2.3	-0.1	+0.3	+5.9	+19.1	5.2	62
2025	+8.9	-2.3	+1.4	+0.4	+1.8	-3.0	+2.1	+0.4	+4.3	+1.0	-1.2	-0.4	+13.7	4.2	62

Exhibit 78. Annual performance table of the strategy restarted from 1985, with number of instruments N per year. RF: annualized return of the 13-week T-bill (IRX) on 100% of NAV. N : number of distinct instruments having traded at least one day in the year (post burn-in 1,280 days OOS).

Note: The strategy is restarted from 1985 for this table, with the N column being the number of instruments existing in our portfolio. The performances are similar with all our backtests starting strictly in 1990. We seek to be transparent and to demonstrate that the 1990 start date is not chosen for its performance, but arbitrarily and similar to the approach of Andreas Clenow.

The strategy performs during stress periods producing macro-economic dislocations and suffers during periods of rapid reversals:

-1987 (Black Monday): +18.7%

- 1990-1991 (Gulf War): +52% and +3.3%
- 1998 (Russian crisis, LTCM implosion): +12.7% over the year, but -4.9% in September and -4.4% in October, months during which LTCM collapses.
- 2000-2002 (Dot com crisis): +36.4%, +23.6% and +38.0%.
- 2007-2009 (Subprime crisis): +19.5%, +34.6% and -9.7%.
- 2020 (COVID crisis): +23.6% (decomposed: crash 19 Feb–23 Mar 2020, conditional SR -3.80 *annualized* over the short window of market drop (approx. 23 trading days, a value whose annual extrapolation is mechanically amplified by the brevity of the window); rebound March–December 2020, SR +1.99; positive annual balance, but the strategy suffers the crash itself before recovering it.)
- 2022 (Inflation shock): +15.6%

The most profitable episodes are those that produce *progressive and persistent* dislocations: the bursting of the Dot com bubble (2000–2002, +36 to +38% each year), the subprime crisis (2007–2008, +19 and +34%) and the inflation shock (2022, +15%) all last several quarters and allow the trend component to align its positions in the same direction. The weakly profitable or negative episodes are on the contrary those characterized by brutal reversals: the Russian crisis and the LTCM implosion (1998, -3%) compress spread gaps over a few weeks; the COVID crash (Feb-March 2020, conditional SR -3.80 over this short window) drops too quickly for the trend signals to update, the reverse rebound being recovered in the following months; the post-crisis reversal (2009, -9.7%) takes the strategy on the wrong foot when equity markets sharply recover after the *short* positions accumulated in 2008. This asymmetry corresponds exactly to the description of *long gamma / short theta* discussed in chapter 1 §3: the strategy collects long trends like a *premium* and pays for brief reversals as a cost.

The real performance table of the strategy:

Monthly Performance Table — Total Returns (%)

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Annual	RF
1990	-3.4	+2.4	+4.2	+6.7	-2.1	+7.7	+6.8	+3.3	+4.4	+5.4	+2.6	+4.2	+50.5	7.8
1991	+1.9	-1.2	-5.2	+5.2	+3.6	+4.3	-5.9	+2.6	+0.2	-1.5	-3.2	+3.3	+3.3	5.5
1992	-5.6	+2.0	+1.0	+2.4	-1.0	+6.7	+10.6	+4.3	-5.8	-5.2	-0.6	-0.0	+7.6	3.5
1993	+2.5	+1.8	-2.7	+0.9	+1.6	+2.6	-0.9	+0.5	-3.8	+3.0	-3.7	+1.8	+3.5	3.1
1994	+0.2	-3.4	+0.7	-1.6	-1.1	-2.3	+1.2	+2.2	+2.8	+2.8	+6.9	+1.8	+10.5	4.4
1995	-0.8	+0.3	-6.0	-0.6	+9.7	+4.8	+3.9	-1.3	+1.5	+1.1	+2.1	+2.8	+18.0	5.7
1996	-0.1	-3.8	+1.0	+13.5	-1.7	+1.3	-2.9	+0.5	+6.9	+4.5	+5.6	-0.5	+25.5	5.2
1997	+4.0	-2.3	+1.9	+3.3	-1.5	+2.0	+8.6	-4.2	+3.4	+4.2	+1.6	+4.3	+27.5	5.2
1998	+2.3	-1.1	+5.7	-0.4	+2.1	+2.8	+2.3	+0.2	-4.9	-4.4	+5.4	+2.6	+12.6	4.9
1999	-2.6	+5.1	-1.7	-0.2	+2.2	+1.0	-3.4	+0.7	+5.1	+3.0	-0.4	+0.1	+8.8	4.8
2000	+2.1	+6.9	-2.9	+9.5	+3.6	-0.7	+2.9	+3.1	+0.6	+2.6	+4.2	-0.0	+36.4	6.0
2001	+4.0	+0.4	+6.6	-4.8	-0.3	+3.9	+2.2	+4.1	+8.1	+2.9	-3.0	-2.1	+23.3	3.5
2002	+0.4	+0.5	-3.9	+8.3	+2.3	+6.7	+3.3	+3.5	+2.2	+0.4	-1.9	+11.6	+37.8	1.6
2003	+7.3	+3.6	-2.6	+2.4	+8.8	-4.4	-7.9	+0.2	+4.5	-0.2	-0.8	+2.4	+12.9	1.0
2004	+1.8	+7.4	-2.6	-7.4	+2.7	+1.0	-1.6	+2.1	+6.5	+3.0	+0.6	-3.0	+10.1	1.4
2005	+0.4	-1.3	+5.4	+0.7	+2.4	+3.4	+0.1	+1.3	+8.3	+0.1	+2.0	-0.9	+23.6	3.2
2006	-7.5	+2.1	+0.6	-1.8	-0.8	+0.8	-4.5	+1.6	-0.3	-1.3	-0.5	+3.2	-8.6	4.9
2007	+1.2	-1.1	+3.9	+5.9	+0.2	-0.0	-2.0	-0.5	-0.6	+6.2	+3.1	+1.3	+18.8	4.5
2008	+5.7	+8.9	+5.3	+0.3	+2.3	+10.2	+1.2	-5.0	+2.0	-3.2	+1.7	+1.8	+34.7	1.4
2009	-3.5	+0.1	-2.2	-1.8	-6.6	+1.0	+1.3	+3.5	+2.6	-4.2	+10.8	-9.6	-9.7	0.1
2010	-2.3	-2.0	+6.0	+4.8	+4.4	+0.2	+1.3	+6.0	+5.3	+4.3	-0.7	+6.8	+39.3	0.1
2011	+0.1	+11.4	+3.7	+11.1	-5.4	-1.9	+7.2	+2.4	-4.7	+3.1	+1.9	-2.2	+28.3	0.0
2012	+1.7	+1.9	-1.5	-1.9	+4.4	-4.2	+1.0	-0.5	+1.5	-2.7	+0.3	-0.4	-0.7	0.1
2013	+0.1	+8.5	+4.9	+7.0	-6.6	-3.4	+5.0	-4.6	+4.3	+5.0	+3.1	-0.1	+24.2	0.0
2014	-2.0	+1.9	-1.7	+2.7	+6.3	+1.7	-0.7	+7.7	+5.0	+7.2	-0.2	-0.9	+29.7	0.0
2015	+4.1	-1.7	+3.3	-3.8	+3.5	-4.7	+8.6	-4.6	+2.9	+3.2	+1.2	-0.5	+11.2	0.0
2016	+5.7	+2.7	-4.6	-3.2	+1.2	+2.5	+3.6	+0.7	+1.9	+0.4	-0.0	+3.7	+15.0	0.3
2017	-2.5	+5.0	+1.1	+4.2	-0.1	-5.0	+0.3	+3.1	-2.1	+2.1	-0.9	-2.5	+2.1	0.9
2018	-0.9	-2.2	+1.1	+4.2	+5.7	+3.0	+3.4	+2.1	-0.5	-2.8	-4.1	-1.4	+7.4	2.0
2019	-3.0	+3.2	+2.4	+1.9	-1.2	+1.0	+3.2	+2.5	+1.4	-2.8	+5.0	-5.3	+8.0	2.1
2020	+4.1	-1.4	+5.7	+2.3	+1.8	+1.8	+4.6	-0.2	-2.6	-3.2	+2.3	+6.7	+23.7	0.3
2021	-2.1	+6.5	+2.9	+3.8	+3.8	-3.8	+3.9	-0.5	-6.0	+0.5	-2.1	+1.2	+7.5	0.0
2022	+1.3	+1.2	+6.0	+3.6	+3.4	-6.2	-3.3	+8.4	+6.9	+2.5	-8.6	+0.5	+15.2	2.0
2023	-8.5	+10.0	-10.0	+2.8	+5.7	+6.2	+0.5	+1.6	+11.3	+0.7	-13.2	-3.5	+0.2	5.3
2024	+2.5	+8.1	+1.4	+3.0	-3.1	-1.9	-1.6	+1.5	+2.3	-0.2	+0.3	+5.6	+18.7	5.2
2025	+9.1	-2.7	+1.4	+0.6	+2.0	-3.0	+2.0	+0.5	+4.2	+1.0	-1.2	-0.4	+13.7	4.2

Exhibit 79. Complete annual performance table of the strategy (1990–2025).

Simplicity and the ability to survive when markets experience crises form an attractive shop window for these funds, which indeed constitute a hedge against the risk of economic catastrophe. The psychological aspect is here important: everyone prefers to make profits when the rest of the participants lose. Paying commissions to a CTA fund to have this exposure is a second major argument.

The third and last argument follows from the continuity of the hedge against crisis risk: the low correlation to the equity market.



Exhibit 80. Rolling correlation of the strategy to the equity market.

Let us recall that the portfolio universe is composed of a growing share of equity indices as they become accessible. The few correlation peaks are therefore not surprising despite the effectively null mean. The strategy can obviously be pushed to be even more decorrelated by removing the equity instruments from the portfolio:

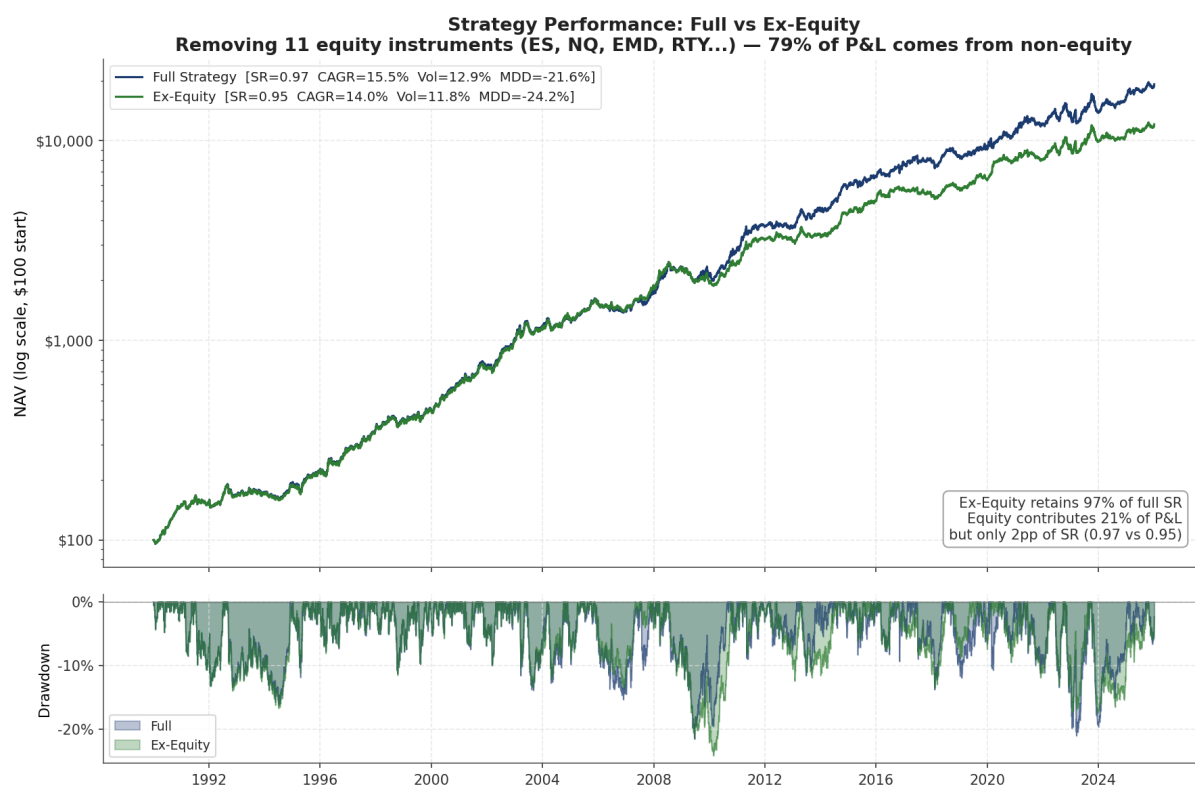


Exhibit 81. NAV and performance of the strategy without equity instruments.

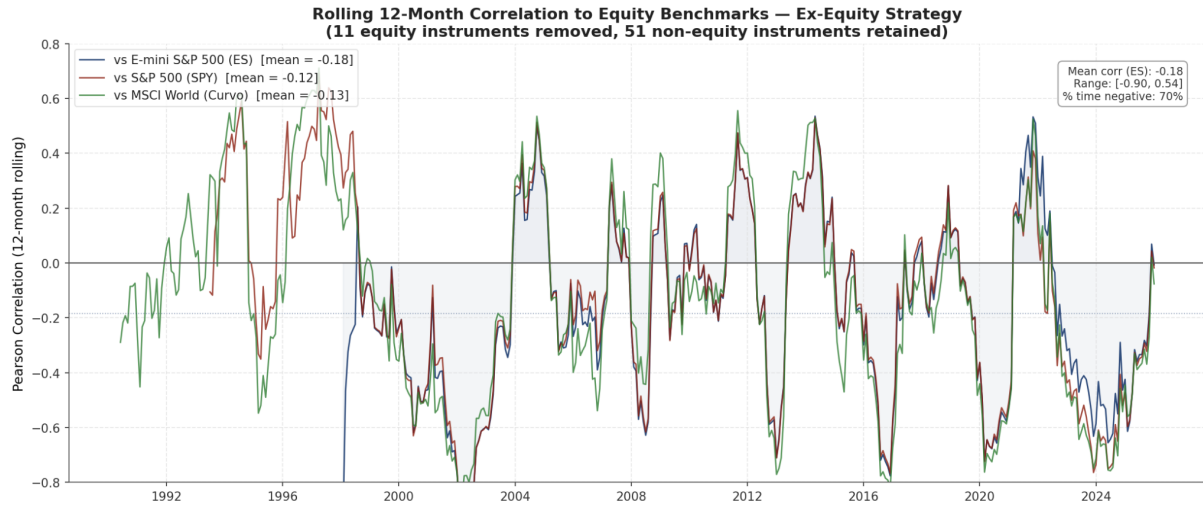


Exhibit 82. Rolling correlation to the equity market of the strategy without equity instruments.

Thus, the strategy is easily promoted to investors by its simplicity, the hedge it provides in periods of crisis and its zero correlation with an equity portfolio. These 3 arguments alone allow us to conclude the object of our study: CTA strategies still have their place in the post-subprime crisis environment. Moreover, a slight optimization of these strategies with the addition of several basic alpha sources and a strict management of the portfolio also allows us to demonstrate that these strategies can also propose entirely interesting returns post-2010. Furthermore, we argue that as it stands, this proposed strategy can be considered as a complete investment and not as an addition to a more classic portfolio.

In fine, we wish to make a few remarks on the implementation capacity of this strategy before moving toward the conclusion of the study. The appendix is entirely devoted to the statistical validity of our results; they will therefore only be briefly discussed here. The backtest SR is 0.97; after multi-test Harvey–Liu–Zhu (2016) correction at $N_{\text{trials}} = 256$, it falls back to 0.75 — still statistically significant ($p_{\text{BHY}} = 3.3 \cdot 10^{-6}$, cf. Annex §4, Exhibit 92). The Sharpe ratio put forward on the full sample is therefore flattering but unlikely to materialize in the future; obtaining only 50% of this SR, i.e., roughly 0.5, would already be a massive performance out of sample, with the corrected SR of 0.75 as an upper bound. Let us recall that this strategy can easily support a capital ranging from 10 million to 10 billion without major change, instrument selection and obvious liquidity issues aside. Moreover, this strategy can be entirely automated, and the addition of the sigmoid functions allows virtually adding any instrument to the universe without even checking whether it is profitable in such a strategy. Furthermore, fearing the risk of *overfitting*, our *backtests* trade the whole of our available sample without exception, even the instruments that are systematically harmful for the strategy over the whole sample:

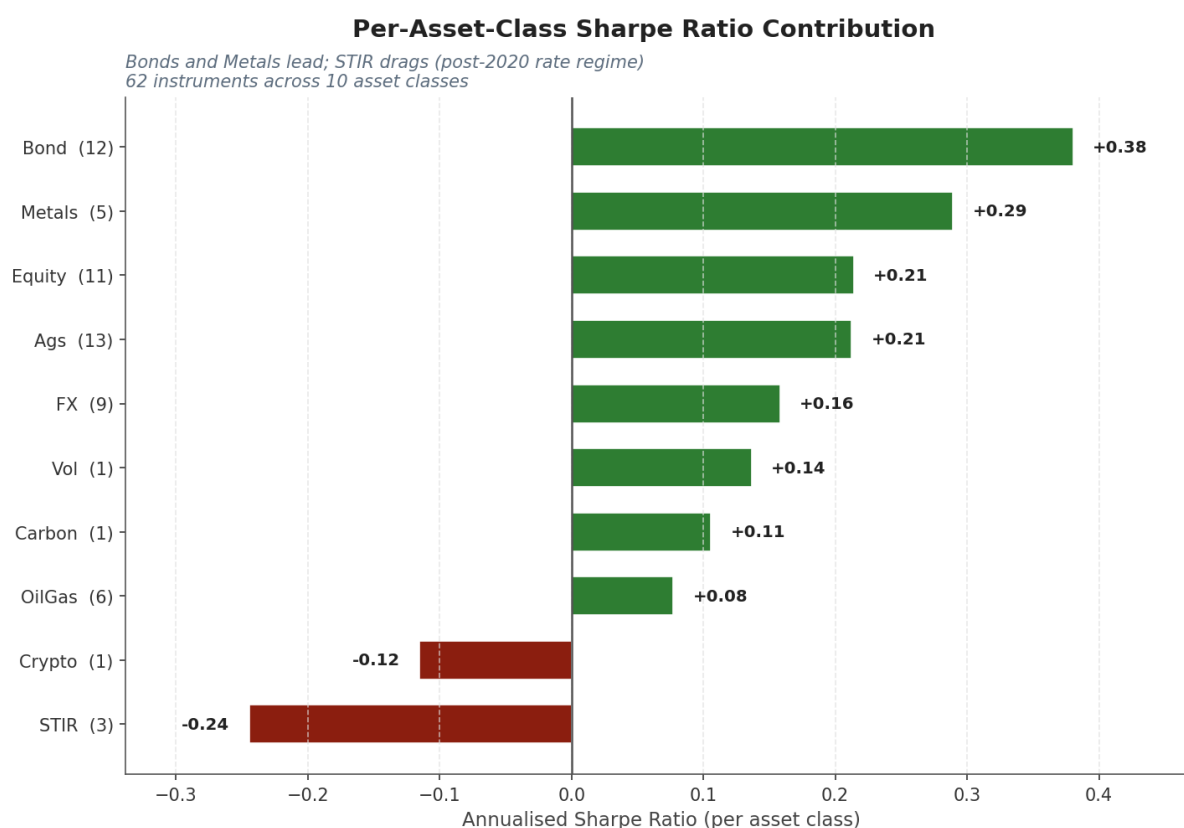


Exhibit 83. Individual performance per instrument of the strategy (Sharpe over the full sample).

A real implementation would first check the instruments or sectors that fundamentally do not work with this strategy and exclude them, for example the 3-Month SONIA (UK Rates) and the 3-Month Euribor (EU Rates), both with negative individual Sharpe on the full sample (cf. Exhibit 84 for the distribution instrument by instrument). A first analysis and exclusion of this kind dramatically improves performance, but directly introduces a *sample fitting* bias.⁸⁴

Per-Instrument Sharpe Ratio Contribution

11 instruments with negative SR (red) vs 51 positive (green)
62 instruments across 10 asset classes

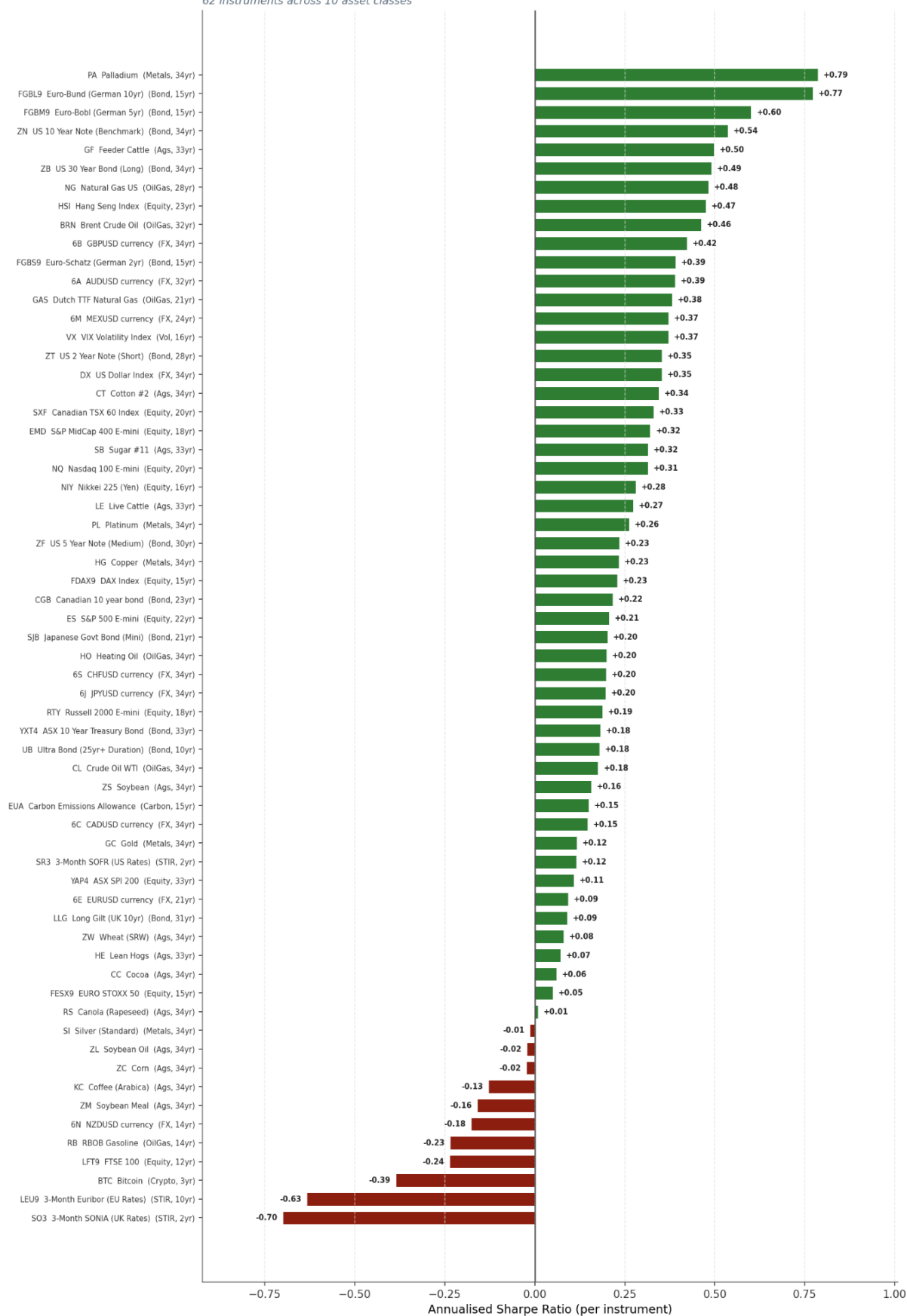


Exhibit 84. SR of each instrument on the strategy.

Moreover, we have not addressed the returns of the short side of the strategy because these properties are already widely cited in the literature, but let us simply recall that making the strategy *long-only* massively improves performance, at the risk of becoming strongly correlated with the equity market and of losing all the differential advantage that these strategies really embody.

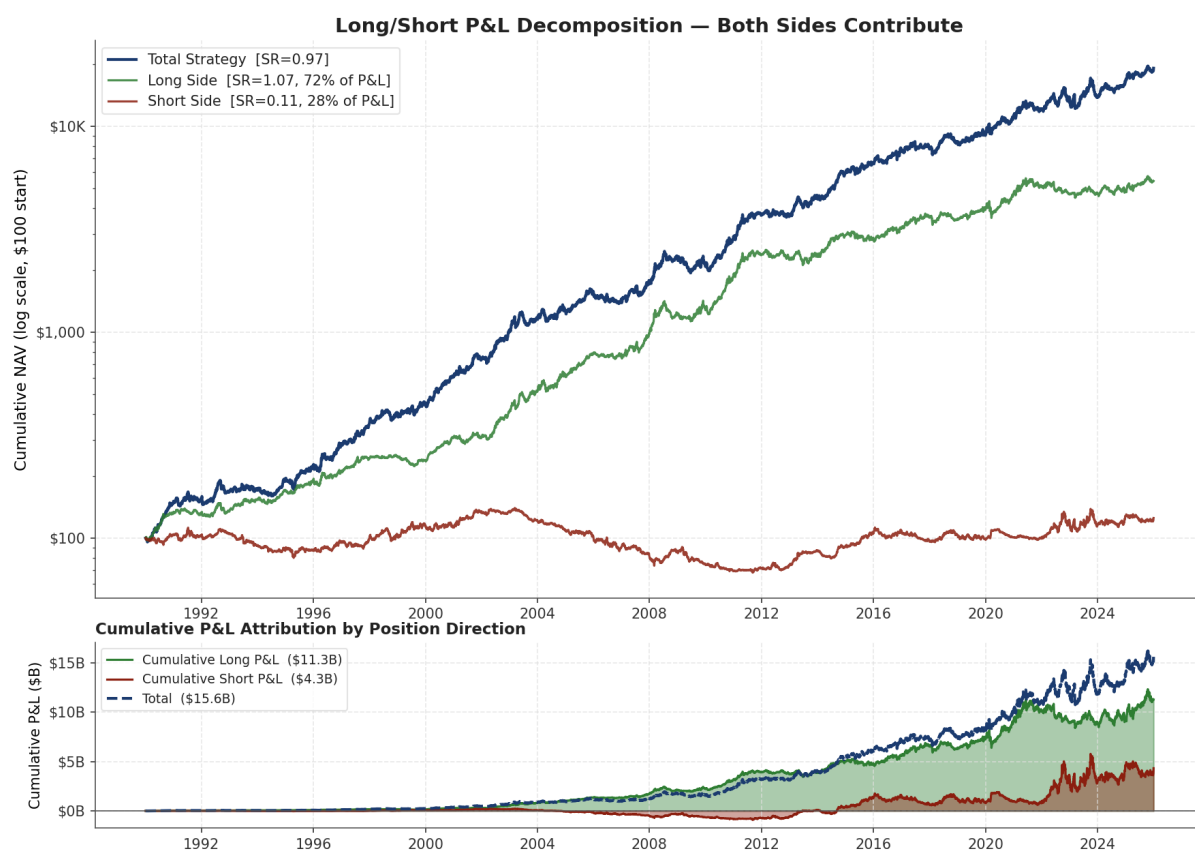


Exhibit 85. Comparison of the long-short vs long-only strategy.

We briefly discuss the implementation elements essential for this strategy and more generally those of the register. The main source of gap between *backtest* and real execution lies in order management: at the inflection points of signals, liquidity tends to disappear, all CTA funds with similar strategies simultaneously entering the same positions. Transaction costs can then increase strongly and erode performance.

Three solutions are conceivable. First, spread the order over time: an execution program decomposes the total order into sub-orders executed over a window typically of 30 minutes, compared *ex post* to the TWAP and VWAP benchmarks; Almgren & Chriss (2000) formalize the optimal scheme in arbitrage between market impact and *timing* risk. Next, multi-venue routing: for instruments listed on several exchanges, a program measures the depth of the order book on each and splits the order so as to reduce our marginal footprint on liquidity. Finally, dynamic analysis of transaction costs: our simplistic model must be replaced by a TCA that measures *ex post* the gaps between arrival price, decision price and period VWAP; instruments

systematically too costly are underweighted in real time. This type of program is notably a recommendation of MiFID II. Furthermore, the capital that the strategy can absorb depends on the management mandate; we however recommend limiting the footprint of orders to a maximum of 10% of the daily liquidity of an instrument. With our weighting of instruments at $1/N$, this leads to limitations around \$500 M of AUM. Beyond this threshold, we are no longer takers of liquidity but providers on certain markets, exposing us directly to undesirable risks. An alternative weighting by liquidity of instruments is discussed in the appendix.

CONCLUSION

In fine, have market trends disappeared since 2010, and do the CTA funds that exploit them retain their *raison d'être*? To answer this, we have reconstructed the very concept of trend; then its economic properties and its exploitability.

On this theoretical basis, we approached the industrial embodiment of the phenomenon: CTA funds, whose systematic, *long-short* and continuous structure articulates three distinctive institutional properties: conceptual simplicity, crisis alpha and zero correlation with the equity market. Our review established two converging facts: the weak intrinsic differentiation between most implementations and the material weakening of post-2010 performance, documented both in the literature and in our own reproductions of trend strategies.

The open question "have trends died" has structured all our empirical work. The contribution of the *trend* component in our strategy goes from +0.272 SR pre-2010 to +0.181 SR post-2010, and becomes actively negative (−0.381 SR) on the adverse window 2024–2026, quantified confirmation of the weakening described in the literature. But this weakening is neither the extinction of the phenomenon nor the end of CTAs; it is compensated by other alpha sources that the same architecture can accommodate without structural modification. The strategy we propose adds to the *trend* signal three complementary alpha sources: *carry*, *skew*, *vol-of-vol*, each derived from the literature, structurally parameterized and combined by equal weighting. The full-sample SR of 0.97, the *block-bootstrap* CI of [0.649, 1.312] which excludes zero with margin, the retained SR of 0.75 after Benjamini-Hochberg-Yekutieli correction over $N_{\text{trials}} = 256$ audited variants, and the post-2010 *bootstrap* median in excess of IRX of +0.999 (statistically indistinguishable from the 0.97 full-sample) constitute the four inferential statistics that defend this thesis.

The specific contribution of our study is not the invention of new premiums, each of the four components predates us in the literature, but the synthesis of a parsimonious architecture and its statistical audit over 36 years of sample, formalized by the battery of 33 tests presented in Annex 1. Multi-alpha diversification does not only work by mechanical addition of four weakly correlated premiums; it also works by rotation: the dominant component changes with the macroeconomic regime, *trend* positively loaded on equity volatility, *carry* on wide credit *spread* and contained volatility, *skew* parallel on the credit axis, and the weighting architecture (*vol-targeting* and *pooled* FDM) absorbs this rotation without additional parameter. The identification of this rotation, formalized by regression of the Sharpe share on three state variables (real Fed funds rate, ES

volatility, *spread* BAA–DGS10), constitutes our novel technical contribution to the literature.

In the end, pure trends have weakened post-2010, but a multi-alpha architecture combining *trend*, *carry*, *skew* and *VoV* restores these performances. CTA funds, read through this enlarged architecture, therefore retain their place as crisis alpha and source of diversification.⁸⁵

ANNEX 1: STATISTICAL TESTS AND RESEARCH REPORT

Summary

The battery of 33 tests presented in this annex converges on a single verdict: the strategy delivers a performance that is statistically significant, structurally justified and stable over the entire 1990–2026 sample.

On the inferential side, the historical Sharpe ratio of 0,97 exceeds zero with a $t_{\text{Mertens}} = 5,79$ and a *block-bootstrap* confidence interval (Politis-Romano, $B = 2\,000$, 21-day blocks) of $[0,649, 1,312]$ which excludes zero by a comfortable margin (T1, T5); corrected for $N_{\text{trials}} = 256$ audited variants via the Benjamini-Hochberg-Yekutieli procedure, the SR falls back to 0,75 and remains significant (T10, T11); extension of the *block-bootstrap* to the post-2010 sub-period, the least favourable to *trend*, delivers a median of +0,999 in excess of IRX, statistically indistinguishable from the full-sample SR of 0,97 (T5, A.3.1).

On the architectural side, the ablation matrix (§7) confirms that the four alphas (*trend*, *carry*, *skew*, *vov*) contribute in a complementary manner and that none is superfluous; the simplex optimisation (T25) places equal-weighting on a plateau of quasi-equivalent optima; the 64-day VoV window (§8) and the set of other parameters are derived structurally (§31). The regression against the SG CTA index (T20) isolates an annualised alpha of +10,17%; the regressions against Fama-French 5 and Fung-Hsieh 7 (T15) deliver +10,64% and +12,97% respectively, that is, an alpha not absorbed by the market beta nor by CTA-native factors.

On the stability side, the 5-year rolling SR has a median of 1,01, never becomes negative (T19); the 41 *rolling-start* cohorts all display a strictly positive SR (T22); the universe frozen at 1990 recovers $\approx 93\%$ of the full-sample Sharpe (§26), demonstrating that post-1990 additions do not explain the alpha; the sensitivity to the volatility target τ is flat on $[0,05, 0,35]$ (§27), the SR varying by less than 0,05 (1,013 at $\tau = 5\%$, 0,964 at $\tau = 35\%$) while the underlying leverage varies by a factor of seven – the CAGR and realised volatility scale linearly in τ , the SR remains invariant by construction.

On the operational side, the strategy absorbs a latent execution at $t+2$ without material degradation (T29: -2% of SR), withstands a multiplicative *stress-test* of costs up to $3\times$ the *baseline* (T18: SR 0,641 at $3\times$, still above 0,5), and remains implementable on capital of 500 M\$ – 1 G\$ without modification of the universe; beyond that, the capacity test (§29) identifies the SJB as a *binding* constraint. The synthetic paths generated by GAN (§33) do not provide an exploitable answer,

the generator does not reproduce the term structure of contracts, but this limit is expected and the generator-free validation provided by the *block-bootstrap* of T5 remains the main proof.

Annex 2 presents the paper trading test from January 2026 to June 2026, which is the last test we can perform.

Structure

1. Performance of the strategy

Recap of the performance of the strategy over the entire sample.

2. Period stability

Q: Is the sample SR driven up by a single period?

3. Sharpe ratio inference (T1-T5-T6-T7-T8-T9)

Q: Is the SR statistically different from 0 under the assumption of a realistic return distribution?

Politis & White (2004), Politis & Romano (1994), Ledoit & Wolf (2008), Jarque & Bera (1987), Ljung & Box (1978), Engle (1982), Lo (2002), Mertens (2002), Bailey & Lopez de Prado (2012).^{86, 87, 88, 89, 90, 91}

4. Correction tests (T10-T11)

Q: Does the SR survive a correction for the number of variants audited during the research?

Harvey, Liu & Zhu (2016), Bailey, Borwein, Lopez de Prado & Zhu (2016)^{92, 93}

5. Deflated Sharpe ratio against a passive benchmark

Q: Does the strategy provide a significant advantage against a simple buy-and-hold approach?

Bailey & Lopez de Prado (2014).^{94, 95}

6. Model confidence set (T12)

Q: Is the strategy a sharp optimum and therefore overfit, or does it sit on a plateau?

Hansen, Lunde & Nason (2011)⁹⁶

7. Ablation matrix

Q: Which components of the strategy are important and which are less so?

Jobson & Korkie (1981), Memmel (2003).⁹⁷

8. Parametrisation of the directional VoV window

Q: Is the choice of a 64-day window for the directionality of the signal overfit?

9. Parameter iteration via Monte Carlo (T24)

Q: Is the performance of the strategy a mere artefact that a simulation with random parameters can reproduce?

10. Simplex weighting optimisation. (T25)
Q: Is equal-weighting optimal?
11. Execution timing. (T29)
Q: Does the strategy survive a $t + 2$ execution of signals?
12. Stability in an adverse regime. (T16)
Q: Does the strategy survive in the worst regime of the post-2015 backtest?
13. Data snooping audit. (T21)
Q: Does the architecture cheat by looking into the future?
 Lo & MacKinlay (1988), Lo (1991).⁹⁸
14. Informational content of the signals. (T17)
Q: Do the signals have real predictive power?
 Efron & Tibshirani (1993), Politis, Romano & Wolf (1999).⁹⁹
15. Transaction costs linked to volatility
Q: Does a significant increase in transaction costs during periods of high volatility threaten the viability of the strategy?
16. Regression against a CTA index (T20)
Q: Does the strategy have real alpha or does it simply follow a classical CTA index?
17. Market timing. (T14)
Q: Is the alpha real or simply a timing of the market-linked beta?
 Henriksson & Merton (1981), Treynor & Mazuy (1966).^{100, 101}
18. Classical regressions (T15)
Q: Does the alpha survive against equity factors and CTA models?
 Fama & French (2015), Fung & Hsieh (2001), Newey & West (1987).^{102, 103, 104, 105, 106}
19. Crisis performance (T13)
Q: Does the strategy really have a crisis alpha?
20. Transaction cost stress test. (T18)
Q: Does the strategy survive more aggressive static transaction costs?
21. Signal smoothing test (T-P3)
Q: Is smoothing the signal at the portfolio rebalancing pace a local optimum?
22. Weighting instruments by liquidity
Q: Is liquidity-based weighting of instruments a better alternative to equal-weighting ?
23. Alpha decay. (T-P2)
Q: Is the alpha durable or decaying
24. Rolling Sharpe ratio. (T19)

- Q: Is performance stable over 5-year rolling periods ?*
25. Cohort test (rolling-start). (T22)
Q: Does historical performance hold for each launch year between 1985 and 2025?
26. 1990 frozen universe (availability bias test)
Q: Does the historical performance hold if the investment universe is frozen at the backtest start date?
27. Sensitivity to τ (vol-target)
Q: Does the Sharpe depend on the "single free variable" $\tau = 20\%$ of the architecture?
28. Gross exposure / NAV
Q: What is the level of gross notional that the strategy holds on average, and what is its peak?
29. Capital size
Q: From what capital size does the alpha of the strategy erode?
30. Asset class contribution
Q: Which markets produce performance and which are negative?
31. Structural derivation of the parameters
Q: Is the parametrisation arbitrary or the product of sample-fitting?
 Robert Carver, (2015) & (2023), Ari Levine, Lasse Heje Pedersen (2016), Moskowitz & Pedersen (2013), Koijen et al. (2018), Fernandez-Perez, Frijins, Fuertes & Miffre (2018), Baltussen, van Bakkum & van der Grient (2018).¹⁰⁷
32. Walk-forward analysis
Q: Do the parameters of the strategy hold under a walk-forward re-estimation?
33. Synthetic paths
Q: Does the strategy withstand a test on synthetic paths generated by GAN, and what are the structural limits of this approach?

A.1 Performance of the strategy

Recap of the performance of the strategy over the entire sample.

Exhibit 86. Summary of the performance of the strategy over the full sample.

Metric	Value
Sharpe ratio (excess of IRX)	0.9724 [IC 95% <i>block-bootstrap</i> Politis–White, $B = 2\,000$, block=21 : [0,649, 1,312]]
CAGR	15.49%
Annualised volatility	12.90%
Maximum drawdown	-21.59%
Max DD duration	585 days (2.3 years)
Calmar ratio	0.718
Sample	1990-01-03 to 2026-01-02 (9,340 days, 36 years)
Instruments	62 global futures

The SR is computed in excess of the daily IRX risk-free rate.

A.2 Period stability

Q: Is the sample SR driven up by a single period?

We seek to determine whether the results are the product of a period that could be considered an anomaly distorting the final results.

Exhibit 87. Sharpe ratio by decade.

Decade	Sharpe ratio
1990s	0.89
2000s	1.02
2010s	1.18
2020s	0.73

The Sharpe is greater than 0.5 during every decade, with the post-2020 period reflecting the inclusion of STIRs which compress the ratio.

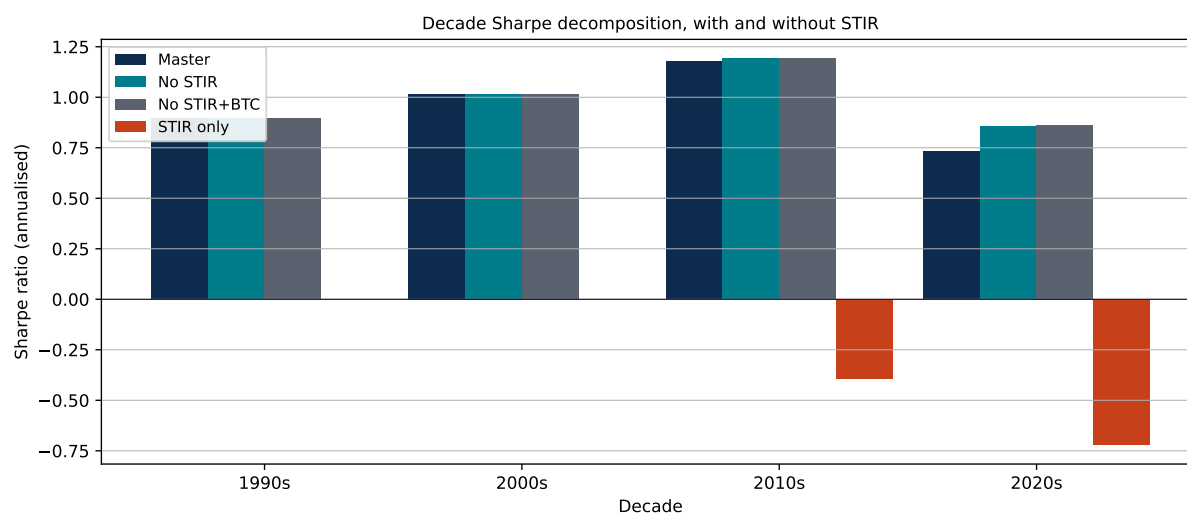


Exhibit 88. Decade-by-decade Sharpe decomposition, with and without STIR. The 2020 decade goes from $SR = 0,73$ to $SR = 0,86$ when the three STIRs are excluded, confirming that STIRs explain about half of the recent underperformance. STIRs alone: $SR = -0,72$ over the 2020 decade.

Decade	Master	No STIR	No STIR + BTC	STIR only
1990s	0.89	0.89	0.89	nan
2000s	1.02	1.02	1.02	nan
2010s	1.18	1.19	1.19	-0.39
2020s	0.73	0.85	0.86	-0.72
Full sample	0.97	1.00	1.00	-0.33

Exhibit 89. Sharpe ratios by decade: the strategy, ex-STIR, ex-STIR and Bitcoin, STIR only.

A.3 Sharpe ratio inference (T1-T5-T6-T7-T8-T9)

Q: Is the SR statistically different from 0 under the assumption of a realistic return distribution?

We run a set of tests to verify the properties of the historical Sharpe ratio.

Exhibit 90. Synthesis of the inference tests on the Sharpe ratio (T1, T5, T6, T7, T8, T9).

Test	Result
T1 — Politis-White optimal block length	RT_BLOCK = 21 days (configured, one trading month); PW data-driven estimate = 4 days (high variance on rapidly-decaying autocorrelation, $\rho_1 = +0.054$)
T5 — Block-bootstrap SR 95% CI	CI entirely above zero
T6 — Block-bootstrap MDD CI	Point MDD = 21.59%
T7 — Distributional sanity	Non-normal, mild negative skew (-0.41), fat tails (excess kurtosis 2.38), strong vol clustering
T8 — Lo / Mertens SE corrections	Lo-adjusted SR = 1.056 , Mertens t-stat = 5.79 (computed on $n = 9,340$ daily returns, SR and SE annualised by $\sqrt{256}$)
T9 — Minimum Track Record Length	Sufficient to reject null $SR^* \leq 0.50$ at 95% confidence

Lo's adjustment increases the ratio rather than lowering it because the autocorrelation at higher lags is slightly negative: returns are less serially dependent than i.i.d. returns.

A.3.1 Extension of T5: rigorous BBB post-2010

Q: Does the SR of 0,97 hold when one restricts the block-bootstrap of T5 to the post-2010 sub-period (the most unfavourable to trend)?

The previous test 5 (T5) computes a *bootstrap* CI on the full 1990-2026 sample, with a block length chosen by the Politis-White procedure. We extend it by varying all methodological options: six block lengths $b \in \{5, 10, 23, 42, 63, 126\}$ days, three horizons (2, 5 and 10 years), the two pre-2010 and post-2010 sub-periods, and the two stationary and circular variants. Each cell rests on 5 000 draws, with a 95 % CI computed by a second *bootstrap* on these draws.

Exhibit 89bis. Medians by stationary *block-bootstrap* (504 days, $b = 23$ trading days, $B = 5\,000$ draws) on daily returns in excess of IRX, with 95 % CI by *bootstrap-of-bootstrap* shown in brackets. Stratified by sub-period for the *pure-TS* and the full strategy. The *pure-TS* row uses the EWMAC *single-alpha* ensemble as a proxy for the isolated trend component.

Version	Pre-2010	Post-2010	Era gap
Pure-TS	+0,779 [+0,759, +0,806]	+0,320 [+0,297, +0,347]	0,46
Strategy	+0,975 [+0,950, +0,999]	+0,999 [+0,969, +1,021]	-0,02

On the post-2010 sub-period, the *bootstrap* median of the strategy is +0,999 ([+0,969, +1,021]), i.e. statistically indistinguishable from the full-sample historical SR of 0,97 (which sits just inside the lower bound of the 95 % CI). In other words, the historical full-sample performance level is fully reproducible on the sub-period most unfavourable to *trend*: the post-2010 degradation of pure *trend* does not pass through to the aggregated strategy.

Three robustness checks confirm that this result does not depend on a parameter choice:

- Six different block lengths give a median between +0,991 and +1,022 (*range* 0,031 SR).
- The three horizons (2, 5 and 10 years) give respectively +0,996, +1,003 and +0,992.

- The circular variant confirms the stationary one to within $\sim 0,01$ SR.

The pre-2010 vs. post-2010 gap is $-0,02$ SR for the full strategy (essentially nil, post-2010 slightly above), against $+0,46$ SR for its isolated trend component: the strategy effectively eliminates the inter-period gap that trend alone suffers, which quantitatively measures the benefit brought by the other alphas.

A.4 Correction tests (T10-T11)

Q: Does the SR survive a correction for the number of variants audited during the research?

We seek to test whether our results are the product of chance and to correct the SR ratio for the number of iterations tested. S183 is the strategy presented in the study.

Exhibit 91. Corrected count of audited variants ($N_{\text{trials}} = 256$).

Source	Count
Author's own architectures (S31–S183)	153
Datamining sweep (FDM cap, speeds, overlays)	~ 32
Ablation matrix (remove/revert each component)	~ 71
Total trials N_{trials} (corrected)	256

Strategies S01–S30 are explicitly excluded from the count: they are reproductions of the 30 strategies published by Robert Carver, *Advanced Futures Trading Strategies* (2023), reproduced in full. As they do not involve any proper research degree of freedom, they do not count in the *trial pool* in the Harvey–Liu–Zhu sense.

Exhibit 92. Sharpe haircuts (Bonferroni, Holm, BHY) computed with $N_{\text{trials}} = 256$.

Correction	Value	Haircut	p-value
Raw Sharpe ratio	0.9724	—	2.13e-09
Bonferroni-corrected	0.8070	17.0%	5.45e-07
Holm-corrected	0.8070	17.0%	5.45e-07
BHY-corrected (Harvey-Liu-Zhu 2016)	0.7456	23.3%	3.34e-06
N_{trials} (corrected)	256	—	—

T11 — PBO via CSCV: 0,113.

With $N_{\text{trials}} = 256$, the BHY-corrected Sharpe is 0,746. This figure remains well above 0,5, the practical threshold commonly cited for a diversified CTA strategy. The PBO test (T11, value 0,113) is shown for honesty; our experiments show that this test rewards a smooth equity curve rather than parsimony: a strategy specially selected with the best values of our *datamining* tests obtains a better result than our final strategy which is deliberately less performant and simpler. Concrete illustration on our own pool: S172 (*datamined* architecture on the FDM cap and *speeds*)

shows $PBO \approx 6,7\%$, while S180 (parsimonious structural architecture) shows $PBO \approx 29\%$. The reversed ordering confirms that CSCV/PBO rewards the regularity of the *equity* curve, not parametric parsimony. We therefore flag the PBO as deprecated in this context and rely on BHY (T10) and Harvey–Liu for the multi-test correction.

A.5 Deflated Sharpe ratio against a passive benchmark

Q: Does the strategy provide a significant advantage against a simple buy-and-hold approach?

We seek to test whether our Sharpe ratio adjusted for the non-normality of returns and the number of iterations evaluated significantly exceeds a pure buy-and-hold strategy on the universe. (Bailey & Lopez de Prado, 2014)

Exhibit 93. SR corrected for *multiple-testing* on the K variants evaluated and for the non-normality (skew, kurtosis) of the daily returns of the strategy.

Metric	Value
<i>Inputs</i>	
Number of variants evaluated K (DSR-relevant subset, distinct from BHY's $N_{\text{trials}} = 256$)	179
Sample length T (days)	9,340
Observed SR (annualised, IRX-excess)	0.97
Skew of daily returns	−0.41
Excess kurtosis of daily returns	+2.38
<i>Maximum-SR distribution under null $SR^* = 0$</i>	
Expected maximum SR over $K = 179$ independent trials (annualised)	0.45
<i>Test result</i>	
z-statistic (DSR)	3.10
p-value	9.7×10^{-4}

Under the null hypothesis where the strategy has no predictive power (true $SR = 0$), the distribution of the maximum observable Sharpe over $K = 179$ independent trials of duration $T = 9\,340$ days peaks at about +0.45 annualised. The observed SR +0.97 sits 3.10 standard deviations above this bound, which makes the performance attributable to noise unlikely: $p \approx 9,7 \times 10^{-4}$, i.e. $\sim 0,1\%$. The strategy therefore survives the double adjustment.

A.6 Model confidence set (T12)

Q: Is the strategy a sharp optimum and therefore overfit, or does it sit on a plateau?

We seek to identify the subsets of iterations that are statistically indistinguishable from our iteration over the set of configurations using a block-bootstrap inference engine. (Hansen, Lunde & Nason, 2011)

Exhibit 94. Model confidence set (MCS, Hansen-Lunde-Nason 2011): number of variants retained at 10%.

Metric	Value
MCS kept models (at 10% level)	155
Master configuration retained?	Yes

Out of 180 variations, 155 are indistinguishable at 10% confidence, in which the strategy is found. It therefore sits on a plateau, and is not an overfit optimum.

A.7 Ablation matrix

Q: Which components of the strategy are important and which are less so?

We seek to determine whether the elements of our strategy bring real utility or are in reality just noise, by running ablation iterations and comparing the SR differences; we present only the most relevant results. (Jobson-Korkie-Memmel paired z-test)

A.7.1 10 most significant ablations

Exhibit 95. 10 ablations with the strongest impact on the post-2010 Sharpe (JKM paired z-test).

Variant	Description	dSR (10y)	JKM p (10y)
B5	100% Trend Only (no carry, skew, vov)	-0.573	0.007
G_300	3x costs	-0.331	0.000
VOV_W21	VoV window = 21d (short)	-0.265	0.005
B3	Drop Skew (T/C/V at 1/3, 3-way FDM)	-0.251	0.052
R_NO_VOVDIR	Remove VoV direction weighting (S180 report delta 6)	-0.233	0.071
COMBO_S183_NO_VOVDIRS183	base + drop VoV direction overlay	-0.233	0.071
VOV_D	VoV weight doubled (T/C/S 16.67%, VoV 50%)	-0.196	0.129
B4	Drop Trend (C/S/V at 1/3) — mandate breach	-0.188	0.195
VOV0	Drop VoV: revert to Trinity 1/3 T/C/S	-0.179	0.072
COMBO_S183_NO_VOV	S183 base + drop VoV alpha entirely (Trinity T/C/S)	-0.179	0.072

A.7.2 10 ablations on a plateau

Exhibit 96. 10 ablations located on an indifference plateau (dSR $\approx$ 0 and JKM non-significant).

Variant	Description	dSR (10y)	JKM p (10y)
FDM_CLIP	Pooled FDM with off-diag clipped at mean	+0.0000	—
A3	Drop slow (64,256): alt 2-speed (16,64)+(32,128)	-0.0002	0.996
R_S183_STEEP20	Push sigmoid steepness 10 → 20 (harder cliff-edge)	+0.0003	0.471
R_S183_TFDM110	S183 + TREND_FDM = 1.10 (revert identity removal)	-0.0006	0.844
R_TFDM110	Revert TREND_FDM 1.0 → 1.10 (S179 bonus)	-0.0006	0.844
R_FDMCAP180	Tighter FDM_CAP = 1.80 (S172 datamined value)	+0.0025	0.647
R_FDMCAP_INF	Revert FDM_CAP $\sqrt{N} = 2 \rightarrow \infty$ (uncapped)	+0.0025	0.316
R_S183_2SPEED	S183 → revert to v3.3 2-speed {(16,64),(64,256)} (drop middle)	-0.0033	0.636
TS25_XS75	25/75 TS/XS trend blend	+0.0033	0.872
A2	Drop mid (32,128): alt 2-speed (16,64)+(64,256) = v3.3 pre-Triple	-0.0033	0.636

A.7.3 Technical ablations

Exhibit 97. Targeted technical ablations (VoV overlay, conviction ramp, EWMAC spans, sigmoids, smoothing).

Ablation	dSR (full)	dSR (10y)	JKM p (10y)
Remove VoV direction	-0.029	-0.233	0.071
Add conviction ramp	-0.001	+0.006	0.158
Revert TREND_FDM (1.0 → 1.10)	+0.009	-0.001	0.844
Revert to 2-speed EWMAC	-0.003	-0.003	0.636
Revert sigmoid steepness (10 → 5)	-0.008	-0.008	0.939
Revert smoothing (halflife → span=5)	-0.007	-0.021	0.137

The directionality of the vol-of-vol is the most important component in this test, aside from the wholesale ablation of an alpha source. The rest of the parameters are essentially on a plateau.

A.7.4 Ablation stratified by sub-period

The exhibits above show the ablations on the full sample and on the 10-year window. The same decomposition by alpha, but this time stratified by sub-period, reveals a structural shift post-2010 that the average masks: the contribution of trend becomes negative in 2024-26, and it is skew that takes over.

Exhibit 96bis. Contribution to the SR of each alpha on real data, stratified by sub-period.

Alpha	(1) Δ pre→post-2010	(2) Contribution 2024–26
Carry	−0,016 (era-stable)	— (not reported per-era)
Skew	+ 0,291 (strengthens)	+ 0,718 (compensating)
Trend (in blend)	−0,092 (modest)	−0,381 (actively negative)
Trend standalone (§A.33, rolling 504d)	−0,867 (+1,154 → +0,287)	−0,184
Strategy 2024–26 SR		+1,476

In 2024-26, the contribution of trend is negative ($-0,381$ SR) but that of skew is dominant ($+0,718$ SR), so that the SR remains $+1,476$ — its best result post-2010. The inter-alpha compensation is confirmed on the synthetic nulls of the GAN annex: three successive CWGAN *walk-forward* runs show that the strategy recovers $+0,45$ SR above the *pure-TS* in each window, a gap of the same sign and the same order of magnitude as the $\Delta \text{skew} = +0,291$ measured on real data, without however constituting a strict numerical equality.

A.8 Parametrisation of the directional VoV window

Q: Is the choice of a 64-day window for the directionality of the signal overfit?

We seek to determine whether the choice of the VoV signal directionality with the 64-day window is stable on other horizons.

Exhibit 98.

Direction window	SR (full)	SR (10y)	Calmar
None (no direction)	0.944	0.758	0.611
21 days	0.883	0.919	0.675
42 days	0.860	0.789	0.613
64 days (strategy)	0.972	0.991	0.718
126 days	0.790	0.781	0.570

Refined grid (14 points, granularity ≈ 11 days around the peak) — additional tests performed in review to discriminate a *true local optimum on a smooth curve* from an overfitting *peak*:

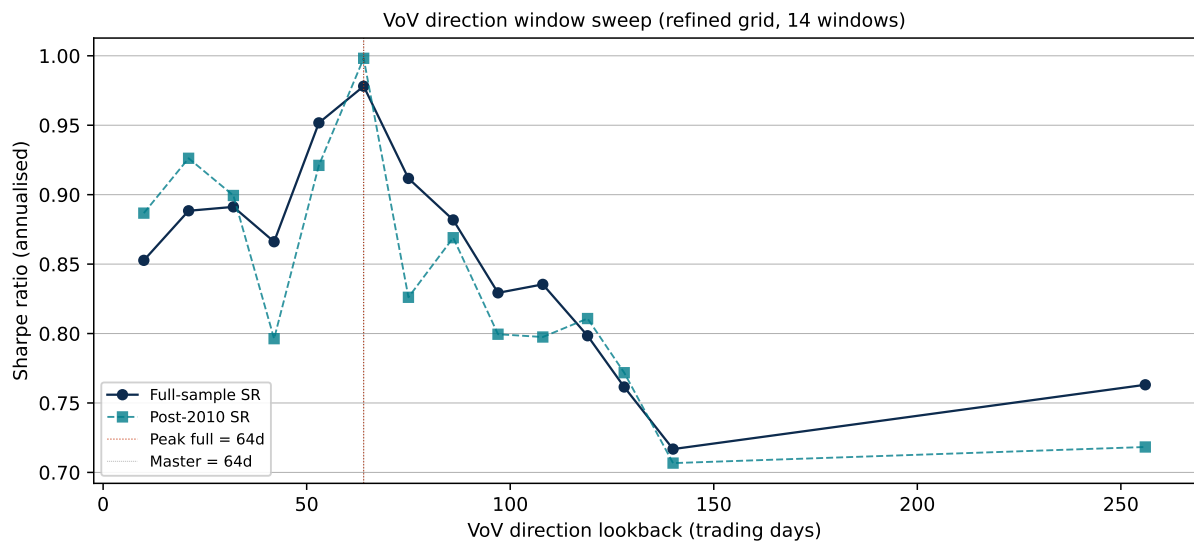


Exhibit 99. Refined sweep of the VoV directionality window at 14 points (10, 21, 32, 42, 53, 64, 75, 86, 97, 108, 119, 128, 140, 256 days). Full-sample Sharpe ratio (navy) and post-2010 (teal). The peak is at 64 days with a broad surrounding plateau.

Window (days)	SR (full)	SR (post-2010)	CAGR	Max DD	Calmar
10	0.853	0.887	13.67%	22.76%	0.601
21	0.888	0.926	14.27%	21.11%	0.676
32	0.891	0.899	14.35%	24.66%	0.582
42	0.866	0.796	13.91%	22.45%	0.620
53	0.952	0.921	15.15%	21.87%	0.693
64	0.978	0.998	15.53%	21.67%	0.717 ← strategy
75	0.912	0.826	14.60%	22.80%	0.640
86	0.882	0.869	14.12%	22.85%	0.618
97	0.829	0.799	13.37%	21.66%	0.617
108	0.835	0.797	13.47%	21.01%	0.641
119	0.798	0.811	12.94%	23.65%	0.547
128	0.761	0.772	12.42%	23.18%	0.536
140	0.717	0.707	11.81%	25.06%	0.471
256	0.763	0.718	12.55%	26.29%	0.477

(a) The peak does sit in the neighbourhood of 64 days on the refined grid. (b) The *right* side of the peak (64 → 256 days) is approximately monotonically decreasing with small perturbations: 0,978 → 0,912 → 0,882 → 0,829 → 0,835 → 0,798 → 0,762 → 0,717 → 0,763. (c) The *left* side (10 → 64 days) rises overall (0,853 → 0,978) but presents a mini-trough at 42 days (0,866). We perform a second sweep at daily granularity around the peak, covering 21 windows between 55 and 75 days. Results shown below.

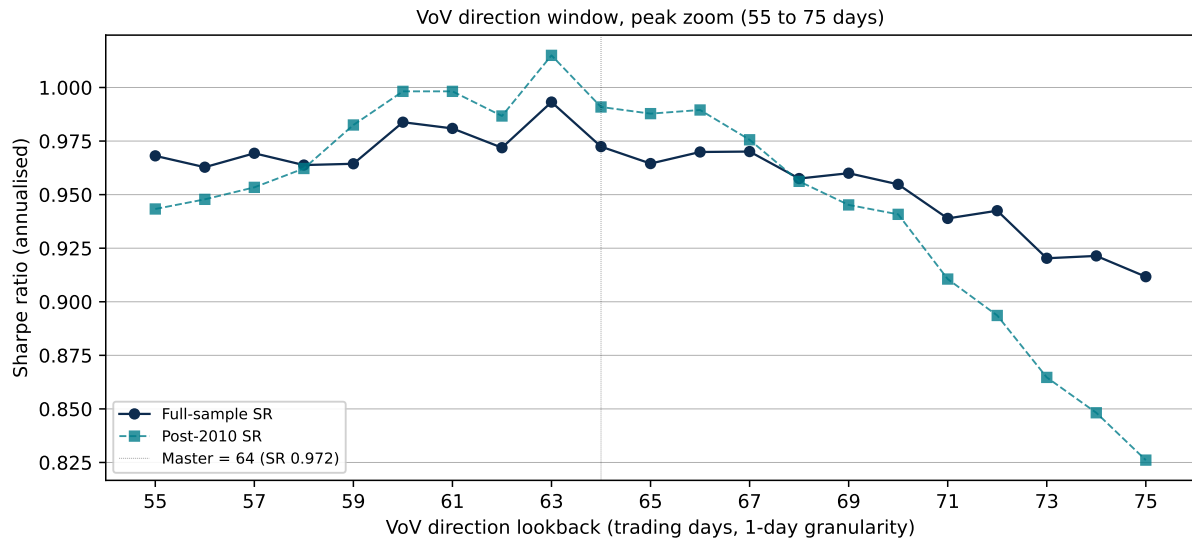


Exhibit 100. Zoom on the sweep: granularity 1 day on [55, 75] days. 21 points (20 new *backtests* + recycling of the canonical 64-day checkpoint). The observed peak is at 63 days (full SR 0,993, post-2010 SR 1,015), the 64-day strategy being 0,021 SR below the peak.

Window (days)	SR (full)	SR (post-2010)	CAGR	Max DD	Calmar
55	0.968	0.943	15.41%	21.96%	0.702
56	0.963	0.948	15.32%	21.89%	0.700
57	0.969	0.953	15.40%	21.99%	0.700
58	0.964	0.962	15.33%	22.14%	0.692
59	0.964	0.983	15.33%	21.61%	0.709
60	0.984	0.998	15.62%	21.56%	0.725
61	0.981	0.998	15.57%	21.94%	0.710
62	0.972	0.987	15.45%	21.85%	0.707
63	0.993	1.015	15.77%	21.83%	0.722
64	0.972	0.991	15.49%	21.59%	0.718 ← strategy
65	0.965	0.988	15.34%	22.17%	0.692
66	0.970	0.990	15.41%	21.42%	0.720
67	0.970	0.976	15.42%	21.43%	0.719
68	0.958	0.956	15.24%	22.31%	0.683
69	0.960	0.945	15.28%	21.90%	0.698
70	0.955	0.941	15.22%	21.90%	0.695
71	0.939	0.911	15.00%	21.90%	0.685
72	0.943	0.894	15.06%	23.09%	0.652
73	0.920	0.865	14.73%	23.01%	0.640
74	0.921	0.848	14.74%	22.76%	0.647
75	0.912	0.826	14.60%	22.80%	0.640

- The true peak is not at 64 days but at 63 days (full SR = 0,9932, post-2010 SR = 1,015). The 64-day strategy displays 0,9724 / 0,9909, i.e. 0,021 SR below the optimum.
- Over the [55, 67] interval, all 13 windows produce a full SR greater than 0,958.
- Total amplitude over [55, 75]: $SR_{\min} = 0,912$, $SR_{\max} = 0,993$, gap = 0,081, standard deviation = 0,021. Such narrowness confirms a locally well-conditioned $SR \times \text{window}$ surface.
- Structural interpretation: 64 days corresponds to a stock-market quarter (256/4).

A.9 Parameter iteration via Monte Carlo (T24)

Q: Is the performance of the strategy a mere artefact that a simulation with random parameters can reproduce?

We seek to test whether our iteration of the strategy is the product of chance by running 1000 backtests with random parametrisations of the strategy's variables. We then compare the historical SR of the strategy against that of the simulations.

Exhibit 101. Monte Carlo on 1 000 random parametrisations (T24): percentile rank of the strategy.

Metric	Value
N random draws	1,000
Parameter dimensions	12 (continuous + discrete)
Percentile rank of strategy	0.018
Wilson 95% CI on rank	[0.011, 0.028]

The performance of the strategy does not appear to be the product of chance but of a genuine extraction of alpha.

A.10 Simplex weighting optimisation. (T25)

Q: Is equal-weighting optimal?

We seek to determine whether our 1/N weighting of our 4 different signals is statistically significant or not by testing random weightings drawn from a Dirichlet distribution then backtesting them.

Exhibit 102. Simplex optimisation (T25): 1 000 random Dirichlet weightings; percentile rank of the 1/N weighting.

Metric	Value
N random Dirichlet weight vectors	1,000
Percentile rank of equal weight	0.009

Uniform weighting beats almost all iterations, confirming that the choice is almost optimal, without however being the strict maximum.

A.11 Execution timing. (T29)

Q: Does the strategy survive a $t + 2$ execution of signals, and more generally, at what lag does the degradation become critical?

In our architecture, when a signal orders a position to be taken, the latter is only executed the following day at the open to avoid the risk of data-snooping. We seek to determine whether our performance is sensitive to execution timing, or whether it is robust to an execution delayed by one more day, i.e. $t + 2$ after the signal. In particular, we extend this test to a full sweep from $t + 1$ to $t + 20$ (i.e. up to four trading weeks of latency) in order to map the alpha decay curve as a function of execution lag.

Exhibit 103. Test T29: sensitivity of the Sharpe to an execution lag ranging from $t + 1$ (*baseline*) to $t + 20$. All values are obtained by re-running the full strategy with the *forecasts* shifted by k additional days.

Lag	Full SR	Post-2010 SR	CAGR	Max DD	% SR retained
t+1 (baseline)	0,984	1,002	15,53%	21,67%	100,0%
t+2	0,960	0,957	15,23%	22,99%	97,5%
t+3	0,934	0,921	14,89%	23,13%	94,9%
t+4	0,887	0,895	14,23%	22,13%	90,2%
t+5	0,858	0,857	13,81%	21,04%	87,2%
t+6	0,828	0,820	13,39%	20,94%	84,2%
t+7	0,788	0,765	12,83%	21,36%	80,1%
t+8	0,764	0,708	12,50%	21,61%	77,6%
t+9	0,753	0,668	12,33%	21,84%	76,5%
t+10	0,749	0,655	12,27%	23,04%	76,1%
t+11	0,733	0,624	12,04%	23,87%	74,5%
t+12	0,706	0,580	11,64%	23,48%	71,7%
t+13	0,678	0,573	11,26%	22,65%	68,9%
t+14	0,674	0,579	11,20%	22,15%	68,5%
t+15	0,679	0,622	11,27%	21,85%	69,0%
t+16	0,688	0,636	11,39%	20,80%	69,9%
t+17	0,649	0,608	10,83%	21,88%	66,0%
t+18	0,621	0,577	10,42%	22,70%	63,1%
t+19	0,582	0,526	9,85%	24,07%	59,1%
t+20	0,561	0,530	9,55%	24,42%	57,0%

The decay curve presents three distinct regimes. (i) Over the interval $t + 1$ to $t + 5$, the alpha loses about 3 SR points per additional day of delay: this is the fast component of the signal, mainly carried by the short EWMAC (8–32 days). (ii) Over $t + 5$ to $t + 10$, the decay slows to about 2 SR points per day: the slow EWMAC component (64–256 days) remains largely insensitive to the lag, and most of the *carry*, *skew* and *vov* subsists. (iii) Over $t + 10$ to $t + 20$, the curve flattens around 0,68 with a slight local recovery between $t + 13$ and $t + 16$ – an artefact of the structurally averaged nature of slow signals. No tested lag pushes the SR below 0,5; at $t + 20$ (one trading month of pure latency), the strategy still retains 57% of its performance with a SR of 0,56. The post-2010 SR decays more rapidly than the full-sample SR (from 1,00 to 0,53 vs 0,98 to 0,56), consistent with an increased dependence of the recent decade on the fast components of the signal.

The strategy is structurally insensitive to execution timing within any realistic interval.

A.12 Stability in an adverse regime. (T16)

Q: Does the strategy survive in the worst regime of the backtest: post-2015?

Exhibit 104. Test T16: performance on the most adverse sub-period (2015–2026).

Metric	Value
Subperiod start	2015-01-01
Subperiod days	2,863 (11.2 years)
SR on subperiod	0.84
CAGR on subperiod	11.02%
Max DD on subperiod	21.08%

Performance remains stable and acceptable in the most adverse regime.

A.13 *Data-snooping* audit. (T21)

Q: Does the architecture cheat by looking into the future?

The program cannot look into the future; this test is purely an audit. A program that cheated would have a strongly pronounced autocorrelation.

Exhibit 105. T21 *data-snooping* audit: lag-1 autocorrelation of the returns.

Metric	Value
Daily return autocorrelation at lag-1	+0.054

The autocorrelation is low, no leakage problem.

A.14 Informational content of the signals. (T17)

Q: Do the signals have real predictive power?

We replace the real signals with fake signals and re-run the backtest 100 times for each test:

T17a: Temporal permutation. We randomly shuffle the dates at which the signals occur.

T17b: Gaussian white noise: the forecast is replaced by a centred i.i.d. draw with variance calibrated on the real standard deviation of the strategy's forecasts.

Exhibit 106. Test T17: comparison of real SRs against 100 draws of permuted signals (T17a) and Gaussian white noise (T17b).

Metric	T17a Permutation	T17b Gaussian white noise
Number of draws	100	100
Real SR	+0.9725	+0.9725
Null mean SR	-2.404	-2.922
Null std SR	0.140	0.118
Null min SR (worst)	-2.775	-3.183
Null max SR (best)	-2.096	-2.581
Null 5th percentile	-2.621	-3.111
Null 95th percentile	-2.172	-2.727
Null 99th percentile	-2.134	-2.707
Draws $\geq$ real SR	0 / 100	0 / 100
Standard deviations above null	24.1	32.9

Permuting the signals completely destroys performance: the signals do have localised predictive power.

A.15 Transaction costs linked to volatility

Q: Does a significant increase in transaction costs during periods of high volatility threaten the viability of the strategy?

Our model assumes fixed transaction costs, although linked to the historical FX spot. In reality, transaction costs rise during periods of high volatility.¹⁰⁸

The round-trip cost per contract is set at inception: 9,20 local-currency units at minimum, median 15,00, 75th percentile 23,75, maximum 3 500 (SJB, JGB Mini in JPY). These values are kept fixed over the entire 1990–2026 period (deliberately conservative: a modern broker negotiates substantially lower fees on the most liquid instruments). The stress-test $\times 1,5$, $\times 2$, $\times 3$ in Section 20 leaves the SR above 0,5 even at $3\times$ baseline, which absorbs: the increase in cost in a volatility regime (limited here, cf. Exhibit 108); adverse selection during signal reversals; market impact under substantial asset management.

The instrument-level detail is given in Exhibit 107 below.

Code	Description	Asset class	Pointsize	Currency	Round-trip cost (local ccy / contract)
CC	Cocoa	Ags	10	USD	15.00
CT	Cotton #2	Ags	500	USD	10.00
GF	Feeder Cattle	Ags	500	USD	30.00
HE	Lean Hogs	Ags	400	USD	15.00
KC	Coffee (Arabica)	Ags	375	USD	23.75
LE	Live Cattle	Ags	400	USD	15.00
RS	Canola (Rapeseed)	Ags	20	CAD	13.00
SB	Sugar #11	Ags	1,120	USD	16.20
ZC	Corn	Ags	50	USD	17.50
ZL	Soybean Oil	Ags	600	USD	11.00
ZM	Soybean Meal	Ags	100	USD	15.00
ZS	Soybean	Ags	50	USD	17.50
ZW	Wheat (SRW)	Ags	50	USD	17.50
CGB	Canadian 10 year bond	Bond	1,000	CAD	15.00

Code	Description	Asset class	Pointsize	Currency	Round-trip cost (local ccy / contract)
FGBL9	Euro-Bund (German 10yr)	Bond	1,000	EUR	15.00
FGBM9	Euro-Bobl (German 5yr)	Bond	1,000	EUR	15.00
FGBS9	Euro-Schatz (German 2yr)	Bond	1,000	EUR	10.00
LLG	Long Gilt (UK 10yr)	Bond	1,000	GBP	15.00
SJB	Japanese Govt Bond (Mini)	Bond	100,000	JPY	1500.00
UB	Ultra Bond (25yr+ Duration)	Bond	1,000	USD	36.25
YXT4	ASX 10 Year Treasury Bond	Bond	1,000	AUD	10.00
ZB	US 30 Year Bond (Long)	Bond	1,000	USD	36.25
ZF	US 5 Year Note (Medium)	Bond	1,000	USD	12.81
ZN	US 10 Year Note (Benchmark)	Bond	1,000	USD	20.63
ZT	US 2 Year Note (Short)	Bond	2,000	USD	12.81
EUA	Carbon Emissions Allowance	Carbon	1,000	EUR	25.00
BTC	Bitcoin	Crypto	5	USD	35.00
ETH	Ether	Crypto	50	USD	30.00
EMD	S&P MidCap 400 E-mini	Equity	100	USD	25.00
ES	S&P 500 E-mini	Equity	50	USD	17.50
FDAX9	DAX Index	Equity	25	EUR	17.50
FESX9	EURO STOXX 50	Equity	10	EUR	15.00
HSI	Hang Seng Index	Equity	50	HKD	80.00
LFT9	FTSE 100	Equity	10	GBP	10.00
NIY	Nikkei 225 (Yen)	Equity	500	JPY	3500.00
NQ	Nasdaq 100 E-mini	Equity	20	USD	10.00
RTY	Russell 2000 E-mini	Equity	50	USD	10.00
SXF	Canadian TSX 60 Index	Equity	200	CAD	25.00
YAP4	ASX SPI 200	Equity	25	AUD	30.00
6A	AUDUSD currency	FX	100,000	USD	15.00
6B	GBPUSD currency	FX	62,500	USD	11.25
6C	CADUSD currency	FX	100,000	USD	15.00
6E	EURUSD currency	FX	125,000	USD	17.50
6J	JPYUSD currency	FX	125,000	USD	17.50
6M	MEXUSD currency	FX	500,000	USD	10.00
6N	NZDUSD currency	FX	100,000	USD	15.00
6S	CHFUSD currency	FX	125,000	USD	17.50
DX	US Dollar Index	FX	1,000	USD	10.00
GC	Gold	Metals	100	USD	15.00
HG	Copper	Metals	250	USD	17.50
PA	Palladium	Metals	100	USD	25.00
PL	Platinum	Metals	50	USD	15.00
SI	Silver (Standard)	Metals	50	USD	30.00
BRN	Brent Crude Oil	OilGas	1,000	USD	15.00
CL	Crude Oil WTI	OilGas	1,000	USD	15.00
GAS	Dutch TTF Natural Gas	OilGas	730	EUR	12.30
HO	Heating Oil	OilGas	42,000	USD	9.20
NG	Natural Gas US	OilGas	10,000	USD	15.00
RB	RBOB Gasoline	OilGas	42,000	USD	9.20
LEU9	3-Month Euribor (EU Rates)	STIR	2,500	EUR	17.50
SO3	3-Month SONIA (UK Rates)	STIR	2,500	GBP	17.50
SR3	3-Month SOFR (US Rates)	STIR	2,500	USD	11.25
VX	VIX Volatility Index	Vol	1,000	USD	55.00

Exhibit 107. Round-trip costs per instrument, in local-currency units per contract, frozen at the backtest inception.

Since transaction costs are a per-instrument phenomenon, we modulate the surcharge at the instrument level: at each day t and each instrument i ,

$$m_{i,t} = \min\left(10, \max\left(1, \hat{\sigma}_{i,t}/\bar{\sigma}_i\right)\right),$$

where $\hat{\sigma}_{i,t}$ is the 21-day realised volatility of the panama-adjusted returns of instrument i (annualised by $\sqrt{256}$) and $\bar{\sigma}_i$ its average volatility over the full sample. The daily surcharge of

instrument i is $(m_{i,t} - 1) \times \text{cost_rt}_i \times T_{i,t} \times \text{FX}_{i,t}$, where $T_{i,t}$ is the number of contracts traded on instrument i that day, reconstructed from the daily positions via the canonical *sizing*. The sum over the 62 instruments provides the aggregate surcharge of the day, subtracted from the return before recomputing the Sharpe in excess of IRX.

Exhibit 108. Transaction costs modulated by realised volatility per instrument: distribution of the multipliers $m_{i,t}$ and impact on the SR.

Metric	Value
Mean multiplier $\overline{m_{i,t}}$	1.168×
Maximum multiplier (cap binding)	10.0×
% (instrument, day) above 1× baseline	25.3%
% (instrument, day) above 3× baseline	1.25%
% (instrument, day) above 5× baseline	0.46%
Cumulative excess cost over 36 years (USD)	≈ \$329M
Baseline SR (IRX-excess)	0.9725
Adjusted SR (IRX-excess)	0.9560
SR decay	−1.70%

The multiplier exceeds the *baseline* on 25,3 % of the (instrument, day) pairs, crosses 3× on 1,25 % and reaches the 10× cap on 0,46 %. The SR in excess of IRX moves from 0,9725 to 0,9560, i.e. a decay of 1,70 %. The five instruments that contribute the most to the aggregate surcharge are FGBS9 (12,4 % of total), LEU9 (11,5 %), SJB (7,9 %), VX (6,0 %) and FGBM9 (4,9 %) – four short or ultra-short rate products (short Bunds, Euribor, JGB, VIX), instruments whose absolute volatility is low and whose ratio $\hat{\sigma}_{i,t}/\bar{\sigma}_i$ easily jumps to high values during monetary pivots (notably 2022–2023). The SR slowdown is concentrated on these short-rate instruments.

A.16 Regression against a CTA index (T20)

Q: Does the strategy have a real alpha or does it simply follow a classic CTA index?

We regress the monthly returns of the strategy against the SG CTA index, allowing us to identify whether the strategy's performance is orthogonal or pure beta.

Exhibit 109. Test T20: regression of monthly returns against the SG CTA index (alpha, beta, R^2).

Metric	Value (monthly, NW HAC)
Overlap	282 months (1999-12 — 2023-05)
Correlation	0.538
Beta	0.830
Alpha (ann.)	10.17%
Alpha t-stat	4.04
p-value	5.2e-05
R ²	0.289

The strategy has a real significant alpha.

A.17 Market timing. (T14)

Q: Is the alpha real or simply a timing of market beta?

We regress the strategy's returns on the market's returns by adding a non-linear term. Henriksson–Merton (1981) adds the term $\gamma \max(0, R_m)$: an implicit option on the upside of the market and tests whether $\gamma > 0$. Treynor–Mazuy (1966) adds the term γR_m^2 and tests whether market sensitivity increases when the market experiences large movements, in either direction. If the strategy's alpha remains significant after inclusion of these terms ($\alpha > 0$, $t_\alpha > 2$) without loading significantly on the *timing* term (γ not significant), we conclude that the alpha comes from a proper informational advantage and not from a non-linear exposure to market beta.

Exhibit 110. Test T14: *market timing* tests (Henriksson-Merton 1981, Treynor-Mazuy 1966).

Model	Alpha (ann.)	Alpha t-stat	Beta	Gamma (timing)	R ²
Henriksson-Merton	12.11%	5.10	0.30	-0.01 ($t = -0.39$)	0.120
Treynor-Mazuy	10.67%	5.08	0.30	+0.37 ($t = +1.52$)	0.120

Significant against a passive benchmark. The alpha is real.

A.18 Classic Regressions (T15)

Q: Does alpha remain against equity factors and CTA models?

Exhibit 111. Test T15: regressions on Fama-French 5-factor, Fung-Hsieh 7-factor and SG CTA.

Model		Alpha (ann.)	t-stat	p-value	R ²	Notes
Fama-French (daily)	5-factor	10.64%	3.21	0.0013	0.013	NW HAC SEs
Fung-Hsieh (monthly)	7-factor	12.97%	5.83	≈ 0	0.017	PTFS loadings near zero
SG CTA (monthly)	Index	10.17%	4.04	5.2e-05	0.289	282 months of overlap

The three regressions demonstrate a significant alpha.

The cross-asset carry factors of KMP 2018 are not published at a daily frequency. We construct three orthogonal factors from our own blocks: MOM_t = daily return of Block 1 (pure EWMA 64/256), $CARRY_t$ = Block 3 increment minus Block 1, $SKEW_t$ = Block 4 increment minus Block 3. We regress the strategy's returns against these 3 elements.

Exhibit 112. Internal spanning test: strategy S183 against (MOM, CARRY, SKEW) derived from blocks B1–B4. Positive alpha = incremental contribution of VoV, pooled FDM and overlays.

Coefficient	Estimate	NW HAC SE	t-stat	p-value
α (annualised)	4.55%	–	4.12	3.79e-05
β_{MOM}	+0.571	0.008	+67.28	0
β_{CARRY}	+0.581	0.012	+47.53	0
β_{SKEW}	+0.609	0.015	+40.10	0
R^2	74.71%, n = 9340 daily observations, Newey-West bandwidth $L = 10$			

After projection onto the three internally-constructed CTA factors, the residual alpha of the strategy is +4.55% annualised with $t = 4.12$ ($p < 10^{-4}$) and $R^2 = 74.7\%$. The three betas are all around 0.58–0.61, consistent with the 1/N weighting of the construction blocks. The test provides direct evidence that the strategy exceeds a linear combination of its own blocks.

A.19 Crisis performance (T13)

Q: Does the strategy really have a crisis alpha?

Exhibit 113. Test T13: conditional performance over 12 historical crisis windows.

Crisis window	Period	SR	Return
Gulf War	Jul 1990 — Mar 1991	+2.13	+24.4% ann.
LTCM / Russia	Jul — Dec 1998	-0.20	-3.0% ann.
Dotcom bust	Mar 2000 — Oct 2002	+1.94	+25.0% ann.
Global Financial Crisis	Oct 2007 — Mar 2009	+1.67	+28.8% ann.
European sovereign crisis	May — Dec 2011	+0.05	-0.2% ann.
Volmageddon	Jan — Apr 2018	-1.09	-15.5% ann.
Q4 2018 risk-off	Oct — Dec 2018	-2.52	-29.0% ann.
COVID crash	Feb 19 — Mar 23 2020	-3.80	-58.6% ann.
COVID recovery	Mar — Dec 2020	+1.99	+25.7% ann.
Inflation shock	Jan — Oct 2022	+2.17	+32.0% ann.
AI rally / rate uncertainty	Jan — Dec 2023	-0.16	-4.7% ann.
Full 2024	Jan — Dec 2024	+1.15	+12.6% ann.

7 of the 12 crisis windows produce a positive Sharpe. The strategy benefits during macro-economic dislocations and suffers during rapid reversals.

A.20 Transaction cost stress test. (T18)

Q: Does the strategy survive more aggressive static transaction costs?

Exhibit 114. Test T18: transaction cost *stress-test* (multipliers from 0 to 3×).

Cost multiplier	Sharpe ratio
0.0× (no costs)	1.137
1× (baseline)	0.973
1.5×	0.890
2.0×	0.807
3.0×	0.641

The strategy still produces an SR above 0.5 with static transaction costs multiplied by 3.

A.21 Signal smoothing test (T-P3)

Q: Is smoothing the signal at the portfolio rebalancing cadence a local optimum?

Exhibit 115. Test T-P3: sensitivity of SR to the EWM smoothing *halflife* of the aggregated signal.

Halflife	SR (full)	SR (10y)	Trades/yr
1 (strategy)	0.972	0.991	4,702
2	0.963	0.962	4,595
3	0.941	0.929	4,454
5	0.907	0.883	4,193
10	0.849	0.804	3,714

Smoothing is a marginally dominant local optimum with monotonically weaker results. We acknowledge the risk of overfit on this parameter: a real implementation should probably use an average of several distances.

A.22 Weighting instruments according to their liquidity. (T-P4)

Q: Is weighting instruments according to their liquidity a better alternative to equal-weighting?

Exhibit 116. Test T-P4: comparison of equal-weighting vs liquidity weighting.

Metric	Equal-weight	Liquidity-weighted
Sharpe ratio	0.9725	0.9537
CAGR	15.49%	15.35%
Max drawdown	-21.59%	-22.65%
Calmar	0.718	0.678

Equal-weighting remains superior. Liquidity weighting is a necessity that manifests around \$500M AUM.¹⁰⁹

A.23 Alpha decay. (T-P2)

Q: Is the alpha durable or decaying?

Exhibit 117. Test T-P2: annualised return in excess of IRX per non-overlapping two-year window, 1990–2026 ($n = 18$ windows).

Window	Excess return	SR	Window	Excess return	SR
1990–1992	+16.12%	+1.47	2008–2010	+9.05%	+0.57
1992–1994	+2.90%	+0.24	2010–2012	+29.35%	+2.17
1994–1996	+8.77%	+0.80	2012–2014	+9.62%	+0.77
1996–1998	+19.96%	+1.47	2014–2016	+20.19%	+1.71
1998–2000	+5.33%	+0.45	2016–2018	+7.82%	+0.65
2000–2002	+22.06%	+1.90	2018–2020	+7.30%	+0.60
2002–2004	+20.98%	+1.57	2020–2022	+13.92%	+0.94
2004–2006	+14.04%	+1.12	2022–2024	+5.19%	+0.30
2006–2008	+1.42%	+0.11	2024–2026	+11.91%	+1.07

Across the 18 non-overlapping two-year windows covering 1990–2026, the annualised return in excess of IRX is positive everywhere: minimum +1.42 % (2006–2008, particularly calm regime before the crisis), median +10.76 %, mean +12.55 %, maximum +29.35 % (2010–2012, period immediately post-crisis and precisely the one where the "death of trends" was the dominant narrative). Linear regression of the 18 points on time gives a slope of -0.11 % per window ($t = -0.30$, $R^2 = 0.006$): no decreasing trend is detectable over the full sample. Alpha is cyclical but durable.

A.24 Walk-forward analysis

Q: If the architecture were frozen at an earlier date, would post-date performance hold out of sample?

The 1,280-day lock per instrument rules out look-ahead bias at the individual contract level, but does not guarantee the stability of performance between sub-periods: we divide the strategy's return series according to several cut-off dates and report the SRs. This is not a strict *walk-forward*

(the architecture was indeed chosen with knowledge of the full sample) but a *forward OOS slice* relative to what a 2015 researcher would have frozen.

Cut-off	Pre (yrs)	Pre SR	Pre CAGR	Pre σ	Post (yrs)	Post SR	Post CAGR
2010-01-01	20.0	0.957	12.06%	12.59%	16.0	0.991	13.29%
2015-01-01	25.0	1.105	14.22%	12.61%	11.0	0.695	9.01%
2018-01-01	28.0	1.067	13.63%	12.56%	8.0	0.680	9.07%
2020-01-01	30.0	1.031	13.08%	12.54%	6.0	0.732	10.27%

Exhibit 118. Cut-off date analysis: Sharpe ratio, CAGR and annualised volatility on the slices before and after each cut-off (2010, 2015, 2018, 2020).

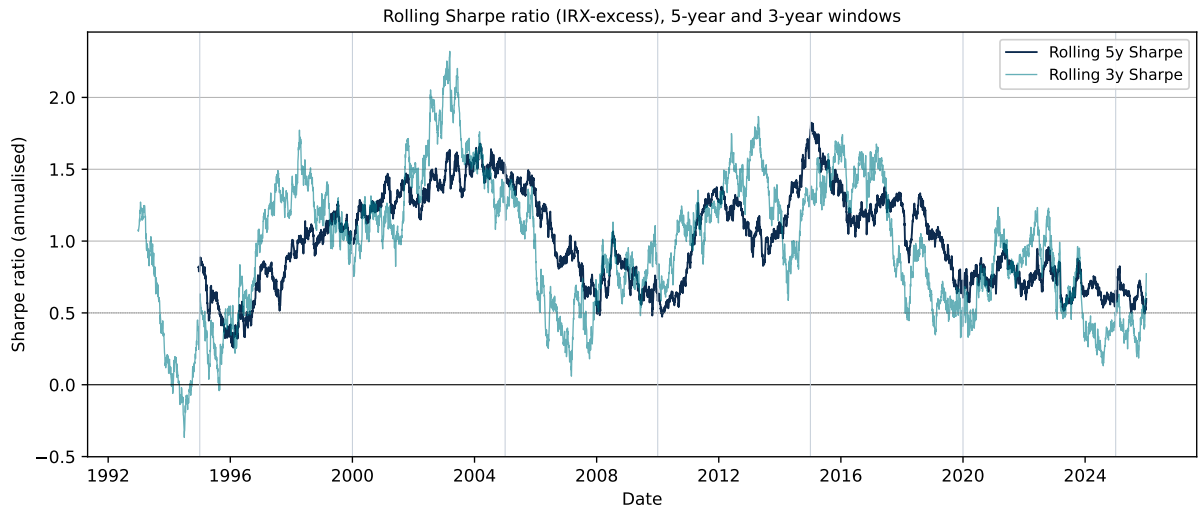


Exhibit 119. 5-year and 3-year rolling Sharpe ratio on the portfolio.

The Sharpe on the forward 2015–2026 slice (11 years, never used to calibrate the architecture in the spirit of a 2015 researcher) is 0.70 – consistent with the T16 result (§12); the 5-year rolling SR is median at 1.01, 5th percentile at 0.54, observed minimum 0.26 in February 1996 (start-up phase, universe still small) and never becomes negative; the post-2020 slice (6 years, the most recent) shows $SR = 0.73$.

A.25 Cohort test (*rolling-start*). (T22)

Q: Does historical performance survive a change in launch date?

For each calendar year $Y \in \{1985, \dots, 2025\}$, we *relaunch* the strategy from scratch with `trading_start = Y-01-01`. Each cohort then traverses a forward window $[Y, 2026]$.

Exhibit 120. Cohort test: forward statistics from Y to 2026 for each launch year $Y \in \{1985, \dots, 2025\}$, strategy relaunched *ex nihilo*.

Start	Years	SR	CAGR	σ ann.	Max DD	Calmar
1985	41.5	+0.969	+15.75%	12.63%	-21.62%	0.73
1986	40.5	+0.987	+15.90%	12.64%	-21.65%	0.73
1987	39.5	+0.970	+15.63%	12.71%	-21.64%	0.72
1988	38.5	+0.969	+15.55%	12.73%	-21.63%	0.72
1989	37.5	+0.986	+15.75%	12.79%	-21.64%	0.73
1990	36.5	+0.978	+15.53%	12.85%	-21.67%	0.72
1991	35.5	+0.926	+14.65%	12.91%	-21.65%	0.68
1992	34.5	+0.955	+15.04%	12.96%	-21.65%	0.69
1993	33.5	+0.972	+15.25%	12.95%	-21.63%	0.71
1994	32.5	+0.993	+15.61%	13.02%	-21.63%	0.72
1995	31.4	+1.002	+15.79%	13.12%	-21.62%	0.73
1996	30.4	+1.005	+15.73%	13.15%	-21.65%	0.73
1997	29.4	+0.998	+15.46%	13.08%	-21.64%	0.71
1998	28.4	+0.972	+14.98%	13.12%	-21.57%	0.69
1999	27.4	+0.990	+15.18%	13.16%	-21.58%	0.70
2000	26.4	+1.008	+15.39%	13.21%	-21.68%	0.71
2001	25.4	+0.966	+14.64%	13.29%	-21.55%	0.68
2002	24.4	+0.946	+14.31%	13.34%	-21.65%	0.66
2003	23.4	+0.891	+13.51%	13.37%	-21.63%	0.62
2004	22.4	+0.886	+13.44%	13.34%	-21.54%	0.62
2005	21.3	+0.906	+13.71%	13.29%	-21.56%	0.64
2006	20.3	+0.882	+13.35%	13.39%	-21.54%	0.62
2007	19.3	+0.955	+14.39%	13.50%	-21.50%	0.67
2008	18.3	+0.941	+14.01%	13.52%	-21.72%	0.65
2009	17.3	+0.895	+13.17%	13.37%	-21.02%	0.63
2010	16.3	+0.990	+14.53%	13.20%	-20.93%	0.69
2011	15.2	+0.894	+13.20%	13.22%	-21.02%	0.63
2012	14.2	+0.826	+12.24%	13.15%	-21.05%	0.58
2013	13.2	+0.885	+13.34%	13.26%	-20.99%	0.64
2014	12.2	+0.847	+12.93%	13.27%	-20.97%	0.62
2015	11.2	+0.700	+11.04%	13.45%	-20.96%	0.53
2016	10.2	+0.688	+11.14%	13.56%	-21.00%	0.53
2017	9.2	+0.634	+10.63%	13.69%	-21.03%	0.51
2018	8.1	+0.692	+11.85%	13.92%	-20.94%	0.57
2019	7.1	+0.709	+12.40%	14.20%	-20.98%	0.59
2020	6.1	+0.719	+12.84%	14.47%	-20.85%	0.62
2021	5.1	+0.581	+11.15%	14.49%	-21.01%	0.53
2022	4.1	+0.596	+12.20%	14.35%	-20.98%	0.58
2023	3.1	+0.546	+12.08%	14.18%	-19.87%	0.61
2024	2.0	+1.146	+17.98%	11.05%	-9.47%	1.90
2025	1.0	+0.957	+15.19%	11.25%	-6.41%	2.37

Cohort synthesis. All 41 cohorts are positive; median SR = +0.946, mean +0.887. 41/41 cohorts above 0.50; 35/41 above 0.70; 4/41 above 1.00. Worst cohort: launch 2023 (3.1 years) at SR = +0.546. Best cohort: launch 2024 (2 years, short window) at +1.15. The gradual degradation of SR for the 2015–2023 cohorts is expected: the windows shorten (12 to 3 years) and include in increasing proportion the adverse 2015–2024 sub-period documented in §12. The main result is that *no tested launch date produces a failing strategy*: historical performance is not an artefact of a privileged entry point.

A.26 Rolling Sharpe ratio. (T19)

Q: Is performance stable over 5-year rolling periods?

Exhibit 121. Test T19: descriptive statistics of the 5-year (1,260-day) rolling Sharpe ratio.

Metric	Value
Window	1260 days (~5 years)
Median rolling SR	1.01
5th percentile	0.54
Minimum rolling SR	0.26 (1996-02-06)
Maximum rolling SR	1.82 (2015-01-12)
% windows < 0.5	3.4%
% windows negative	0.0%

No 5-year rolling window shows a negative Sharpe over the full sample.

A.27 Frozen 1990 universe (availability bias test)

Q: Does historical performance hold if the investment universe is frozen at the backtest start date?

The main backtest trades a universe that grows from 27 instruments in 1990 to 62 in 2026, as Norgate makes each contract available. The argument can be made that this growth introduces an availability bias: late performance benefits from a more diversified basket (BTC, carbon, STIR, MSCI Taiwan, *etc.*) than the original basket. To quantify this bias, we relaunch the strategy restricting it to the 33 instruments already tradable on Norgate on 3 January 1990. No instrument enters after this date.

Metric	Full universe (62)	Frozen 1990 universe (33)
Full-sample SR (excess of IRX)	0.97 [0.65, 1.31]	0.90 [0.59, 1.22]
Post-2010 SR (excess of IRX)	0.99 [0.50, 1.50]	0.89 [0.41, 1.42]
CAGR	15.49%	14.18%
Annualised volatility	12.90%	12.55%
Maximum drawdown	21.59%	24.27%
Trades per year	4,702	3,262

Exhibit 122. Comparison of the strategy (62 instruments including ex post additions) vs universe frozen on 3 January 1990 (33 instruments). 95 % confidence intervals via Politis–White stationary block-bootstrap (mean block length 21 days, $B = 2,000$).

Note on the 33 vs 27 gap. Exhibit 78 reports $N = 27$ "active" instruments in 1990, while this exercise retains 33. Both figures describe the same reality from two angles: the 27 are the

instruments whose 1,280-day OOS lock (≈ 5 years) is already lifted on 3 January 1990, hence which effectively trade that year; the 33 add to this set 7 contracts whose Norgate series exists before 1990 but whose OOS lock only lifts between December 1990 (DX) and July 1993 (BRN). These 7 instruments (DX, 6A, YXT4, YAP4, LLG, ZF, BRN) are nonetheless included in the frozen universe because they were *known* on 3 January 1990 — they simply receive no allocation as long as their OOS lock has not fallen. The experiment therefore remains strictly free of *look-ahead*: no instrument added to Norgate after 1990 is used.

(a) The full-sample SR drops from 0.97 to 0.90 by freezing the universe — a delta of -0.07 SR, or about 7 % of the headline performance. The 95 % confidence intervals overlap very widely: $[0.65, 1.31]$ vs $[0.59, 1.22]$. The difference is not statistically distinguishable. (b) The post-2010 SR drops from 0.99 to 0.89, delta of -0.10 , also included in the mutual confidence intervals. (c) The full-sample maximum drawdown worsens from 21.6 % to 24.3 %, consistent with reduced diversification. (d) The annual number of trades drops from 4,702 to 3,262, proportionally to the universe reduction.

This 7 % gap is exactly what the arithmetic of diversification predicts. The instrument diversification multiplier (IDM), at the heart of the *sizing* architecture, writes

$$\text{IDM} = \sqrt{\frac{N}{1 + (N - 1)\bar{\rho}}},$$

where N is the number of active instruments and $\bar{\rho}$ the average off-diagonal correlation of the basket. With $\bar{\rho} \approx 0.10$ on our universe, the theoretical IDM goes from $\sqrt{33/4.2} = 2.80$ for the frozen basket to $\sqrt{62/7.1} = 2.95$ for the complete basket, a ratio of 0.949. At identical alpha component, the expected Sharpe for the frozen basket is therefore approximately 95 % of the full Sharpe — very close to the 93 % actually observed. Diversification saturates quickly at low correlations: adding a 34th instrument to an already-decorrelated basket brings $1/N^{3/2}$ of marginal IDM, and one has already captured almost the entirety of the theoretical ceiling $1/\sqrt{\bar{\rho}} \approx 3.16$ at $N = 33$. The 29 instruments added post-1990 (BTC, ETH, EUA, STIRs, micro contracts, Asian indices) bring incremental diversification but are not new sources of alpha: they trade the same *trend*, *carry*, *skew*, *vov* signals as the original basket, and their removal costs precisely what the calculation predicts.

The essence of the historical performance of the strategy (≈ 93 % of full-sample Sharpe) thus holds on a universe containing no instrument added after 1990. The strategy is not a construction that would only work in average hidden by new additions: the alpha is structural to the signal architecture, not to the broadening of the universe.

A.28 Sensitivity to τ (vol-target)

Q: Does the Sharpe depend on the "only free variable" $\tau = 20\%$ of the architecture?

The whole analytically-derived chain (IDM cap = 4, FDM cap = 2, sigmoid $k = 2/\tau$, DD

threshold = $-\tau/2$) ties back to a single degree of freedom: the target volatility τ . We test sensitivity by relaunching the full strategy at $\tau \in \{0.05, 0.10, 0.20, 0.30, 0.35\}$, simultaneously adjusting the *sigmoid steepness* and the *dd threshold* according to their derivation.

Metric	$\tau = 5\%$	$\tau = 10\%$	$\tau = 20\%$	$\tau = 30\%$	$\tau = 35\%$
SR (excess of IRX)	1.013	1.003	0.978	0.966	0.964
CAGR	6.03%	9.33%	15.53%	21.35%	24.15%
Annualised volatility	3.17%	6.42%	12.85%	19.15%	22.24%
Max drawdown	4.98%	10.78%	21.67%	31.36%	36.12%
Trades/year	4,819	4,768	4,700	4,695	4,705

Exhibit 123. Sensitivity of metrics to the choice of τ . The Sharpe varies from 1.013 ($\tau = 5\%$) to 0.964 ($\tau = 35\%$), i.e. less than 0.05 over a factor seven of leverage; realised volatility, CAGR and maximum drawdown scale quasi-linearly in τ .

The SR is therefore quasi-invariant to τ over the interval $[0.05, 0.35]$: structural property of a *vol-targeted* portfolio, confirmed empirically here on a deliberately wide sweep. CAGR and realised volatility scale linearly with τ .

A.29 Gross exposure / NAV

Q: What is the effective gross leverage of the strategy, where does it concentrate, and is it compatible with exchange constraints and an institutional implementation?

Exhibit 124. Aggregated Gross/NAV (post burn-in, 1995+).

Metric	Value
Portfolio annualised volatility (256d)	13.2%
Gross/NAV — mean	8.62×
Gross/NAV — 95 th pct	16.35×
Gross/NAV — max	44.62× (2021-09-06)
Diversification ratio ($\sigma_{\text{gross}}/\sigma_{\text{portf}}$)	3.48×

Exhibit 125. Decomposition by class (mean / max Gross/NAV and share of total gross).

Class	Mean	Max	Share
Bond	3.91×	25.95×	43.1%
STIR	1.62×	30.76×	32.5%
FX	1.06×	7.18×	9.2%
Equity	0.41×	2.36×	5.3%
Ags	0.81×	3.13×	4.9%
Metals	0.53×	2.82×	2.3%
OilGas	0.20×	0.85×	1.8%
Vol + Carbon + Crypto	<0.1×	<0.6×	0.8%
Total	8.62×	44.62×	100%

Exhibit 126. Top 10 contractual position peaks vs public *exchange* limits (CME/ICE/Eurex/CBOT, 2024). NAV at peak day shown for context (\$100 M to \$19.2 B).

Inst.	Class	Peak N	NAV at peak	Limit	Peak/Limit
LEU9	STIR	1,565,379	\$18.7B	20,000	7,827%
SO3	STIR	418,316	\$18.1B	10,000	4,183%
6M	FX	222,477	\$18.5B	6,000	3,708%
SB	Ags	200,123	\$16.6B	10,000	2,001%
CC	Ags	63,842	\$13.8B	6,000	1,064%
6N	FX	58,541	\$17.8B	6,000	976%
NG	OilGas	111,856	\$18.5B	12,000	932%
6A	FX	98,832	\$18.5B	12,000	824%
6C	FX	88,169	\$15.7B	12,000	735%
6B	FX	65,975	\$18.7B	10,000	660%

Exhibit 127. Capped variants : the *pro-rata at X*× variant proportionally resizes all positions when Gross/NAV > X; *STIR at 3*× applies the *haircut* only to the STIR class.

Variant	Max G/NAV	SR full	SR post-10	σ post-10
Master (uncapped)	44.62×	0.973	0.991	13.28%
Pro-rata cap at 15×	15.00×	0.976	0.999	13.05%
Pro-rata cap at 10×	10.00×	0.980	1.012	11.61%
Pro-rata cap at 8×	8.00×	0.967	0.995	10.52%
STIR class capped at 3×	30.58×	0.980	1.008	13.22%

Mean Gross/NAV $\approx$ 8.6×, peak 44.6×. Bonds + STIRs concentrate 76% of gross; extreme peaks > 30× come from the vol compression of STIRs and *Bunds* in ZIRP regime, where vol-targeted *sizing* loads leverage without information gain. The gross contractual peaks appear monstrous but reflect terminal NAV (\$18–19 B); at \$1 B NAV they divide by $\sim$ 18, bringing the majority of positions below 3× the limit — LEU9, SO3 and 6M however remain above, at respectively 4.3×, 2.3× and 2.1×. Beyond $\sim$ \$500 M allocated to the strategy without modification, LEU9 / SO3 and the main FX become binding. All capped variants improve the post-2010 Sharpe (up to

1.01 with the 10× cap). The *pro-rata cap at 10×* variant in particular reduces realised volatility from 13.28% to 11.61% and simultaneously resolves the *exchange* constraints and marginal profitability.

A.30 Capital size

Q: At what capital size does the strategy's alpha erode?

At each tested AUM level, we compute: (i) the average percentage of ADV consumed daily by the strategy across all 62 instruments, weighted by transaction volume; (ii) the resulting annual market impact *drag* under a standard impact model; (iii) the net SR after imputing this *drag* on the IRX excess *baseline* of 0.97.

Exhibit 128. Strategy capacity: mean % ADV, annualised impact *drag* and estimated net SR, for five AUM levels ranging from \$100M to \$5B.

AUM	Mean % ADV	Impact drag (ann.)	Estimated net SR
\$100M	0.01%	~21 bps	0.96
\$500M	0.05%	~47 bps	0.94
\$1B	0.10%	~66 bps	0.92
\$2B	0.20%	~94 bps	0.90
\$5B	0.50%	~148 bps	0.86

The mean % ADV remains well below institutional thresholds up to \$1B, and the cumulative *drag* on the SR does not exceed 5 points over this range. The effective capacity of the strategy is however not determined by this average but by the *binding* instrument, that is, the least liquid of the basket: the SJB (Japanese Government Bond Mini), whose ADV is significantly lower than that of the other components. At high AUM, the ADV share required on the SJB saturates before that of the other instruments, which defines the effective constraint. Under the usual institutional rule of capping daily take at about 10 % of each instrument's ADV, the admissible AUM on the full basket is on the order of \$500M to \$1B. Beyond that, industrial deployment requires either excluding the SJB, or a re-weighting favouring the most liquid instruments — at moderate cost on the SR (on the order of −0.05 to reach \$2B, −0.11 for \$5B).

A.31 Asset class contribution

Q: Which markets produce performance and which are negative?

Exhibit 129. Marginal contribution to the annualised Sharpe of the diversified portfolio, by asset class (to be distinguished from the *standalone* Sharpe of each class taken in isolation, reported in Exhibit 89 for the STIR class: −0.33 *standalone* vs −0.24 marginal on the diversified strategy).

Asset class	N instruments	Ann. Sharpe contribution
Bonds	12	+0.38
Metals	5	+0.29
Equity	11	+0.21
Agriculturals	13	+0.21
FX	9	+0.16
Volatility	1	+0.14
Carbon	1	+0.11
Oil & Gas	6	+0.08
Crypto	1	-0.12
STIR	3	-0.24

The Bond market dominates,¹¹⁰ followed by all commodities (Metals, Agriculturals, Oil & Gas combined at +0.58 SR cumulative),¹¹¹ STIRs strongly dragging post-2020 performance down.

The negative contribution of STIRs (−0.24 SR) is more than an anecdote: it is directly pertinent to the central question of the study. STIRs (Euribor, SO3, SR3) trade massively on the smoothing of monetary policy expectations, a behaviour structurally *mean-reverting* on quarterly to annual horizons, precisely the horizons of our trend signals. In other words, on this asset class, the proposition "*trends are dead*" is literally true, and that independently of 2010. The STIR drag is therefore not an accident but information: it delimits the domain of validity of the strategy, which works on asset classes where directional momentum exists (equities, long bonds, metals, *agri*) and loses on those where mean reversion dominates (STIR and, to a lesser extent, crypto at −0.12 SR on a single very volatile instrument).

A.32 Structural derivation of parameters

Q: Is the parametrisation arbitrary or the result of sample-fitting?

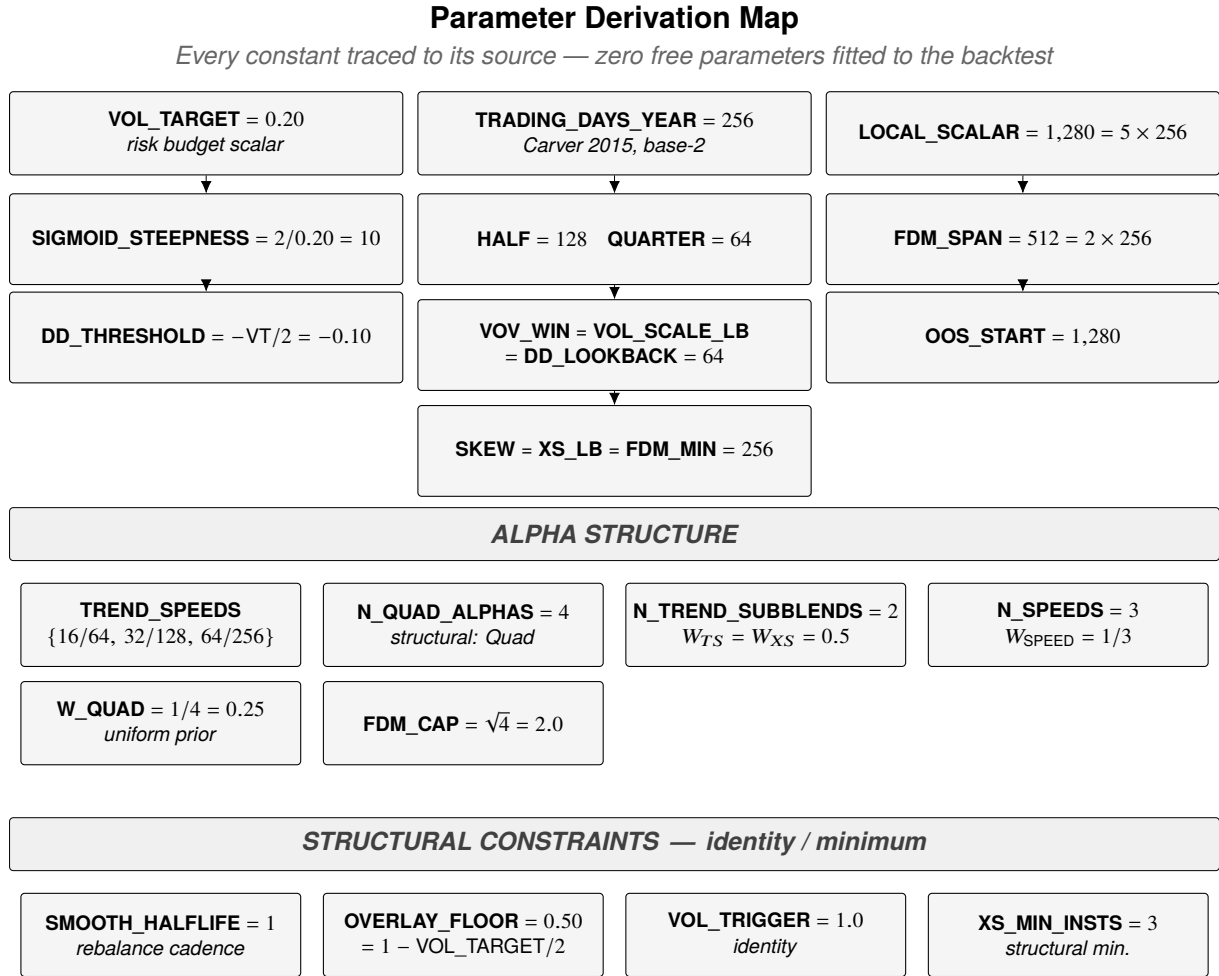


Exhibit 129bis. Parameter Derivation Map.

All parameters are either derived, or directly copied from the literature.

A.33 Synthetic paths

This section reports an alternative validation approach to T5 (*block-bootstrap*, A.3) that we explored but that we did *not* retain as primary evidence.

A.33.1 Tests on the full strategy

Q: Does generating synthetic paths allow verifying the results of the strategy?

Methodology. A conditional *Wasserstein GAN* network (CWGAN, Arjovsky et al. 2017; Gulrajani et al. 2017) is trained on the daily returns of 11 core instruments, then extended to the 51 other instruments by a Gaussian copula (Sklar 1959). The model generates 5,000 alternative market paths over 2-year windows (504 days). The strategy is then *backtested* on each path with its real transaction costs and a T-bill income on the NAV on both sides.

Two *backtest* protocols coexist in the literature: 1) frozen positions: historical exposures rescaled on synthetic returns; 2) full re-simulation: the signal engine re-run on each path to produce new

positions. We report both.

However, the GAN only synthesises daily returns: it does not reproduce the term structure of futures contracts (the slope of the *carry/contango* curves). Yet the *carry* component of the strategy (25 % of signal weighting) depends precisely on this structure. Part of the strategy is therefore structurally not identifiable by construction on this type of path.

Exhibit 130. Full strategy on the CWGAN synthetic paths (5,000 paths per horizon, median SR over 504 days, net of commissions, with *cash* return on NAV). The first four rows use the *frozen-positions* protocol; the last two rerun the strategy’s signal engine on each path. Per-alpha contribution = median of full strategy – median of the ablation (real reference SR: +0.97). The *Drop VoV* ablation is not reported separately; by additive consistency with the other three rows, the implicit VoV contribution is $\sim +0.04$ (residual not significant at the table’s *bootstrap*).

Variant	504d median (SR)	Per-alpha contribution
Full strategy (4 alphas)	+0,258	—
Drop carry (3 alphas)	+0,381	−0,12 (carry)
Drop skew (3 alphas)	+0,028	+0,23 (skew)
Drop trend (3 alphas)	+0,148	+0,11 (trend)
<i>Signal engine re-run on each path:</i>		
Skew alone	+0,622	—
Full strategy, 3 walk-forward windows	+0,13 / +0,00 / +0,02	—

The synthetic median of the full strategy is +0.26, far from the real SR +0.97. The decomposition by ablation identifies the cause: among the four alphas, only the *skew* carries the performance on synthetic markets (measured contribution of +0.23 SR); the *carry* shows a slightly negative contribution (−0.12 SR), so much so that removing it even improves the result. This is exactly the expected consequence: the generator does not reproduce the term structure on which *carry* depends, which becomes inoperative on the synthetic paths. The *walk-forward* re-simulation of the full strategy on three successive windows gives medians +0.13, +0.00 and +0.02, with no clear trend.

A.33.2 Tests on the pure trend component

Q: Does generating synthetic paths allow verifying the results of a pure-trend version of the strategy?

The pure trend component (EWMAC alone, without *carry*, *skew*, *VoV* or *cross-sectional* momentum) only depends on daily price paths, which the generator is supposed to reproduce correctly. The test is therefore, in principle, in the zone of competence of the generator. Same 5,000 CWGAN paths, same costs. To allow triangulation, we also report: the results of a second generator (DDPM, Ho et al. 2020, diffusion model, paradigm distinct from GAN), trained on the same panel; five alternative definitions of the trend signal (EWMAC at different windows,

Donchian, TSMOM) on the CWGAN paths, to verify that the result does not depend on the precise signal used by the strategy; the stationary *block-bootstrap* (BBB) of A.3.1, as a benchmark free of any generative assumption.

Exhibit 131. *Pure-TS* component on synthetic paths — comparison across five protocols (horizon 504 days, net of commissions, with *cash* return on NAV ; SRs computed in excess of IRX). The BBB row is reproduced from Exhibit 89bis as a benchmark independent of any generative model. The medians of the real rolling SR are indicated above the median line for context.

Protocol	504d median (SR)
Real rolling pre-2010 (reference)	+1,154
Real rolling post-2010 (reference)	+0,287
BBB post-2010 (generator-free, cf. A.3.1)	+0,320
CWGAN, frozen-positions protocol	+0,457
DDPM (single seed shown), frozen positions	+0,582
CWGAN, signal re-run on each path	−0,470
5 alternative trend signals on CWGAN (range)	[−0,59, −0,35]

The estimates spread between -0.47 and $+0.58$ depending on protocol, a gap of more than one point of Sharpe ratio. Under frozen-positions protocol, the synthetic paths (CWGAN $+0.46$, DDPM $+0.58$) deliver a systematic over-estimate of the BBB in excess of IRX ($+0.32$): the generators reproduce a dynamic close enough to the real statistic to preserve the sign, but overestimate the depth of the trend component by about 0.1 to 0.3 Sharpe points. Under full re-simulation, the CWGAN swings to -0.47 and the five alternative signals confirm this level in the range $[-0.59, -0.35]$.

First, nearly half of the drop in the re-simulated CWGAN is attributable to an autoregressive drift specific to this architecture: the mean SR over the first half of a path (-0.09 on days 1–252) drops to -0.49 on the second half (days 253–504). The same measure on the DDPM architecture is statistically null. Next, the DDPM itself is strongly unstable from one random initialisation (*seed*) to another: five successive *seeds* give medians $+0.58, -0.50, -0.06, +0.05, -0.45$ (range of 1.09 SR). The $+0.58$ reported above corresponds to the most favourable *seed*. Moreover, two diametrically opposed interpretations of the -0.47 result are compatible with the generator’s diagnostics: the real post-2010 was particularly unfavourable within a distribution of plausible markets whose median is $+0.50$ (BBB); or, the generator does not correctly reproduce the persistence that the trend component exploits, and -0.47 would be close to the true structural post-2010 SR. The GAN’s validation diagnostics do not allow deciding between the two.

The synthetic-path test fails to provide an exploitable answer. On the full strategy, the generator cannot represent the *carry* component; the observed gap is attributable to this structural limitation rather than to a weakness of the strategy. On the pure trend component, the estimates spread over more than one point of SR depending on the protocol and architecture, and at least one of these

sources of dispersion (DDPM *seed* instability) is documented as methodologically disqualifying in the recent literature on generative tests (Bowman et al. 2016 on training pathologies; subsequent *seed-stability* literature).

For these reasons, we report these tests for information and rely, as the primary evidence of the statistical robustness of the strategy, on the Politis-Romano (1994) stationary *block-bootstrap* restricted to the post-2010 sub-period (subsection A.3.1): methodology already used for T5, free of any generative assumption, and giving for the full strategy a post-2010 median in excess of IRX of +0.999 ([+0.969, +1.021]) — statistically indistinguishable from the full-sample historical SR of 0.97.

ANNEX 2: PAPER TRADING

B.1 Out-of-sample performance (3 January 2026 – latest available data)

The last test we can perform is a paper-trading run, meaning we run the strategy live with fake money. While it could have been possible to actually do it with a broker, it is way easier to run it ourself by downloading from NorgateData the new data available from 2026-01-03 to the latest available date before the paper is due. As of writing, the 2026-05-26, the purely of our sample data amounts to 145 days. Realistically, the sample is too small to make any significant conclusion, but this experiment serves as a proof of concept and good faith. We present the performance with a couple exhibits that we quickly summarize: the strategy performs as expected with a realized volatility compressed due to our structurally conservative doctrine of "better safe than sorry". Compared to historical performance and benchmarks, we underperform with a medium margin of both fronts, the agricultural sector heavily dragging the P&L with a strong reversion of trends. Still, the SR of 0.44 and the raw return of annualized return of 7.33% are acceptable, whatever that may mean on such a short period. An updated version of this paper-trading will be held in september for the paper defence, allowing us to better see if the model actually works out of sample, although we would probably need at least 2-3 years of out of sample data to be able to produce any significant conclusion.

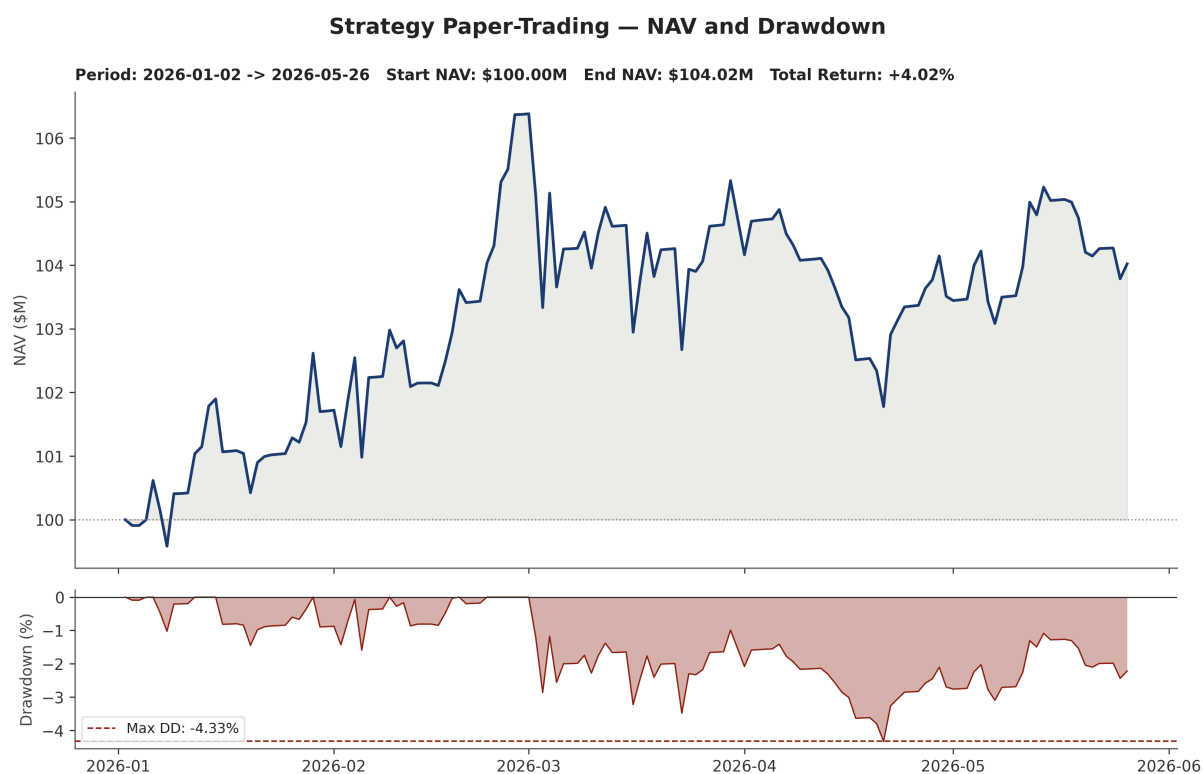


Exhibit 132. NAV and drawdown of the strategy in *paper-trading* from \$100 M on 2 January 2026: end NAV \$104.02 M (+4.02%), max drawdown -4.33% .

Exhibit 133. Summary statistics of the *paper-trading* performance over the 145-day sample.

Metric	Value
Window	2 January 2026 – 26 May 2026
Trading days	145
Sharpe ratio (excess of IRX)	+0.44
Annualised return (arithmetic)	+7.33%
Annualised volatility	8.56%
Max drawdown	-4.33%
Trades	3,499

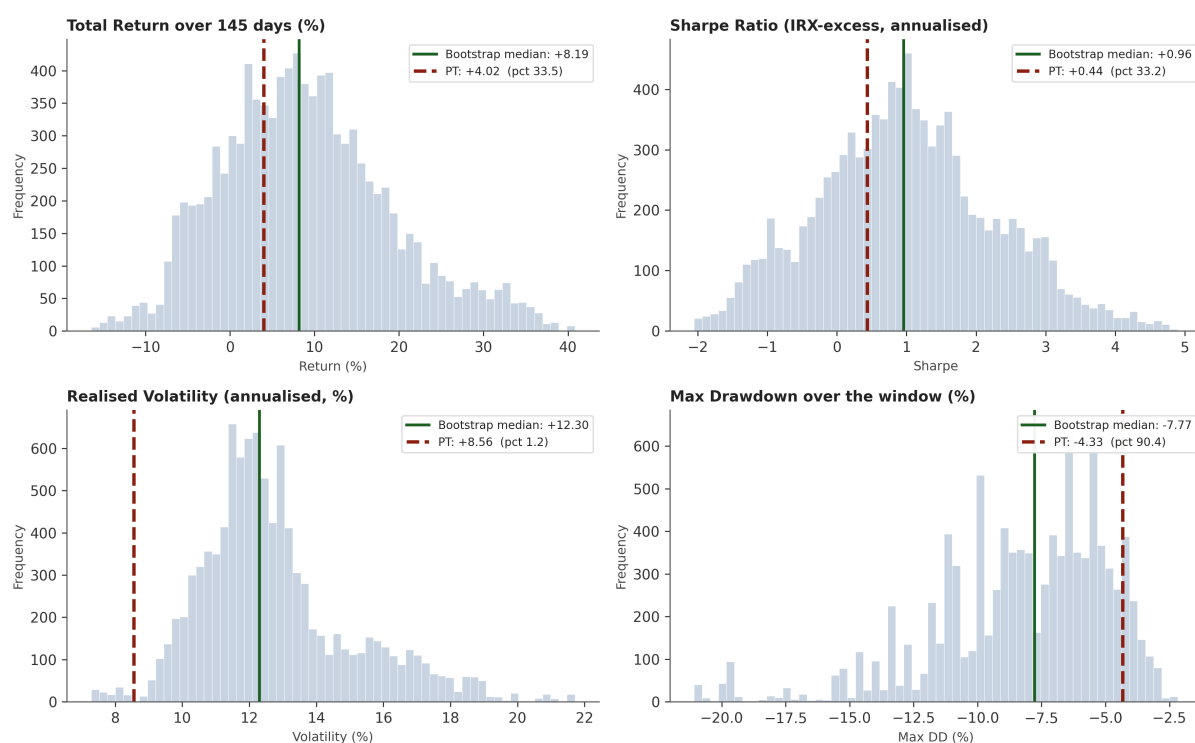
Paper-Trading 145-Day Window vs Canonical Strategy Bootstrap (N=10,000 overlapping 145-day windows, 1990-2026)

Exhibit 134. 145-day window placed against 10,000 overlapping 145-day windows bootstrapped from the canonical 1990–2026 history: total return and Sharpe at the 33rd percentile, realised vol at the 1st (very low), max drawdown at the 90th (very mild).

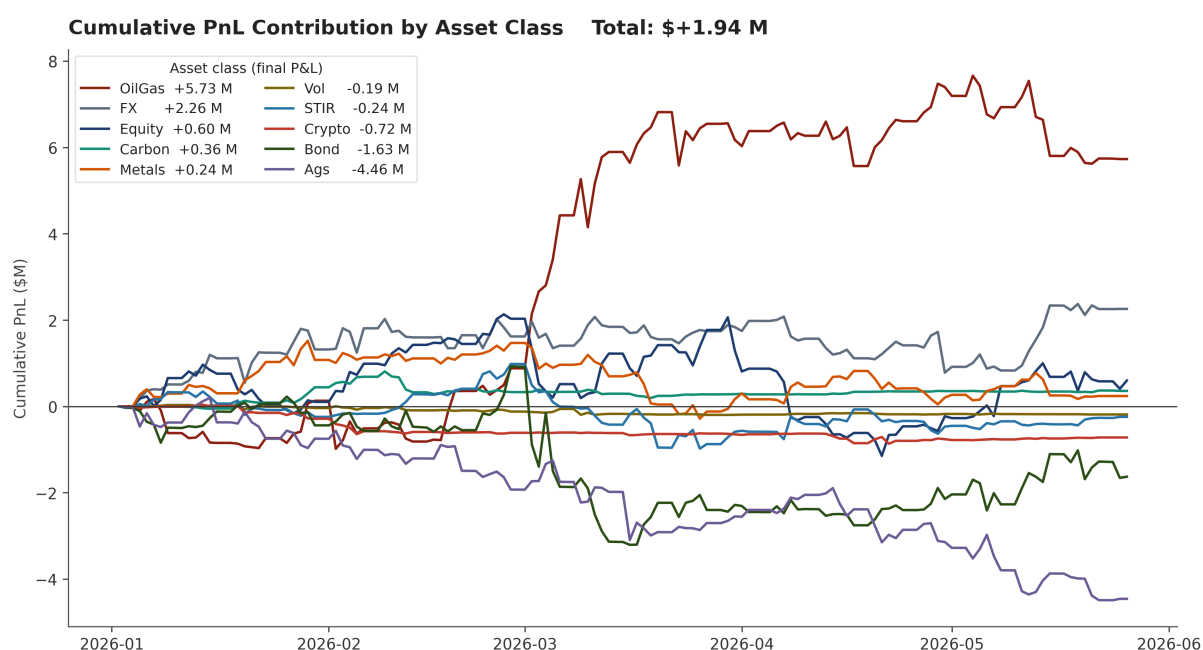


Exhibit 135. Cumulative dollar PnL contribution by asset class: OilGas and FX lead the positive side, Ags the negative (−\$4.46 M from a synchronous trend reversal across grains, softs and livestock).

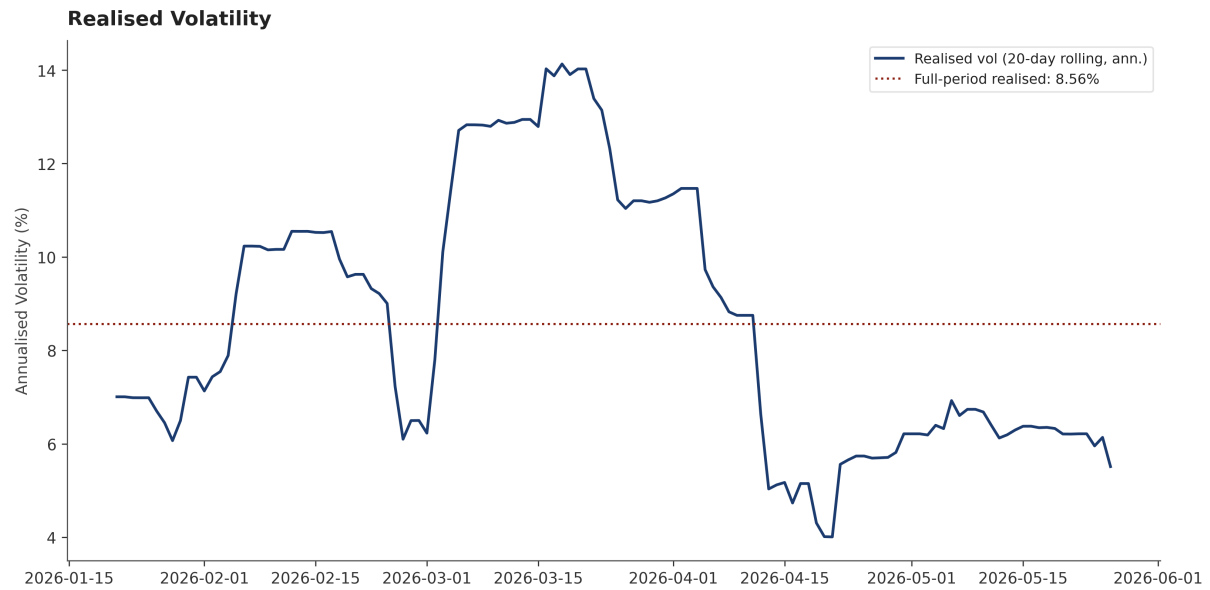


Exhibit 136. 20-day rolling realised annualised volatility; oscillates around the 8.56% full-period value, well below the 20% nominal target by structural design of the §VIII vol/DD sigmoid overlay.

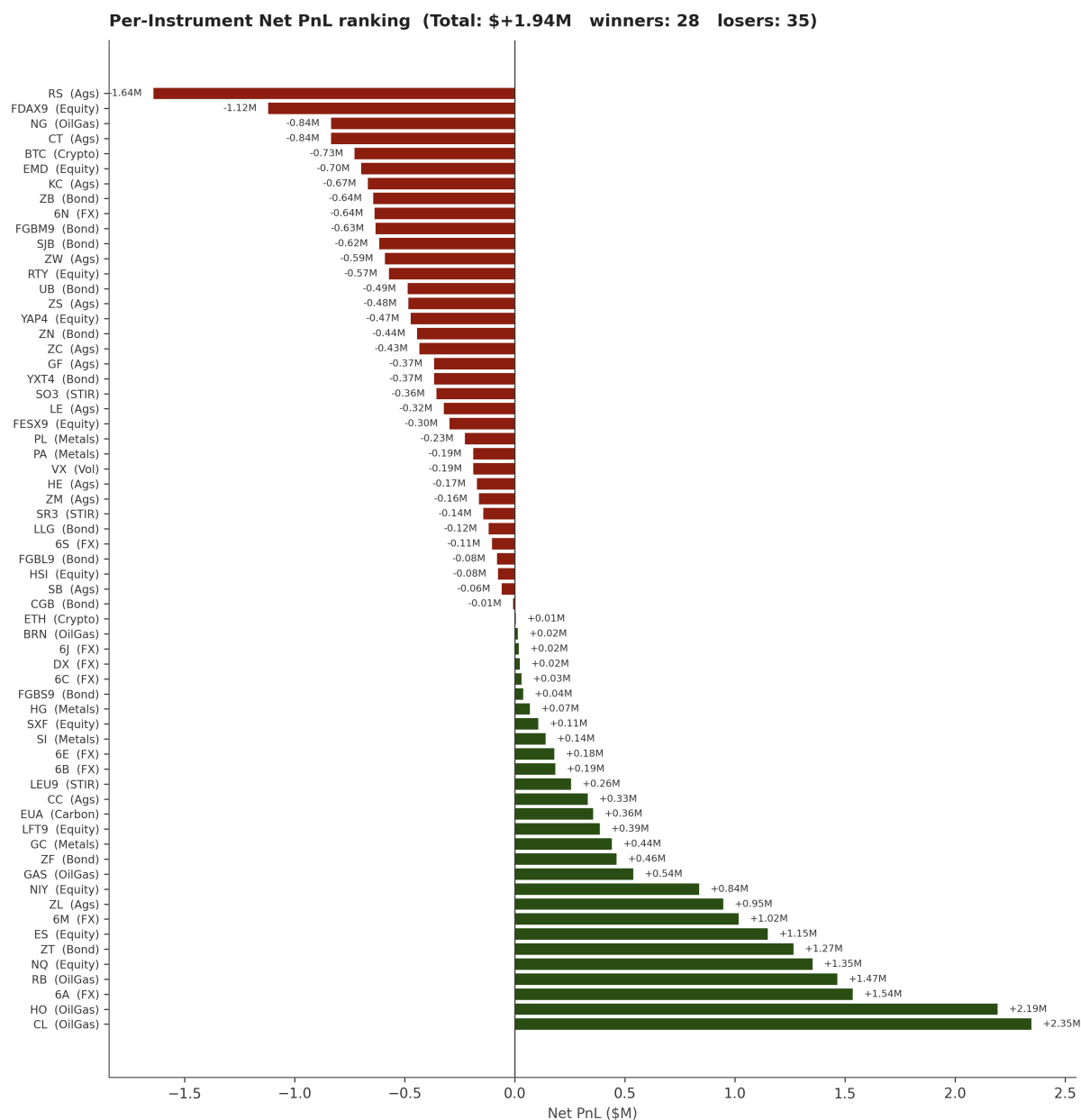


Exhibit 137. Per-instrument net PnL, ranked: top contributors CL, ZL, NQ, ES; largest detractors RS, CT, KC, ZW (Ags whipsaw).

Monthly Compounded Returns (%)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD
2026	+1.71	+4.58	-1.53	-1.18	+0.49								+4.02

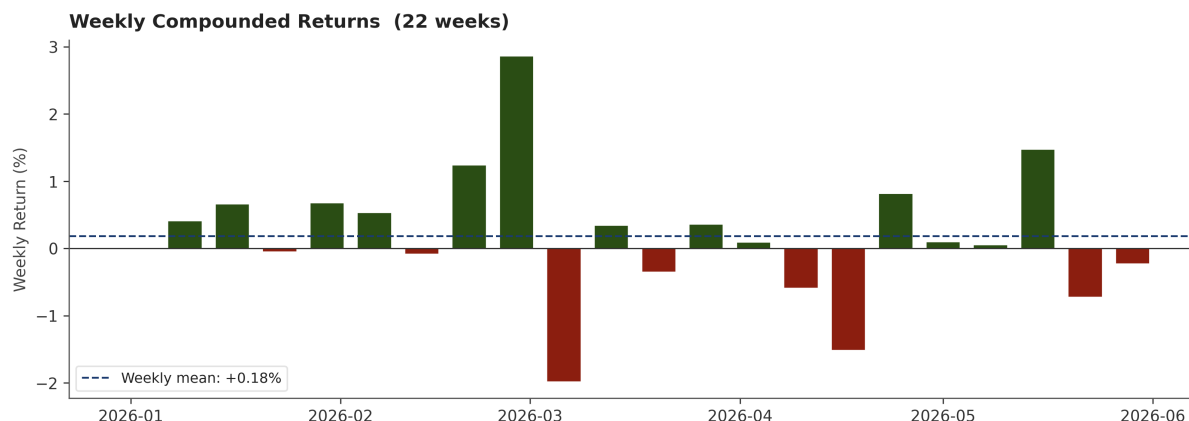


Exhibit 138. Monthly compounded returns (top grid) and weekly compounded returns (bottom bar chart) over the paper-trading window.

Paper-Trading vs Canonical Daily Returns (PT N=145, Canonical N=9,341)

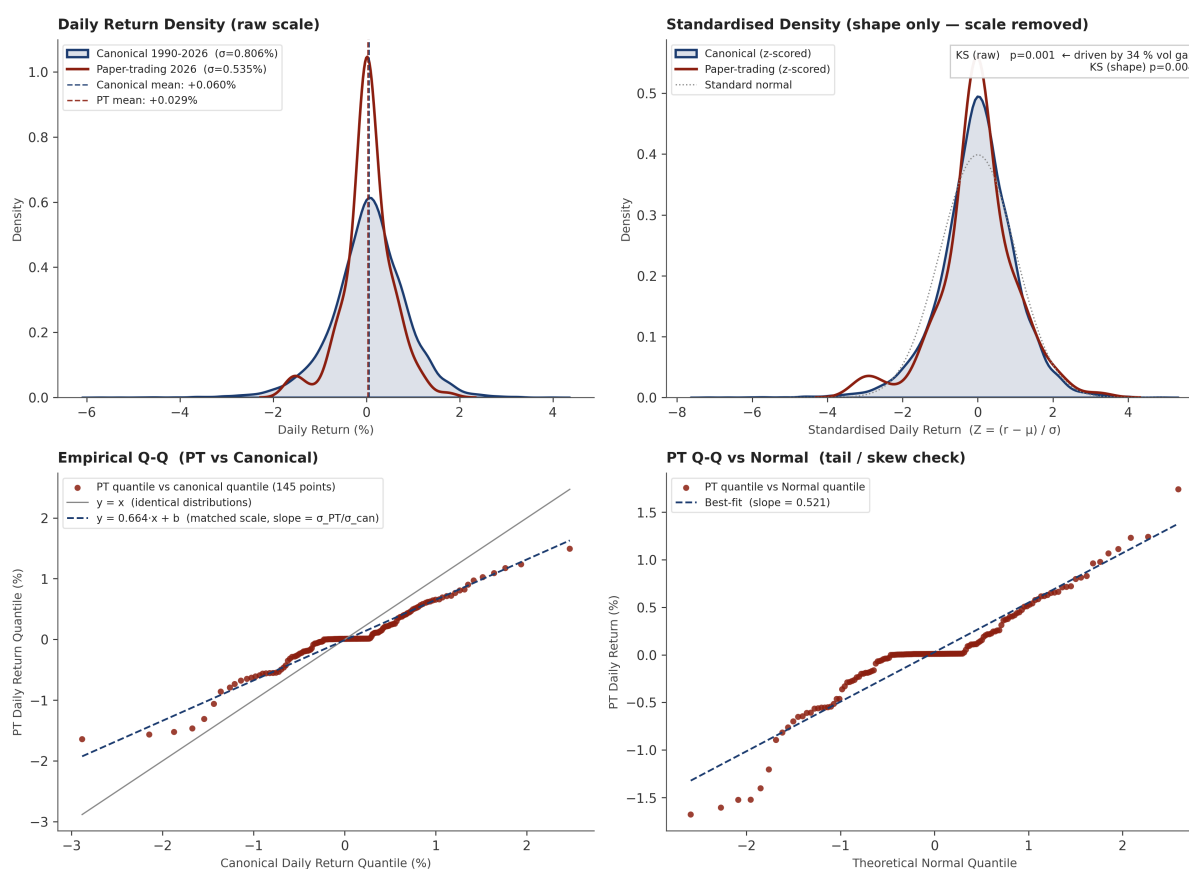


Exhibit 139. Daily return density and Q-Q plots versus the 1990–2026 canonical backtest: skew and kurtosis match almost exactly; the only meaningful difference is a 34% lower scale, consistent with the realised-vol shortfall.

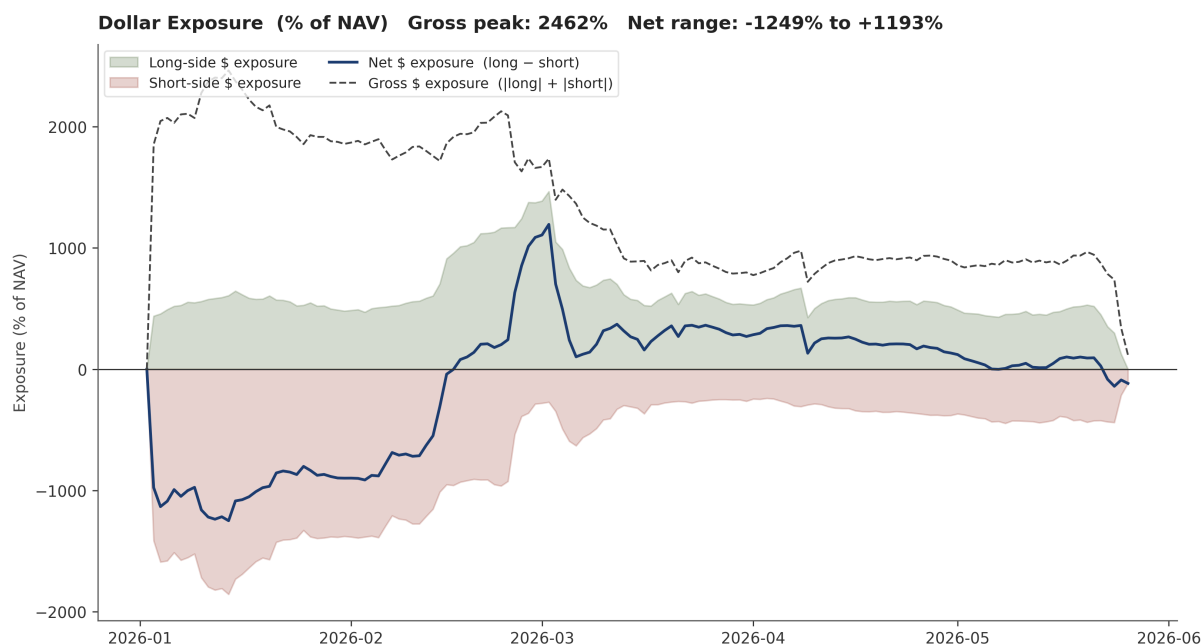


Exhibit 140. Gross ($\sum |\$pos|$) and net ($\sum \pos) dollar exposure as a percentage of NAV; the gross peaks near 25 $\times$ leverage (rates-dominated) while the net oscillates around zero, confirming a genuinely long-short architecture.

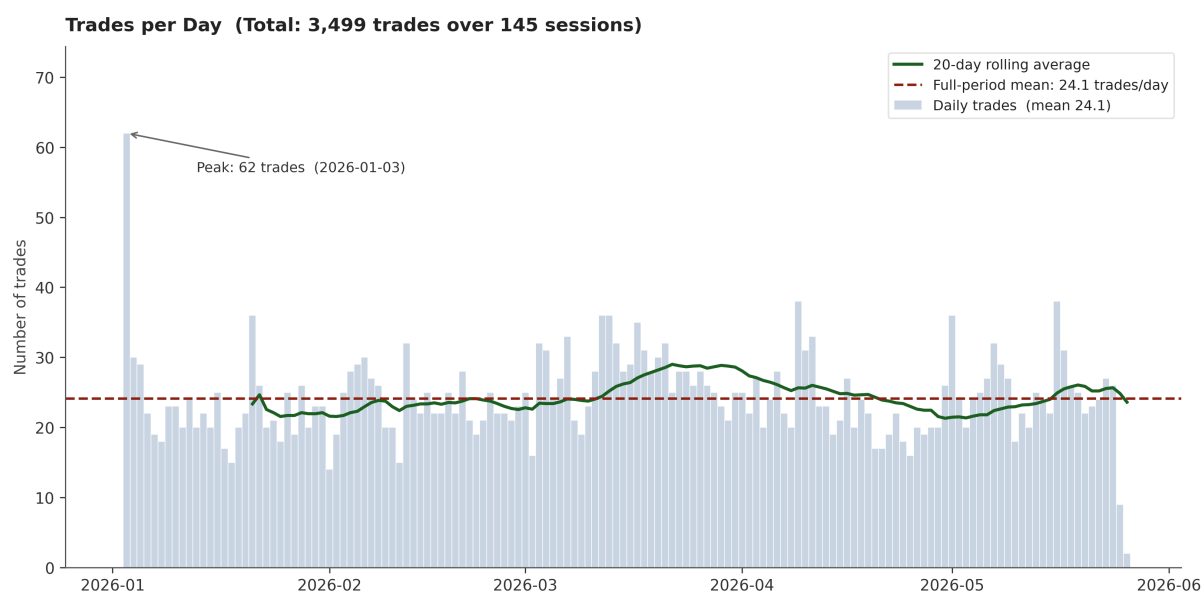


Exhibit 141. Daily trade count; mean 24 trades/day, peak of 62 on the initialisation day (3 January 2026) when positions were built from zero across the universe.

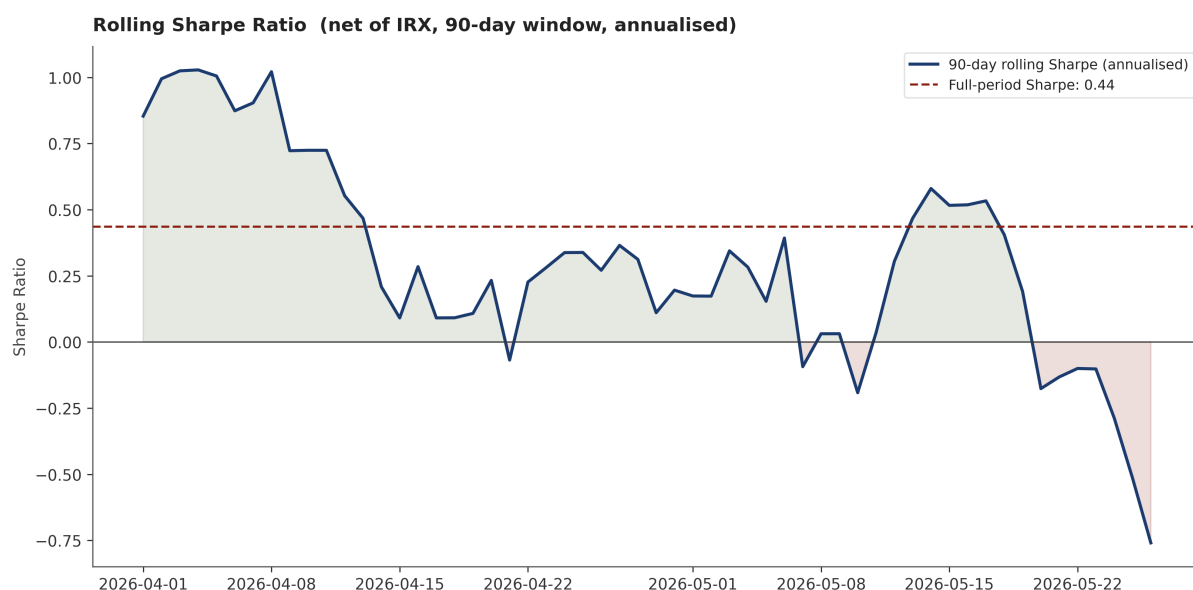


Exhibit 142. 90-day rolling Sharpe ratio (IRX-excess, annualised). On a 145-day sample the rolling estimate has $SE \approx \pm 1.8$, so the chart is descriptive only.

Strategy vs Benchmarks — Paper-Trading 2026-01-02 to 2026-05-26

Strategy / Benchmark	N	Total Return	Ann. Return	Ann. Vol	Sharpe (vs IRX)	Max DD
Strategy (Paper-Trading)	145	+4.02%	+7.33%	8.56%	+0.44	-4.33%
S&P 500 (Total Return)	98	+10.13%	+26.17%	13.90%	+1.62	-8.89%
MSCI World ETF	98	+9.38%	+24.49%	14.69%	+1.42	-9.06%
60/40 Balanced (US)	98	+6.14%	+16.01%	9.45%	+1.31	-5.84%
CTA Peer Proxy (KMLM)	98	+10.69%	+27.65%	14.90%	+1.61	-4.24%

Exhibit 143. Comparison versus S&P 500 TR, MSCI World, 60/40 Balanced and a CTA peer (KMLM) over the paper-trading window; the strategy underperformed all four in total return and Sharpe during this strongly risk-on period, but delivered the smallest max drawdown of the five.

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Notes on academics: there's a lot more studies on the subject that we are aware of but lacked the time to read. We acknowledge the ones we are aware of for future research and hope that what we've done here has not been already done in the same exact manner by them: Asness-Liew-Stevens (1997), Griffin-Ji-Martin (2005), Bhojraj-Swaminathan (2006), Erb-Harvey (2006), Ribeiro-Normand-Loeys (2006), Gorton-Hayashi- Rouwenhorst (2007), Miffre-Rallis (2007), Pukthuanthong-Levich-Thomas (2007), Blitz-van Vliet (2008), Schneeweis-Kazemi-Spurgin (2008), Fuertes-Miffre-Rallis (2010), Shen-Szakmary-Sharma (2010). We've also encountered Jacobus Arie (Arco) van Oord (2016) thesis on the subject which was a fascinating study and highly relevant to our subject, but his experiment only focused on equities.

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(The CFA I and II curriculums are also heavily used, but having only access to the digital versions which makes proper referencing difficult, we refrain from citing them. As an example, when we refer to “classical” equations or formulations, they usually come from those curriculums.)

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Notes on practitioners: We’ve surprisingly only met a couple of charlatans in our readings, whom we obviously didn’t include nor continue reading. George Kleinman’s book is excellent up to the 200~, we didn’t continue reading after this point due to the emphasis on technical analysis, which we frown upon. We’re sure other good practitioners must exist who have published excellent books, but time is of the essence and we are of the mind that we’ve read the best. “(Rob’s) latest book is not just the best coverage of the topic of leveraged trading of small portfolios I’ve read, it’s actually the only good one I’ve come across. . . This is a great book.”

Financial history

Mallaby, S. (2010). *More money than god: Hedge funds and the making of a new elite*. The Penguin Press.

Poundstone, W. (2005). *Fortune's formula: The untold story of the scientific betting system that beat the casinos and Wall Street*. Hill and Wang.

Taleb, N. N. (2004). *Fooled by randomness: The hidden role of chance in life and in the markets* (2nd ed.). Random House.

Notes on financial historians: by a landslide the most fascinating of our research, albeit the less practical. We invite further research on the subject to include more of this kind of knowledge, which ultimately shapes the perception of the markets.

Philosophy

Pascal, B. (1670). *Pensées* [Thoughts].

Notes on philosophy: Nassim Nicholas Taleb could have easily been put in this category due to his heavy emphasis on philosophy. Orders might be placed by algorithms, but the latter are still programmed by men. Better understanding their mind is an essential part of financial knowledge. Expanding this subject could easily double the length of the study and the references, but we fear this would not have been to our jury's taste.

NOTES

This chapter lists the ideas, formulas and results from the body of the thesis whose origin is found in the bibliography. Each note includes a short description of the annotated idea and the corresponding bibliographical reference. The entries are numbered in the reading order of the thesis, in order to facilitate cross-references in discussion.

1. **Efficient frontier.** Harry Markowitz, *Portfolio Selection*, 1952, Journal of Finance.
2. **Random walk of prices.** Louis Bachelier, *Théorie de la Spéculation*, 1900, Gauthier-Villars.
3. **Panama method for rolls of continuous contracts.** The complete methodology used for the adjustment of time series and corrected returns is that of Robert Carver, *Advanced Futures Trading Strategies*, 2023, Harriman House Ltd (initial chapters on series construction). The first edition of Andreas F. Clenow, *Following the Trend: Diversified Managed Futures Trading*, 2013, John Wiley & Sons, also describes the principle and historically inspires this approach.
4. **List of studies "most similar to our works".** Antti Ilmanen, *Expected Returns*, 2011, Wiley ; Kent Daniel & Tobias J. Moskowitz, *Momentum Crashes*, 2016, JFE ; Pedro Barroso & Pedro Santa-Clara, *Momentum has its moments*, 2015, JFE ; T. Moskowitz, Y. H. Ooi & L. H. Pedersen, *Time-series momentum*, 2012, JFE ; C. S. Asness, T. J. Moskowitz & L. H. Pedersen, *Value and momentum everywhere*, 2013, JF ; A. Levine & L. H. Pedersen, *Which trend is your friend ?*, 2016, FAJ ; Brian Hurst, Yao Hua Ooi & L. H. Pedersen, *A Century of Evidence on Trend-Following Investing*, 2017, JPM. More precisely, we borrow from each a specific element. *Ilmanen (2011)* provides the conceptual framework *alpha / alternative beta / beta* that we take up in chapter 2 to classify the role of CTAs. *Moskowitz, Ooi & Pedersen (2012)*, founders of *time-series momentum*, provide the EWMAC signal and the empirical justification of the positive serial autocorrelation on 12 months ; we replicate it in Exhibit 35 and reuse it as lower bound of our *trend* component in chapter 3. *Asness, Moskowitz & Pedersen (2013)*, founders of *value + momentum everywhere*, provide the cross-sectional Z-score and the systematic documentation of cross-asset trends; the Z-score is reused directly in our XS component in chapter 3. *Daniel & Moskowitz (2016)* provide the documentation of post-2009 *momentum crashes* and the dynamic WML *overlay*, which motivate our sigmoid *overlay* on the latent drawdown. *Barroso & Santa-Clara (2015)* provide the *constant-volatility scaling* that analytically justifies our volatility target. *Levine & Pedersen (2016)* provide the multi-speed EWMAC combination from which we take the two additional speeds (16, 64) and (32, 128) on top of the Carver speed. *Hurst, Ooi & Pedersen (2017)* provide the centennial evidence that confirms our post-2010 observation: the 2010s decade is the weakest of their sample of more than 100 years for *trend-following*.
5. **Panama method and adjusted return.** Robert Carver, *Advanced Futures Trading Strategies*, 2023,

Harriman House Ltd.

6. **Annualization conventions.** Andrew W. Lo, *The statistics of Sharpe ratios*, 2002, Financial Analysts Journal ; Robert Carver, *Advanced Futures Trading Strategies*, 2023.
7. **Reference Sharpe definition (primary formula).** William Sharpe, *The Sharpe ratio*, 1994, Journal of Portfolio Management.
8. **Lower tail / upper tail methodology.** Robert Carver, *Advanced Futures Trading Strategies*, 2023.
9. **Hypothesis of serial autocorrelation of returns and 12-month rolling sign test.** T. Moskowitz, Y. H. Ooi & L. H. Pedersen, *Time-series momentum*, 2012, JFE.
10. **Importance of moving-average subtractions as trend measure.** Valeriy Zakamulin, *Which Trend Is Your Friend ?*, 2015, University of Agder.
11. **Markov processes and Markov regimes applied to prices.** Chris Brooks, *Introductory Econometrics for Finance*, 2019, 4th ed., Cambridge University Press, p. 588. On the same theme: J. D. Hamilton, *A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle*, 1989, *Econometrica* 57(2) ; A. Ang & G. Bekaert, *International Asset Allocation With Regime Shifts*, 2002, *Review of Financial Studies* 15(4) ; J. Bulla, S. Mergner, I. Bulla, A. Sesboué & C. Chesneau, *Markov-switching asset allocation: Do profitable strategies exist?*, 2011, *Journal of Asset Management* 12(5) ; P. Nystrup, H. Madsen & E. Lindström, *Stylised facts of financial time series and hidden Markov models in continuous time*, 2015, *Quantitative Finance* 15(9). This topic was one of our first research avenues, which we abandoned as ineffective, we summarize briefly: on our sample, a Gaussian HMM with 2/3/4 states applied in *walk-forward* with 36 specifications combining states, *features* and re-estimation cadence, and confronted with three distinct definitions of the "trending" character: the posterior probability of the trending regime aligns on average with the forward trend SR at $\rho \approx +0.14$ (95 % of positive cells), but does not beat the *variance ratio* indicator of Lo-MacKinlay (1988) in 46 % of cases. The application of the published recipes of Bulla (2011) to the complete panel only replicates in 2/62 instruments (3.2 %; mean deviation -0.07 SR against *buy-and-hold*), the two "successes" being instruments in structurally bearish markets where the transition to *cash* mechanically avoids the loss. The detection of regimes by HMM does not produce any significant result on our sample.
12. **Two-regime regression (bullish / bearish) for trends.** Kent Daniel & Tobias J. Moskowitz, *Momentum Crashes*, 2016, *Journal of Financial Economics*.
13. **Distinction data-mining / sample-fitting / overfitting.** D. Bailey & M. Lopez de Prado, *The deflated Sharpe ratio*, 2014, *Journal of Portfolio Management*.
14. **Efficient markets.** E. F. Fama, *Efficient Capital Markets: A Review of Theory and Empirical Work*, 1970, Vol. 25(2), *Journal of Finance*.
15. **Risk of data-snooping / sample-fitting .** D. Bailey, J. M. Borwein, M. Lopez de Prado & Q. J. Zhu, *The probability of backtest overfitting*, 2016, *Journal of Computational Finance* ; framework taken from C. Harvey & Y. Liu, *Backtesting*, 2015, JPM.
16. **Futility of t-tests on raw returns.** Antti Ilmanen, *Expected Returns*, 2011, Wiley ; Robert Carver, *Advanced Futures Trading Strategies*, 2023.

17. **Importance of grounding parameters out of sample.** Kent Daniel & T. J. Moskowitz, *Momentum Crashes*, 2016, JFE.
18. **Commodities and inflation.** G. Gorton & K. G. Rouwenhorst, *Facts and fantasies about commodity futures*, 2006, FAJ ; C. B. Erb & C. R. Harvey, *The strategic and tactical value of commodity futures*, 2006, FAJ.
19. **Macroeconomic framework.** Olivier Blanchard, *Macroeconomics*, 2017, 7th ed., Pearson ; Charles I. Jones, *Macroeconomics*, 2017, 4th ed., W. W. Norton. On inflation, policy rates and sovereign debt (Chapter 1, §2).
20. **Presence of trends in all asset classes unexplained by economic fundamentals.** C. S. Asness, T. J. Moskowitz & L. H. Pedersen, *Value and momentum everywhere*, 2013, Journal of Finance.
21. **Asymmetry of trends and role of behavior in stress periods.** Kent Daniel & T. J. Moskowitz, *Momentum Crashes*, 2016, JFE.
22. **Unifying model newswatchers / momentum traders (slow diffusion of information).** explicit: H. Hong & J. C. Stein, *A Unified Theory of Underreaction, Momentum Trading, and Overreaction in Asset Markets*, 1999, Journal of Finance.
23. **Black-Scholes model (implicit reference to 1973).** Standard exposition taken up in John C. Hull, *Options, Futures, and Other Derivatives*, 2022, 11th ed., Pearson.
24. **Similarities trend strategy / call selling and buying, convexity in crisis.** Kent Daniel & T. J. Moskowitz, *Momentum Crashes*, 2016, JFE.
25. **Convexity of returns (gamma-like) of trend-following strategies.** C. S. Asness, T. J. Moskowitz & L. H. Pedersen, *Value and momentum everywhere*, 2013, JF ; taken up and extended by Robert Carver, *Advanced Futures Trading Strategies*, 2023.
26. **Globally vega-neutral character of long-scale trend-following strategies.** Brian Hurst, Yao Hua Ooi & L. H. Pedersen, *A Century of Evidence on Trend-Following Investing*, 2017, JPM.
27. **Notions of contango / backwardation and carry in futures.** John C. Hull, *Options, Futures, and Other Derivatives*, 2022 (futures chapter) ; applied development: Robert Carver, *Advanced Futures Trading Strategies*, 2023.
28. **Taylor series development** $dV \approx \Delta dS + \frac{1}{2} \Gamma (dS)^2 + \Theta dt$. John C. Hull, *Options, Futures, and Other Derivatives*, 2022, 11th ed., Pearson, chapter on the greeks.
29. **List of fund collapses (LTCM, Amaranth, Bear Stearns, Lehman...).** Sebastian Mallaby, *More Money Than God: Hedge Funds and the Making of a New Elite*, 2010, The Penguin Press.
30. **Direct quote:** Pascal, *Pensées*, 1670, Port-Royal ed. (bibliographical entry: Pascal, *Pensées*, 1670), translated from French.
31. **List of CTA funds followed (Millburn, Dunn, Chesapeake, Campbell, Eckhardt, Estlander, TICOM, Transtrend, Mulvaney, Alpha Quest) and chart of performances (Exhibit 24 reproduced).** Andreas F. Clenow, *Following the Trend: Diversified Managed Futures Trading*, 2023, 2nd ed., John Wiley & Sons Ltd, p. 46, fig. 3.1. We thank Mr. Clenow again.
32. **Historical studies of CTAs and strategy replications.** Abrams, Bhaduri & Flores, *Lintner revisited*:

A quantitative analysis of managed futures..., 2010, Chicago CME Group.

33. **Modern portfolio theory and formulas of expected return / variance / simplified Sharpe (Exhibits 25–27).** Harry Markowitz, *Portfolio Selection*, 1952, JF ; William Sharpe, *The Sharpe ratio*, 1994, JPM.
34. **MSCI World + trend efficient frontier (Exhibit 28 reproduced).** Andreas F. Clenow, *Following the Trend*, 2013, 1st ed., John Wiley & Sons Ltd, p. 81, fig. 5.2 (explicit reproduction with permission).
35. **Rise of correlations in crisis and "flight-to-quality".** Kent Daniel & T. J. Moskowitz, *Momentum Crashes*, 2016, JFE ; *flight-to-quality* and margin-call explanations: Antti Ilmanen, *Expected Returns*, 2011, Wiley.
36. **Extreme events more frequent than intuition imagines.** Nassim Nicholas Taleb, *Fooled by Randomness: The Hidden Role of Chance in Life and in the Markets*, 2004, 2nd ed., Random House.
37. **Autocorrelation bias in declared returns of hedge funds.** Andrew W. Lo, *The statistics of Sharpe ratios*, 2002, FAJ.
38. **Hedge fund benchmarks based on factors (implicit for the argument of biases).** W. Fung & D. Hsieh, *The risk in hedge fund strategies*, 2001, RFS ; *Hedge fund benchmarks: a risk-based approach*, 2004, FAJ.
39. **Undeniable high-return stories in the hedge fund industry.** Sebastian Mallaby, *More Money Than God*, 2010 ; William Poundstone, *Fortune's Formula*, 2005, Hill and Wang.
40. **"Alpha, alternative beta or beta" framework.** Antti Ilmanen, *Expected Returns: An Investor's Guide to Harvesting Market Rewards*, 2011, Wiley.
41. **On Russian roulette (Fooled by Randomness).** Nassim Nicholas Taleb, *Fooled by Randomness*, 2004, 2nd ed., Random House, p141.
42. **Quote "...enough margin to take the heat when it heats up".** George Kleinman, *Trading Commodities and Financial Futures: A Step-by-Step Guide to Mastering the Market*, 2013, 4th ed., FT Press, p. 105.
43. **Affinity with Shannon's information theory.** William Poundstone, *Fortune's Formula: The Untold Story of the Scientific Betting System That Beat the Casinos and Wall Street*, 2005, Hill and Wang.
44. **Replication of long-short momentum strategies (Exhibit 30–35).** Sources of the six replicated strategies: C. S. Asness, T. J. Moskowitz & L. H. Pedersen, *Value and momentum everywhere*, 2013, JF (AMP 2013) ; Kent Daniel & T. J. Moskowitz, *Momentum Crashes*, 2016, JFE (DM 2016, Dynamic WML) ; Brian Hurst, Y. H. Ooi & L. H. Pedersen, *A Century of Evidence...*, 2017, JPM (Hurst 2017) ; T. Moskowitz, Y. H. Ooi & L. H. Pedersen, *Time-series momentum*, 2012, JFE (Moskowitz 2012, TSMOM) ; P. Barroso & P. Santa-Clara, *Momentum has its moments*, 2015, JFE (Barroso 2015) ; A. Levine & L. H. Pedersen, *Which trend is your friend ?*, 2016, FAJ (LP 2016, MACROSS).
45. **Return normalization formula at target volatility** $r_{\text{scaled}} = r \cdot \sigma_{\text{target}} / \sigma_{\text{realized}}$. Classical convention taken from Robert Carver, *Systematic Trading*, 2015, Harriman House Ltd.
46. **Direct quote:** Andreas F. Clenow, *Following the Trend*, 2013, 1st ed., John Wiley & Sons Ltd, p. 43.
47. **Direct quote:** George Kleinman, *Trading Commodities and Financial Futures*, 2013, 4th ed., FT

Press, p. 168.

48. **Quant crash of August 2007 and overcrowding.** Sebastian Mallaby, *More Money Than God*, 2010, The Penguin Press.
49. **Disappointing CTA returns post-2010 / returns over more than 100 years.** Andreas F. Clenow, *Following the Trend*, 2023, 2nd ed., John Wiley & Sons Ltd ; Brian Hurst, Y. H. Ooi & L. H. Pedersen, *A Century of Evidence on Trend-Following Investing*, 2017, JPM.
50. **Solution by diversification of alpha sources.** Andreas F. Clenow, *Following the Trend*, 2023, 2nd ed. ; and Robert Carver, *Advanced Futures Trading Strategies*, 2023.
51. **Direct quote:** Chris Brooks, *Introductory Econometrics for Finance*, 2019, 4th ed., Cambridge University Press, p. 502.
52. **On short-term autocorrelation / long-term mean reversion.** Antti Ilmanen, *Expected Returns*, 2011, Wiley, p. 476.
53. **Ljung-Box Q test for return autocorrelation.** G. M. Ljung & G. E. P. Box, *On a measure of lack of fit in time series models*, 1978, Biometrika.
54. **Newey-West estimator with truncation parameter $L = \text{INT}[4(T/100)^{2/9}]$, Bartlett weights, HAC regression.** Whitney K. Newey & Kenneth D. West, *A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix*, 1987, Econometrica ; *Automatic Lag Selection in Covariance Matrix Estimation*, 1994, Review of Economic Studies ; truncation formula taken from Chris Brooks, *Introductory Econometrics for Finance*, 2019, Cambridge University Press.
55. **Information Coefficient / Information Ratio framework, $IR \approx IC\sqrt{BR}$.** R. Grinold & R. Kahn, *Active Portfolio Management*, 2000.
56. **Intrinsic impossibility of having signals statistically significant in isolation.** Robert Carver, *Advanced Futures Trading Strategies*, 2023.
57. **Trend signal — EWMAC 16/64, 32/128, 64/256 and equal aggregation.** The 64/256 and the EWMAC philosophy come directly from Robert Carver, *Systematic Trading*, 2015, Harriman House Ltd, as well as *Advanced Futures Trading Strategies*, 2023. The faster spans (16/64 and 32/128) are taken from A. Levine & L. H. Pedersen, *Which trend is your friend ?*, 2016, FAJ.
58. **Cross-sectional component (Z-score of 256-day returns) in f_{trend} .** T. Moskowitz, Y. H. Ooi & L. H. Pedersen, *Time-series momentum*, 2012, JFE.
59. **Carry signal — F1/F2 differential annualized by maturity.** R. Koijen, T. Moskowitz, L. Pedersen & E. Vrugt, *Carry*, 2018, JFE ; Robert Carver, *Advanced Futures Trading Strategies*, 2023.
60. **Skewness signal — insurance selling (inverted skew on 256 days).** A. Fernandez-Perez, B. Frijns, A. Fuertes & J. Miffre, *The skewness of commodity futures returns*, 2018, Journal of Banking & Finance.
61. **Volatility-of-Volatility (VoV) signal — regime of higher-order uncertainty.** G. Baltussen, S. van Bakkum & B. van der Grient, *Unknown unknowns: vol-of-vol and the cross section of stock returns*, 2018, JFQA ; related framework: P. Carr & L. Wu, *Variance risk premiums*, 2009, RFS.
62. **Direction of VoV: "buy the winners, sell the losers" at the origin of cross-sectional momentum.**

- The sign applied to the VoV signal ($\text{sign}(\Delta P_{64})$) takes, in spirit, the seminal idea that recent winners continue to outperform and losers to underperform – a central mechanism of TSMOM and, more generally, of any trend-exploitation strategy. N. Jegadeesh & S. Titman, *Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency*, 1993, Vol. 48(1), Journal of Finance.
63. **On alpha diversification.** Synthesis of the works Koijsen *et al.* (2018), Fernandez-Perez *et al.* (2018), Baltussen *et al.* (2018) and Moskowitz *et al.* (2012) already cited.
 64. **1/N equal-weighted signal weighting.** V. DeMiguel, L. Garlappi & R. Uppal, *Optimal versus naive diversification: how inefficient is the 1/N portfolio strategy?*, 2009, RFS.
 65. **Normalization to target $\Phi = 10$ and cap at ± 20 (forecast-scaling and forecast-cap).** Robert Carver, *Systematic Trading*, 2015, Harriman House ; taken up in *Advanced Futures Trading Strategies*, 2023.
 66. **Forecast Diversification Multiplier (FDM) and theoretical upper bound under zero correlation.** Robert Carver, *Systematic Trading*, 2015, and *Advanced Futures Trading Strategies*, 2023.
 67. **Volatility targeting (sigmoid around a target $\tau = 20\%$).** Risk-management framework by volatility target formulated in Robert Carver, *Systematic Trading*, 2015, and further developed in A. Moreira & T. Muir, *Volatility-managed portfolios*, 2017, JF.
 68. **Sigmoid on 64-day drawdown at threshold -10% .** Own derivation but we base ourselves on the logic of *volatility scaling* of Robert Carver (2015/2023).
 69. **Management of drawdowns and sigmoid attenuation: understanding of the risk structure of momentum strategies.** The choice of a 50% reduction ceiling on both sigmoids (volatility and drawdown) reflects the will to avoid the asymmetric *momentum crashes* identified in the literature. B. D. Grundy & J. S. Martin, *Understanding the Nature of the Risks and the Source of the Rewards to Momentum Investing*, 2001, Vol. 14(1), Review of Financial Studies.
 70. **Instrument Diversification Multiplier (IDM) bounded $[1, 4]$.** Robert Carver, *Advanced Futures Trading Strategies*, 2023.
 71. **Mixed volatility estimator $0.70 \text{ EWM}_{32} + 0.30 \text{ rolling } 2560$.** Robert Carver, *Advanced Futures Trading Strategies*, 2023 (exact formula).
 72. **EWM halflife = 1 day smoothing and 10% buffer on position changes.** Robert Carver, *Systematic Trading*, 2015 (buffer) and *Leveraged Trading*, 2019, Harriman House.
 73. **"three types of lies: backtests"** This adaptation is an adage, the original quote is different.
 74. **Block 1 — EWMAC 64/256 (Carver's iteration 6).** Robert Carver, *Advanced Futures Trading Strategies*, 2023, strategy six (EWMAC slow).
 75. **Block 2 — 3-Speed EWMAC + XS.** Fast and medium spans from A. Levine & L. H. Pedersen, *Which trend is your friend ?*, 2016, FAJ ; XS component from T. Moskowitz *et al.*, *Time-series momentum*, 2012, JFE.
 76. **Block 3 — addition of Carry.** R. Koijsen *et al.*, *Carry*, 2018, JFE.
 77. **Block 4 — addition of Skew.** Fernandez-Perez, Frijns, Fuertes & Miffre, *The skewness of commodity futures returns*, 2018, JBF.
 78. **Block 5 — addition of VoV.** Baltussen, van Bakkum & van der Grient, *Unknown unknowns:*

- vol-of-vol...*, 2018, JFQA ; Carr & Wu, *Variance risk premiums*, 2009, RFS.
79. **Uncertainty vs risk framework.** F. H. Knight, *Risk, Uncertainty and Profit*, 1921, Hart, Schaffner & Marx.
 80. **Block 6 — FDM pooled + halflife 1.** Robert Carver, *Advanced Futures Trading Strategies*, 2023 (re-interpretation of FDM as pool on the universe).
 81. **Block 7 — volatility and drawdown sigmoids.** Own derivation but modeled on forecast-to-position Carver architecture (2015).
 82. **"not comfortable with black-box trading".** Michael W. Covell, *Trend Following: How to Make a Fortune in Bull, Bear and Black Swan Markets*, 2017, 5th ed., Wiley, p. 137.
 83. **Argumentation on the 3 properties of CTAs (simplicity, crisis alpha, zero correlation).** Synthesis of Andreas F. Clenow, *Following the Trend* (2013 and 2023), John Wiley & Sons.
 84. **Liquidity impact and instrument exclusion.** Y. Amihud, *Illiquidity and stock returns: cross-section and time-series effects*, 2002, Journal of Financial Markets.
 85. **"pure trends have weakened post-2010; multi-alpha restores performance".** Brian Hurst *et al.* (2017, JPM) and Andreas F. Clenow (2023) for the weakening ; and on Kojien *et al.* (2018), Fernandez-Perez *et al.* (2018), Baltussen *et al.* (2018) for the multi-alpha sources.
 86. **T1 — Politis-White optimal block length.** D. Politis & J. Romano, *The stationary bootstrap*, 1994, JASA ; automatic version in D. Politis, J. Romano & M. Wolf, *Subsampling*, 1999, Springer.
 87. **T5 — Block-bootstrap 95 % CI of Sharpe.** B. Efron & R. J. Tibshirani, *An Introduction to the Bootstrap*, 1993, Chapman & Hall ; robust Sharpe CI: O. Ledoit & M. Wolf, *Robust performance hypothesis testing with the Sharpe ratio*, 2008, J. Empirical Finance.
 88. **T6 — Block-bootstrap of MDD.** Same Efron & Tibshirani (1993) framework, extended to the drawdown statistic.
 89. **T7 — Distributional sanity (normality, skew, kurtosis, vol clustering).** Jarque & Bera, *A test for normality...*, 1987, International Statistical Review ; vol clustering test: R. F. Engle, *Autoregressive conditional heteroscedasticity...*, 1982, Econometrica ; cf. also Ljung & Box (1978).
 90. **T8 — SE corrections of Lo and Mertens.** Andrew W. Lo, *The statistics of Sharpe ratios*, 2002, FAJ ; E. Mertens, *Comments on variance of the IID estimator in Lo (2002)*, 2002, working paper, University of Basel.
 91. **T9 — Minimum Track Record Length.** D. Bailey & M. Lopez de Prado, *The Sharpe ratio efficient frontier*, 2012, Journal of Risk.
 92. **T10 — Bonferroni, Holm, BHY haircuts.** C. Harvey, Y. Liu & H. Zhu, *...and the cross-section of expected returns*, 2016, RFS.
 93. **T11 — PBO via CSCV.** D. Bailey, J. M. Borwein, M. Lopez de Prado & Q. J. Zhu, *The probability of backtest overfitting*, 2016, Journal of Computational Finance.
 94. **Deflated Sharpe Ratio.** D. Bailey & M. Lopez de Prado, *The deflated Sharpe ratio*, 2014, JPM.
 95. **Superior Predictive Ability.** P. Hansen, *A test for superior predictive ability*, 2005, JBES.

96. **MCS at 10 %.** P. Hansen, A. Lunde & J. Nason, *The Model Confidence Set*, 2011, Econometrica.
97. **Jobson-Korkie-Memmel paired z-test.** J. Jobson & B. Korkie, *Performance hypothesis testing with the Sharpe and Treynor measures*, 1981, JF ; correction: C. Memmel, *Performance hypothesis testing with the Sharpe ratio*, 2003, Finance Letters.
98. **Random walk tests and autocorrelation.** A. Lo & A. C. MacKinlay, *Stock market prices do not follow random walks*, 1988, RFS ; A. Lo, *Long-term memory in stock market prices*, 1991, Econometrica.
99. **Temporal permutation and Gaussian white noise.** B. Efron & R. J. Tibshirani, *An Introduction to the Bootstrap*, 1993 ; D. Politis, J. Romano & M. Wolf, *Subsampling*, 1999, Springer.
100. **Henriksson-Merton (binary variable).** R. D. Henriksson & R. C. Merton, *On market timing and investment performance. II*, 1981, Journal of Business.
101. **Treynor-Mazuy (quadratic variable).** J. L. Treynor & K. K. Mazuy, *Can mutual funds outguess the market ?*, 1966, Harvard Business Review.
102. **Hedge fund benchmarks framework.** W. Fung & D. Hsieh, *The risk in hedge fund strategies*, 2001, RFS ; *Hedge fund benchmarks: a risk-based approach*, 2004, FAJ.
103. **Fama-French 5 factors.** E. F. Fama & K. R. French, *A five-factor asset pricing model*, 2015, JFE.
104. **Fung-Hsieh 7 factors.** W. Fung & D. Hsieh, *The risk in hedge fund strategies*, 2001, RFS.
105. **HAC NW standard errors.** W. Newey & K. West, 1987, Econometrica.
106. **Covariance-matrix estimation framework** (common context to NW and Ledoit-Wolf regressions): O. Ledoit & M. Wolf, *A well-conditioned estimator for large-dimensional covariance matrices*, 2004, J. Multivariate Analysis.
107. **Parameters "either derived, or copied from the literature".** Explicit synthesis of the end-of-annexe header: Robert Carver, *Systematic Trading* (2015) and *Advanced Futures Trading Strategies* (2023) ; A. Levine & L. H. Pedersen, *Which trend is your friend ?*, 2016, FAJ ; T. Moskowitz & L. H. Pedersen (sic: the annexe text cites "2013"; we refer to T. Moskowitz, Y. H. Ooi & L. H. Pedersen, *Time-series momentum*, 2012, JFE) ; R. Koijen *et al.*, *Carry*, 2018, JFE ; Fernandez-Perez, Frijns, Fuertes & Miffre, *The skewness of commodity futures returns*, 2018, JBF ; Baltussen, van Bakkum & van der Grint, *Unknown unknowns: vol-of-vol...*, 2018, JFQA.
108. **Conservative cost / slippage framework.** Robert Carver, *Leveraged Trading*, 2019, Harriman House ; and *Advanced Futures Trading Strategies*, 2023.
109. **Liquidity weighting, Amihud framework.** Y. Amihud, *Illiquidity and stock returns*, 2002, Journal of Financial Markets.
110. **Bonds / Equities / STIR framework.** Rate factors: R. Litterman & J. Scheinkman, *Common factors affecting bond returns*, 1991, Journal of Fixed Income.
111. **Commodities.** G. Gorton & K. G. Rouwenhorst, *Facts and fantasies about commodity futures*, 2006, FAJ ; C. B. Erb & C. R. Harvey, *The strategic and tactical value of commodity futures*, 2006, FAJ ; G. Bakshi, X. Gao & A. Rossi, *Understanding the sources of risk underlying the cross section of commodity returns*, 2019, Management Science.

112. **Data sources for the analysis of component rotation (Exhibit 73).** The regressions of the Sharpe share on the macroeconomic state variables and the *walk-forward* predictability test use the following series:

- Effective Fed funds rate (DFF), FRED: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=DFF>.
- Consumer price index, urban set (CPIAUCSL), FRED: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=CPIAUCSL>.
- Moody's BAA corporate bond yield, FRED: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=BAA>.
- Constant-maturity 10-year Treasury yield (DGS10), FRED: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=DGS10>.
- ES (S&P 500 E-mini), NorgateData panama-adjusted series: `Data/PanamaMethod/ES_continuous.csv` (additive panama applied according to the method described in note 3).